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Germany

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2001-2002

Germany

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Table of contents

Assessment and recommendations 9

I. Macroeconomic developments 23

Why did Germany's economic growth slow in the 1990s? 23
Recent economic developments and prospects 33

II. Fiscal policy 57

General government finances in 2001 58
The 2002 budget 59
The 2003 budget 62
Medium-term objectives 64

III. Issues in public sector spending reform 67

Federal fiscal relations and main forces shaping public spending 68
Managing public sector spending 73
Selected spending programmes 82
Summary of recommendations 98

IV. Implementing structural reform: a review of progress 103

Labour market reforms to tackle high structural unemployment 103
Competition enhances potential growth 119
Financial markets 139
Sustainable development in Germany 152

Notes 172

Bibliography 189

Annexes

I. Chronology of main economic events 197
II. Financial market developments in Germany 200

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Boxes

1. The economic performance of the new states 48
2. Basic legal principles for budget formation, execution and control 70
3. The Mainz Model for employment promotion in the low-wage sector 86
4. Basic features of the German health care system 95
5. Recommendations for enhancing public sector spending reform 99
6. Recent legislation and commission proposals for labour market reform 104
7. Features of the Public Employment Service 118
8. The German system of chambers of craft trades and product market competition 120
9. Securing the construction of competing UMTS networks 128
10. Should the opening of domestic markets be conditioned on the opening of foreign markets? 136
11. The Act on the Acquisition of Securities and Take-overs 142
12. A review of progress on structural reform 145
13. The integration of environmental concerns into government policy 153

Annex

A1. Venture capital support programmes 204

Tables

1. Sources of growth in real GDP per capita 24
2. Private consumption and earnings 30
3. Demand and output 35
4. Trade by region 38
5. Current account of the balance of payments 39
6. The German labour market 46
7. Public sector financial balances 58
8. The Federal Budget 60
9. Appropriation account for general government Including social security 62
10. Medium-term fiscal objectives for the general government 65
11. Functional spending by layer of government, 1998 69
12. Federal transfers to the *Länder* 2002 75
13. Opening clauses for wages and hours worked in selected sectors 107
14. Orientation of national supervisors 144
15. Performance indicators: air pollution 156
16. Selected international commitments related to air quality 158
17. Performance indicators: waste 162
18. Waste treatment 162
19. Key indicators for sustainable retirement income 167

Annex

A1. Selected bank statistics, 1999 201
A2. Stock market capitalisation and corporate debt 203

Figures

1. Real GDP growth in Germany and the EU 25
2. Employment by sector in Germany and the EU 27
3. Employment in the 1990s 28
4. Labour compensation and taxation 29
5. Real GDP growth rates between 1991 and 2001 30
6. Germany's international competitiveness 32
7. Macroeconomic performance 34

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8. A comparison of economic cycles 36
9. A comparison of recoveries in the 1990s 40
10. The business climate 42
11. Energy prices and real house prices in European countries 43
12. Decomposition of change in total employment 44
13. Beveridge curve 45
14. The registered unemployment rate 47
15. The inflation environment 49
16. Interest-rate developments 51
17. Yield curves 52
18. Credit growth to private sector 53
19. Real interest rates 54
20. General governement balances 63
21. Trends in general government spending 71
22. General government spending by type, 2001 72
23. The Solidarity Pact II with the new states 79
24. The labour market status of social assistance recipients, end of year 2000 83
25. Highest educational attainment of social assistance recipients, end of year 2000 83
26. Effects on net income of the tax and transfer system 85
27. Costs and duration of tertiary studies, 1999 91
28. Health care spending and contributions 94
29. Part-time employment in international comparison 110
30. Employment protection within the OECD 111
31. Telecommunications prices in EU countries 125
32. Electricity prices in the EU 131
33. Gas prices in the EU 132
34. Emission trends for air polluants 155
35. Relative diesel and petrol taxes 160
36. Trends in packaging consumption and recycling 164
37. Age of withdrawal and labour force participation 168
38. Net replacement rates and contribution rates 169
39. Incentives for early retirement: replacement rate and change in pension wealth 170

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BASIC STATISTICS OF GERMANY, 2001

THE LAND

Area (thousand sq. km)	357.0	Major cities, 31.12.2000 (thousand inhabitants)	
Agricultural area (thousand sq. km)	195.4	Berlin	3 382
Forests (thousand sq. km)	104.3	Hamburg	1 715
		Munich	1 210
		Cologne	963
		Frankfurt	646
		Essen	595
		Dortmund	589
		Stuttgart	584
		Düsseldorf	569
		Bremen	539

THE PEOPLE

Population (thousands)	82 340	Labour force (thousands)	40 232
Number of inhabitants per sq. km	231	Employment (thousands)	38 918
Net natural increase in population (thousands), 2000	–72	*of which*:	
Net migration (thousands), 2000	167	Agriculture, forestry, fishing and hunting	960
		Industry (including construction)	11 122
		Private services	26 837

THE PRODUCTION

GDP (billion of euro)	2 071	Origin of GDP (per cent)	
GDP per capita (US$)	22 527	Agriculture, forestry, fishing and hunting	1.2
Gross fixed investment		Industry (including construction)	29.1
Per cent of GDP	20.1	Services	69.7
Per capita (US$)	4 528		

THE GOVERNMENT

Public consumption (per cent of GDP)	19.0	Composition of Parliament:	Seats
General government current revenue (per cent of GDP)	43.0	Social Democratic Party (SPD)	251
Public debt (ratio to general government current revenue)	140.1	Christian Democratic Party (CDU)	190
		Christian Social Union (CSU)	58
		Greens	55
		Free Democratic Party (FDP)	47
		PSD	2
		Last general election: September 2002	
		Next general election: 2006	

FOREIGN TRADE

Exports of goods and services (per cent of GDP)	33.8	Imports of goods and services (per cent of GDP)	33.4
Main exports (per cent of total merchandise exports)		Main imports (per cent of total merchandise imports)	
Food live animal, beverages and tobacco	4.0	Food live animal, beverages and tobacco	6.3
Crude materials, inedible, oils and fat	1.6	Crude materials, inedible, oils and fat	3.3
Mineral fuels, lubrificants and related materials	1.4	Mineral fuels, lubrificants and related materials	8.5
Chemicals and manufactured goods	41.1	Chemicals and manufactured goods	44.4
of which: manufactured goods	15.1	*of which*: manufactured goods	14.0
Machinery and equipment	52.0	Machinery and equipment	37.5
Total	100.0	Total	100.0

THE CURRENCY

Irrevocable conversion rate	1.95583	Currency units of euro per US$, average of daily figures:	
		Year 2001	1.1166
		November 2002	0.9984

Note: An international comparison of certain basic statistics is given in an Annex Table.

This Survey is published on the responsibility of the Economic and Development Review Committee of the OECD, which is charged with the examination of the economic situation of member countries.

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The economic situation and policies of Germany were reviewed by the Committee on 4 November 2002. The draft report was then revised in the light of the discussions and given final approval as the agreed report by the whole Committee on 28 November 2002.

•

The Secretariat's draft report was prepared for the Committee by Eckhard Wurzel, Jens Høj, Christoph Menzel and Richard Herd under the supervision of Andreas Wörgötter.

•

The previous Survey of Germany was issued in May 2001.

Assessment and recommendations

An economy in need of fiscal consolidation and comprehensive labour market reform to strengthen growth

While in the beginning of the 1990s economic growth in Germany was stronger than in most other OECD countries on account of the unification boom, growth has faded since then. This reflects in part imbalances in labour markets and in the construction sector that themselves arose from unification, as well as a variety of adverse external shocks. But the difficult adjustment process and the prolonged weakness of domestic demand also point to the need to overcome structural rigidities that impair the resilience of the economy and in particular to strengthen its capacity to create employment. Insufficient incentives to take up work and hire labour together with a high tax wedge driving up labour costs, in combination with adverse macroeconomic factors, contributed to weak job creation which bears on consumer demand. This combination of factors has weakened Germany's growth potential and resilience to outside shocks. Burdened with a legacy of high social expenditures and a shrinking tax potential, the government is now confronted with high structural deficits. There is thus a pressing need for structural measures to strengthen the budget over the medium term. At the same time, with growth too low and overly dependent on external demand fundamental reforms in labour markets and continued reform in product markets are necessary and urgent to strengthen growth and reduce unemployment. The German government is committed to addressing these problems; it is putting forward a fiscal consolidation programme and has adopted the findings of a labour market reform commission with many proposals going in the right direction. Nevertheless, more will be needed.

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Slow growth in 2001-02 has been due to weak domestic demand and the global slowdown

The German economy has barely grown in 2001-02 and employment has fallen. Domestic demand has been particularly weak and fell in 2001 for the first time since the post-reunification recession in 1993. Despite earlier tax cuts, consumer confidence declined steeply and private consumption weakened as labour market prospects deteriorated and disposable incomes were depressed by a number of temporary inflationary shocks. At the same time, fixed investment in machinery and equipment collapsed – reflecting a deteriorating international environment, falling capacity utilisation and stagnating profits – and the downsizing in construction continued. Although growth for 2002 as a whole is likely to be even weaker than in 2001, some recovery is occurring and growth may pick up to 1½ per cent in 2003 and 2½ per cent in 2004, so long as the global situation improves and real incomes grow again on the back of low inflation. The output gap will nevertheless not close over the projection horizon. There are significant risks in these projections, most importantly as regards the development of global demand and hence German exports. Domestically growth may turn out lower, if deficit reduction is implemented mainly via revenue increases. Forceful reform actions on the other hand could invigorate confidence and strengthen growth.

Monetary conditions

Monetary policy was eased during 2001 as policy rates were lowered by an accumulated 150 basis points by the European Central Bank. Subsequently, there was a significant downward shift of the yield curve, so that by autumn 2002 market rates across all maturities were down to their lowest level since 1999. On the other hand, the euro has appreciated somewhat. OECD projections build in a further 50 basis points cut in short-term interest rates. In sum, monetary conditions should remain supportive for GDP growth. However, the recent sharp fall in German inflation to among the lowest in the euro area means that real short-term interest rates risk dampening the recovery of demand in Germany to an extent, which may not be compensated by the corresponding gain in competitiveness.

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The general government balance deteriorated substantially and consolidation will require expenditure cuts

Fiscal policy was expansionary in 2001, as new business and income tax reductions were phased in. Since the series of tax reductions over the last couple of years was not financed by corresponding reductions in government spending, the structural deficit increased by nearly a percentage point and weak economic growth lifted the general government deficit to 2.8 per cent of GDP. Accordingly, the deficit overshot the federal government's consolidation target by a wide margin. With economic activity still weak the general government deficit is likely to exceed 3½ per cent in 2002 largely on account of unforeseen tax shortfalls. Structural balances will improve in 2003, aided by cuts in government employment and increases in indirect taxes. The government is designing a consolidation package, whose volume, if fully realised, would contribute to further reducing the structural deficit. While the government has reinforced its commitment to continue fiscal consolidation and balance the general government budget, this objective is unlikely to be met in the short run. One-off measures to eliminate the deficit over a short horizon would risk being self-defeating by further weakening domestic demand. There is a need, rather, for lasting consolidation measures on the spending side so as to redress durably present budget imbalances and to create room for further tax cuts. Over the medium term structural surpluses necessary to pre-fund ageing related spending increases should be aimed at.

Programme selection should rely more on outcome-oriented evaluation of government spending in general

For fiscal consolidation to be successful, streamlining social spending and improving resource allocation within the public sector are unavoidable. At present social welfare spending accounts for more than 50 per cent of general government outlays, while compensation for public sector employees represents 17 per cent of spending and gross investment and subsidies account for 4 and 3 per cent, respectively. Against this background, a more rigorous approach to evaluating public sector spending projects is called for in areas like health care, education, active labour market measures and infrastructure investment. Across the board cuts may be simpler to implement, but risk to reduce the growth potential of the economy. The introduction of a cost accounting and controlling framework for public sector services at various levels of government should be reinforced. To enhance spending effectiveness,

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outcome-oriented evaluation methods should be applied, and this entails estimating the full costs and benefits of the action taken over a long-term horizon.

The allocation of funds across states needs to be made more effective and the domestic stability pact should be sharpened…

While the German fiscal federal system – largely based on inter-state co-operation and revenue sharing – has been successful in promoting a uniformly high standard of living, it generates incentives to overspend. As well, the process itself is in part administratively heavy while shared responsibilities are not supportive of proper project evaluation. The volume of co-funding should therefore be scaled back, and linked more closely to proper cost-benefit criteria. Progress has been made involving all layers of government in a domestic stability pact with agreed spending caps for 2003 and 2004. However, the pact is lacking an enforcement mechanism for budgetary discipline. Moreover, its budgetary objectives should be cyclically adjusted and based on national accounting standards.

… and special funding for the new states in particular needs to be carefully evaluated

Aggregate growth in the new states is stagnating at a low level and the economic convergence with the old *Länder* is stalling, despite high-growth pockets in the industrial sector. Against this background, special funding of considerable magnitude is foreseen to facilitate infrastructure investment in Eastern Germany until 2019. Incentive mechanisms are needed to ensure that such infrastructure investments are directed to areas that contribute most strongly to growth: making federal funding conditional on the new states producing comprehensive evaluations of the longer-term benefits and the full costs of infrastructure investment projects, would be one such mechanism.

More progress is necessary to improve public sector efficiency in general

Some changes have recently been introduced in the government sector to improve the quality and cost efficiency of public services. Working time flexibility has increased in recent years, and an element of performance-related remuneration has been established in some administrations. Reform needs to be continued, however, so as to further improve work incentives and give the public authorities more flexibility in adapting the appropriate level and composition of the labour force. In particular, more flexibility of public sector pay schemes with respect to function and region should be allowed and the application of tenured contracts limited.

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Health sector reforms should go further to make competition more effective

As in most OECD countries, health expenditures have shown a long-term tendency to increase in relation to GDP. Within the OECD, Germany is one of the countries with a relatively high spending-to-GDP ratio. While the access to health services is excellent, various indicators of health outcomes are close to the OECD average, suggesting scope for efficiency improvements. Recent steps to improve incentives in the health care system include the introduction of a diagnosis-related payment system (DRG) for hospital care, which is presently in preparation. While this could imply a significant source for savings by cutting the average length of hospitalisation, it needs to be insured that efficient systems of cost information and for monitoring the quality of services are in place once the DRG comes into operation. The freedom for health funds to selectively contract with health care providers has been widened and more freedom given to consumers in choosing the insurer. Up to now, however, health funds have offered relatively homogenous products and have competed mainly on the basis of risk selection strategies. Recent reforms introduced financial incentives to implement disease management programmes for selected chronical conditions. While this is a positive step, further action is required to establish a basis for the health funds to evolve into active purchasers of health services on behalf of the patients, and to offer differentiated products to their subscribers. For this purpose the current arrangements favouring collective bargaining between the funds and the organisations of health providers should be re-examined. A more comprehensive risk equalisation system would mitigate risks of cream skimming while supporting the desirable trends outlined above.

Policy action is required to improve the performance of secondary schooling and tertiary education

Germany is spending more than most other OECD countries for secondary education while performance of students has been found to be poor in the recent international PISA study. The study also suggests that successful education systems are those that combine standardised targets for educational outcomes with decentralised flexibility and responsibility on how to achieve them. The German school system is characterised by tight school regulation, segmentation of educational paths and the absence of uniform educational standards. Hence, a thorough change of the institutional landscape of secondary education towards more outcome-

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oriented regulatory framework is required. Nation-wide standards for schooling attainments that are regularly evaluated should be introduced, while leaving schools more freedom in determining suitable ways to reach their targets. Tertiary education is likewise costly for public budgets, while suffering from long study duration and high dropout rates. Recent initiatives to shorten tertiary education and improve incentives for university teachers to deliver high quality teaching need to be widened. Tertiary studies with a shorter duration should be introduced on a broader scale. Furthermore consideration should be given to allowing universities the right to levy tuition fees while distributional aspects could be addressed through a proper adjustment of student support schemes. Moreover, reform should move on quickly implementing a higher degree of competition between universities. Funding of universities out of the general government budget should be linked to the universities' performance.

Sustainability of public pensions will require increased labour force participation of pre-pensioners and may need further measures for the longer term

After a series of partial reforms to the public pension system in the 1990s, reforms in 2001 created the foundations for a multi-pillar pension system. The reforms targeted a reduction in the public pension replacement rate and a cap on the contribution rate, revised the indexation of the pay-as-you-go (PAYG) pension, the phase-out of early retirement programmes and the creation of incentives for individuals to opt for compensating occupational and individual pensions. These measures should maintain the relatively high overall pension incomes in the next decade. However, as demographic pressures mount, there is considerable uncertainty whether the targeted reduction in replacement rates will ensure financial balance of the public pension system up to 2020 without additional measures. The uncertainty, as to whether the system is sustainable in the later decades on present parameters may be even higher. Due to the long time needed to phase in pension reforms, there is urgency in addressing these issues. The government should review the statutory retirement age in the light of life expectancy increases and ensure that the phasing out of early retirement programmes does indeed raise the effective age of retirement. Public opinion might also be influenced by considerably lengthening

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the time period of official projections, in order to better highlight the risk of falling replacement rates in the PAYG system unless further action is taken.

Fundamental steps are necessary to increase the flexibility of labour markets

Achieving stronger growth in the context of population ageing requires, in particular a substantial improvement in labour market functioning, and fundamental reforms across a broad front are warranted. Consideration should be given to widening the scope for wage determination at the company level, so as to better align collective wage contracts with labour market conditions. Germany's employment protection procedures appear to be complex and should be reviewed in order to promote employment while striking a proper balance between flexibility and security. An example could be to offer an option to open work contracts to specify severance pay in case of redundancy in return for a less stringent dismissal protection.

Public employment service reform is important to improve placement efficiency…

In 2002 new legislation came into force designed to improve the activation of the unemployed and to increase the efficiency of the public employment service (PES). Several initiatives are steps in the right direction, but as yet incomplete. Furthermore, the federal government set up a commission (*Hartz Kommission*) to make suggestions for a reorganisation of the PES and labour market reform more generally. Placement activities of the PES have been found to be inefficient. An option for more fundamental reform should allow more private sector delivery of PES services, as is already usual in other OECD countries. Furthermore, profiling should be made standard practice and active labour market measures should be used in the first place to facilitate job search activities. Also, the employment-generating potential of temporary work agencies (TWAs) should be better exploited. Timing restrictions on temporary work contracts (*Synchronisationsverbot*) should be abolished for TWAs. Furthermore, consideration should be given to abolishing the compulsory equalisation of work and pay conditions between temporary employed and regular staff after the thirteenth month of employment. However, TWAs should not be subsidised and no preferential terms should be given to public TWAs. Other legislation that has been introduced, such as the legal right to claim part-time employment

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and the widening of the workers' co-determination, need to be monitored with respect to its impact on labour costs, and modified if necessary.

… and social assistance and unemployment related transfers need to provide better incentives for job search

The German tax-benefit system discourages low-wage earners from taking up work. Effective tax rates for low-wage earners need to be reduced, job search requirements strengthened and incentives to leave the labour force removed. Such reform would entail *inter alia* establishing verifiable criteria for whether benefit recipients are actively engaged in job search as has been proposed by the *Hartz Kommission*. Currently potential sanctions are not very effective because the proof of misconduct has to be provided by the PES. Hence, benefit recipients should be required not only to produce evidence that they are actively searching but also why they are unable to accept a valid job offer and benefits should be reduced or withdrawn in case of non-compliance with the rules. Unemployment assistance should be combined with social assistance into one single means-tested social assistance benefit.

The general framework conditions for competition are highly effective…

Strengthening of competition and efficiency in product markets can also contribute to improved growth performance. Over the last years Germany emerged amongst the pioneers in product market reform and market opening in Europe. The power of the German competition authority (*Bundeskartellamt*) to sanction violations of the competition law is well developed, but some scope for improvement remains. In view of the increasing complexity of competition cases the authority's investigative power could be strengthened by increasing the level of admissible fines imposed in cases of non-compliance with information requests. The effectiveness of the recently introduced leniency programme should be monitored with a view of possibly increasing the associated costs of non-participation and benefits of participation. Furthermore, it should be examined whether resources available to the Federal Competition Authority suffice, both to cover merger activity and promote competition in newly deregulated areas. Moreover, competition can be further enhanced by stepping up privatisation efforts, including at local levels of government.

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… while past success of regulatory reform needs to be followed by decisive policy action…

As has been outlined in the last Survey, considerable progress has been made in recent years in introducing more competition in network industries and this has had positive effects on the economy through lower prices, better-quality services for consumers, and increased competitiveness for enterprises. But major challenges remain.

– In the *telecommunications* sector a number of pro-competition measures have been introduced. Nevertheless tight sector-specific regulation continues to be necessary as the incumbent still dominates the telecommunication infrastructure. In the mobile phone sector developments in joint market dominance, roaming charges and charges for connecting fixed and mobile lines need to be closely monitored by the regulator. More generally, ways to facilitate the emergence of competing networks such as cable TV and UMTS networks should be explored.

– The approach of opening the markets for *electricity and gas* without establishing an independent sector regulator has led to uneven progress in promoting competition. Eliminating entry barriers remains a challenge. In this respect framework agreements (Association Agreements) have been expanded and the regulatory framework is to be strengthened with the adoption of a new energy law proposal. However, if the remaining access problems cannot be resolved in the electricity sector, the government should consider introducing a sector regulator. In the gas sector only a limited number of consumers have obtained more favourable contracts and, given the scale of access problems, the government should reconsider its decision not to install a sector regulator. Vertically-integrated conglomerates are an impediment to non-discriminatory network access. Hence, further unbundling of energy generation, distribution and transmission beyond accounting and management separation would reduce regulatory challenges, but is currently hampered by widespread public ownership in local utilities.

– In the *railways* sector the regulatory framework has been improved as the Railways Office was granted additional regulatory powers to remove technical barriers for

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network access. Monitoring is necessary to see if these changes are sufficient. The decision of the authorities to separate infrastructure and the provision of transport services by strict organisational unbundling is supposed to secure non-discriminatory access to the network in general. However, ownership unbundling should be considered in the course of privatising *Deutsche Bahn*.

– Liberalisation of *postal services* is relatively advanced in comparison with many other EU countries. Postponement of complete market liberalisation in the *postal sector* implies higher costs for consumers and reduced competitiveness of enterprises. Hence, the termination of the monopoly rights of the Post Office should be accelerated. Moreover, the privatisation of the *Deutsche Post* should be completed so as to exclude conflicts of interest between ownership and regulation.

… and further regulatory reform in retailing would support a more friendly consumer climate

There is still scope to enhance competition in retailing. While legal provisions restraining discounts and promotional offers were abandoned in 2001 the effects of this reform were more modest than foreseen since related provisions are still in place. To remedy this situation the prohibition of time-limited discounts outside the sales periods should be given up. Moreover, the current regulation against setting prices below costs should be reconsidered. These measure should be complemented by further deregulation of shop opening hours as emphasised in previous *Surveys*. By giving consumers more choice, such reforms might contribute to improving the general consumption climate in Germany.

In several respects the framework conditions for financial markets and corporate governance have improved significantly

Worldwide turbulence on increasingly integrated financial markets has reinforced the importance of a sound regulatory framework in this area. The integration of financial market supervision provides a more coherent framework for the supervision of institutions and products across financial sectors and supports international co-operation in the field. The government should ensure that all benefits of a unified financial supervisor materialise. The recent establishment of a legal framework for take-overs with stringent information requirements and time schedules is an improvement over the previous voluntary approach. It also improves the protection

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of minority shareholders. It can be improved further by ensuring strict management neutrality. Also, relaxing certain information requirements related to future business plans should be considered if they are economically costly. Corporate governance will be strengthened by the Fourth Financial Market Promotion Act. New provisions for security trading enhance investor protection via better detection and prevention of market manipulation. This includes the obligation for the corporate management and the supervisory board to immediately report share trades. Such reporting requirement should be extended to stock options and employee share schemes.

Sustainable development will be aided by the integrated policy approach that increases the use of economic instruments

Sustainability is vital for long-term economic, social and environmental development. It is receiving a lot of attention in Germany through the mechanism of the national sustainability strategy. Considerable resources are being provided for measures aiming at ecological improvements and risk prevention. There have been substantial reductions in air pollution and greenhouse gases in the past decade, leading to improved health and environmental quality, but the costs of further improvements may be rising, depending on the instruments that are applied. Command-and-control measures aimed at point sources of air pollution are likely to result in increased costs relative to benefits. In this light, there should be a further alignment to externalities of the taxes on vehicles and fuels (both in the transport and electricity-generation sectors) in order to ensure a cost-effective approach. Any such moves would need to be revenue neutral as in the ecological tax reform. An expansion of road pricing could also contribute to a reduction in localised pollution problems. Ideally, abatement costs should be equalised across sectors and balanced with benefits. In the area of waste management, waste production has been reduced but the balancing of costs and benefits is a matter of concern. The government has set ambitious and arbitrary targets and transferred the responsibility for its implementation to manufacturers and retailers. However, costs are high and one company dominates the recycling market. The government's commitment to review costs and improve competition in recycling is welcome but it is still necessary to ensure that price-based mechanisms result in a better cost-benefit

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ratio. This would ensure that the packaging tax payments of producers and consumers do not exceed the total of the private costs and the environmental externalities of other methods of waste disposal. Indeed, in both air and waste policy, it is necessary to ensure that the costs they impose on society are better related to the benefits they provide to society.

Summary

In the wake of the unification boom Germany has been for a number of years an economy characterised by weak domestic demand and lower growth than in many other OECD countries. Low employment growth and a high share of structural unemployment are highlighting the need for comprehensive labour market reform. These developments have been shaped by long drawn-out adjustment to the exceptional unification shock as well as increased vulnerability to adverse external shocks. Growth has been very low since 2001, and the projected recovery relies strongly on the expansion of world trade. The budget deficit is likely to exceed 3½ per cent of GDP in 2002. The high structural deficit needs to be addressed by structural measures over a number of years, but a start has to be made already in the 2003 budget. The federal government has committed itself to strict fiscal consolidation and is now designing a consolidation package. Furthermore all levels of government have agreed to establish a domestic stability pact. The recent pension reform is a step in the right direction, but more may have to be done to pre-fund ageing related expenditure increases and increase the labour force participation of older workers. A more rigorous approach to evaluating public sector spending projects should be implemented to target expenditure cuts, which are necessary to put the budget on a sustainable consolidation path and to create room for lowering taxes. Boosting potential growth and employment requires deeper structural reform, especially on the labour market. Further measures to improve job placement – including some of those proposed by the *Hartz Kommission* – need to be implemented quickly. The positive momentum for reform should be used to implement measures that are likely to improve labour utilisation. This requires, *inter alia*, reducing the tax wedge, promoting a higher degree of wage responsiveness to market conditions, and reviewing

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the complex employment protection procedures. Over the last years Germany emerged amongst the pioneers in product market reform and market opening in Europe. Nevertheless further steps in opening up markets should be considered. Consumers could benefit from a further liberalisation of retail legislation and a consolidation of past reform successes in the general competition framework and network industries. Unless more comprehensive progress is made in promoting competition on energy markets the government should consider installing independent sector regulators. Access to the railtrack needs to be guaranteed on a non-discriminatory basis and the postal monopoly rights need to be terminated. Considerable progress has been made reducing air pollution, greenhouse gas emission and limiting waste production. However, further environmental progress will come with sharply rising economic costs, if not offset by more consistent application of economic instruments. Progress on all these scores is essential to facilitate the changes required for Germany to strengthen economic growth, support social cohesion and maintain an attractive environment.

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I. Macroeconomic developments

In 2001 growth was subdued in Germany and employment fell in the course of the year. Output has barely grown in 2002 but growth should pick up in 2003, driven by strengthening exports. As activity broadens in 2004, GDP is projected to grow above potential, at some 2½ per cent. While the conjunctural slowing of activity over the last couple of years largely affected the OECD area overall, since the middle of the 1990s economic growth in Germany has been lower than in most other OECD countries. Indeed, in the Secretariat's assessment German potential GDP growth stands at some 1½ per cent, well below the average in the European Union and the OECD as a whole. While activity over more than a decade has been shaped by long drawn-out adjustment to the exceptional unification shock, subdued labour utilisation appears as the primary contributor to the weakness of economic growth since the second half of the 1990s, and weak domestic demand increases the dependence of the economy on international trade and makes it vulnerable to external shocks. The first part of this chapter reviews important aspects of German economic growth in the 1990s. The second part considers more recent macroeconomic developments and forces acting, along with the OECD's assessment of short-term economic prospects.

Why did Germany's economic growth slow in the 1990s?

Average per capita growth of real GDP in Germany, Europe and the United States over the 1980s were roughly equal, totalling around 2 per cent. Thereafter, averaged over a decade, growth in Germany slowed down substantially, significantly more than in other European countries (Table 1). To a considerable extent, this reflects a desynchronisation of economic activity between Germany and the other EU countries in the early 1990s on account of the economic shock associated with reunification. Indeed, between 1990 and 1991 German growth steeply accelerated – driven by buoyant demand on account of the unification boom – while real growth in the EU had already peaked in 1988.[1] German growth exceeded the European average by a significant margin until 1992. In 1993 Germany went into recession, as did most other EU countries, and the growth differential reversed

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Table 1. **Sources of growth in real GDP per capita**[1]

Averages of annual changes in per cent

		Germany				France, Italy and Spain				United States			
		1980-90	1991-01	1991-95	1995-01	1980-90	1991-01	1991-95	1995-01	1980-90	1991-01	1991-95	1995-01
Real GDP	**1**	2.3	1.5	1.3	1.6	2.4	1.9	1.1	2.5	3.1	3.2	3.1	3.2
Population	**2**	0.3	0.3	0.5	0.2	0.3	0.3	0.2	0.3	0.9	1.2	1.0	1.3
Real GDP per capita	**3**	2.0	1.2	0.8	1.4	2.0	1.7	0.9	2.2	2.2	2.0	2.0	2.0
Labour input (hours per population)	**4**	–0.4	–0.8	–1.6	–0.4	–0.7	0.2	–1.5	1.1	0.9	0.1	0.8	–0.1
Working-age population (share of total population)	**5**	0.5	–0.3	–0.5	–0.2	0.5	0.0	0.1	–0.1	0.3	–0.2	0.0	–0.2
Labour force participation (share of working-age population)	**6**	0.5	0.3	–0.1	0.5	–0.2	0.5	–0.2	0.9	0.4	0.1	0.2	0.1
Employment (share of labour force)	**7**	–0.3	–0.2	–0.7	0.1	–0.3	0.0	–1.0	0.6	0.2	0.2	0.3	0.2
Hours worked per employed	**8**	–1.1	–0.6	–0.4	–0.9	–0.7	–0.3	–0.4	–0.3	0.1	0.0	0.3	–0.1
Labour productivity	**9**	2.4	2.0	2.4	1.9	2.7	1.5	2.4	1.0	1.3	1.9	1.3	2.0
Multi-factor productivity	**10**	1.4	1.1	1.1	1.2	1.4	0.5	1.0	0.4	1.0	1.4	1.2	1.3
Capital intensity of labour	**11**	1.0	0.9	1.3	0.6	1.3	0.9	1.5	0.6	0.3	0.5	0.1	0.7
For information:													
Terms of trade	**12**	0.3	–0.4	0.1	–0.6	0.1	–0.3	–0.7	–0.1	–0.2	–0.1	–0.2	–0.1

1. Growth in real GDP per capita is decomposed into growth in labour input variables and growth in labour productivity using the identity [gdpv/pop] = [popt/pop][lf/popt][et/lf]H[gdpv/(et X H)] in logarithmic differences, and growth in labour productivity is decomposed into multi-factor productivity and capital intensity of labour using the identity [gdpv/(et X H)] = [gdpv/(kbv$^{(1-a)}$ (et X H)^{a})][kbv/(et X H)]$^{(1-a)}$ in logarithmic differences where *gdpv* is real GDP, *pop* is total population, *popt* is working-age population, *lf* is labour force, *et* is employment, H is average annual hours actually worked by persons en employment, *kbv* is capital stock and *a* is the share of labour compensation in GDP. So row 3 is the sum of rows 4 and 9, row 4 is the sum of rows 5 to 8 and row 9 is the sum of rows 10 and 11.
Terms of trade are [GDP deflator]/[[private consumption deflator]. Calculations include western Germany up to 1990 and Germany from 1991 onwards.

Source: OECD Secretariat.

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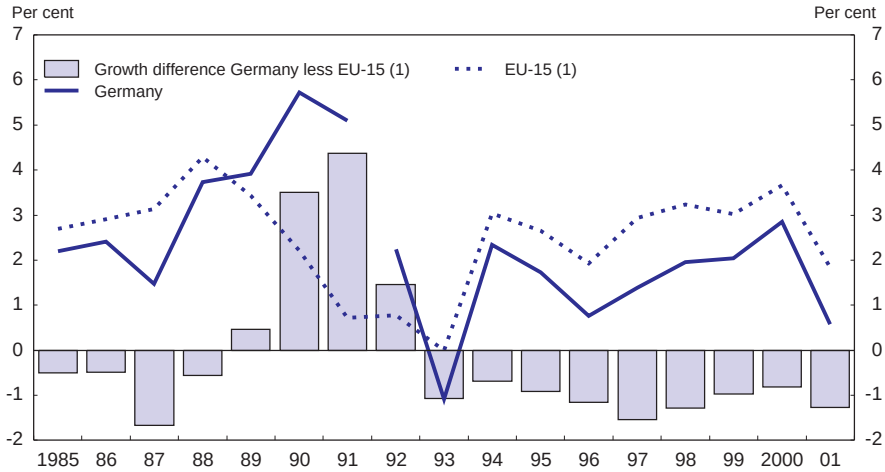
Figure 1. **Real GDP growth in Germany and the EU**



1. Sum of EU countries excluding Germany.
Source: OECD.

(Figure 1). However, the persistence of the negative growth spread since 1993 points to both the size of the structural adjustment challenge Germany is facing and difficulties in adapting to the changed environment after unification.

Weak employment generation largely accounts for reduced growth…

A decomposition of real per capita growth into the growth contributions of labour – in terms of total hours worked per inhabitant – and labour productivity – in terms of real GDP per hours worked – indicates that the weakening of Germany's economic performance in the 1990s is mainly reflected in weak employment generation. Indeed, over the last decade total hours worked per inhabitant declined, accounting *ceteris paribus* for a weakening in real GDP growth per capita by some ¾ percentage point per year. By contrast, employment contributed slightly positively to growth in other European countries and the United States (Table 1, third and fourth rows). Although at the same time, labour productivity rose faster in Germany than in other European countries (Table 1, row 9) this differential was not high enough to offset the adverse effect on GDP growth of weak labour input.

Declining labour input in Germany reflects several elements. Germany has been subject to a somewhat more pronounced reduction in the working age population than the United States and most other European countries (Table 1,

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row 5). More important, however, has been the impact of lower employment rates and reduced hours per employed worker. The increase in labour force participation has also been weaker than in other European countries (Table 1, sum of rows 6 to 8). While many European countries shifted to employment-intensive growth in the second half of the 1990s, this did not occur in Germany. In France, Italy and Spain higher employment in terms of total hours worked contributed over 1 percentage point to per capita GDP growth since 1995, while in Germany it remained negative. The sectoral evolution of German employment is differentiated, though. While employment in the manufacturing, construction and government sectors declined markedly, employment in the service sector has expanded over the last decade (Figure 2).

Perhaps the most striking aspect of the declining labour input in Germany – particularly since 1995 – has been the sharp decline in average hours worked per employee (Figure 3). A full explanation for this phenomenon is difficult to arrive at, but several factors seem to be at play. The share of part-time employees – in particular those working very few hours per week – has increased substantially in Germany over the last years (see Chapter IV below). With part-time jobs allocated primarily in the service sector, the sectoral distribution of job growth has been important for this development. Jobs with very few hours worked have benefited from a more favourable tax treatment than regular employment, and provide an extra degree of flexibility to employment adjustment. The potentially high costs of adjusting the regular workforce to labour demand conditions, implicit in Germany's relatively strict employment protection regulation, favour non-standard forms of employment (see Chapter IV below). At the same time, enterprises frequently avoid redundancies by shifting to "short-time work" (*Kurzarbeit*), *i.e.* part-time unemployment associated with income replacing transfers from the Federal Labour Office. Introduction of short-time work has played a significant role over the last couple of years. In addition to the factors shaping labour developments for Germany as a whole, there is no doubt that the German development has been substantially influenced by the ongoing restructuring in the new states, and notably the downsizing of the construction and government sectors. But, overall, the fact that these negative employment shocks have not yet been offset by stronger employment growth elsewhere within the new states or in Germany as a whole – so that total labour input remains lower today than in 1995 – indicates the need to improve adjustment mechanisms in the German economy.[2]

Low employment generation and high effective taxation of labour have interacted in reducing disposable incomes and retarding private consumption growth. Within the OECD, Germany stands almost at the top with respect to the tax wedge between wages paid by employers and received by the employees (Figure 4).[3] The tax wedge widened rapidly in the first half of the 1990s when social transfers to the east were financed via social charges on earnings. This retarded both labour demand as well as disposable income growth per employee, despite

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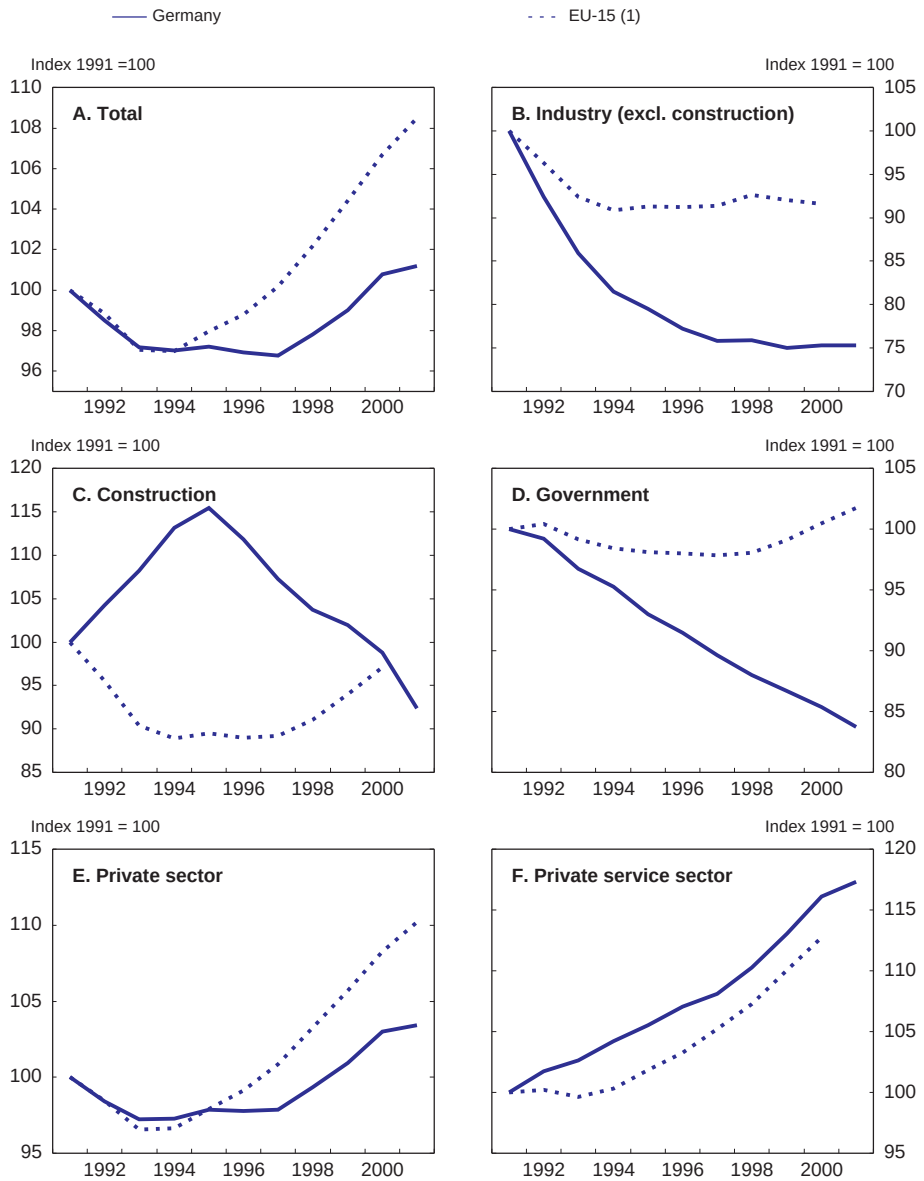
Figure 2. **Employment by sector in Germany and the EU**



1. EU countries excluding Germany.
Source: OECD.

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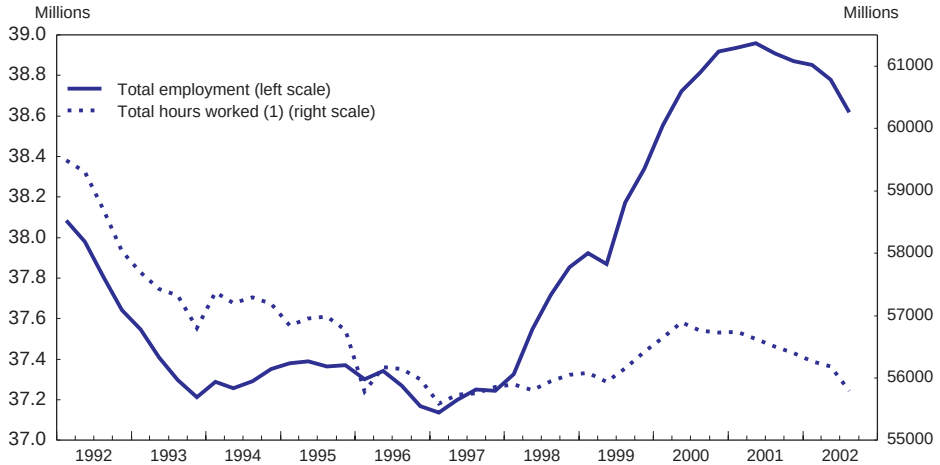
Figure 3. **Employment in the 1990s**



1. Annualised figures.
Source: Deutsche Bundesbank, Federal Labour Office and OECD.

buoyant increases in wage rates. Indeed, between 1991 and 1995 after-tax real wages per worker rose by only 0.3 per cent annually although real compensation per dependent employee increased by 2.0 per cent. In the second half of the 1990s high labour costs and the steep decline in Germany's international competitiveness put pressure on wages. While wage growth moderated, so far this did not translate into stronger employment growth. These elements combine to reinforce weak income and consumption growth. Between 1991 and 2001 annual real consumption growth in Germany totalled merely 1½ per cent, as opposed to 2¼ per cent in the EU net of Germany (Figure 5). Perhaps the most striking fact is that consumption growth in Germany remained virtually flat between the first and the second half of the decade while it more than doubled in the EU (Table 2).[4]

... while labour productivity has been restrained

Against the background of Germany's significant labour shedding in the last decade, the performance of labour productivity – in terms of real GDP per total hours worked – appears quite low. Although German labour productivity grew faster than in other European countries (dropping less after the 1980s), this did not suffice to offset the adverse impact on economic growth of weak employment generation (Table 1, row 9). A number of factors appear to have prevented

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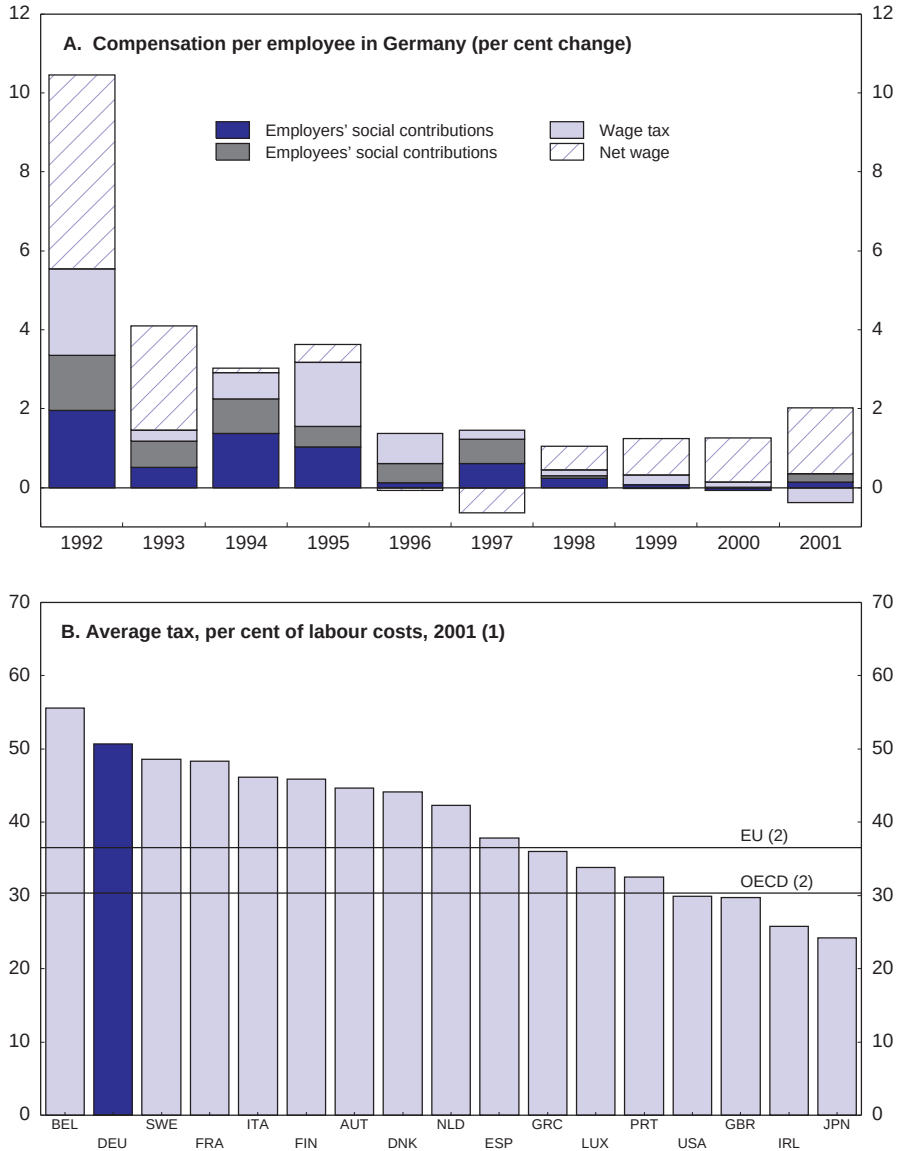
Figure 4. **Labour compensation and taxation**



1. For a single individual at the level income of the average of production worker.
2. Weighted average using 1995 GDP.

Source: National accounts and OECD, *Taxing wages*.

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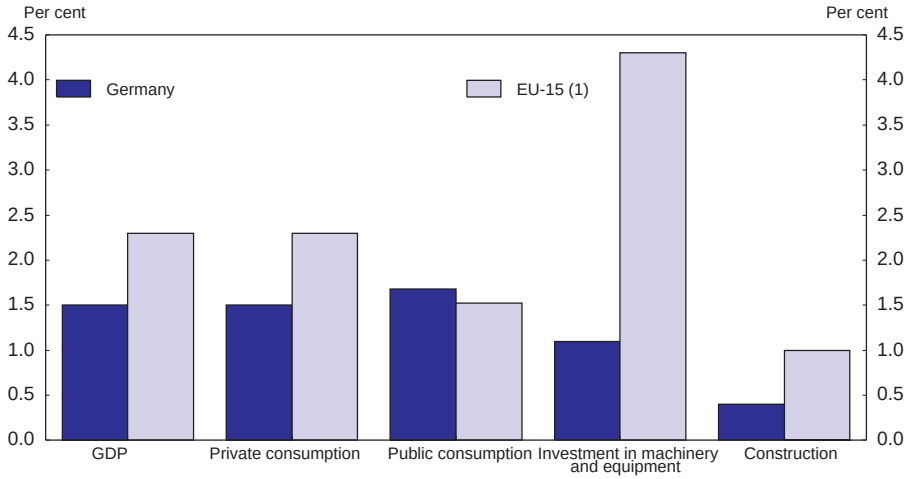
Figure 5. **Real GDP growth rates between 1991 and 2001**



1. Sum of EU countries excluding Germany.
Source: OECD.

investment in Germany from generating stronger productivity growth, and this concerns both the quality and the level of investment. To a considerable degree fixed investment has been diverted by support policies for the new states. The reallocation of resources has helped rebuilding the industrial and infrastructure

Table 2. **Private consumption and earnings**

Annual change in per cent	Germany		EU-15 excluding Germany[1]	
	1991-95	1995-2001	1991-95	1995-2001
Real private consumption	1.5	1.5	1.2	3.0
Real compensation per dependent employee	2.0	–0.2	0.3	1.2
Real gross wages and salaries per dependent employee	1.8	–0.2	0.2	1.3
Real net wages and salaries per dependent employee	0.3	–0.3	–	–
Real net wages and salaries	–0.7	0.4	–	–
Employment (persons)	–0.7	0.6	–0.5	1.7
Real disposable income of private households[2]	1.0	1.4	1.1	2.3

1. All real variables deflated by the Private Consumption Deflator.
2. Excluding Luxembourg.
Source: OECD.

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capital stock in the east, but implied a loss in allocative efficiency in some areas. As has been argued in the last *Survey*, subsidies in favour of capital intensive production caused distortions in the manufacturing production structure in eastern Germany that appears to have retarded productivity.[5] The boost of construction investment in the first half of the 1990s, which was largely driven by state aid, also implied a significant diversion of investment away from areas associated with higher productivity growth. To some extent the weaker productivity growth might also have been held back by the fact that the size of Germany's ICT (information and communication technology) sector is relatively small, and the use of ICT in Germany is likely to fall short of that in some other OECD countries.

Regarding investment volume, equipment investment (including other assets) boomed in Germany at the beginning of the 1990s, but thereafter a wide negative growth differential has opened up in comparison to the other EU countries (Figure 5).[6] Indeed, since the middle of the 1990s subdued investment in machinery and equipment accounts for roughly one-quarter of the GDP growth differential between Germany and the EU.[7] The steep increase in labour cost and the decline in Germany's international competitiveness in the first half of the 1990s, as well as the persistent weakness of private consumption probably all contributed to restraining the propensity to invest. Also, while convergence of real interest rates to lower German levels in the run-up to European Monetary Union implied some investment stimulus for several European countries, such as Italy and France, Germany did not benefit from this convergence effect. The weakness in equipment investment and the phased downward adjustment of over-capacities in the eastern German construction sector – recently amplified by receding residential construction in the west – interact with low employment growth, thereby adding to the persistent weakness in total domestic demand.[8]

... and growth largely depends on exports

The competitiveness of the German export sector declined until the middle of the 1990s when unit labour costs increased steeply and the DM appreciated in nominal terms against the currencies of Germany's major trading partners (Figure 6, panel A). In addition, at the beginning of the 1990s a large share of west German production was diverted to eastern Germany rather than being exported.[9] These factors contributed to a substantial loss in Germany's export market share in the first half of the last decade, and the growth contribution of net exports became significantly negative for Germany as a whole (Figure 6, panel B). Both the loss in international competitiveness and weak domestic demand increased the pressure on Germany's industry to price goods closely to the world market in order to maintain export market share. This combined with more moderate wage growth after 1995 to cause an improvement in Germany's international competitiveness in the second half of the decade (Figure 6, panel A). Reflecting this process, Germany's terms of trade

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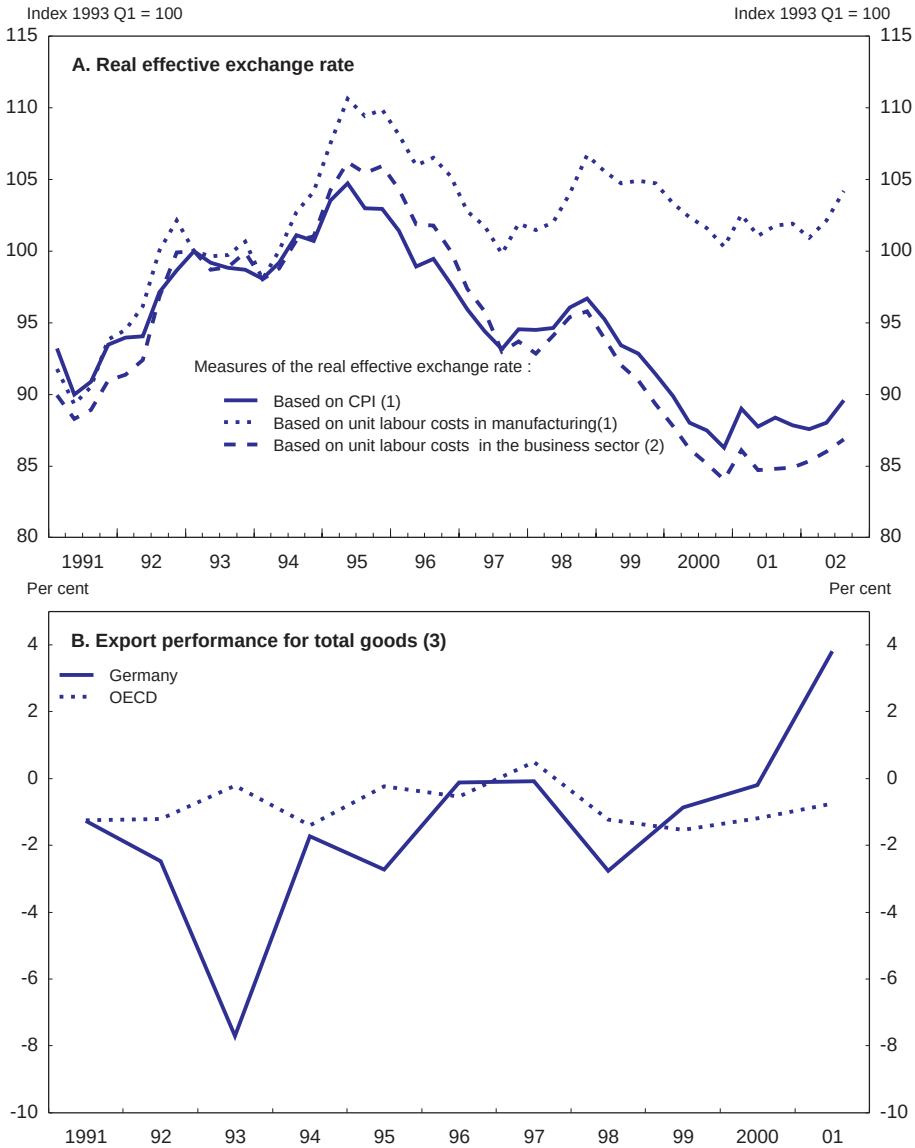
Figure 6. **Germany's international competitiveness**



1. Weighted external value of the DM against currencies of 42 industrial countries.
2. Calculated as trade-weighted average of 19 german trading partners (industrial countries).
3. Percentage change in ratio between export volume and export market of total goods.

Source: Deutsche Bundesbank and OECD.

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deteriorated after the middle of the 1990s (Table 1, row 12), while its export market share improved (Figure 6, panel B). For several years now, German GDP growth has been accounted for largely by net exports as exports have been maintained while imports have weakened substantially reflecting the weakness of the country's domestic demand.

Recent economic developments and prospects

The underlying forces shaping the German economy since 1990 continue to influence the present cyclical situation, most notably the economic situation in East Germany and the continued restructuring of the construction sector, hampering domestic demand growth, which in 2001 declined for the first time since 1993 (Figure 7). Low capacity utilisation and the continued downsizing of the construction sector were major factors depressing investment. While income taxes have been reduced, modest employment creation, low real wage growth and higher indirect taxation, have hampered the growth of disposable income, contributing to the slow expansion of private consumption. In sum, the economy contracted in the second half of 2001 and has been growing by less than one per cent between the third quarter 2002 and the last quarter in 2001 and with little prospect of markedly faster growth over the next months (Table 3).

Sources of the economic slowdown

The slowing of economic growth in Germany was more pronounced than in most other countries (Figure 8). The lower growth can hardly be attributed to a particular severity of external shocks. The ICT producing sector, whose recession has been an important factor in explaining the present economic cycle, has a relatively small weight in the German economy. The stock market correction was somewhat more pronounced than in other European countries, but wealth effects on consumption appear to be relatively modest. The marked slowdown in world trade naturally had a dampening effect on German export growth, but was partly cushioned by high stocks of foreign orders. German export market performance was better than in most other EU countries. In contrast, the relatively low GDP growth is attributable to weak domestic activity.

Exports of goods and services slowed, seasonally adjusted, in 2001 before accelerating during the first three quarters of 2002. Nevertheless, the annual growth rate in 2001 was still around 5½ per cent, reflecting a large overhang from the doubled digit growth rates in 2000. As German export markets decelerated during 2001, there was a marked gain in export market shares, which continued in 2002 as world trade strengthened somewhat. While the competitiveness of the German export sector improved since the middle of the 1990s the gain in market share also reflects a positive effect of the goods composition. Exports to North America maintained some momentum not least due to the relatively strong

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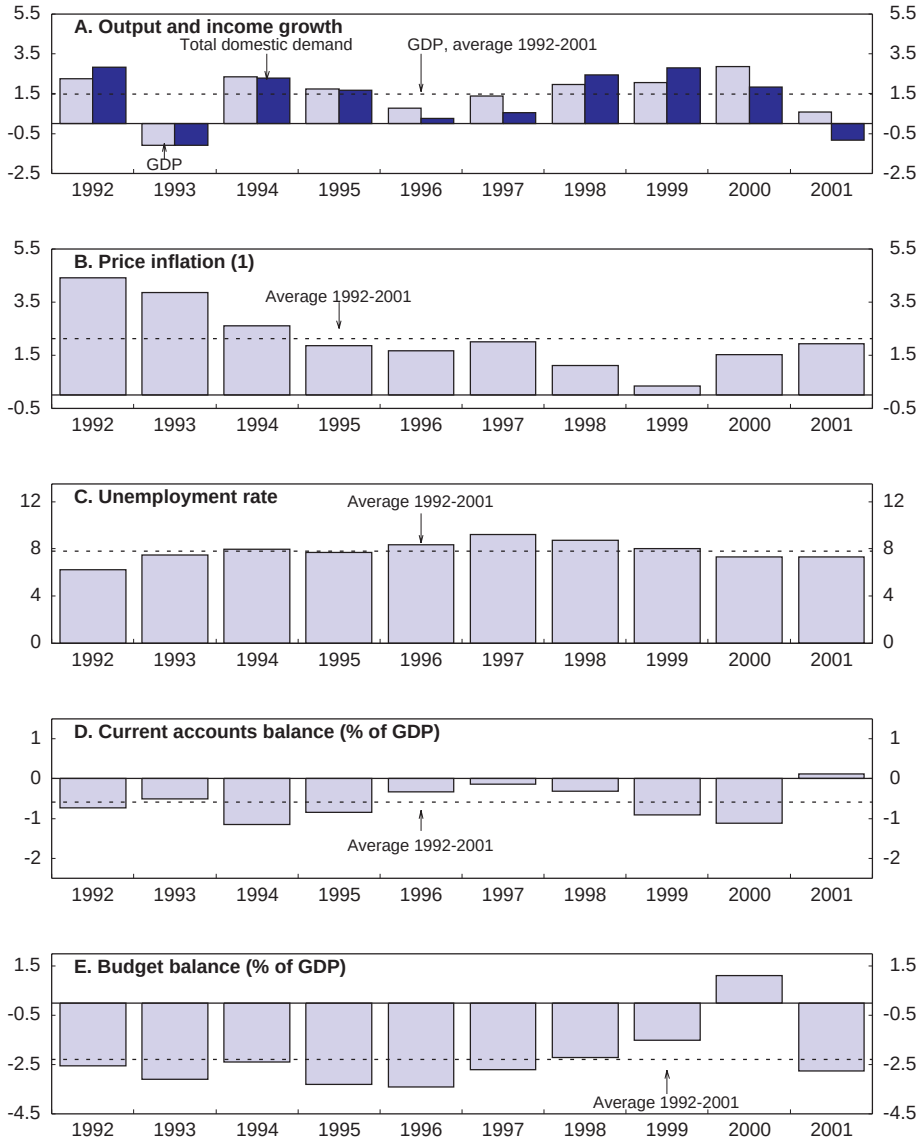
Figure 7. **Macroeconomic performance**

Per cent



1. Growth in private consumption deflator.

Source: OECD.

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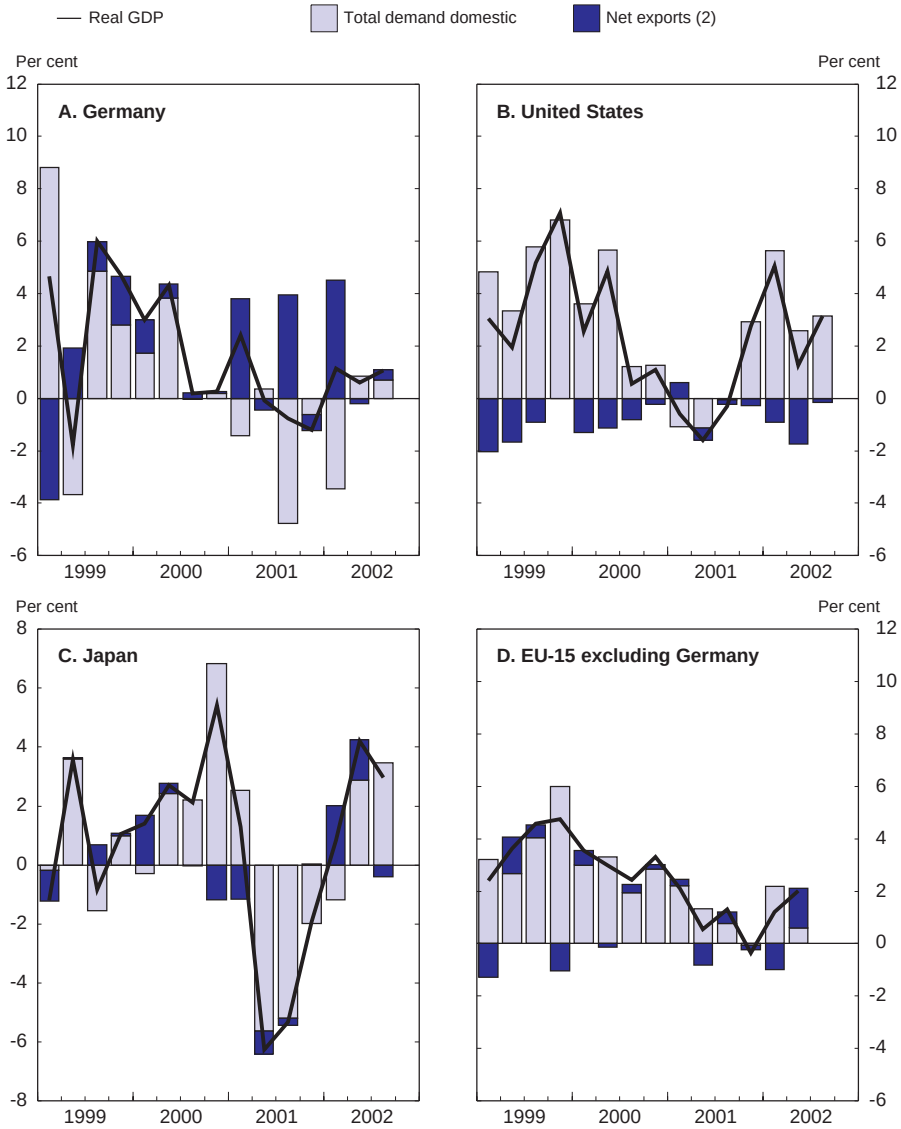
Table 3. **Demand and output**

	1995	1996	1997	1998	1999	2000	2001	2002[1]	2003[1]	2004[1]
Private consumption	2.1	1.0	0.6	1.8	3.7	1.4	1.5	–0.5	1.1	2.2
Government consumption	1.5	1.8	0.3	1.9	1.0	1.2	0.8	1.1	0.8	0.7
Total fixed investment	–0.6	–0.8	0.6	3.0	4.1	2.5	–5.3	–4.7	0.6	1.3
Construction	–1.8	–2.9	–1.5	–1.0	1.4	–2.6	–6.0	–3.6	0.4	–2.3
Machinery and equipment	1.5	2.8	4.0	9.4	7.9	9.3	–4.4	–6.0	0.9	5.7
Stockbuilding[2]	0.3	–0.5	0.0	0.3	–0.4	0.2	–0.6	0.0	0.4	0.4
Total domestic demand	1.7	0.3	0.6	2.4	2.8	1.8	–0.8	–1.1	1.4	2.1
Net exports[2]	0.1	0.5	0.8	–0.4	–0.7	1.0	1.4	1.4	0.2	0.5
Exports of goods and services	5.7	5.1	11.2	7.0	5.6	13.7	5.0	1.8	5.3	8.0
Imports of goods and services	5.6	3.1	8.3	9.1	8.5	10.5	1.0	–2.5	5.4	7.7
GDP at market price	1.7	0.8	1.4	2.0	2.0	2.9	0.6	0.4	1.5	2.5
Memorandum items :										
Total employment[3]	0.2	–0.3	–0.2	1.1	1.2	1.8	0.4	–0.5	–0.1	1.0
Household savings rate	11.2	10.8	10.4	10.3	9.8	9.8	10.1	10.4	10.1	10.2
Relative ULC in manufacturing	8.0	–2.7	–4.6	2.4	0.8	–6.3	0.0	..	..	..
Export market growth in manufacturing	10.1	7.3	11.2	9.0	7.6	13.6	0.9	1.5	7.2	8.9
GDP deflator	2.0	1.0	0.7	1.1	0.5	–0.3	1.4	1.6	1.2	1.1
Private consumption deflator	1.9	1.7	2.0	1.1	0.4	1.5	1.9	1.6	1.4	1.1
Unemployment (in millions)[3]	3.1	3.4	3.8	3.6	3.3	3.1	3.1	3.3	3.4	3.2
Unemployment (per cent of labour force)[3]	7.7	8.4	9.2	8.7	8.0	7.3	7.3	7.8	8.1	7.7
Current account (per cent of GDP)	–0.8	–0.3	–0.1	–0.3	–0.9	–1.1	0.1	2.0	2.3	2.8

1. Projections.
2. Contributions to GDP growth.
3. According to the labour force survey.

Source: OECD.

Figure 8. **A comparison of economic cycles**[1]
Per cent changes at annual rates



1. Seasonally adjusted.
2. Contributions to real GDP growth.
Source: OECD.

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demand for cars in the United States (Table 4), and the comparatively low content of ICT goods implied that German exports were relatively moderately affected by the international downswing in ICT investment.

Import developments mirrored the domestic economic situation in 2001. Import levels remained roughly unchanged in 2001, despite a positive overhang from the previous year, because of a sharp contraction in imports during the year and into 2002. More recently, import growth recovered. This pattern reflects a pronounced downwards adjustment of stocks, which reduced GDP growth by more than ½ percentage point in 2001. Destocking ceased accelerating in early 2002, which has contributed to the recent recovery in import growth, and might indicate that the de-stocking cycle is reaching an end.[10] This picture is also reflected in the structure of imports, where the decline in imports of intermediate goods and raw materials was much more pronounced than the contraction in imports of finished goods.

The diverging patterns in exports and imports, combined with roughly unchanged terms-of-trade resulted in an improvement of the trade balance of about € 30 billion in 2001 (Table 5). The invisible balance deteriorated with a worsening of the service and investment income balances, the former having been affected by higher insurance payments following the terrorist attacks in the United States. In sum, the current account improved from a deficit of about 1 per cent of GDP in 2000 to a small surplus in 2001 – the first one since German reunification. The current account surplus widened in 2002, mostly on the back of a further improvement in the goods balance as exports grew faster than imports.

Investment in equipment and machinery contracted by 5.3 per cent in 2001 despite a strong positive overhang from the previous year. The contraction continued in the first three quarters of 2002 (bringing the ratio of investment to GDP to its lowest level since the beginning of the 1990s). This occurred against the background of weak aggregate demand, and uncertainty in profit expectations. Taking into account the relatively larger fall in capacity utilisation, the contraction in equipment and machinery investment is comparable with other economic downturns in the 1990s (Figure 9). Capacity utilisation fell to its lowest level since the mid-1990s, before increasing mildly in 2002. An additional – difficult to quantify – factor was the introduction of less favourable depreciation rules in connection with the 2001 step of the business tax reform, which may have brought investment forward into 2000. For example, there was a strong growth in vehicle investment in 2000, which contracted rather sharply the year after. Additional downward pressure in the construction sector came from reduced housing construction subsidies in 2000 and tighter capital gain taxation rules for investors in 2001, although the sector is likely to receive a short-term boost from repairs and replacement investments following the severe flooding during the summer of 2002.[11]

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Table 4. **Trade by region**

	1998	1999	2000	2001	01 I	01 II	02 I	01 I	01 II	02 I
	Percentage change from previous year							Share		
Exports to[1]										
EU	9.6	6.5	14.9	4.3	7.8	0.7	–0.9	55.7	54.0	55.8
of which: Euro area	10.3	7.3	15.0	3.9	8.0	–0.4	–1.6	43.8	41.9	43.3
EFTA	10.0	1.1	13.4	9.4	18.2	0.0	–6.6	5.3	5.1	4.9
Central and Eastern Europe	11.4	–5.6	21.9	16.5	21.0	13.0	7.0	10.6	11.7	11.4
of which: Russia	–13.4	–31.8	33.3	52.2	53.9	52.1	20.2	1.4	2.0	1.7
North America[2]	16.1	11.2	18.9	8.9	13.7	4.4	–2.7	11.1	11.0	10.9
Japan	–11.0	11.0	27.0	–1.6	4.7	–7.5	–9.5	2.0	2.0	1.8
Asia[3]	–25.1	1.3	28.8	1.9	14.6	–8.8	–2.5	3.9	3.8	3.8
Central and South America	24.7	0.2	9.5	8.6	18.2	0.2	–17.8	2.9	2.8	2.4
Other	2.3	0.9	21.8	9.5	15.6	6.6	8.4	8.5	9.7	9.3
Total	7.8	4.6	17.1	6.7	11.3	2.3	–0.5	100.0	100.0	100.0
Memorandum item:										
Share of machinery and transport equipment in total exports	52.0	52.0	51.2	52.0	52.3	51.6	51.9			
Imports from[1]										
EU	6.5	4.0	14.3	5.1	13.2	–5.0	–6.3	51.8	51.6	52.3
of which: Euro area	7.3	3.8	12.7	5.9	14.4	–4.5	–6.0	41.6	41.2	42.1
EFTA	3.2	4.2	20.9	3.7	11.9	–6.4	–0.1	5.2	5.3	5.6
Central and Eastern Europe	13.3	14.0	27.9	9.3	15.0	4.1	2.8	12.0	12.8	13.3
of which: Russia	..	26.3	51.7	–0.5	14.7	–12.7	–13.0	2.6	2.6	2.4
North America[2]	13.8	7.7	26.4	–5.6	5.6	–16.0	–13.9	9.2	8.3	8.6
Japan	5.7	3.3	24.1	–16.4	–7.3	–25.5	–23.7	4.4	3.8	3.6
Asia[3]	1.0	5.8	35.1	–9.0	0.6	–17.2	–8.9	4.9	5.0	4.8
Central and South America	27.5	3.8	27.0	–5.3	6.6	–16.6	–34.4	2.5	2.2	1.8
Other	2.9	12.4	30.6	–13.3	–9.0	–2.0	–6.9	10.0	11.0	10.0
Total	7.3	6.1	20.5	0.5	8.6	–6.7	–7.2	100.0	100.0	100.0

1. F.o.b., including adjustment.
2. United States and Canada.
3. Dynamic Asian economies.

Source: Deutsche Bundesbank, *Zahlungsbilanzstatistik*, OECD, *Foreign Trade Statistics Series* A.

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Table 5. **Current account of the balance of payments**

Euros billion, not adjusted

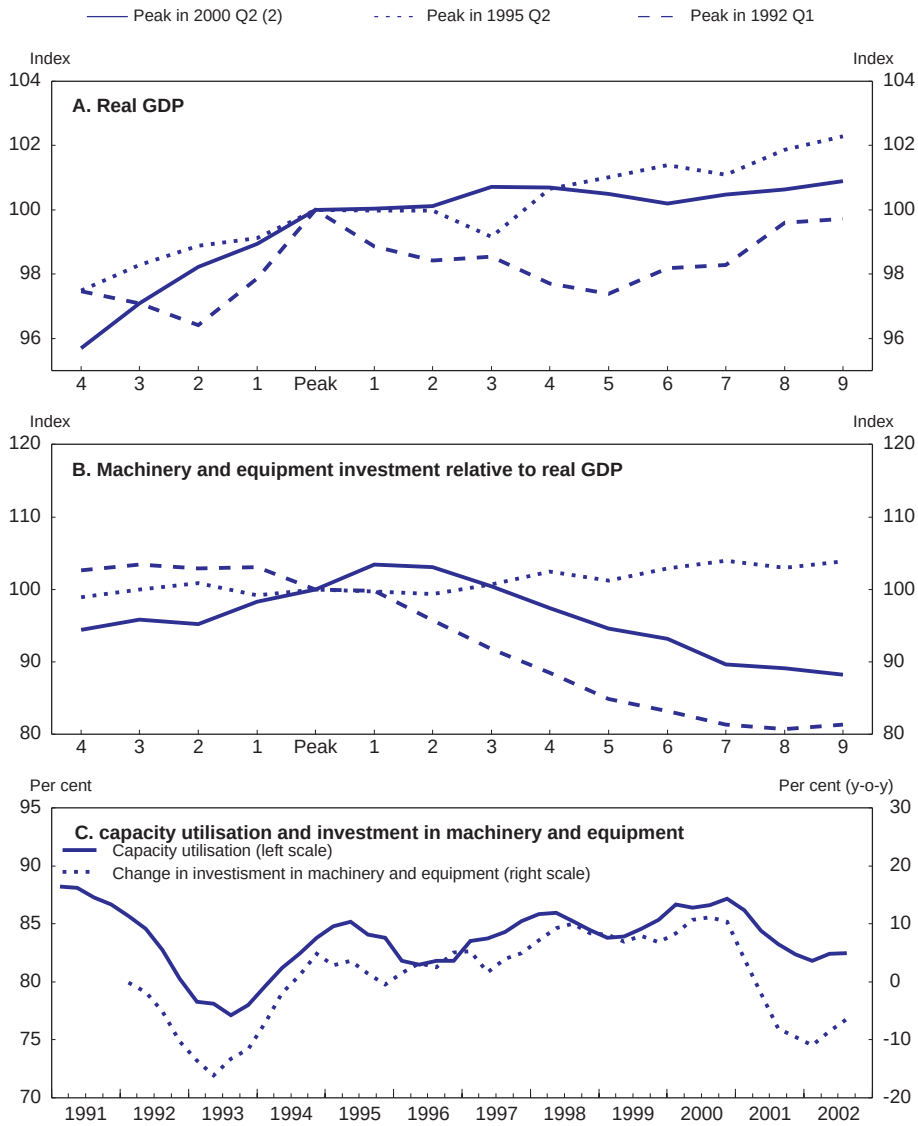
	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002[1]
Goods	34.8	42.3	47.6	54.4	63.4	69.9	66.6	63.0	99.9	98.6
Exports (f.o.b)[2]	323.5	356.0	383.4	402.1	452.3	487.5	509.7	596.9	636.7	480.1
Imports (f.o.b)[2]	288.7	313.8	335.8	347.8	389.0	417.6	443.1	533.9	536.8	381.4
Services	–28.6	–34.0	–34.4	–35.0	–37.6	–41.4	–49.8	–55.1	–58.0	–36.6
of which: Tourism	–22.6	–26.4	–26.3	–27.1	–27.0	–27.5	–30.6	–31.6	–32.4	–23.9
Factor income	14.0	2.5	0.1	0.7	–1.2	–6.8	–9.0	–3.3	–12.6	–9.4
of which: Investment income	13.6	2.3	0.8	1.6	–0.3	–6.0	–8.1	–2.4	–11.1	–8.4
Transfers	–28.5	–30.6	–28.5	–26.2	–27.0	–27.3	–25.7	–27.1	–26.7	–19.0
of which :										
Net contribution to the EU	–13.9	–16.2	–15.3	–14.1	–14.6	–15.5	–13.8	–15.4	–12.8	–8.2
Other public transfers	–7.2	–6.6	–5.3	–4.0	–4.2	–3.5	–3.6	–3.6	–4.2	–2.9
Current account	–8.3	–19.8	–15.2	–6.1	–2.4	–5.6	–17.9	–22.6	2.7	33.7
As a percentage of GDP	–0.5	–1.1	–0.8	–0.3	–0.1	–0.3	–0.9	–1.1	0.1	6.4

1. Sum of the first nine months.
2. Special trade, according to the foreign trade statistics. Including supplementary trade items; imports exclude freight and maritime transport insurance costs, which are included under services.

Source: Deutsche Bundesbank, *Zahlungsbilanzstatistik*.

Figure 9. **A comparison of recoveries in the 1990s**

Index "peak"[1] = 100



1. The "peak" is defined as the highest GDP growth.
2. The nineth period corresponds to 2002 Q3.

Source: OECD, *National accounts*.

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Inflowing domestic orders came down from the beginning of 2001 onwards, but stabilised in 2002 at nearly its lowest level in more than three years. This was largely reflected in a deterioration of the business climate. Export orders also declined in early 2001 and fell sharply following the September 11 terrorist attacks in the United States. In 2002, export orders recovered during spring and appear to have stabilised. The recovery instigated a temporary improvement in business expectations. But since summer expectations deteriorated markedly (Figure 10).

Conditions for internal financing of investment projects should have benefited from the reduction of business taxes in 2001. External financing became more expensive as real long-term interest rates increased between spring 2001 and summer 2002, although they have since come down somewhat. More importantly, equity prices continued to fall, nationally as well as internationally, until autumn 2002, when they appeared to be stabilising.[12] Apart from the impact on equity financing this might also have led to a tightening of loan conditions.[13] The increase in foreign direct investment and the integration of international capital markets have strengthened the demand for German firms to produce internationally comparable rates of return on capital. This implies that the effects of slower international growth would include demands to restore rates of return, which increases the pressure on firms for larger and faster restructuring when capacity utilisation is declining.

Private consumption expanded fairly briskly in the first half of 2001, at a rate well above that of the economy, boosted by the 2001 step of the income tax reform, which increased disposable income by more than 1 per cent. However, consumer sentiment deteriorated in the second half of the year in reaction to the September 11 terrorist attack and as real incomes were eroded by an accelerating inflation rate and increasing unemployment. This triggered a contraction of private consumption and an increase in the households' savings rate.[14] At the start of 2002, consumer sentiment bottomed out at low levels and private consumption continued to contract. Higher indirect taxes and concerns about higher prices in connection with the changeover to the cash euro contributed to this effect. As inflation subsequently came down and wages accelerated following the spring wage negotiations, private consumption strengthened. There is a possible replacement cycle for cars emerging, where later years' low level of new car registrations has triggered a substantial ageing of the car fleet.[15] The lower growth of German private consumption relative to other EU countries should to some extent be related to increases in energy taxes. Also, while in some European countries increasing real prices for owner-occupied housing implied positive wealth effects for consumers, this did not occur in Germany (Figure 11, Panels A and B).[16] While net financial wealth in terms of GDP declined significantly in 2001 (some 2½ per cent), the wealth effect on private consumption appears to be lower than in other European countries.[17, 18]

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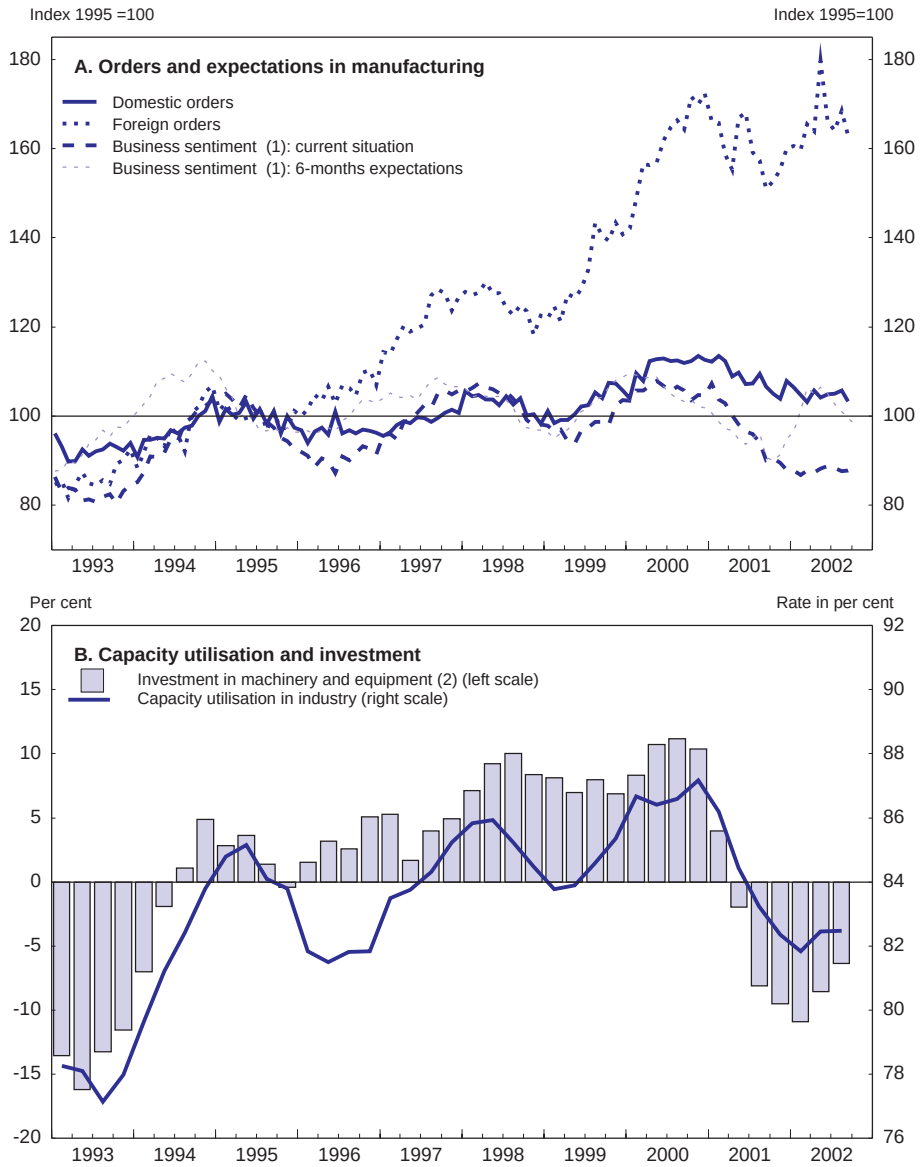
Figure 10. **The business climate**



1. Industry, west Germany.
2. At 1995 prices. Percentage change over previous period.

Source: Ifo Institut für Wirtschaftsforschung; Deutsche Bundesbank and OECD, *Main Economic Indicators*.

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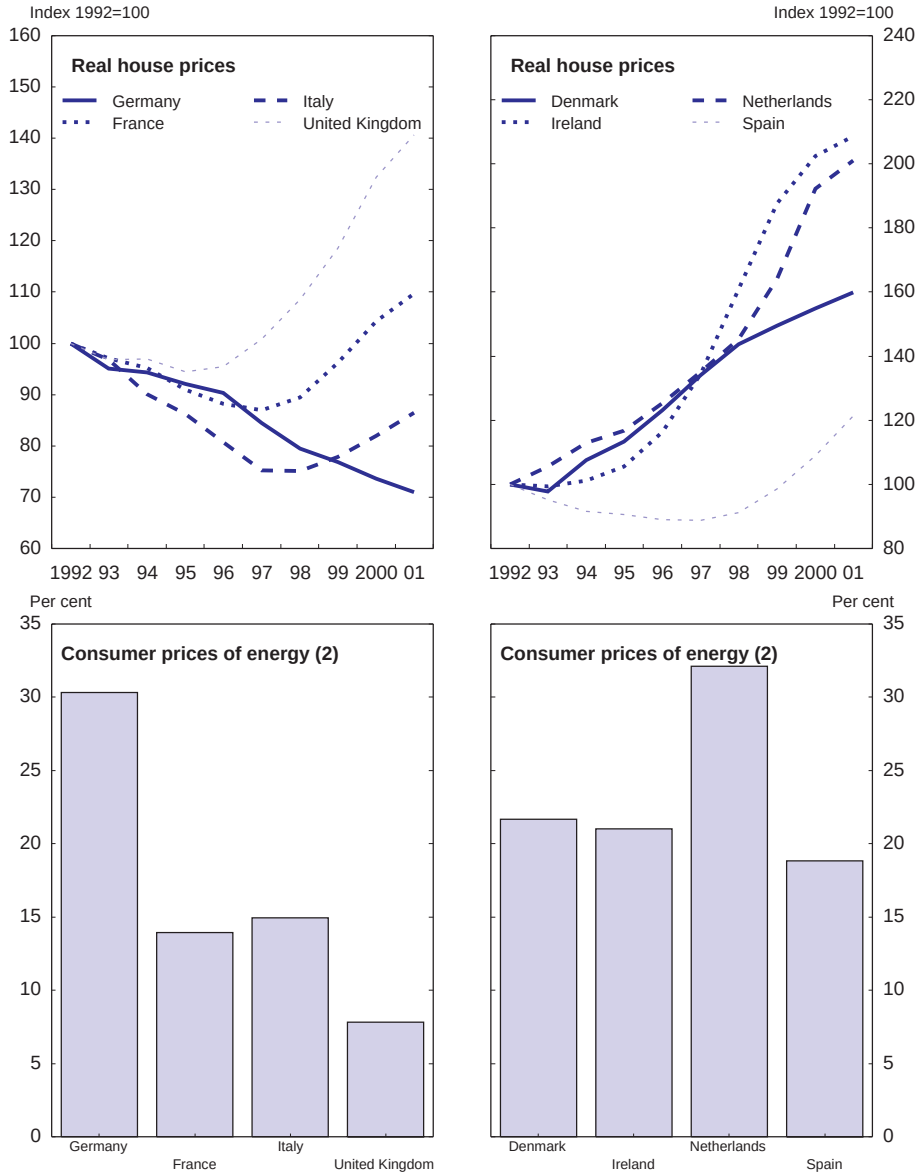
Figure 11. **Energy prices and real house prices in European countries**[1]



1. Nominal house prices deflated by CPI.
2. Growth over March 1999 – September 2002.

Source: BIS (2002), *Annual Report* and OECD, *Main Economic Indicators*.

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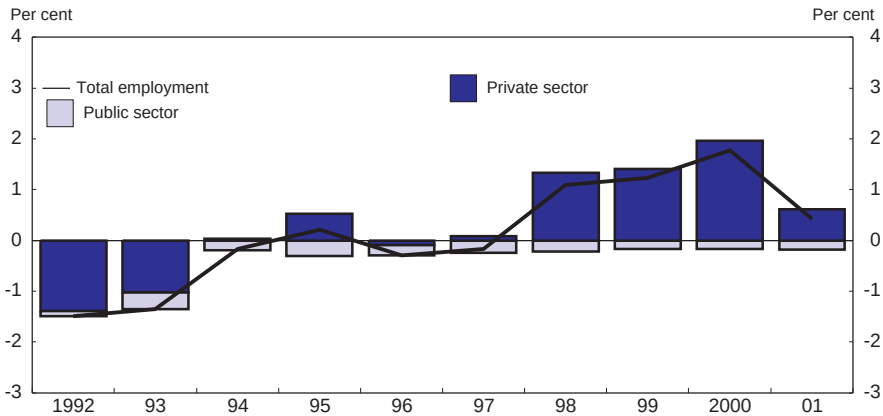
Figure 12. **Decomposition of change in total employment**[1]

Percentage change



1. The decomposition of employment growth has been approximated by a weighted sum of employment growth in the private and the government sectors, with the weigths being the share of each type of employment in the total recalculated each year. Employment in Treuhand companies in eastern Germany is classified as the private sector.

Source: Federal Statistical Office and OECD.

Employment peaked in the first half of 2001. It then fell as economic activity deteriorated, undercutting the peak by about ½ per cent in the third quarter of 2002 (Figure 2). Reflecting the continued contraction of the construction industry and a deteriorating outlook for the manufacturing sector, employment in these sectors declined. On the other hand, the private service sectors continued to expand employment until summer 2002. Thereafter it began to fall as well (Figure 12). The labour force began to contract in 2002. Almost all sectors of the economy have expanded small hours jobs, with particularly noticeable increases within the retail and business related service sectors.[19]

The total volume of hours worked has decreased twice as much as per-person employment since the first semester of 2001, as average working time came down across all sectors. Against the background of slow GDP growth, this partly reflects the fact that employers – perhaps motivated by earlier increases in labour mismatch – have exploited increased possibilities for flexible working hour arrangements and reduced overtime (Figure 13). Subsidised short-time work almost doubled in the first half of 2002 (Table 6). The fall in average working time should imply a more sluggish response in the number of employed when the economy recovers. The registered unemployment rate (seasonally adjusted) increased from early 2001 by more than half a percentage point to nearly 10 per cent in autumn 2002 (Figure 14).

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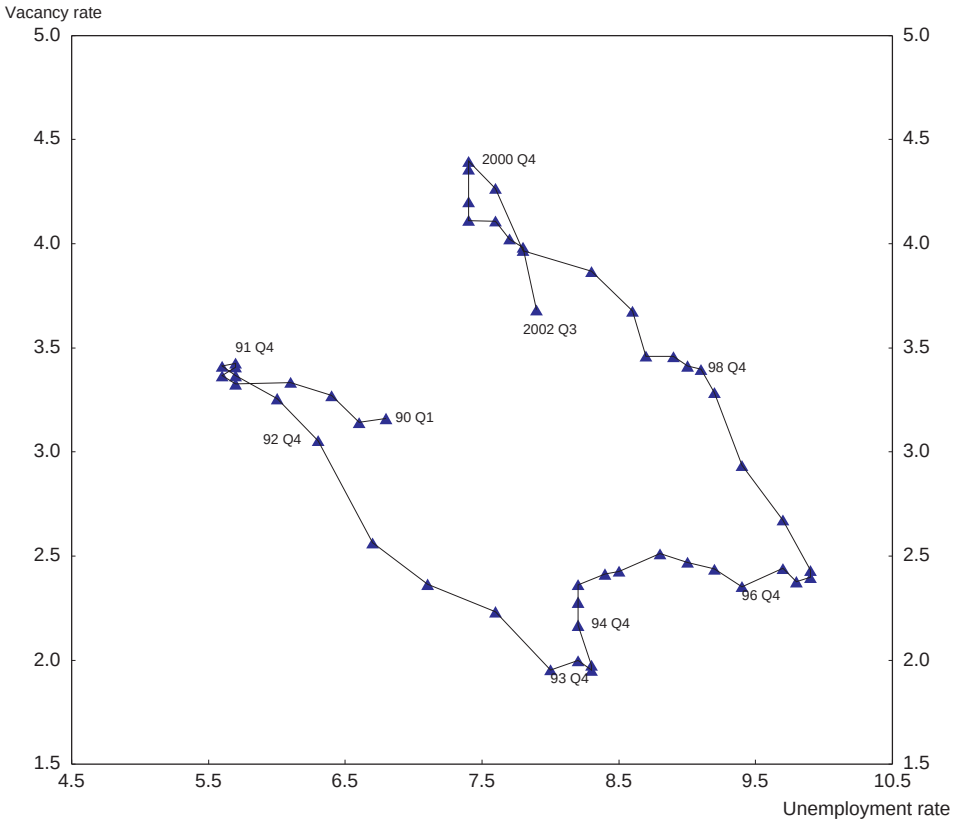
Figure 13. **Beveridge curve**



Source: OECD.

Despite the worsening labour market outlook, there was in 2001 and early 2002 a decrease in long-term unemployment, which reflects among other things a more focused use of active labour market programmes (ALMPs) as per head enrolment in the programmes halved since 1999, although the number of long-term unemployed have since started to increase. Also, increased placement efforts built into the *Job-Aqtiv* law, in place since the beginning of 2002 (see Chapter IV), has induced an increasing number of registered unemployed to abstain from re-registering, and this contributed to the decline in the labour force. In particular, since the act was implemented 45 000 unemployed aged 58 years or over declared themselves as non-available for the labour market (while maintaining unemployment benefit). More generally, the number of people retiring before the standard retirement age of

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Table 6. **The German labour market**

Thousands of persons

	Germany				Eastern Germany			
	1999	2000	2001	Change Jan.-April 2002 *vs.* 2001	1999	2000	2001	Change Jan.-April 2002 *vs.* 2001
Unemployment								
Registered unemployment[1]								
Total	4 100	3 889	3 852	205	1 344	1 359	1 374	36
Aged below 25 years	429	428	444	55	138	156	159	13
Aged above 55 years	948	842	714	–112	291	258	221	–37
Unskilled	1 482	1 396	1 386	–	288	292	284	–
Long term	1 417	1 375	1 285	24	428	456	474	45
Handicapped	193	184	171	–15	36	38	38	–3
Foreigners[2]	610	548	530	36	52	52	52	7
Effectively retired older unemployed[3]	200	192	228	232	89	82	86	87
Early retirement	4 728	5 043	5 281	–	1 160	1 253	1 322	–
of which:	1 859	1 846	1 818	–	504	502	494	–
Unemployed Handicapped/reduced employability	110	109	106	–	18	19	20	–
Women	1 962	2 206	2 406	–	555	641	710	–
40 years work[4]	797	882	951	–	83	91	98	–
	Levels				Levels			
Labour market programmes[5]								
Subsidised jobs[6]	364	325	273	251	267	238	197	175
Training courses	386	381	368	364	152	151	147	148
Short-time work	36	33	42	80	11	11	12	22
Job seekers	5 268	4 901	4 817	5 020[9]	1 833	1 789	1 767	1 787[9]
Receivers of unemployment benefits	1 938	1 694	1 725	–	722	647	625	
Receivers of unemployment assistance	1 563	1 457	1 478	–	584	607	648	
Employment								
Labour force	41 343	41 752	41 930	41 916[10]				
of which: Employed	38 081	38 706	38 773	–	6 451	6 396	6 287	–
of which: Small hour jobs[7]	3 805	4 064	4 100	–	398	447	474	–
Fixed-term contracts[8]	2 842	2 744	2 740	–	–	–	–	–
Part-time contracts	3 773	3 939	4 103	–	610	618	633	–

1. Definition of Federal Labour Office (Bundesanstalt für Arbeit).
2. Including immigrants of German origin.
3. Covering unemployed older than 58 years, receiving unemployment support without being registered as job seekers (paragraph 421 SGB III); 2002 figures based on budget plans of the Federal Office.
4. Retirement due to sufficient years of contribution.
5. Programmes subsidised by the Federal Labour Office; number of persons calculated as equivalent to unemployment; 2002 figures based on budget plans of the Federal Labour Office.
6. Including general ALMP, structural adjustment measures (*Strukturanpassungsmassnahmen*).
7. Jobs with not more than 15 hours per week ("325-Euro-Jobs").
8. Excluding apprentices, internships, persons in vocational retraining.
9. Covering the period January to October 2002.
10. Covering the period January to September 2002.

Source: Institute for Labour Market Research; Federal Labour Office, Federal Statistical Office.

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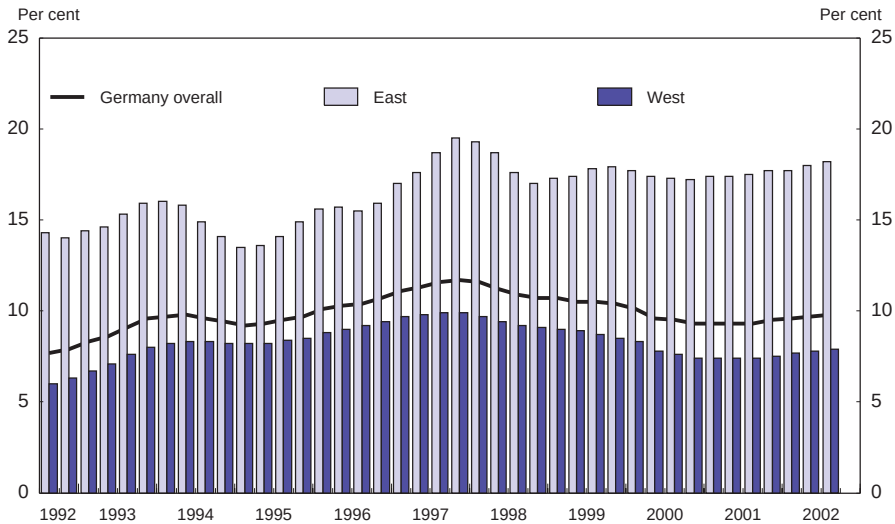
Figure 14. **The registered unemployment rate**



Source: Federal Labour Office.

65 years has risen, by more than 1 million between 1999 and 2001. A strong demographic effect pushed the number of employees with full retirement rights into the age brackets where they could enter retirement before the standard retirement age of 65 years, increasing early retirement for these groups by nearly 600 000.[20] Moreover, unemployed and to a lesser extent handicapped have taken up various early retirement possibilities, accounting for nearly ½ million people.[21] Furthermore, the subsidised part-time employment programme for older workers ("*Altersteilzeitgesetz*") has reduced the effective labour market participation of older employees. The programme can either be taken as a reduction of daily or weekly working time or as early withdrawal from the labour market.[22] Annual take-up of the programme has increased strongly since 1999 and should have passed 40 000 in 2001.[23]

Wage developments in 2001 were to a large extent determined by the two-year wage agreements concluded the previous year. Collectively bargained wages rose by about 2 per cent and mainly as the result of the stagnation in the growth of small hours jobs, the negative wage drift was small. Average wage growth fell short of consumer price inflation by about ½ per cent. With productivity growth slowing in 2001, hourly unit labour costs increased by 1½ per cent after having fallen the year before. The new wage round, concluded during spring 2002, implies wage increases of between 3 and 3½ per cent in the first year and somewhat lower increases in the second year of the contract period. As a result, wages accelerated in the second

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half of 2002. There should be a considerable negative wage drift, however, in that companies are crediting voluntary bonus payments against the bargained pay rise.

Monetary conditions, inflation and competitiveness

Consumer prices (as measured by the harmonised index of consumer prices) increased by nearly 2½ per cent in 2001, up by about ¼ percentage point as compared with the year before (Figure 15). The higher inflation reflects a sizeable overhang and increases in indirect taxes in the beginning of 2001 that added some 0.4 percentage point to the inflation hike.[24] Further upward pressures on prices

Box 1. **The economic performance of the new states**

The convergence process between the new and old states in terms of GDP per capita continued to stall in 2001, as economic growth in the new states (excluding Berlin) was basically flat. The feebleness of economic expansion can to a large extent be explained by the continued contraction of the construction sector. If construction investment had remained unchanged, the eastern German economy would have expanded *ceteris paribus* by some 3 per cent as compared with a similar estimate of about 1¼ per cent for the total German economy. The fastest growing sectors in east Germany were manufacturing and agriculture, while the service sector expanded at a more modest pace, reflecting mainly continued reduction of public employment. The manufacturing sector benefited from strong external demand for goods as well as a 2¼ per cent decline in unit labour costs, on account of modest wage increases and high productivity growth. Labour productivity increased 68 per cent of the west German level. Exports grew by double-digit rates to almost all export destinations and appear to have benefited from the high content of automobiles. The strong export performance lifted the export-to-GDP ratio from 11¼ to 12¾ per cent.

Reflecting the weakness of aggregate economic growth in 2001, the labour market situation in the new states remained subdued with employment contracting by 1.7 per cent. Non-subsidised employment including commuters to the old states decreased by nearly 1 per cent, while subsidised employment fell more sharply as the use of ALMPs was reduced by more than 10 per cent. The expansion of small hours jobs continued, but at a slower rate. Since 1999, small hours jobs in the new states have grown twice as fast as in the old states, although their share in overall employment is still about a quarter below that in the old states. In 2001 the labour force declined by about 2 per cent as early retirement increased and the weak labour market situation led to a discouraged worker effect. Reflecting these trends the registered unemployment rate remained at around 17½ per cent. In 2001 and into 2002, the distribution of unemployment changed somewhat as unemployment among older workers declined and the number of the long-term unemployed increased.

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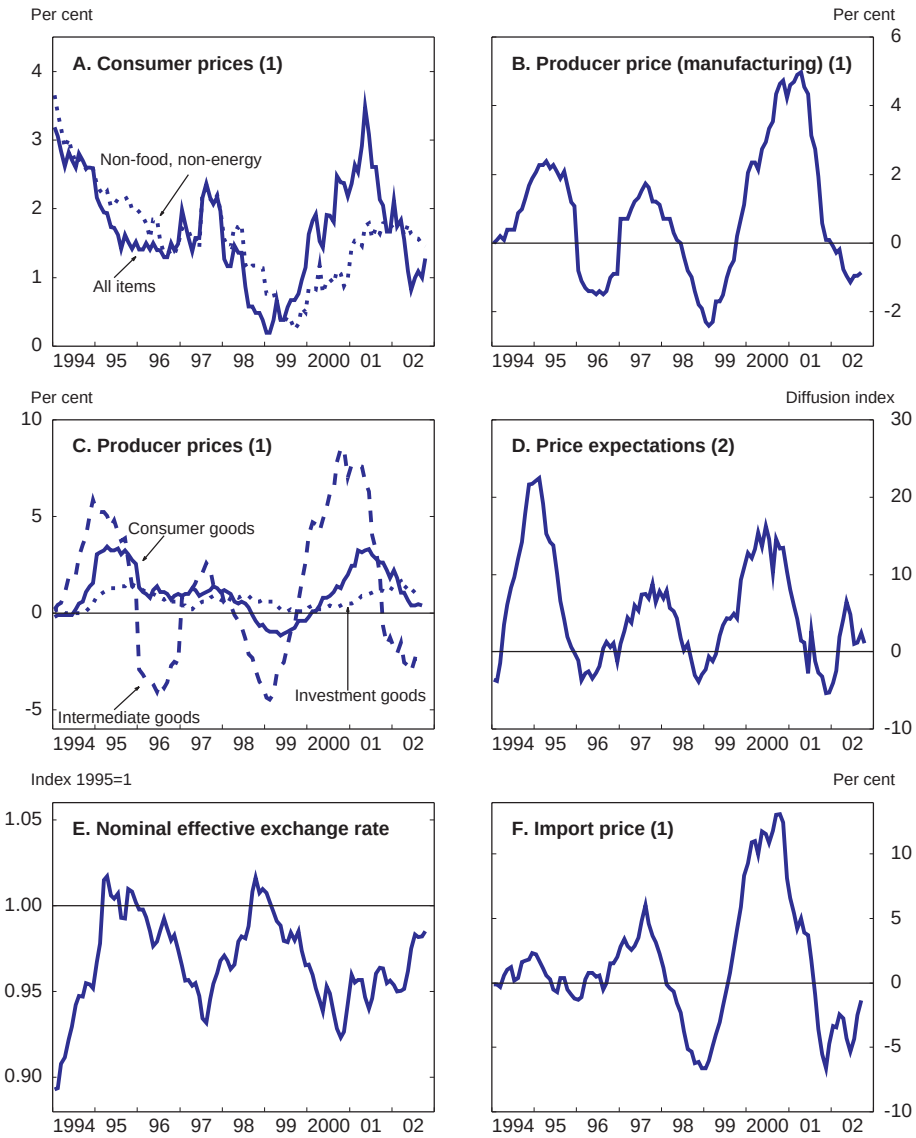
Figure 15. **The inflation environment**

Percentage change



1. Percentage change over 12 months.
2. Balance of higher and lower expectations. Seasonally adjusted.

Source: Ifo Institut für Wirtschaftsforschung; Deutsche Bundesbank and OECD, *Main Economic Indicators*.

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came during the spring from higher food prices in the wake of the outbreaks of the foot-and-mouth disease and BSE, leading to a peak in inflation of just above 3½ per cent in May.[25] However, the inflation rate came down to around 1 per cent over the summer 2002 as oil prices came down in the second half of the year, food prices stabilised and the base effects of previous year's indirect tax increases disappeared. Since then it has risen to about 1¼ per cent. Core inflation (excluding food and energy items) was more stable and subdued, increasing from about 1¼ per cent in early 2001 to just below 1½ per cent after mid-2002. German headline inflation declined from a level above the EU average in 2001 to among the lowest in mid-2002, partly on account of more subdued core inflation.

Amid substantial fluctuation of the exchange rate in 2001, the euro depreciated some 3 per cent compared with its average value the previous year *vis-à-vis* the US dollar. This was more than offset during 2002 as the euro appreciated by about 10 per cent (Figure 6). On the other hand, the real effective exchange rate (based on the CPI) in autumn 2002 equalled roughly the level prevailing in early 2001. This reflects the fact that the appreciation of the nominal effective exchange rate was offset by somewhat higher consumer price inflation among Germany's trading partners. A similar picture arises in terms of relative unit labour costs.

Monetary conditions eased between spring and autumn 2001 as the European Central Bank lowered its policy rates by an accumulated 150 basis points in response to the weakening of economic activity in Europe and subsequently as part of the international relaxation of monetary policy following the September terrorist attacks in the United States (Figure 16, Panel A). The European Central Bank lowered its policy rates by another 50 basis points in late 2002. Market rates came down until end-2001, when they began to rise, but these increases were more than reversed during the summer of 2002. This left long-term rates, as measured by ten-year government bonds, in autumn 2002 about ¼ percentage point below the level at the beginning of 2001 (Figure 16, Panel B). In Autumn 2002 the yields of ten-year US government bonds fell to ¾ percentage points below the yield of German instruments, revising the sign of the spread prevailing in 2001. Rates on shorter market instruments have declined somewhat faster than long-rates. Consequently, there was a significant downward shift of the yield curve, leaving market rates across all maturities at their lowest level since 1999 (Figure 17).

In line with the slow-down in economic activity there has been a sharp reduction in real credit growth, affecting lending to all sectors of the economy (Figure 18, Panels A and B). Some concerns have been raised as to whether this development constitutes a situation of credit rationing. Indeed, banks appear to assess risks more carefully, responding to the increase in bankruptcies by 20 per cent between 1999 and 2001. On the other hand, a weakening credit demand corresponds to low economic growth.[26] Indeed, real interest rates have increased between mid-2001 and 2002 if deflated with the rise in the consumer price index (Figure 19). The

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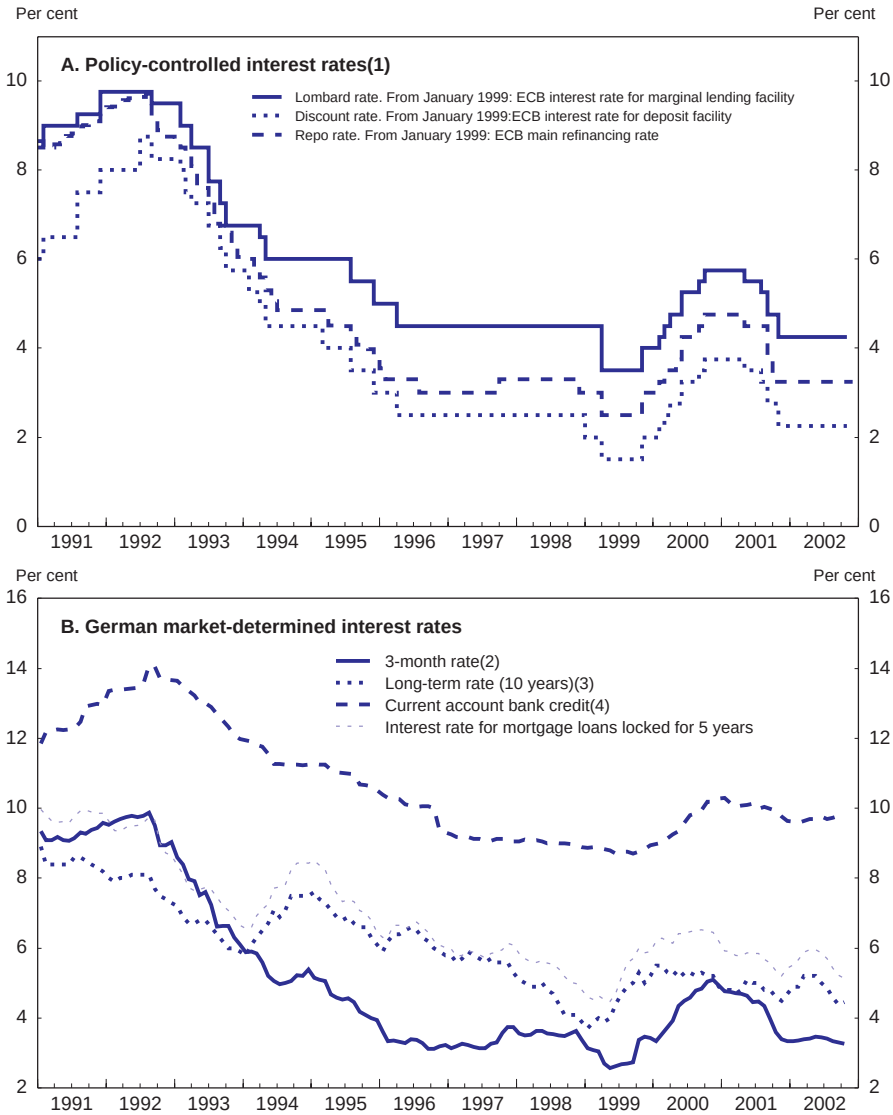
Figure 16. **Interest-rate developments**



1. End of period.
2. Prior to January 1999: FIBOR, thereafter: Euribor.
3. Yield on government bonds outstanding.
4. Credit of less than euros 500 000.

Source: Deutsche Bundesbank.

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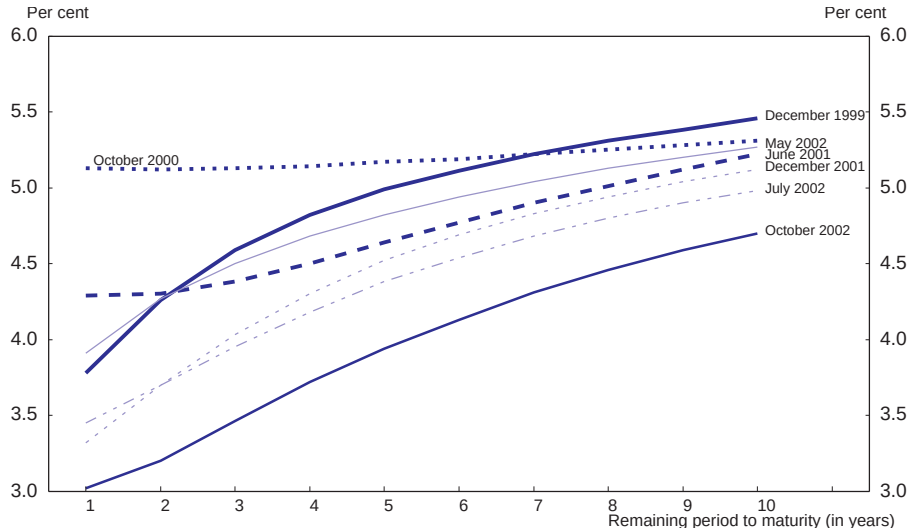
Figure 17. **Yield curves**



Source: Deutsche Bundesbank.

increase is less pronounced if a measure of inflationary expectations is used as the deflator.

Short-term prospects and risks

With activity remaining weak in the second half of 2002, average growth for the year as a whole is likely to be around ½ per cent. Although higher wage growth and lower inflation are supporting real disposable incomes, private consumption will remain subdued well into 2003 as consumer confidence is low and unemployment is projected to remain at high levels through next year. World trade is expected to recover next year, and accelerating exports will be the main driving force for higher growth in both 2003 and 2004. Construction investment will be temporarily boosted in the first half of 2003 on account of the flood relief measures. The relief package is assumed to leave overall activity unaffected, however, because higher spending is to be financed by temporary tax hikes. Consumption will strengthen as employment ceases to fall, and the recovery will be supported by income tax reductions in 2004. Rising foreign and domestic demand and increasing capacity utilisation should lead to strengthening investment in machinery and equipment. All in all, GDP should in 2003 grow at its potential rate (currently estimated at around 1½ per cent). As the upswing broadens in 2004, growth is projected to increase to some 2½ per cent (Table 3). These projections are those of

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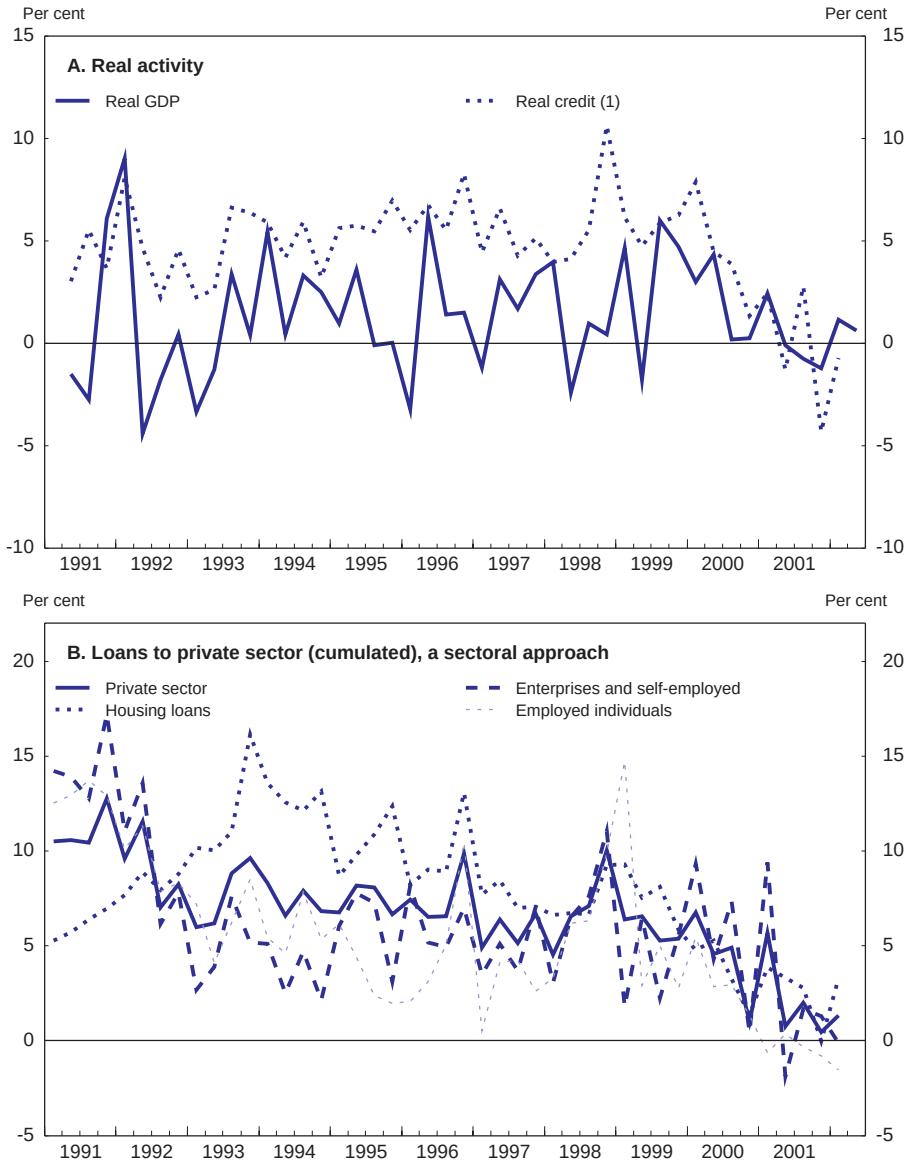
Figure 18. **Credit growth to private sector**
Percentage change over previous period (saar)



1. Loans to private sector (cumulated) adjusted by GDP deflator.
Source: Deutsche Bundesbank and OECD, *National accounts.*

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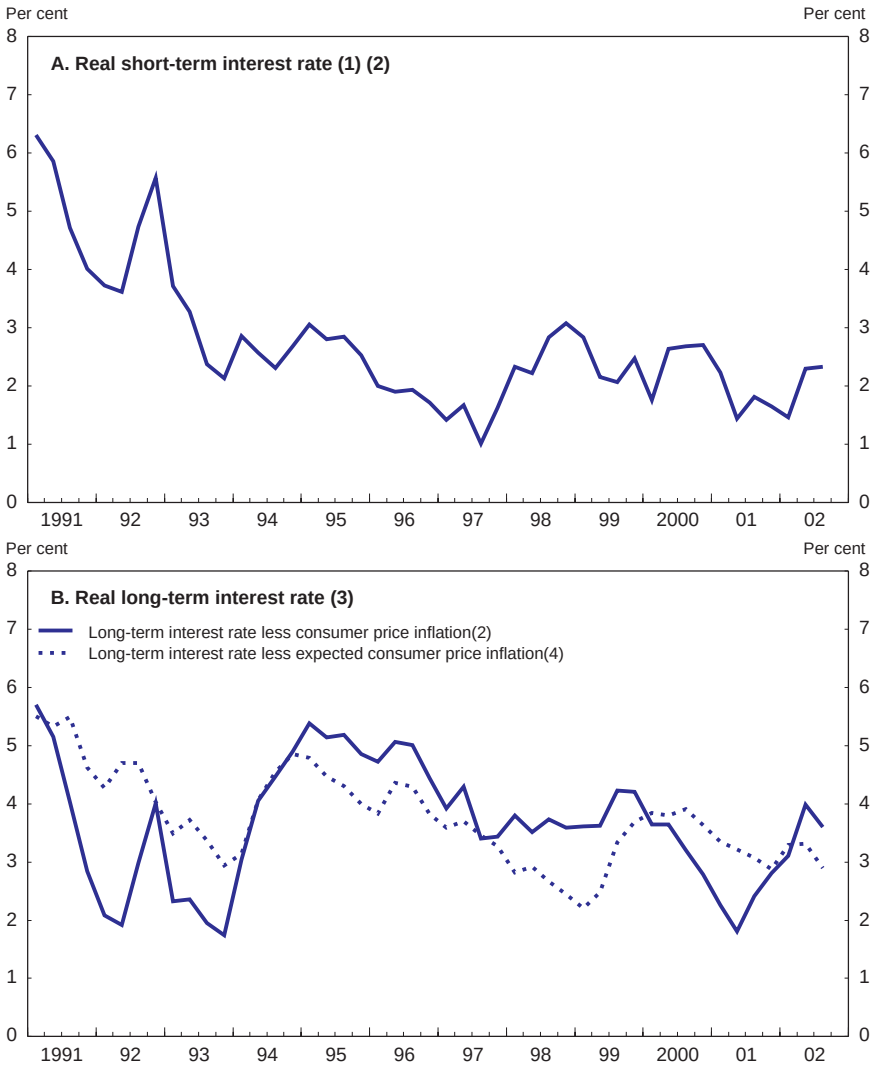
Figure 19. **Real interest rates**
Quarterly average in per cent



1. 3-month interbank rate. Prior to January 1999: FIBOR, thereafter: Euribor.
2. As measured by the 4-quarter percentage change in the consumer price index.
3. The nominal rates refer to yields on listed federal securities with residual maturity of 9 to 10 years.
4. As measured by Consensus expectations for consumer price inflation.

Source: Deutsche Bundesbank and OECD, *Main Economic Indicators*.

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the OECD *Economic Outlook* 72, and do not incorporate the tax increases of the government's recent consolidation programme. Growth may turn out lower if deficit reduction is implemented mainly via revenue increases.

Risks to these projections largely arise from the uncertainty surrounding the pace of the recovery of world trade and the time needed for consumer and investor confidence to return. In this respect, monetary conditions – including a projected further 50 basis points cut in short-term interest rates – should remain supportive for GDP growth. However, the recent sharp fall in German inflation to among the lowest in the euro area means that real short-term interest rates risk dampening the recovery of demand in Germany to an extent, which may not be compensated by the corresponding gain in competitiveness. On the other hand, if labour market reform were to deliver significant results, confidence and activity might recover more strongly than projected.

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II. Fiscal policy

Fiscal consolidation is a key priority on the policy agenda. However, since consecutive tax reductions over the last couple of years were not financed by proportionate cuts in government spending weak economic growth sufficed to lift the general government deficit to 2.8 per cent of GDP in 2001, overshooting the consolidation target by a wide margin and giving almost no scope for the automatic stabilisers to operate. In early 2002 the government revised downwards its GDP growth projection for 2002 to ¾ per cent, from the 1¼ per cent assumed in Germany's Stability Programme to the EU of December 2001. Correspondingly, the government raised the 2002 deficit target for the general government by half a percentage point to 2½ per cent of GDP while maintaining the balanced budget target for 2004. Against this background and to avoid an early warning by the EU Council of Economics and Finance Ministers (ECOFIN) in order to prevent the occurrence of an excessive deficit, the federal government made a commitment in February 2002 to tighten the path of budgetary restraint. This obligation comprised the government's endeavour to ensure that in 2002 the 3 per cent Maastricht deficit limit would not be breached, to closely monitor the budget at all levels of government, to utilise any budgetary scope to reduce the deficit, and to reach a budgetary position close to balance in 2004. In autumn 2002 growth prospects weakened further and the official tax projections were substantially revised downwards. Consequently, the federal government revised upwards its estimate for the 2002 general government deficit to above 3.5 per cent of GDP, and tabled to Parliament a supplementary budget. With the deficit likely to exceed the limit of the EU Stability and Growth Pact by a wide margin, the EU Commission took steps to ascertain whether the deficit is excessive according to the rules of the Pact. At the same time, the government is designing a consolidation package to reduce the deficit-to-GDP ratio to below 3 per cent in 2003 and to balance the budget in the medium term.

This chapter reviews recent budgetary developments and assesses the need for continued fiscal consolidation against the background of slow economic growth and announced tax and spending measures. Balancing the budget in the medium term will need to entail budgetary consolidation involving structural measures, an issue picked up in the next chapter on public sector spending reform.

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General government finances in 2001

Fiscal policy was expansionary in 2001, with new business and income tax reductions having been phased in. The structural deficit increased by almost ¾ per cent of GDP. In total, the general government deficit came in at 2.8 per cent of GDP, some 1½ percentage points above the level of 2000 (net of the receipts from the auctioning of Universal Mobile Telephone Service licences in 2000, which amounted to 2½ per cent of GDP). Accordingly, the deficit overshot the federal government's target laid down in the Stability Programme of October 2000 by 1¼ percentage points. About half of the deterioration of the fiscal balances was unexpected, caused by higher social transfers and revenues foregone on account of slower economic growth and delayed effects of previous tax reforms.[27] The debt to GDP ratio fell, despite the substantial increase in the deficit. This is mainly due to accelerated debt redemption out of the receipts from the auctioning of the (UMTS) licenses in 2000, which has been spread over 2000 and 2001 (Table 7).

General government finances in 2001 have been heavily affected by the 2001 step of the income and business tax reform.[28] Associated revenues foregone are estimated to total more than 1 per cent of GDP. Moreover, there were substantial unforeseen losses in tax revenues on account of both changes in taxation

Table 7. **Public sector financial balances**

Euro billion

	1996	1997	1998	1999	2000	2001[1]
General government (Maastricht basis)	–62.7	–51.0	–42.8	–29.7	22.8	–57.6
(Excluding UMTS proceeds)					–28.0	
Per cent of GDP	–3.4	–2.7	–2.2	–1.5	1.1	–2.8
(Excluding UMTS proceeds)					–1.4	
of which:						
Territorial[2] (authorities)	–62.3	–48.1	–28.8	–26.9	–33.7	–47.3
Federal government	–40.1	–32.4	–28.9	–26.2	–23.9	–22.9
State governments	–23.8	–19.5	–14.3	–8.3	–10.3	–25.9
Local governments	–4.5	–2.8	2.2	2.2	2.0	–3.9
Federal funds[3]	6.4	6.7	12.2	5.4	–1.7	5.0
National accounts adjustments[4]	6.2	–4.0	–16.7	–8.0	6.7	–7.6
Social security	–6.6	1.1	2.7	5.2	–1.0	–2.6
Cyclically-adjusted, per cent of potential GDP	–2.5	–1.7	–1.4	–0.9	–1.5	–2.4

1. Preliminary data and Government estimates.
2. Until 1997 including hospitals. The sum of the individual items differs for some years from the summary entry for the territorial authorities due to discrepancies in statistical reporting.
3. Economic Recovery Programme Fund, Financial Equalisation Fund, Compensation Fund, Railway Fund, Coal Fund, German Unity Fund, Inherited Debt Fund.
4. Including lending operations, privatisation receipts, timing adjustments, and, until 1995, profits paid by the Bundesbank into a debt sinking fund.

Source: Ministry of Finance and OECD.

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parameters stemming from earlier reform and weak economic activity. Corporate income tax receipts collapsed completely. Mainly, this was the result of tax refunds implied by earlier reductions of corporate tax rates for retained profits. The timing when such refunds would materialise was largely uncertain and the effect in 2001 was therefore unforeseen. Roughly one-third of the loss was recovered in terms of higher revenue from the capital revenues tax (*Kapitalertragssteuer*). VAT receipts, representing the largest single tax item in Germany, also came in weakly. Subdued consumption growth should explain most of this shortfall, but VAT fraud also appears to have reduced revenues. On the other hand, revenues from the Bundesbank turned out to be larger than expected.

The deficit of the states and communities more than tripled, equalling roughly the level of the federal deficit (Table 7). Substantial shortfalls in income, value added and business taxes – all of which are shared between the different layers of government – were the primary reason for this development. On the spending side, outlays for personnel – accounting for 38 and 27 per cent of total spending by the *Länder* and communities, respectively – increased only moderately reflecting further reductions in government employment. Spin-offs of units from the budget into the enterprise sector are contributing to this effect. However, outlays for retired civil servants rose substantially. For the communities higher interest payments, associated with rising debt levels, exhibited the largest percentage increase among individual spending items. In the social security system weak growth in revenues in combination with accelerating outlays on some items, in particular spending on pharmaceuticals, let the health funds slide into a deficit amounting to some 0.1 per cent of GDP.

The 2002 budget

The original federal budget for 2002 envisaged an increase in net borrowing by € 1.3 billion (Table 8). While some spending items and tax allowances are set to expand, spending restraint in other fields and revenue raising measures were envisaged to contain the deficit. Higher transfers to the federal railway fund account for a large share of the rise in spending. Transfers to the fund had been exceptionally low in 2001 due to the fund's one-off receipts from real estate sales.[29] Additional expenses on defence and security measures implemented in response to the terrorist attacks in the United States, and higher outlays in favour of road and railways infrastructure, are other major items increasing spending. Rising outlays for pensions are also reflected in the federal budget. They comprise phased transfers to the public pension system, and higher retirement outlays for former civil servants including retired postal workers.[30] Moreover, the phased increase in financial support for newly concluded funded pension contracts sets in this year, although outlays remain subdued in 2002. As in the years before, federal government employment is scheduled to decline by 1.5 per cent, remaining a major contributor

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Table 8. **The Federal Budget**
Euro billion

	2001		2002		2003
	Budget	Outcome	Budget	Supplementary Budget	Draft Budget
Expenditure	243.9	243.2	247.5	252.5	247.9
(Per cent of GDP)	11.8	11.8	11.7[1]	12.0[1]	11.4[1]
of which:					
Personnel	27.0	26.8	27.1	27.1	27.1
Operating expenditure	17.0	18.5	16.1	16.1	17.3
Interest	39.4	37.5	38.9	38.9	38.1
Current transfers	130.9	132.4	38.4	143.3	140.0
Investment	7.2	6.9	6.8	6.8	6.9
Other capital expenditure	23.1	20.9	18.7	18.7	20.3
Revenues	221.5	220.2	223.8	215.2	228.6
(Per cent of GDP)	10.7	10.7	10.6[1]	10.2[1]	10.5[1]
of which:					
Taxes	196.6	193.8	199.2	190.7	202.4
Other current revenue	13.0		12.6	12.5	15.2
Capital revenue	11.8		12.0	12.0	11.0
Financial balance	22.4	22.9	23.7	35.5	19.5
(Per cent of GDP)	1.1	1.1	1.1[1]	1.7[1]	0.9[1]

1. GDP: estimates by OECD.
Source: Ministry of Finance and OECD.

to budgetary consolidation. Other main items of spending restraint comprise reduced subsidies for the coal industry and cuts in operating expenses. At the level of the states and communities, further reductions in personnel can also be expected. Moreover, in several states the ministers of finance have enacted administrative spending caps that require the approval of the minister for increases in discretionary outlays above a certain threshold. Spending restraint is likely to largely manifest itself in a further reduction of public investment.

In autumn 2002 weaker growth than expected and substantial downward revisions in the official tax projections led the government to revise upwards its general government deficit estimate to above 3.5 per cent of GDP and table a supplementary budget to Parliament. According to the latter the federal deficit will be some € 12 billion higher than in the original budget. Tax shortfalls account for almost two-thirds of this increase, and spending for unemployment related benefits for the rest. The damage implied by the flood in late summer 2002 also instigated substantial revisions to both spending and revenues plans of all levels of government. General government spending in 2002 is set to increase by a significant margin that could total some 0.1 per cent of GDP, although most of the measures will become effective only in 2003. The federal government and the states agreed on establishing a

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"reconstruction fund" totalling € 7.1 billion. The fund serves both the reconstruction of infrastructure at the federal, state and community levels as well as providing grants to enterprises and households affected by the flood. The outlays are planned to be matched entirely by additional income tax revenues, which accrue to all levels of government. For this purpose the reductions in statutory income tax rates originally scheduled for 2003 will be delayed by one year. Moreover, the corporate income tax rate – which was lowered in recent years – will be temporarily increased by 1½ percentage points to 26.5 per cent. Total public spending related to the flood is planned to exceed the volume of the fund, involving in addition a re-organisation of other spending items, notably on transportation, and aid from the EU.

While tax shortfalls mainly concern the corporate income tax and the value-added tax, general government revenues are subject to other forces, working in opposite directions. Indirect taxes – on tobacco, insurance, mineral oil and electricity – have been raised in January. Phased revenue-raising measures associated with the business tax reform, which were legislated in 2000, also became effective this year. Moreover, new measures came into force in January designed to combat VAT fraud. They consist of a bundle of administrative and legal provisions designed to narrow the scope for fraud, enhance the investigative powers of the authorities involved and strengthen prosecution.[31] On the other hand, revenues are also being depressed by a further increase in child allowances, implemented at the beginning of the year.

Several health insurers have raised their contribution rates at the beginning of 2002 so as to restore their balance after the rise in the deficit in the year before.[32] Measures became effective in July to dampen outlays for pharmaceuticals (see Chapter III below). However, in the first half of 2002 prices for pharmaceuticals continued to rise rapidly, and with the contribution base growing slowly, there is a risk that further increases in health contribution rates might become necessary to avoid a further rise in the deficit. The revenues from the phased increase in the energy tax – projected to yield some € 2½ billion additional revenues – are to be used almost entirely to finance pensions.[33] While in the past hikes of the same size helped to reduce the pension contribution rate – by 1.2 per cent of the wage base between 1998 and 2001 – in 2002 the additional revenues do not suffice to hold contribution rates constant. The government has therefore cut the required financial buffer stock of the statutory pension system (*Schwankungsreserve*) from one month's spending to 80 per cent, to prevent contribution rates from rising.

Overall, the Secretariat projects a deterioration of the general government structural balance in 2002 by ¼ per cent of GDP. With weak economic activity depressing revenues and higher unemployment putting upwards pressure on spending the overall deficit is likely to increase further to 3¾ per cent of GDP (after 2.8 per cent in 2001). Accordingly, the debt GDP ratio is projected to increase (Table 9 and Figure 20).

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Table 9. **Appropriation account for general government Including social security**
Per cent of GDP

	1999	2000	2001	2002[1]	2003[1]	2004[1]
Currents receipts[1]	44.6	44.4	43.0	42.4	42.8	42.5
of which :						
Direct taxes	12.0	12.5	11.1	10.8	11.1	11.0
Social security contributions	19.0	18.6	18.5	18.6	18.7	18.6
Indirect taxes	12.2	12.0	11.9	11.6	11.8	11.9
Property and entrepreneurial income	0.7	0.5	0.8	0.6	0.4	0.4
Other current transfers, receivable	0.7	0.7	0.7	0.7	0.7	0.7
Sales and property income	1.4	1.2	1.5	1.3	1.1	1.1
Current disbursements[1]	45.0	44.6	44.4	44.9	44.7	43.9
of which :						
Government consumption	19.1	19.1	19.0	19.0	18.9	18.5
Property income payable	3.5	3.4	3.3	3.3	3.3	3.3
Subsidies	1.8	1.7	1.6	1.5	1.5	1.4
Social security outlays to households	18.9	18.8	18.9	19.3	19.2	18.8
Other current transfers	1.6	1.7	1.6	1.7	1.8	1.9
Saving, net	–0.4	–0.2	–1.4	–2.5	–1.9	–1.4
Net capital outlays	1.1	–1.4	1.3	1.2	1.4	1.2
Gross investment	1.9	1.8	1.7	1.7	1.7	1.5
Net capital transfers received	–0.8	1.5	–1.3	–1.1	–1.3	–1.2
Consumption of fixed capital	1.7	1.7	1.6	1.6	1.6	1.5
Net lending	–1.5	1.1	–2.8	–3.7	–3.3	–2.6
Memorandum items :						
Net lending, Maastricht basis	–1.5	1.1	–2.8	–3.7	–3.3	–2.6
Cyclically-adjusted net lending (potential GDP)	–0.9	–1.5	–2.4	–2.7	–2.3	–2.1
Debt, Maastricht definition	60.9	61.2	60.2	59.5	61.7	63.0

1. Projections.
Source: Federal Statistical Office and OECD.

The 2003 budget

The commitment of the federal government to tighten the path of fiscal consolidation was underpinned in February 2002 by a decision of the Financial Planning Council (*Finanzplanungsrat*) – which co-ordinates fiscal policies between the various levels of government – to narrow the admissible expenditure growth of the federal government and the states and communities in 2003 and 2004. The agreement stipulates that over these two years expenditures by the states and communities must not grow by more than 1 per cent per annum while federal government spending needs to fall by 0.5 per cent each year. This would imply a considerable sharpening in comparison to the previous path envisaged. Indeed, Germany's Stability Programme of December 2001 foresaw that expenditures of the three levels of government would expand by 1.5 per cent on average. For the general government as a whole – with the social security system added in – the programme foresaw spending rising by 2 per cent.

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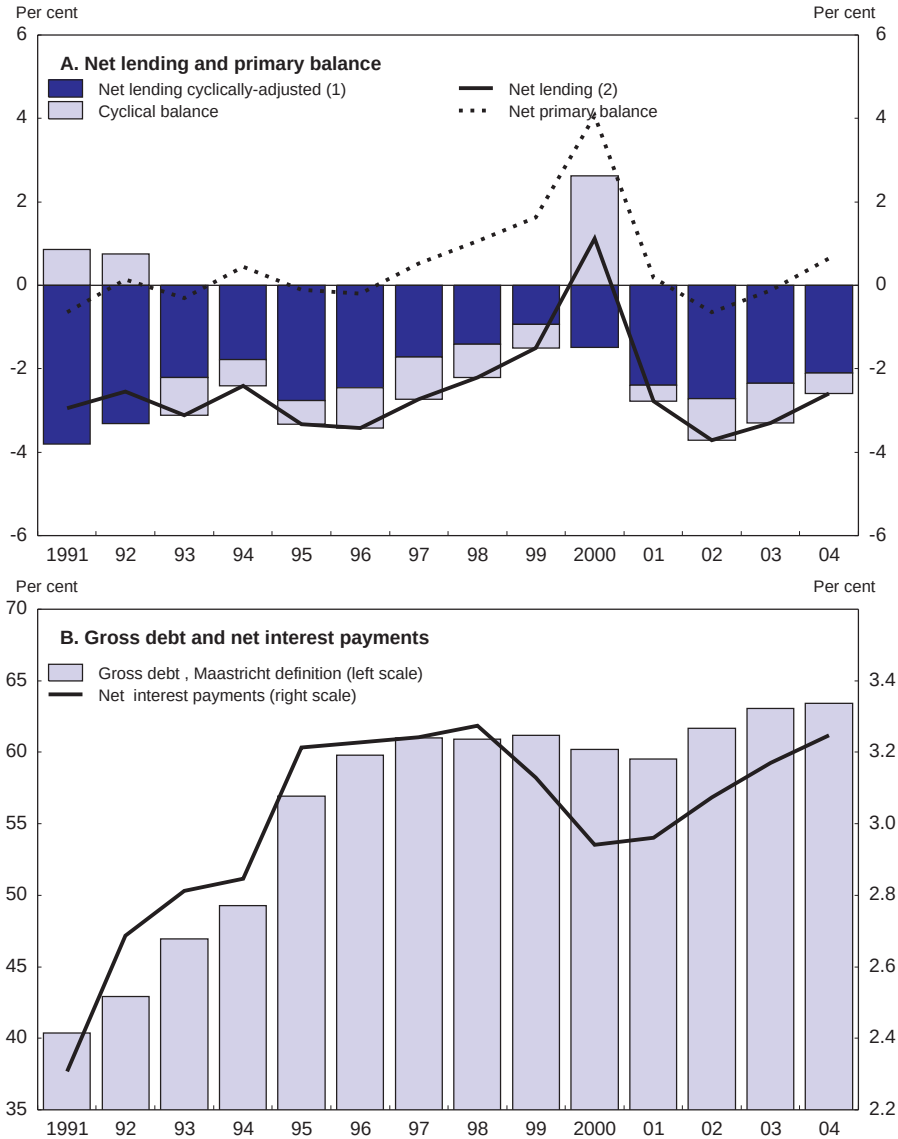
Figure 20. **General governement balances**
Per cent of GDP



1. Per cent of potential GDP.
2. Per cent of GDP.
Source: OECD.

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With the general government deficit in 2002 likely to exceed 3½ per cent of GDP the newly-elected federal government in autumn 2002 is designing a consolidation package that aims at reducing the general government deficit to a level below 3 per cent of GDP in 2003 and reaching balance in the medium term. The package largely consists of various measures generating higher revenues in income, business and indirect taxes. Mainly, this comprises reductions in tax allowances, *e.g.* for the construction of owner-occupied houses or the netting of business losses with profits, and broadening the base for the turnover tax. The 2003 draft budget of the federal government foresees a reduction in the federal deficit by € 16 billion. As in the year before, major spending increases are foreseen for pensions, including the phased rise in financial support for privately-funded pension schemes. Also, the federal government compensates the communities for the financial burden associated with the introduction of a new subsistence income scheme for pensioners, which is less strictly means-tested than social assistance transfers (see Chapter IV below).[34]

Other revenue raising measures, apart from those financing the reconstruction fund, comprise a further hike in the energy tax and the tobacco tax. Even with the additional revenues from the 2003-step of the ecological tax reform channelled into the pension system, pension contribution rates will increase for demographic reasons. Spending restraint on the level of the general government is set to continue in some areas, notably with respect to personnel and subsidies. Investment is also likely to be cut further, net of the flood-related outlays. However, to a considerable extent the expenditure caps agreed in the domestic Stability Pact would still have to be validated by pertinent legislation of the parties involved. While the financial stance of the health care system will partly depend on the effectiveness of cost containment measures implemented over the last couple of years, some increase in health contribution rates is likely to be necessary to keep the system in balance.

Under the assumption that the spending and revenue measures relating to the flood leave the deficit unaffected, and based on the recent trend to contain expenditures on all levels of government, the OECD projects that the general government deficit will fall to 3.3 per cent of GDP in 2003. As GDP is projected to grow at the potential rate in 2003, the drop in the deficit of 0.4 percentage point is fully structural. In 2004 the deficit is projected to drop further to 2.6 per cent of GDP, helped by accelerating growth (Figure 20). These projections – which are those of the OECD *Economic Outlook* 72 – do not incorporate the measures of the recent consolidation programme. If additional measures were fully implemented the 2003 deficit could drop below 3 per cent of GDP.

Medium-term objectives

The federal government has reinforced its commitment to balance the general government budget in the medium term. Further fiscal targets until 2005 concern reductions in the debt, tax and spending-to-GDP ratios (Table 10). As has

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Table 10. **Medium-term fiscal objectives for the general government**
Per cent of GDP[1]

	2001	2002	2003	2004	2005
	Baseline scenario				
Expenditure	48	48	46½	45½	44½
Tax and social charges	41½	42	42	41½	41
Gross debt (Maastricht definition)	60 (59.8)	60	59	57	55½
Budget deficit	–2½ (–2.8)	–2	–1	0	0
at real GDP growth	¾	1¼	2½	2½	2½
	Alternative scenario				
Budget deficit (weaker growth in 2002)	–2½ (–2.8)	–2½	–1½	–½	–½
at real GDP growth	¾	¾	2½	2½	2½
Budget deficit (weaker growth in 2002-05)	–2½ (–2.8)	–2½	1½	–1	–1
at real GDP growth	¾	¾	2¼	2¼	2¼

1. Targets according to Germany's Stability Programme of December 2001. Figures in brackets: outcomes in 2001.
Source: Ministry of Finance; OECD Secretariat.

been outlined in previous Surveys, several measures have been implemented to restrain spending. These include *inter alia* phased reductions of personnel at all levels of government, restraining the growth of pension outlays and reductions of subsidies. Also, the domestic Stability Pact, agreed in 2002, is an important step towards securing a higher degree of fiscal discipline across different levels of government (see Chapter III below). However, the objective of balancing the budget is unlikely to be met in the short run. In the same vein, with a projected deficit of around 3¾ per cent of GDP there is little scope for the automatic stabilisers to operate, and further negative growth shocks could easily push the deficit up further.[35] Also, scheduled tax increases are likely to dampen economic activity. While construction spending will be boosted, the deferral of the income tax reductions originally scheduled for 2003 and the increase in the corporation tax rate might introduce some degree of uncertainty about future tax policies with potentially adverse implications for consumption and investment.

Caution is warranted regarding the sustainability of a zero balance once reached. Substantial further income tax cuts are already legislated to be phased in from 2005. While such reductions are likely to be self-financing to some degree in the medium term, cuts in expenditures are necessary to stabilise the budget. The rapid increase of the deficit in 2001 and 2002 illustrates the budgetary risks associated with tax reductions that are not financed via cuts in expenditures. Similarly, the effectiveness of tax cuts with respect to stimulating economic activity is likely to be reduced if cuts are effectively deficit-financed or subsequent tax increases are to be expected to prevent the deficit from rising. Hence, further tax reductions – in addition to those that have already been legislated in the recent past –

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should only be implemented if they are embedded in a public sector spending reform reducing government outlays. Furthermore, larger-than-expected revenue losses and smaller revenue gains associated with already implemented or forthcoming income tax reductions and tax increases, respectively, cannot be excluded, given the large degree of uncertainty surrounding tax estimates. The unforeseen collapse of corporate income tax revenues in 2001 and the lack of its recovery in 2002 is an example. It is also not yet clear to what extent certain policy initiatives – such as the measures against VAT tax fraud and cost containment measures in the health care system – will be effective in generating savings. Also, some of the recent policy initiatives to restraint spending are one-off. This is true for administrative spending caps imposed by the ministers of finance of the B*und* or the L*änder*. In a similar vein, investment spending has been cut back over-proportionately at lower levels of government, raising questions about the adequacy of the structure of spending. Overall, one-off measures to eliminate the deficit over a short horizon would risk being self-defeating by further weakening domestic demand. Instead, further consolidation measures on the spending side are necessary to secure a balanced budget in a sustainable way and to create room for further tax cuts. Such measures need to be linked to structural reform. There is also scope to improve the efficiency of federal fiscal relations. Some of these issues are considered in greater detail in the next chapter.

Further efficiency gains, aiding the sustainability of public sector finances and increasing the growth potential of the economy, can also be realised on the revenue side of the general government budget. The tax reform measures implemented since 1999 mark progress in several respects. Statutory tax rates have been lowered and, at the final stage of the reform in 2005, the burden of personal and corporate income taxes will have been reduced substantially. Some measures have also helped to improve the neutrality of the tax system. This is notably true for the reduction in depreciation rates, which reduce the bias of the tax system in favour of capital-intensive production. Simulations indicate that the spread of taxing the returns of corporate investment, averaged over different types of investment and modes of financing, has declined from almost 40 per cent of the lowest rate before the business tax reform to around 30 per cent in 2002. Nevertheless, effective average tax rates for corporate profits remain among the highest within the EU.[36] Also, there remains ample scope to level the "playing field" for taxable activity. For example, the corporate tax system continues to favour debt financing over equity financing, and personal income taxes include preferences such as the deductibility from the tax base of wage premia for night work or commuting expenses. In the same vein, the German tax system remains extremely complex, and making it more transparent might free substantial economic resources.

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III. Issues in public sector spending reform

The German federal fiscal system is based on budgetary autonomy of the different layers of government while simultaneously attaching a high importance to realising broadly equal living conditions across the federation. The latter is reflected in both a high degree of co-operation and revenue sharing between the authorities and a generous transfer system to promote social cohesion. While the system has been successful in promoting a high standard of living even in regions whose economic capacity is low, tensions have emerged. F*irstly*, it has been difficult to ensure a sustainable fiscal policy, and the size of the overall deficit makes it difficult for the automatic stabilisers to operate freely. S*econd*, overlapping responsibilities between governments and the financing of public spending programmes via inter-governmental transfers provide individual governments with little incentives to contain costs and demand. T*hirdly*, social spending often discourages a higher utilisation of labour. Furthermore, public expenditure is going to have to cope with increasing pressure due to the substantial deterioration of the old-age dependency ratio projected for the next years and decades and the continuing fiscal burden associated with reunification.

After highlighting in the second section major structural forces shaping public spending, the remainder of this chapter presents options for public sector spending reform. The third section reviews policy initiatives designed to improve the effectiveness of resource allocation within the public sector. These include new public management methods, mechanisms of funding across government levels, and Germany's domestic fiscal stability pact. The last section examines important aspects of improving efficiency within selected areas, and in particular those that are attracting increasing attention in the current policy debate: improving the incentive structure in the social assistance system, raising the effectiveness of schooling and fostering efficiency within the health care sector. The section proposes options for further enhancing public spending effectiveness.

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Federal fiscal relations and main forces shaping public spending

Federal fiscal relations promote a high degree of equality in living conditions

Germany is a federal state, with the federal government (*Bund*), the states (*Länder*) and the communities (*Gemeinden*) in principle enjoying budgetary autonomy. In practice, however, fiscal autonomy of sub-central governments is highly restricted, and complex institutional arrangements are in place designed to promote inter-governmental co-operation, co-ordination and revenue sharing. While federal fiscal relations are largely based on the desire to create broadly equal living conditions across the federation, resource allocation often lacks discretionary flexibility and is subject to inefficiencies.

The federal government is assigned by the Constitution responsibility for matters considered to be of relevance for the whole country. The *Länder* are responsible for all tasks which are not explicitly attributed to the federal government. Major competencies comprise the education and university systems and health services. Also, administrative responsibilities belong to a considerable degree to the states. This holds, for example, for income tax collection, although the legislative power is in the hand of the federal government. The communities are attributed the right to control local matters such as road construction, school building, public transport and the provision of social assistance. However, in practice the situation is not so clear because responsibilities overlap. For example, all three layers of government and the social security system perform tasks with respect to health care: the federal government sets the legal conditions, the health funds finance operating costs for hospitals, the *Länder* finance hospital investment and regulate hospital capacity, and the communities provide local health care services. Table 11 provides a picture of the effective distribution of spending across levels of government.

Sub-central layers of government have very little power in raising own taxes, but in any case revenues from individual taxes are overwhelmingly shared between the various layers of government.[37] Superimposed on this primary system of tax sharing is a secondary redistribution system, deriving from the constitutional objective to broadly equalise living conditions across regions. This equity objective is achieved through a complicated redistribution system (see *Germany Survey* 1998 for details) involving both vertical transfers from the federal government to the states and horizontal transfers between the states. Similar arrangements exist between the states and the communities. Special vertical transfers from the federal government go to the new states and are devoted to foster the catch-up process of eastern Germany ("Solidarity Pact"). Moreover, certain spending projects are co-funded between the federal government and the *Länder* and the *Länder* and the communities. Net of tax sharing, transfers from the federal government to the states alone account for more than 1.7 per cent of GDP.

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Table 11. **Functional spending by layer of government, 1998**[1]
Percentage of total

	Federal government	Social security	States	Communities	All
Public security	10.2	–	60.9	29.0	100
Law enforcement	3.5	–	96.5	–	100
Schools and pre-schools	0.0	–	71.9	28.1	100
Universities	9.6	–	90.4	–	100
Other education	23.4	–	38.8	37.8	100
Research and development outside universities	69.7	–	28.3	2.0	100
Culture	3.4	–	48.2	48.5	100
Social security	18.9	71.0	4.3	5.7	100
Health, sports, recreation	15.3	–	38.5	46.2	100
Housing and area planning	7.5	–	25.1	67.4	100
Nutrition, agriculture	30.7	–	66.1	3.2	100
Energy, water, financial aid to enterprises	66.8	–	26.9	6.3	100
Transport, communication	44.0	–	28.0	28.0	100

1. Intergovernmental transfers netted out.
Source: Federal Statistical Office; OECD.

The states are also directly represented in the second chamber of Parliament (*Bundesrat*), involving them in the federal legislative process (Box 2). The major instrument for policy co-ordination between all levels of government combined is the Financial Planning Council (*Finanzplanungsrat*), comprising the Ministers of Finance of the Federation and the *Länder*, the Federal Minister of Economics and representatives of the communities. The council exchanges assumptions about the economic and financial development and makes recommendations on the budgets of the federal, state and community governments.

Main forces affecting spending

Spending grew rapidly in the first half of the 1990s, largely driven by new demands associated with German reunification (Figure 21). Social spending was boosted when the elaborate western German social security system was fully extended to the east, and work provision and training measures were established on a large scale to cushion labour shedding. The wage bill for civil servants surged, on impact due to a larger government sector in eastern Germany, and over the first years of the 1990s on account of buoyant wage increases.[38] Government investment accelerated sharply immediately after reunification. Inter-governmental transfers surged, both for investment and consumption purposes, and aid was extended to enterprises to cope with the comparative disadvantages associated with the transition of the eastern German economy. Large inflows into retirement, a secular trend to increase health

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Box 2. Basic legal principles for budget formation, execution and control

The federal government tables its draft budget to Parliament on an annual basis. After the first chamber of Parliament has decided on the budget, the Second Chamber (*Bundesrat*), which represents the federal states, has the right to demand changes to the budget or can reject it. Such objections, in turn, can be rejected by the first chamber with a two-thirds majority. As a rule, the states aim at influencing the budget process prior to its adoption by Parliament without rejecting it in the *Bundesrat*. Based on the legislated budget, each federal ministry is responsible for budget execution in its own domain. If the execution deviates from the budget law, the Minister of Finance is obliged to restore congruency, except in exceptional cases. To the extent exceptional spending over-runs exceed certain thresholds the budget committee of the Parliament needs to be informed.

An independent Court of Auditors examines whether the government's budget execution did conform with the pertinent legislation. In particular, the Court investigates whether resource allocation has not been wasteful. In the past the Court has highlighted spending programmes where this condition was not met. The Court's report provides the basis for parliament's decision to discharge the federal government for the respective fiscal year. A similar system applies to the federal states.

care spending, persistently high unemployment rates and repeated massive unforeseen shortfalls in tax revenues added to the stress on fiscal balances. In the process, social charges on wages surged, widening the tax wedge on labour (see Chapter I above). The general government deficit increased, peaking at a ratio of 3.4 per cent of GDP in 1996. Debt levels rose rapidly at all levels of government, in particular in the new states. Inherited debt stemming from the GDR contributed to this process.

Consolidation measures adopted in response by the federal government and the sub-federal authorities comprised spending restraint in a number of fields.[39] In particular, over the last decade government employment was reduced substantially. The strongest reduction occurred in the new states, although on a per capita basis public sector employment in the east is still significantly higher than in the west. This action led to a rough stabilisation of the nominal wage bill after 1995, but did not suffice to reduce it (Figure 21).[40] Governments also responded by cutting down their outlays for investment. This was particularly pronounced by the communities, which account for roughly two-thirds of government investment. Since 1994 the new states and communities are reducing their investment outlays dramatically, although the level per capita still exceeds the one in the west.[41] On the other hand, there have been phased reductions since 1999 in income

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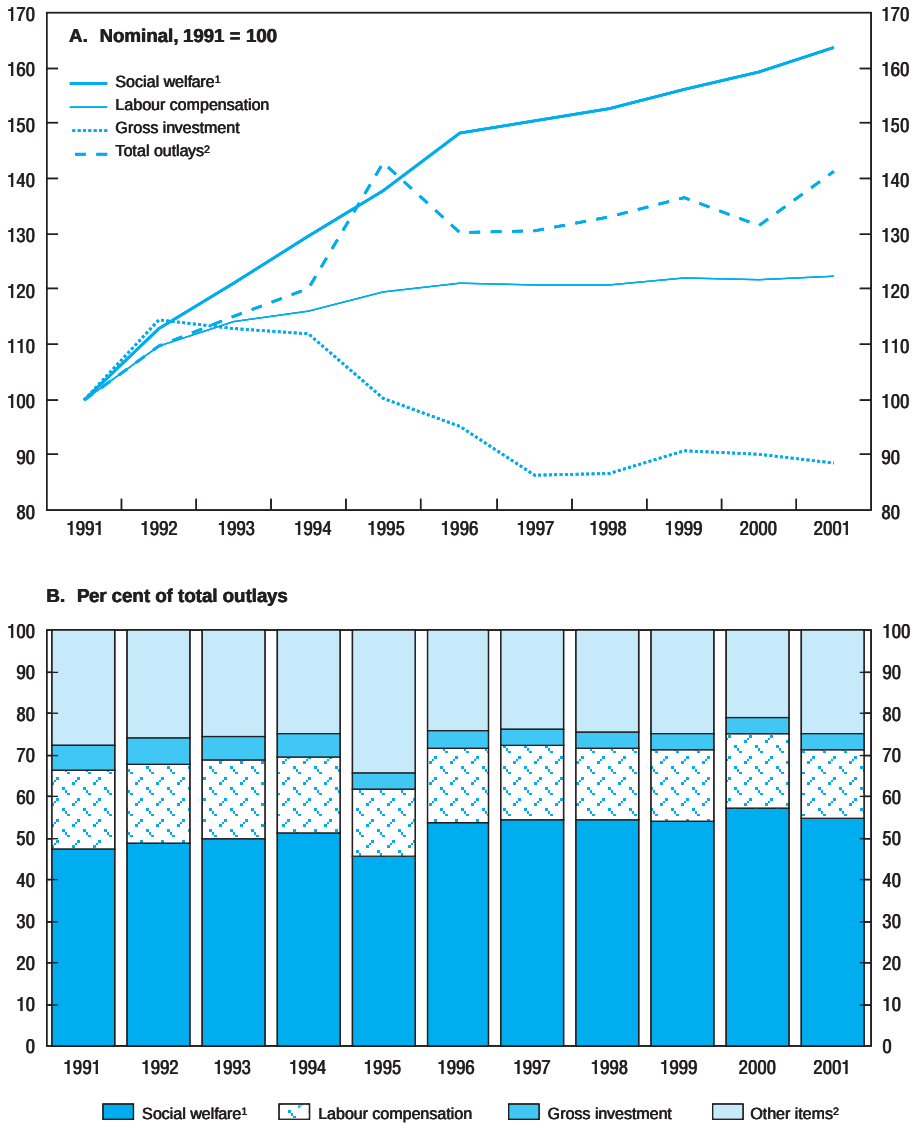
Figure 21. **Trends in general government spending**



1. Both benefits and benefits in kind.
2. In 1995 debt relating to unification was taken over by the federal government. This was recorded as a flow in the national accounts, boosting temporarily general government spending.

Source: Federal Statistical Office; OECD.

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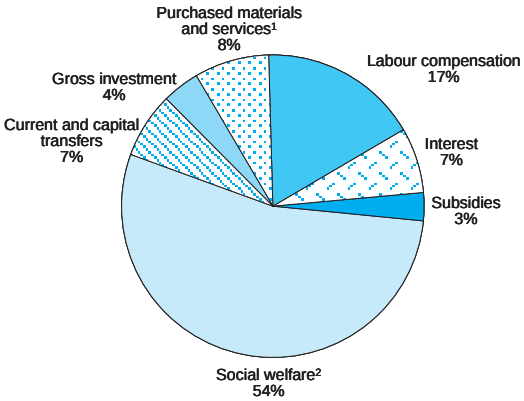
and business taxes, and these add to the budgetary pressure. In sum, at present the general government deficit comes close to the three per cent deficit limit of the Maastricht Treaty. Among the problems this presents is that there is little scope for the automatic stabilisers to work.

Hence, continuing the process of fiscal consolidation and improving the quality of government spending remains one of the most pressing policy challenges in Germany. Spending pressures are likely to persist, due to the continuing fiscal burden associated with reunification and the substantial deterioration of the old-age dependency ratio projected for the next years and decades. At present social welfare spending accounts for more than 50 per cent of general government outlays. Indeed, on a net basis (with taxes on social security transfers taken into consideration) social spending in Germany in terms of GDP ranks second in the OECD, behind Sweden. Some 17 per cent of the general government budget is spent for compensation of public-sector employees. Gross investment and subsidies account for 4 and 3 per cent, respectively (Figure 22). This composition indicates that for fiscal consolidation to be successful, streamlining social spending is unavoidable. In seeking areas for consolidation, public investment is too small to contribute much but, in any case, cutting here could have adverse consequences for the growth potential of the German economy. This is notably the case for the new states, where empirical studies suggest that substantial infrastructure investment is important to improve the framework conditions for sustainable growth.

Figure 22. **General government spending by type, 2001**

Per cent of total outlays



1. Net of the increase in non-produced goods.
2. Both benefits and benefits in kind.

Source: Federal Statistical Office; OECD.

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Managing public sector spending

Public resource management is becoming more flexible

Some changes have recently been introduced in the general government sector to improve the quality and cost efficiency of public services. Working time flexibility has increased in recent years. In the federal administration, working time can be shifted over days, weeks and months, depending on the tasks performed. The share of part-time employment has risen over the last decade. An element of performance-related remuneration has been established in some administrations, paying a premium on top of the basic salary. However, the system governing the remuneration of civil servants still precludes the functional and regional variation of wages, apart from a differential between the new and the old states.[42] These restrictions should be lifted to give the public authorities more flexibility in adapting the appropriate level and composition of the labour force and improve work incentives.[43]

New management principles are also being introduced with respect to cost accounting, although other OECD countries have gone further. Up to now budget plans account for revenue and expenditure flows only, without assessing the costs of administrative services. In the federal administration, an accounting framework for assessing and controlling the costs of services is being developed (*Kosten- und Leistungsrechnung*), and in 1998 new legislation on budgeting principles (*Haushaltsgrundsätzegesetz*) became effective allowing the federal and the states' administrations to record operating outcomes in terms of business-like balance sheets that also take into account the use of assets.[44] The introduction of a cost accounting and controlling framework for public sector services at various levels of government should be reinforced.

The federal court of auditors as well as the *Länder* courts in their annual reports have stated repeatedly that public sector spending has not sufficiently been subjected to evaluation. In response, the Minister of Finance now requires some form of evaluation of all administrative measures. The ambition to increase public sector spending effectiveness would be enhanced when outcome-oriented evaluation methods are applied. Within this framework the costs of a project are assessed against its impact on the targeted population. This entails estimating the costs and the benefits of the action taken over a long-term horizon. The case of applying active labour market measures (ALMPs) as reintegration instruments for the unemployed stands as an example for the potential usefulness of outcome-oriented evaluation methods. Job provision and training schemes have a long tradition in Germany, and are being applied on a large scale in eastern Germany since the beginning of the 1990s.[45] Although the immediate budgetary burden implied by these measures is high, regional labour offices have only recently been obliged to release statistical information on placements that allow some inference about the success of such schemes in terms of re-integrating the unemployed into

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the primary labour market.[46] More comprehensive micro data sets tracking participants in ALMPs are now being prepared that allow econometric evaluation of the effectiveness of different programme schemes. The usefulness of more systematic evaluation is illustrated by the investigations by the Federal Court of Auditors about the effectiveness of placement activities by the Federal Labour Office, which instigated a vigorous public debate about the Office's institutional structure and performance. As a consequence, a consensus has developed that far-reaching reform of the employment service is necessary (see Chapter IV below).

The need to base public sector resource allocations on outcome-oriented *ex ante* and *ex post* evaluation becomes particularly visible in the new states. Key aspects of the infrastructure endowment in eastern Germany have substantially improved, thereby removing bottlenecks to growth. Deficiencies remain, however, which could seriously affect the attractiveness of the new states as a business location.[47] This is true, in particular, for certain regional transport networks, for which the communities are responsible. Apart from establishing new networks there appears to be a need to rehabilitate existing ones that were run down at the time of the GDR and have not yet been replaced.[48] Early action would be necessary to avoid further decay that raises the overall fiscal burden for the communities. However, over much of the 1990s the new states and communities devoted a declining share of their revenues to investment. Moreover, the capital stock of the public sector was over-proportionately expanded in fields whose importance for fostering growth appears to be subdued, such as infrastructure for social and cultural services and the central administration. To remedy this situation, priority schedules for government spending need to be established that are based on assessments of future costs and benefits.

Funding principles do not always promote allocative and cost efficiency

The contribution of the federal government to co-financed activities totals almost 1 per cent of GDP (€ 19.2 billion in 2002).[49] Part of these funds are passed on by the states to the communities. In addition, the *Länder* co-finance projects of their communities with their own resources. Overall, co-financed spending accounts for a sizeable share of total public expenditures. Broadly, three schemes involving the federal government can be distinguished. "Joint Fiscal Tasks" (*Gemeinschaftsaufgaben*) are jointly decided by the federal government and all states, and cover fields such as university construction, subsidies for the purpose of improving the regional economic structure and financial aid for research and development. "Investment Aid" (*Investitionshilfen*) can be granted for major investment projects by the *Länder* or communities for the purpose of averting economic disequilibrium, equalising differing economic capacities or promoting growth. Finally, grants for the "Disbursement of Funds" (*Geldleistungsgesetze*) support social transfers by the *Länder* and may be implemented at the request of the federation (Table 12).

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Table 12. **Federal transfers to the *Länder* 2002**

Million euro

Co-financing schemes								Other compensation	
Joint tasks				Financial aid for investment		Disbursement of funds			
University construction	1 074	Education and research	556	Communal road construction and transport	8 421	Education benefits	997	Federal supplementary transfers	14 579
Regional economic structure	1 001			Construction of social housing	678	Residence benefits	1 707	Other items	4 358
Agricultural structure and coast protection	933			Modernisation of residential buildings	6	Other social benefits	213		
				Area development	399				
Total	3 008		556		9 502		2 917		

Source: Federal government; OECD.

In several respects co-financed schemes do not appear to be well geared to achieve allocative efficiency and to control costs. Sub-central governments are likely to opt for spending projects as long as the benefits perceived exceed their own – partial – budgetary costs, even if the total costs of the project are large relative to its social value. The incentive to act in this way increase the larger a share the federal government or a state assumes. As far as the federal government is concerned, in most schemes its financial contribution amounts to some 50 per cent of the project costs, but in some fields financial participation totals up to 90 per cent of the total. Moreover, joint participation of both the federal government and the *Länder* in local spending projects – notably in the area of city development – implies that the communities' own contribution is reduced to a relatively small fraction. To the extent co-financing is conditioned on a certain minimum size of a spending project, as is often the case, the incentives for overspending increase.

Similarly, splitted responsibilities across different administrations appear to seriously impede suitable programme evaluation and control. Auditors criticised that the different administrations focussed only on narrowly defined aspects, preventing comprehensive programme evaluation.[50] Differing objectives might further impede a coherent evaluation strategy. Simultaneously, co-financing has been found to be administratively costly with significant resources being devoted to the co-ordination process.

While in theory co-funding might be warranted to the extent that the provision of local services generates benefits for residents of other localities, in view of the high administrative costs and potentially serious adverse incentives the volume of co-funding should be scaled down. Funding of lower levels of government might be reached by revising their share in tax revenues. Such reform could include widening the tax raising competencies of local governments so as to achieve a higher degree of congruency between revenue and spending responsibilities. Efforts should also be made to internalise spill-overs of local services via co-operation between local governments or public institutions or the merging of local authorities.

To the extent co-financing is considered to be an important instrument in the future, the full costs associated with spending programmes in the selection, planning and evaluation stages need to be taken into consideration, and this includes financial flows over time. The fact that follow-up and maintenance costs associated with co-financed investment are not co-financed reinforces the importance of this comprehensive approach. For example, in one instance communities have been criticised by the responsible state auditor that their failure to properly maintain bridges implied more expensive renewal costs at a later state. While the former type of spending was not co-financed by the state, the latter was.[51]

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The Revenue Equalisation System and the Solidarity Pact with the new states

Reform of the inter-state equalisation system still leaves scope for further improvements…

New legislation, that will become effective in 2005, reforms Germany's inter-state fiscal equalisation system, which transfers tax revenues from the wealthier to the poorer states.[52] The initiative was a response to complaints by donating federal states that transfer obligations would unduly burden them. Moreover, the Federal Constitutional court in a recent ruling demanded clarification of the link between the degree of agglomeration and financial needs.[53] The system has also been criticised to produce disincentives for regional governments to develop their tax base.[54] Tax receipts of the communities will be given a higher weight in the new system, such that the states with relatively small revenues of their communities receive relatively larger equalising transfers.[55] Also, a number of measures have been legislated so as to reduce the transfer burden of the donating states and associated disincentives for policy makers to develop their tax base. In particular, a "premium" will be introduced designed to reward states whose increase in tax receipts – measured against the year before – exceeds the average revenue increase of all states combined.[56] The volume of vertical transfers by the federal government (*Bundesergänzungszuweisungen*) will be increased, securing that no state will be left with a less favourable position in the redistribution system.

The higher weighing of community taxes appears to be a more realistic assessment of the states' financial endowments. It is not clear, however, whether the introduction of the premium will have the desired incentive effects for regional policy makers. A permanent above-average increase in the level of a state's tax revenues – generated for example by a shift from public consumption to infrastructure investment – would imply a transitory premium for one year only. This effect might be too small to trigger a policy response. Generating lasting rewards would require lifting revenue growth – rather than the level – in a durable way above the average. Moreover, the fact that the assessed discrepancy between the financially strong and weak states has been widened, on account of the higher weight attached to community revenues, tends to increase the system's marginal effective taxation of revenues. Overall, the changes in the incentive structure should be relatively modest, and the system remains highly complex.

Hence, earlier recommendations to increase the incentive structure of the equalisation system remain valid. In particular, the federal government should make more use of equalising lump sum transfers. This would entail designing transfers for each state so as to fill part of the gap between the state's revenues and finance needs for a base year, without adjustment to reflect the actual growth performance over a multi-annual period.[57] This would allow reducing the rates that govern variable transfers offsetting the states' own tax revenues on an annual basis. Moreover, regional governments could be given more scope in generating own tax revenues.

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For example, they could be allowed to incorporate a surcharge into a reformed – and correspondingly reduced – income tax schedule.[58] For more local autonomy in setting tax rates to be meaningful the equalisation system needs to be adapted such that discretionary changes in local tax parameters are not neutralised by offsetting inter-governmental transfers. This suggests computing equalisation transfer positions on the basis of notional rather than actual revenue, disregarding changes in local tax rates. Some countries, such as Denmark, Finland and Spain have incorporated such provisions.

… and so does the new "Solidarity Pact" for the new states

The major share of federal government transfers to the new states is based on the "Solidarity Pact" (*Solidaritätspakt*, SP), which is in force since 1995. The SP has a volume of € 10 billion annually (DM 20.6 billion, 0.5 per cent of GDP in 2001) of which about two-thirds are available unconditionally and the remainder is earmarked for investment expenditures.[59] Initially, the SP was conceived to expire without replacement after ten years, at the end of 2004. But with growth in the new states stagnating at a low level and the economic convergence with the old *Länder* stalling, a consensus developed that special funding for eastern Germany should continue for the purpose of infrastructure investment. Eventually, a new arrangement – the Solidarity Pact II – was negotiated between the federal government and the new *Länder*, which forms a supplement to the inter-state fiscal equalisation system. The SP II will be effective for 15 years, coming into force at the beginning of 2005 and terminating at the end of 2019.

The SP II contains two baskets. Within the first one the new states obtain annual lump sum payments by the federal government (*Bundesergänzungszuweisungen*, BEZ), distributed across the *Länder* on a per capita basis (Figure 23). The purpose of the provision is to fund infrastructure investment and to compensate for the weak financial endowment (*kommunale Finanzkraft*) of the eastern German communities.[60] While the agreement specifies that the transfers be used so as to foster the recovery of the new states, funds are not formally earmarked for investment, in contrast to the corresponding provision of the present Solidarity Pact. Each state is obliged, however, to present for information annual "Progress Reports" to the inter-governmental Financial Planning Council that show the use of the funds and the state's fiscal stance. The second basket of the SP II refers to investment of the federal government out of its own budget. The government committed itself to target a spending volume of some € 50 billion between 2005 and 2019 in favour of the new states in terms of investment or further investment aid to the new states.

In the future, the scope for non-entitlement spending of the new states will be narrowed from both the revenue and the spending side. From the revenue side, the new states need to cope with the fact that transfers originating from the SP II are digressive over time. Also, the admission of new members to the EU

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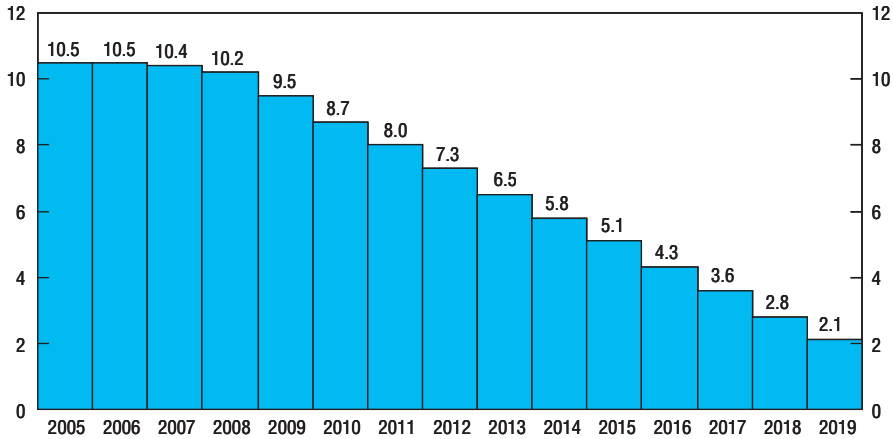
Figure 23. **The Solidarity Pact II with the new states**

Unconditional transfers to the new states, bln euro



Source: Federal government.

makes it questionable that aid from EU structural funds remain unabated.[61] From the spending side, there is a risk that certain categories of entitlement spending will play a larger role in the future than they did in the past. In particular, outlays for retired civil servants – whose share is small yet in comparison to the one of the old states – are set to increase.[62] This reinforces the need to re-orient the spending priorities of the new states and communities in favour of such projects that are most likely to reduce bottlenecks to GDP growth, notably in the area of transport. Immediate action along these lines is important: the capacity of the new states to generate revenues needs to be raised before the digressive phase of the SP II sets in. Hence, steps should be taken to incorporate into the SP II incentive mechanisms in favour of growth-enhancing investment. The following points could help in this respect:

– It is not clear whether the earmarking of transfers in the first Solidarity Pact has improved resource allocation in favour of investment. Giving it up is unlikely to be a loss. On the other hand, the Progress Reports required in the SP II are too general to support the efficient allocation of grants given to the new states for investment purposes. Instead, the new states should be required to make the Progress Reports a detailed instrument for policy analysis. This would entail, as a binding obligation, outcome-oriented *ex ante* and *ex post* evaluations of infrastructure investment projects that make transparent for what reason certain projects are pursued and others not. In this setting, funding by the federal government would be made conditional on the presentation of appropriate evaluation schemes rather than on the type of project chosen.

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– By its nature the SPII is only concerned with determining the transfers to the states as an entity, while a large share of the required public investment in infrastructure falls into the responsibility of the communities. Hence, provisions need to be made to insure that an adequate share of the SP II funds are passed on from the *Länder* governments to the communities. In the same vein, funding of the communities should be linked to outcome-related project evaluations.

– In the past, aid to the new states has repeatedly been extended, and resisting demands for aid appears to be politically difficult. Establishing a reliable medium-term finance framework that is incentive-compatible should embrace that transfers are phased out as agreed in the SP II. This is also important in view of the fact that the additional fiscal burden associated with Germany's demographic "population ageing" is projected to surge over the next 15 to 20 years.

The domestic Stability Pact is an important step towards a sustainable spending path

Given budgetary autonomy of the different layers of government, fiscal consolidation at the general government level requires inter-governmental co-ordination. This necessity is reinforced by the provisions of the Maastricht treaty, relating to the general government, to obey the 3 per cent deficit limit and balance the budget in the medium term. According to the constitution the budgeted deficit of the federal government must not exceed federal investment spending, and similar "golden rules" are also contained in most constitutions of the *Länder*. While the purpose of the rule is to prevent deficit spending that is not considered to be economically useful, it is not designed to cap deficits in absolute terms. Linking the deficit to government investment is also subject to some draw-backs, such as broad budgetary definitions of investment and the fact that more and more infrastructure investment is being made outside the government sector.[63]

The federal government and the states agreed on new legislation, which specifies that both parties, including the communities, should aim at balanced budgets. Moreover, the inter-governmental Financial Planning Council (*Finanzplanungsrat*, FPR) is attributed the tasks to make recommendations on how to accomplish or restore fiscal discipline and monitor whether the authorities' spending and budget balances evolve in line with the targets set and the requirements of the EU Stability Pact. Violations will not be subject to sanctions, however. Based on this legislation the FPR agreed on spending caps for 2003 and 2004, stipulating that the states (including the communities) limit expenditure growth to 1 per cent annually, and the Federal government reduces expenditures by 0.5 per cent (see Chapter II above).

The agreement is a significant attempt to incorporate the EU consolidation targets into Germany's federal fiscal system. Other federalist countries in

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Europe, such as Austria, Spain and Switzerland, have also introduced rules for the purpose to contain the deficits of the central and sub-central governments. It is an open question, however, whether without enforcement mechanisms commitments will be met. In Austria, whose federalist structure resembles Germany's in important respects, all three layers of government recently agreed on sanctions in case of a party's non-compliance with negotiated budgetary targets.[64] Moreover, the provision to aim at balancing the budgets in the medium term might not be specific enough to secure lasting discipline of the parties involved. This includes the risk that fiscal policies over 2003 and 2004 might be biased towards one-off measures, which do not link fiscal consolidation to public sector reform.

Hence, the FPR should promote lasting budgetary improvements based on public sector reform. Moreover, the agreement should be supplemented by a medium-term consolidation rule that comes into operation once the goal of balancing the budgets has been attained. Experience within the OECD indicates that in cyclical downswings a rule requiring budgets to be balanced might not be obeyed, which in turn can impair the credibility of the consolidation strategy. More appropriate would be a rule – as adopted in Switzerland by some Cantons and legislated for the federation – requiring the budgets to be balanced over the cycle. In view of the secular deterioration of Germany's old age dependency ratio structural surpluses to pre-fund ageing related spending increases should be aimed at. Such a rule would allow automatic stabilisers to work freely, but against the background of a reduction in the debt-to-GDP ratio in the longer term.[65]

For the implementation of the domestic stability pact the technical ability of the governments and the Financial Planning Council to monitor and control the evolution of the budgets is of high importance. A complex system of inter-governmental transfers tends to make it more difficult for each individual government to foresee the implications of financial interactions on revenues and spending. Hence, reducing the variety of inter-governmental transfer schemes – notably opaque programmes of co-financing – would raise the success chance of a domestic stability pact.

Efficient and timely controlling of both individual and consolidated budgets of all layers of government also requires that identical and appropriate accounting standards be utilised across all governments involved. While finance plans are based on identical standards, it should be secured that consolidated budgets by government level can be transformed into the national accounts standard, which is the reference for the EU stability pact.[66] This condition is not yet fully met. Provided budgetary accounting frameworks are standardised across the federation generating national accounts compatible information should not pose particular problems, with the possible exception of the phasing of particular revenue items. A base for the continuity of the budget process exists already at the federal level, where medium-term budget projections were introduced many

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years ago. Financial plans covering a horizon of five years (*Finanzplan*) are prepared every year together with the budget and include a considerable amount of detail. Multi-annual finance plans are also prepared by all state and local governments.

Selected spending programmes

This section outlines important pending issues in some selected areas that are increasingly attracting attention in the current policy debate: improving the incentive structure in the social assistance system, raising the effectiveness of schooling and fostering the efficiency of the health care sector.

Provision of social assistance can be made more incentive compatible

The present system of social assistance benefits has been attacked on grounds that it maintains high unemployment of low-qualified workers. The main argument is that high transfer withdrawal rates over a considerable gross income band make work at low wages not worthwhile, thereby preventing the creation of a low-wage sector. Moreover, overlapping responsibilities for the provision of social assistance on the one hand and – likewise means-tested – unemployment assistance on the other hand appear to hamper effective job placement.

Social assistance for income support (*Laufende Hilfe zum Lebensunterhalt*, HLU) is extended to all persons, working or not, whose income from other sources falls below a certain subsistence level.[67] At the end of 2000 some 2.7 million persons received HLU, of which some 37 per cent are not employed but classified as able to work.[68] The classification with respect to work ability rests in the first place on the recipients' own assessment. Hence, the share of persons able to work can be expected to increase if employment conditions improved and incentives for labour force participation were increased (Figure 24).[69]

HLU recipients are relatively low-qualified: More than 50 per cent do not have a vocational degree, and even more than 10 per cent have no school degree. Among those with a formal qualification basic education is prevailing. Hence, to employ social assistance recipients the wage distribution would have to be spread out sufficiently to accommodate their low productivity (Figure 25).

Overlapping responsibilities hamper efficient job placement

Some 10 per cent of social assistance recipients receive simultaneously unemployment assistance (UA), which also provides means-tested income replacement for unemployed persons whose eligibility for receipt of unemployment insurance benefits is exhausted. While both types of benefits are of potentially indefinite duration, UA is more generous and extended at a fixed replacement ratio of previous earnings. UA is administered by the Federal Labour Office and financed by the federal government while the communities are responsible for the administration

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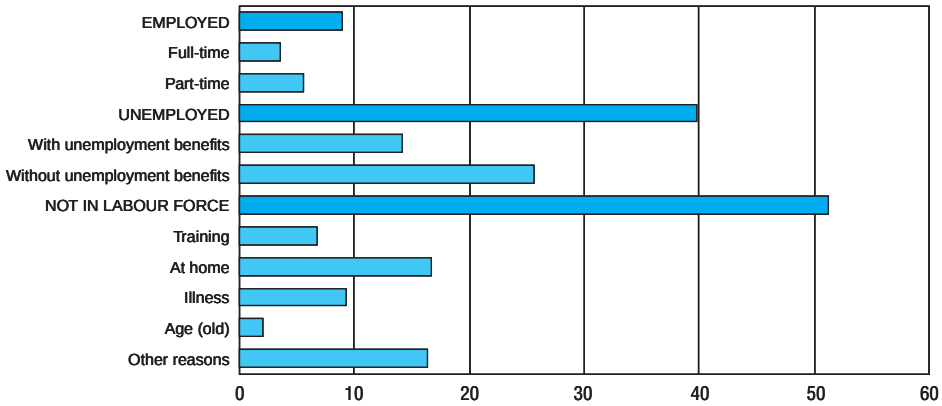
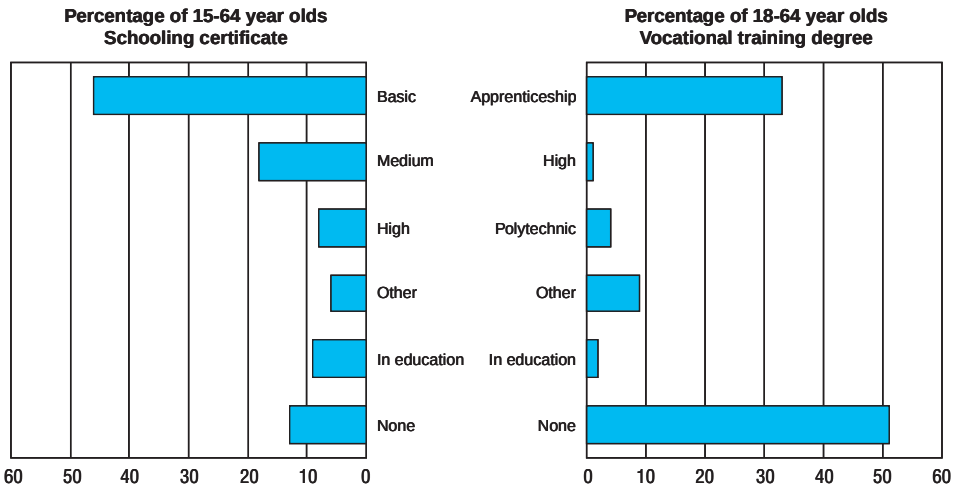
Figure 24. **The labour market status of social assistance recipients, end of year 2000**

Percentage of persons aged 15-64



Source: Federal Statistical Office.

Figure 25. **Highest educational attainment of social assistance recipients, end of year 2000**



Source: Federal Statistical Office.

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and financing of social assistance. The dichotomy of the two systems is associated with inefficiencies. Firstly, the fact that means-tested benefits, based on different rules, are processed by two different layers of administration increases administrative costs. It also raises the bureaucratic exposure of the recipients of both types of benefits, who have to deal with two separate administrations. Secondly, for both types of benefits the budgetary costs for each administration fall short of the full costs, which might have a negative impact on job placement efforts by the labour office on the one hand and the communities on the other hand. Finally, there is an incentive by each layer of government to shift the respective financial burden to the other layer. The communities have an incentive to induce participation of unemployed social assistance recipients in work provision schemes, offered by the federal Labour Office, which re-establish eligibility for unemployment benefits. On the other hand, by lowering unemployment assistance benefits the federal government can effectively shift part of its financial obligations to the communities.[70] The federal government has established a commission, which will make proposals about how the two systems can be integrated.

Overall, this suggests to combine unemployment assistance and social assistance into one single means-tested social assistance benefit, as it has been proposed recently by a task force on labour market reform commissioned by the government (H*artz* Commission, see Chapter IV). In fact, half of the OECD member countries do not have separate unemployment assistance benefits. To the extent that the provision of social assistance is left with the communities, the reform would imply that a large financial burden is shifted from the federal government to the communities, and this would have to be matched by a reallocation of tax receipts to the local level.[71] The mode of funding should preserve incentives for the communities to integrate the unemployed into the primary labour market. This suggests that the social assistance outlays should not be reimbursed via unrestricted transfers from the federal government. Instead, transfers could be put on a predetermined lump sum basis.

Incentives to take up employment are low…

Non-employed recipients of HLU, who are able to work are obliged to accept job offers, and non-compliance can be sanctioned by a reduction in benefits of 25 per cent. However, enforcement of this provision appears to be low while marginal effective tax rates associated with the withdrawal of HLU when earnings rise are high. Indeed, over an earnings range extending over more than € 700 HLU is withdrawn at rates between 85 and 100 per cent. As a consequence, in combination with other relevant provisions of the tax-benefit system, and without subsidised wages, net incomes of different types of families hardly increase with gross earnings over a broad income band relevant for low paid part-time and full-time occupations (Figure 26, solid lines).[72] Also, depending on family composition, the

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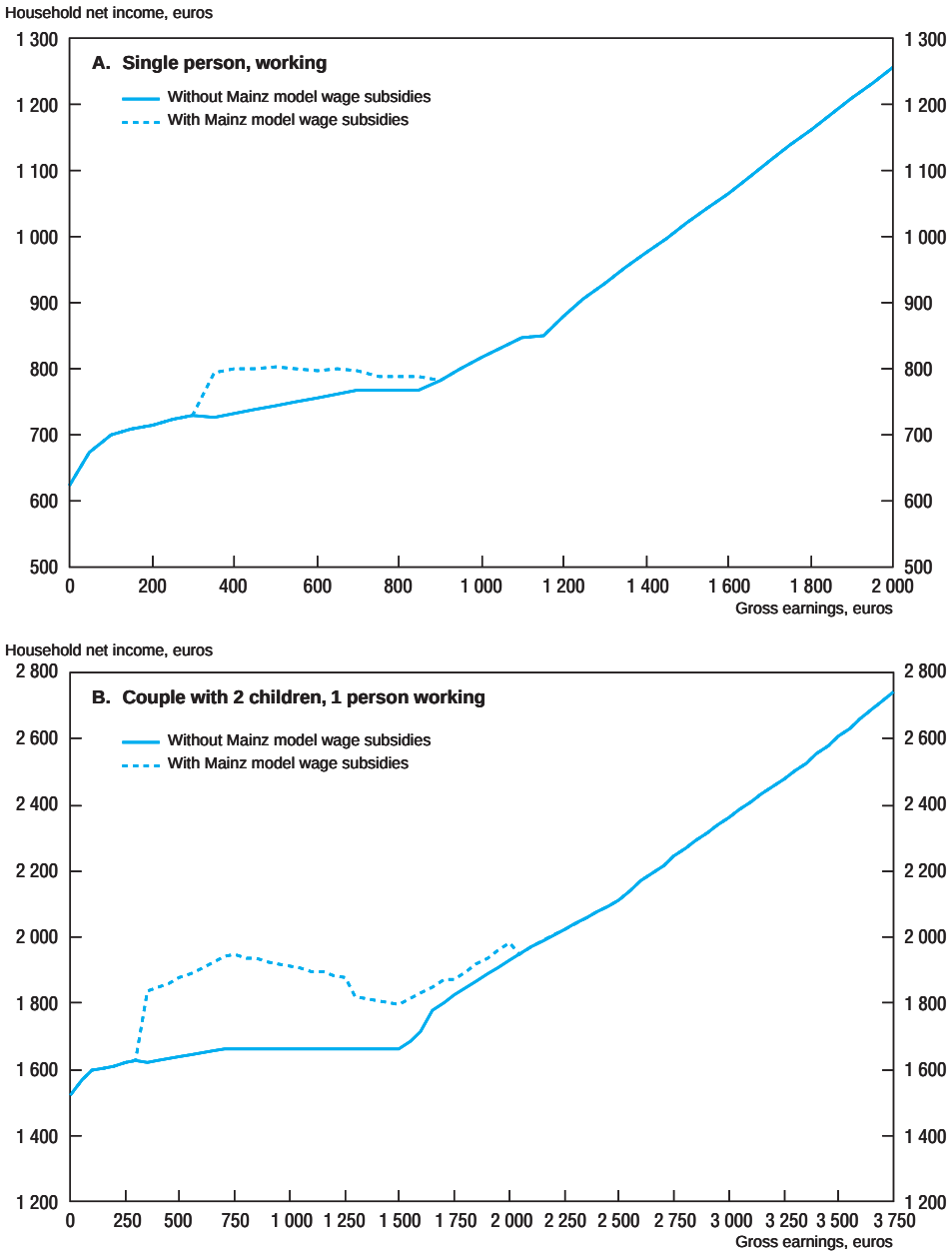
Figure 26. **Effects on net income of the tax and transfer system**



Source: Ifo Institute for Economic Research.

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level of social assistance benefits comes close to collectively bargained wages in certain occupations, notably within the service sector. Hence, there is little incentive to take up low-paid employment. Indeed, employment shares have been found small in collectively bargained wage groups that are relatively close to benefit levels.[73] Overall, these features are likely to define effective wage floors that prevent the wage distribution from spreading so as to generate significant employment opportunities for the low-qualified. Indeed, in contrast to other major economies the German wage distribution has not spread out, despite rising unemployment among unskilled workers.

To stimulate employment in the low wage sector, the government has recently introduced wage subsidies (MM) accruing to employees (see Box 3). The government has estimated that application of the MM could increase employment in the low-wage sector by some 20 000 to 30 000 persons.

For very low gross earnings the MM implies an increase in net pay, which should raise incentives to accept work in this income bracket. However, the scheme does not remedy the fact that the transition into employment of social assistance recipients is associated with very high marginal effective tax rates. On

Box 3. The Mainz Model for employment promotion in the low-wage sector

Since mid-2000 experimental wage subsidisation schemes have been introduced in selected regions in the new and the old states so as to promote employment of the low-qualified in the low-wage sector (see Germany Survey 2001). The "Saar Model", implemented in some regions of the Saarland and Saxonia, subsidises the employers' social security contributions. The "Mainz Model", implemented in some regions of Rhineland-Palatinate and Brandenburg, subsidises the social security contribution of the employees. Eventually, the Mainz Model (MM) was established – in a modified version – in Germany overall in March 2002.

Recipients need to work for at least 15 hours per week, at wages that are collectively bargained or meet the local norm. Employees who have been in regular low-wage employment prior to the introduction of the scheme do not qualify for the benefits, unless they change employer. Persons who have been in small jobs (jobs with less than 15 working hours per week) are eligible for the benefits if they make a transition into regular employment. Singles and married couples receive benefits over a gross earnings band spanning from € 325 to € 897 and € 1 707, respectively. Within this bracket benefits are gradually withdrawn. In addition, there are digressive child allowances of up to € 75 per month. The maximal period of support is 36 months. The scheme is set to be terminated in 2006.

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the contrary, the main shortcoming of the MM is that over a considerable income band it produces marginal effective tax rates that exceed 100 per cent. This is due to both the scheme's own high withdrawal rates and its interaction with the HLU schedule. Regarding the latter, the employees' social security contributions, including MM benefits received, are subject to the social assistance withdrawal schedule. For families with children, the reduction in child benefits adds to this adverse effect (Figure 26, dotted lines).

While persons who were already employed on low-wage jobs prior to the introduction of the MM do not obtain the support, they face an incentive to quit the present job and take up low-wage employment with another employer so as to qualify for the scheme. Past experiences with job substitutions associated with a regime shift in the taxation of minor employment indicate that such effects might be significant.[74] Hence, the introduction of the MM might lead to a re-allocation of labour, which could imply higher budgetary costs of the scheme than assumed by policy makers.

In view of the scheme's high marginal effective tax rates it appears not well suited to foster the transition into low wage employment, and should be modified. Instead, withdrawal rates of social assistance would have to be lowered substantially and/or wage subsidies would have to be introduced such that overall remuneration increases with gross earnings. Simultaneously, the implied fiscal costs would have to be limited. Both conditions are difficult to combine. For example, if the withdrawal rate of social assistance benefits were fixed at 50 per cent over the entire phase-out range, recipients with a guaranteed monthly income of € 1 500 would still receive benefits at monthly gross earnings totalling € 3 000 (€ 1 500 times the inverse of the withdrawal rate). This is more than average gross earnings in Germany.[75] Hence, means to reduce the effective taxation of labour at the low-wage end of the wage distribution need to be complemented by further reform of the tax and benefit system that fosters job search of the unemployed and dismantles barriers to labour force participation.[76] Important elements of such reform would be:

– Social assistance benefits should be cut substantially for recipients who are able to work but reject job offers. The budgetary effect of such a policy is likely to manifest itself not only in terms of lower outlays in case of non-compliance with the obligation to accept employment but also via reductions in the duration of non-employment spells. Indeed, empirical evidence elsewhere suggests that reductions in income replacing benefits imposed in cases of non-compliance with job search requirements can significantly reduce unemployment duration. There is also evidence that the credibility of implementing sanctions in practice plays an important role for the readiness to pick up employment.[77] In line with these findings, some OECD countries, including Denmark, Switzerland and Australia, have tightened eligibility criteria in recent years and closely monitor job search activities.

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- Unemployment assistance should no longer be granted indefinitely. Instead, it could be collapsed into social assistance once the two systems are integrated.
- Eligibility conditions for unemployment insurance benefits should be revisited in favour of fostering job search. At present, unemployed aged 59 years and older can receive unemployment insurance benefits over an extended period of three years without job search requirement. Effectively, this provision constitutes a measure for early retirement, and has been utilised by employers to reduce their labour force at the expense of the Federal Labour Office.[78] It should therefore be abolished. More generally, the appropriateness of extending eligibility periods for unemployment benefits with age needs to be revisited. In a similar vein, subsidies granted if older employees reduce their working time (*Altersteilzeit*) provide an inventive for effective early retirement and should be abolished. Also, the acceptability criteria in the unemployment insurance scheme should be tightened and non-compliance with job search requirements sanctioned (see Chapter IV below).
- The system of active labour market measures (ALMPs) needs to be revised. Empirical evidence indicates that general work provision and training schemes, while being costly, have not been effective in raising re-employment probabilities for the primary labour market and appear to be associated with significant lock-in effects.[79] Other problems include a considerable circularity between different programmes, potentially high dead-weight costs and crowding out of private sector activity. Moreover, efficient placement in training programmes has apparently been impaired by the vested interests of the providers of such schemes (see Chapter IV below). Hence, a large share of the work provision and training schemes presently applied should be phased out, which would generate substantial savings. Experience in other OECD countries indicates, however, that ALMPs can support the transition into the primary labour market if they are designed to foster job search and associated with intensive profiling and counselling of the unemployed as well as the monitoring of search activities.[80] Redesigning ALMPs in Germany along these lines should be combined with the scheduled reform of the Federal Labour Office (see Chapter IV below). Moreover, reform should also establish a high degree of congruency for each public sector authority involved between the financing of unemployment-related transfers and ALMPs. This would create financial incentives for the respective administration to foster programme designs that support best the transition into the primary labour market. In some OECD countries ALMPs are utilised by the communities to test the willingness to work of social assistance recipients. While such an approach might also be an option for Germany,

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it has been suggested that the communities should offer employment opportunities to all HLU recipients who are able to work.[81] However, such comprehensive provision risks creating another ALMP-industry under the umbrella of the communities, with similar adverse outcomes that have been observed with the present system.

Education

Secondary education

German spending per student on upper secondary education exceeds the OECD average by some 60 per cent.[82] However, educational outcomes – as measured in terms of basic competencies of students aged 15 years by the OECD international PISA study – fall significantly short of the OECD average.[83] Moreover, a domestic PISA study, conducted in winter 2001/02, shows a wide variation in students' average competence levels across the *Länder*, which are autonomous in organising the school system and determine the schools' curricula.[84] The study shows that the performance of the school system cannot be equated with the level of resources devoted per pupil. Indeed, the federal state that ranks lowest in all competence fields tested spends most per student. Overall, this suggests considerable scope to raise the efficiency of German secondary education.

The PISA results point to some institutional factors that appear to be relevant for increasing performance overall while limiting the impact on attainments of the students' socio-economic background. Many of the countries that performed well in international comparison have been shifting education policy and practice away from a focus on inputs – the resources, structures and content of schooling – towards a focus on learning outcomes. With this approach schools are attributed the authority for organising their own programmes and are held responsible for them. Also, in several well performing countries the extent of educational differentiation between schools is limited. In Germany schools are subject to tighter regulation than in almost any other country within the OECD. They have no freedom to select teachers and very little discretion in disposing of their funds. As a rule, differentiated schools group students by level of performance, and teaching is subject to detailed curricula leaving little scope to develop educational paths that might improve learning outcomes. Moreover, mandatory country-wide uniform educational targets against performance can be assessed have not been established.[85] The lack of uniform educational standards along with the high variation in average educational performance across states are also likely to impair the mobility of employees with children.

This suggests reform introducing a more outcome-oriented regulatory framework, which sets nation-wide standards for schooling attainments, to be tested on a regular basis. For this end centralisation is needed with respect to the determination of the targets. One option to achieve this would be to reallocate the

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competence to set such targets to the federal level. Simultaneously, schools should be given more freedom in determining their own programmes of how to improve learning outcomes. The flexibility across educational paths should be increased and more weight put on individual coaching of students. Incentives to deliver high quality education could be improved if the salary scheme for public sector employees were opened for performance-related remuneration, and schools obtained the right to choose teachers. Within this framework government funding of schools should be linked to educational outcomes relative to the benchmark.

Other features of the German school system suggest reform. Annual schooling time per year in Germany falls markedly short of the OECD average, in particular over the first years of schooling. This is largely attributable to the fact that the supply for full-day schooling is very limited, in contrast to several other countries, which score higher in the PISA study. For example, scheduled schooling per year for a nine-year-old pupil totals 752 hours in Germany while the OECD average stands at 829 hours. For youths aged 14 years, scheduled annual hours amount to 914 in Germany as opposed to the OECD average of 944.[86] The government recently decided significant financial aid to support the establishment of full-day schooling.[87] Also, the relatively low performance in Germany often occurs on the back of insufficient control of the German language. Regarding the socio-economic composition of the students, supplementary analysis carried out by the German PISA-*Konsortium* indicates that the high heterogeneity of the German population with respect to its immigration status is a primary factor relating to low scores.[88] The language deficiency is also associated with negative externalities for school children of non-immigrant families in the same classes. Compulsory language tests and compensating compulsory language training is therefore necessary to improve the general trainability of students with immigrant background. More generally, efforts need to be made to foster their integration into the school system.

Tertiary education

In recent years Germany's universities have increasingly come under criticism of being expensive while not well adapting to changing demands for new skill profiles. Indeed, cumulative costs per tertiary studies are in the upper range within the OECD. Effective durations of studies are relatively long, explaining most of the cost overhang (Figure 27). This is attributable to both the fact that up to now the statutory minimum duration for Germany's first tertiary ("type A") programme is relatively long and that the system contains few incentives for efficient delivery and utilisation of university services.[89] Competition between universities is hindered by federal and state laws and regulations, public universities are not allowed to charge fees (apart from a minor contribution), and the incentives for university teachers to deliver adequate teaching and research were weak once tenure has been granted. New legislation has been implemented that addresses

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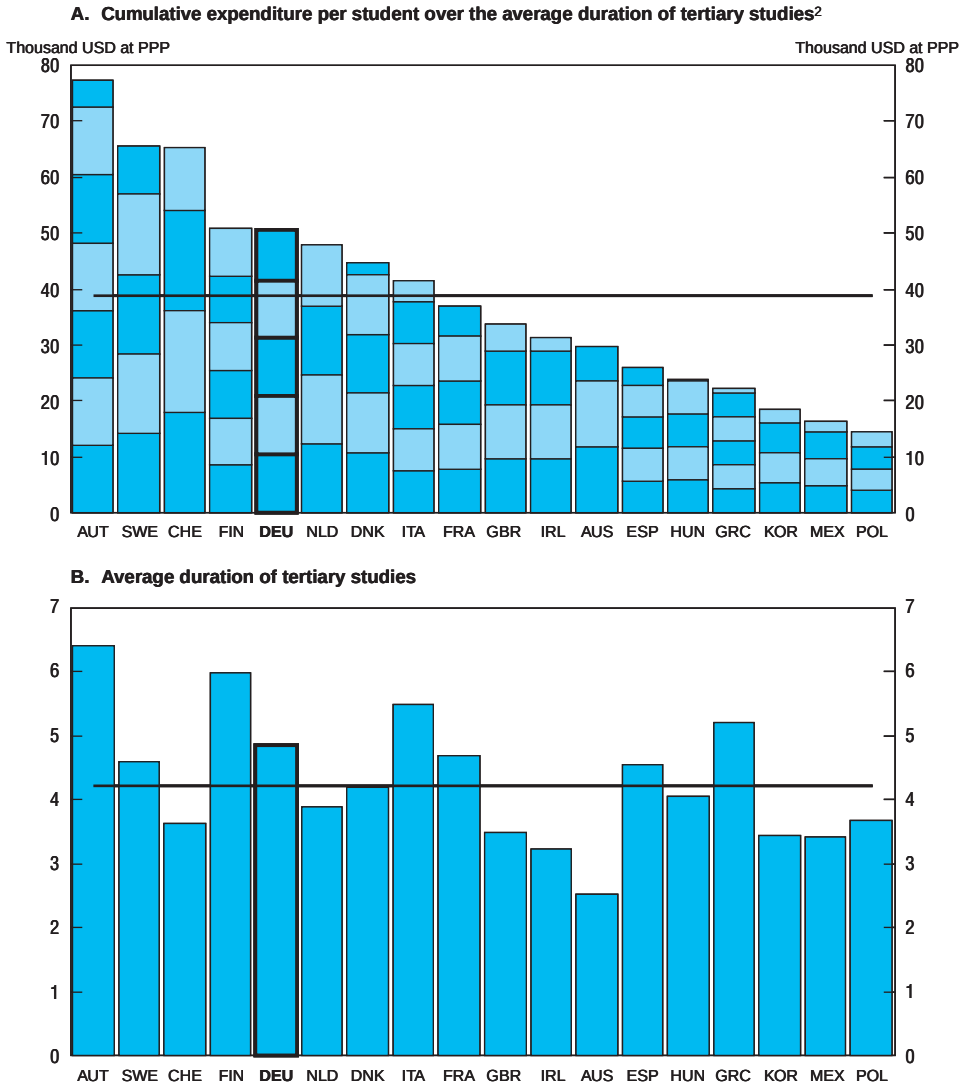
Figure 27. **Costs and duration of tertiary studies, 1999**



1. Countries are ranked in descending order of the cumulative expenditure per student. The horizontal line is the average of the countries in the graph. Public and private institutions except for GRC, HUN, ITA, POL, CHE which are public only. Duration of tertiary studies refers to the 1995 academic year for AUT, DNK, FRA, KOR, MEX, NLD, ESP, CHE.
2. Each segment of the bars represents the annual expenditure per student. The number of segments represents the number of years, on average, that a student stays in tertiary education.

Source: OECD, *Education at a Glance* 2002.

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some of these issues. Universities have obtained the option to offer bachelor and master programmes whose duration match the international standard. Moreover, performance oriented remuneration schemes have been introduced for university teachers, and the new carrier path of a junior professorship has been established.

While these measures are welcome steps in the right direction, further reform is required, to increase the allocational efficiency of the university system. Tertiary studies with a shorter duration should be introduced on a broad scale. Indeed, reducing the minimum length of tertiary education programs can go some way to improve the overall supply of labour with tertiary education. Within the OECD countries which offer only long first tertiary education programmes generally have significantly lower graduation rates than countries that offer shorter tertiary programmes.[90] Long study periods tend to discourage investment in education unless they are matched by proportionate gains in productive capacity that are reflected in subsequent remuneration. In general, the latter condition does not appear to hold, suggesting that studies may become less focussed once they are stretched. Indeed, within the OECD Germany belongs to the group of countries with the lowest (private) return on tertiary education, and this relates largely to the long duration of studies.[91] Moreover, reform should quickly move on implementing a higher degree of competition between universities. In this setting, universities should get the right to levy tuition fees. This would prevent students from considering tertiary education as a free resource and contribute to improving the financial endowment of tertiary education. Student fees would also serve to reduce potentially large subsidies to students with wealthier family background, who tend to choose education of longer duration. To mitigate potentially adverse distributional impacts for students stemming from low-income families, Germany's student support system could be suitably adjusted. Funding of universities out of the general government budget should be linked to the universities' performance, and this requires a re-enforcement of quality control.

Aspects of current health care reform

Health care spending in Germany accounts for some 10½ per cent of GDP, up from 6.3 per cent three decades ago. Outlays surged in the 1990s when the western German health care system was extended to the new states.[92] In addition, some segments of health care provision, such as the pharmaceutical sector, experienced recurring disproportionate growth. The introduction of the long-term care insurance also contributed to raising outlays as spending on some items were previously accounted elsewhere. The increase in spending implied a sharp increase in wage-based contribution rates over the last decades, which was reinforced by the comparatively weak growth of the contribution base in the new states and low growth in earnings since the second half of the 1990s. Between 1991 and 2002 alone contribution rates in the old and the new *Länder* increased from 12.2 and 12.8 per

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cent of the wage base, respectively, to 14 per cent. Since the beginning of the 1990s, the health insurance system exhibited the strongest increase in contribution rates within the social insurance system overall (Figure 28).[93]

Within the OECD Germany belongs to the countries with the highest spending-to-GDP ratio, behind the United States and Switzerland (Figure 28). Public health care spending amounts to eight per cent of GDP, mainly financed by a compulsory insurance system that consists of both public sector and private insurers, with the former covering around 90 per cent of the population (see OECD *Germany Survey* 1997). While the infrastructure of and the access to health services is excellent, various indicators of health status – such as "potential years of life lost", and life expectancies – place the performance of the health care system in the medium range within the OECD.[94] With utilisation of health care services relatively high, the system appears to contain significant efficiency reserves.[95] In the future the demand for health care services is likely to increase further substantially, mainly due to both expanding technological possibilities in medical treatment and population ageing. As many other countries, Germany now faces the difficult political question, whether to accept the disproportionate increase in health spending and to find ways to finance it or to restrict spending growth. This decision might imply quite different policy responses, even if in both cases policies to encourage the efficient use of the allocated resources remain desirable.

The main response of the government to rising cost during the 1990s was the introduction of sectoral budgets that were allowed to grow at the same rate as the assessment base of the social insurance system. While such budget caps are designed to create stable contribution rates, they proved problematic in practice. For example, the sectoral budget for outpatient services caused an increase in health care service units as physicians individually attempted to maintain their income levels under the fee-for-service payment rules.[96] This volume response has raised concerns about quality of care and access of the socially insured. The Physician Association voluntarily replaced the sectoral budget with budgets on the individual practice level, which are also subject, however, to significant shortcomings. Several other measures were taken to hold health contribution rates constant, including steps to introduce cost-reducing elements of competition into the health care system without compromising the system's equity objectives (see Box 4). Reform needs to continue, however, as several new mechanisms are not yet fully in operation and can be improved by adaptations of the institutional setting.

Budget caps for pharmaceuticals proved politically difficult to enforce and have been replaced by collectively negotiated agreements on spending volumes between the regional doctors' associations and the health funds to control the increase in spending.[97] In the absence of sanctions in case of non-compliance the new arrangement proved unsuitable to resist the pressure in spending: pharmaceutical expenditures surged in 2001 growing by some 11 per cent per insured. The

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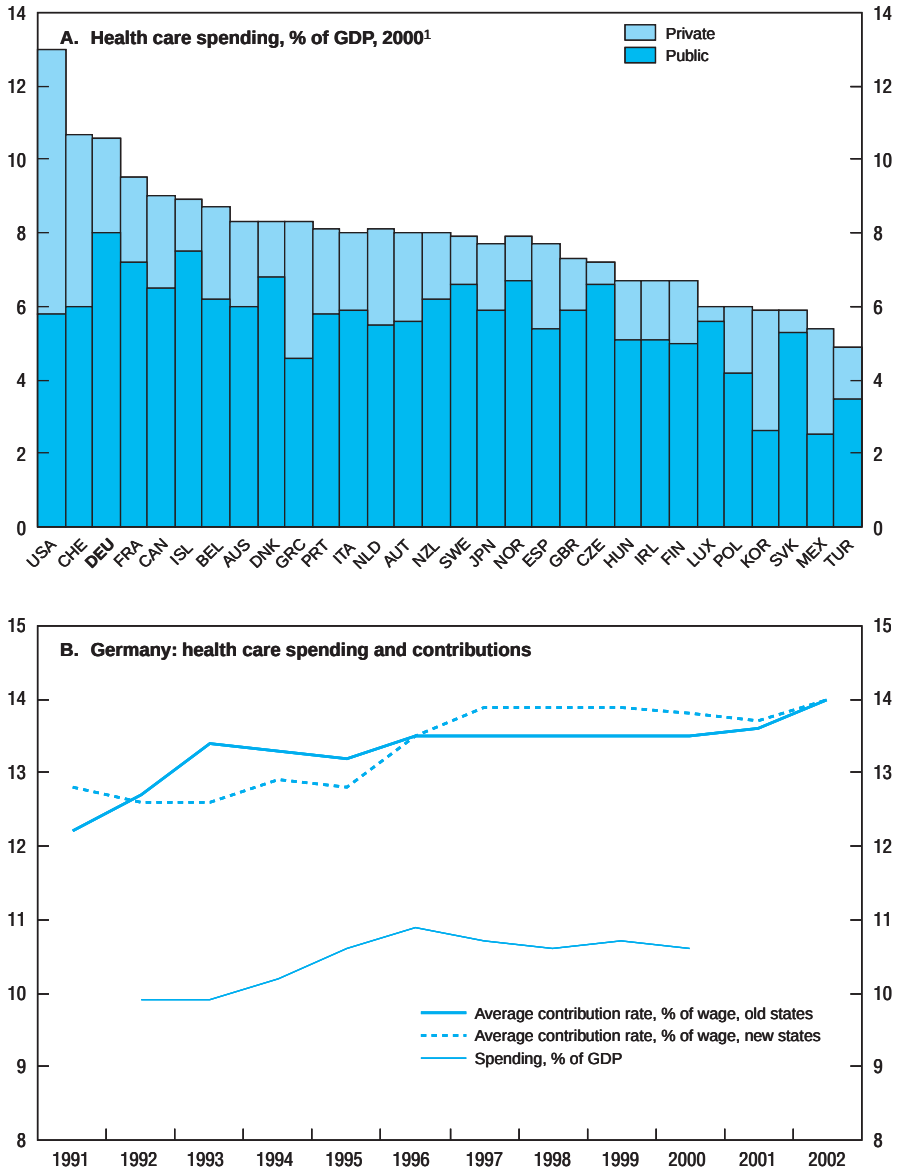
Figure 28. **Health care spending and contributions**



1. 1999 for LUX. 1998 for SWE, TUR. Refer to OECD, *Health Data 2002, Sources and Methods* for details of concept variations across countries.

Source: OECD, *Health Data 2002 (4th edition)*; Federal Government.

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Box 4. Basic features of the German health care system

Germany follows the social insurance model for the provision of health care, which is characterised by:

- *Compulsory coverage* for most employees, apprentices, pensioners and those on social benefits with income beneath a threshold level by non-profit public sector health funds (*Gesetzliche Krankenkassen*, GKV). Dependants are automatically insured so that the system covers about 90 per cent of the population. The funds are financed by contributions from employers and employees and they account for nearly a half of all health expenditures including cash benefits to the sick. Until 1997 consumer choice as to which health fund to join was quite restricted.
- The system is *self-governing*, with incorporated bodies representing health funds, doctors and hospitals negotiating important aspects of health policy. The legislative competence for health care policies is allocated to the federal level. The *Länder*, are mainly responsible for securing inpatient care, and this includes hospital capacity planning and investment.
- The *solidarity principle* is a key aspect of this statutory health system with contributions linked to gross wages (up to a limit) without reference to health risk and with household dependants co-insured without additional contributions.
- *Voluntary health insurance for those with income above a statutory level and for the self-employed*. They may stay in the statutory system or opt out to private health insurance in which case they no longer pay contributions to the system. Once having left the statutory system it is almost impossible to return.
- *Co-existence of private risk-based health insurance with the solidarity-based statutory system*. For those who opt out of the statutory system private health insurance is available with premiums based on risk profiles. Insurance rates for young people are much lower than in the compulsory system and insurance companies compete actively for young clients in particular. Over the last years, private insurers considerably raised premiums for the older insured and increased their old age reserves.
- There is for all intents and purposes *freedom of choice* of physician and hospital in both the compulsory and private systems and rationing of health services has been steadfastly avoided.

For further detail on the German health care system see OECD (1997).

government responded with new legislation (*Arzneimittelausgaben-Begrenzungsgesetz*) requiring pharmacists to supply cheap generic drugs instead of higher priced medicines, provided the prescribing doctor does not exclude such substitution. The doctors' prescription practice is subject to peer review. Generic drugs that

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qualify for substitution need to be priced within the lower third of the price distribution of the generic medicine. Other requirements also need to be met such as comparable packaging size. While inducing a larger use of generic drugs might go some way in containing pharmaceutical expenses the new regulation might not be robust enough against avoidance strategies by pharmaceutical producers and health care suppliers. There is some evidence that pharmaceutical producers are considering marketing responses that aim to reduce the substitutability of medicines by lower priced generics such as the creation of new packaging sizes and the introduction of highly priced generic pharmaceuticals, which shift upwards the lower third of the pharmaceutical price distribution.[98] However, there is no evidence that such responses have indeed been taken. Moreover, shifting to generic drugs might only result in a one-off reduction in the price level of drugs without affecting the underlying growth of expenditures for drugs. Hence, further action is required so as to economise spending on medicine. Pharmacies could be allowed to discount nationally set maximum prices, as is the case in the Netherlands. In this system, allowing chains of pharmacies to develop could stimulate price reductions. Also, the peer review of the physicians' prescribing practice could be strengthened by referring to external benchmarks rather than average prescription practice, as is the case in the present system.

As regards hospitals, new legislation has been passed that foresees replacing the present payment system by DRG related payments, under which hospitals receive a fixed amount per admission for a given diagnosis (diagnosis related groups, *diagnosebasierte Fallpauschalen*) by 2003. The payments are based on national historic cost. Compared to the present system, which is based on a mix of therapy related fees (*Fallpauschalen*, *Sonderentgelte*) and *per diem* rates (*Pflegesätze*) combined with overall budgets based on negotiated activity levels, a DRG system can increase the efficiency of pricing hospital services.[99] In particular, DRG payment creates incentives to cut the average length of hospital stays, and this could become particularly relevant for Germany, which has the second longest average length of hospitalisation within the OECD.[100] The DRG system is also likely to increase competition for patient flows, because it is intended to fully replace the currently pre-negotiated activity levels. To make the transition to the DRG system a success, particular care should be taken to insure that efficient systems of cost information and for monitoring the quality of services are in place.[101]

The 1997 Survey concluded that an efficiency-oriented health care system requires establishing a basis for the health funds to evolve into active purchasers of health services from suppliers on behalf of the patients, rather than remaining passive financers of health care services. Two preconditions have been established that would allow for this development: first, since 1997 the insured have the right to freely select their statutory insurance fund; second, insurance funds are now permitted to negotiate the terms for the provision of medical services selectively (individually) with groups of providers in addition to the collective contracts. In

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practice, however, selective contracting is almost never used, but collective contracting with the doctors' association remains the dominant model. Indeed, effectively exercising the first option would require sizeable investments by the insurance funds in information technology and human capital. To make this option worthwhile the range of admissible product differentiation would have to be widened. In particular, more scope would need to be given to the insurers to offer contracts to patients who accept a limitation of their choice of physicians and hospitals in return for lower insurance premia. Elements of such contracts could be gatekeeping, *i.e.* the requirement that a primary care physician authorises all speciality and hospital care, or preferred provider arrangements, *i.e.* the restriction of coverage to selected physicians, who discount their fees in exchange for a higher certainty in patient flows. Strengthening the gate keeper function of general practitioners could foster the integration of the health care system overall helping to reduce costs. Indeed, allowing for this type of managed care has introduced a certain degree of competition between insurances for statutory health care services in other OECD countries such as the Netherlands. Moreover, the health funds could be given some scope to compete on the basis of health service provision. One option would be to specify a voluntary part of the health care services in which there would be room for manoeuvre by the funds in specifying content and competition. Also, making selective negotiation between individual insurers and health care providers viable suggests dropping the dominant role of the doctors' association in determining the terms of health care provision rather than permitting a coexistence of collective and selective contracting. Supplementary measures should also be taken to encourage more efficient resource utilisation on the demand side. For example, a more systematic approach should be considered to requiring co-payments by health care consumers so as to reduce precipitate demand for services.

The introduction of free choice of health funds in 1997 was intended to encourage the funds to compete for members on the basis of administrative efficiency, but has rather led to competition among funds for better risks.[102] This left the regional health insurance funds (*Allgemeine Ortskrankenkassen*, AOK) with a worsening risk profile and increasing contribution rates, which in turn was considered undesirable by policy makers. Widening the scope for the health funds to offer differentiated services along the lines outlined above would therefore be an important step to improve the consistency of the present regulatory framework of the German health care system. Moreover, in order to avoid adverse selection of the insurance funds a mandatory risk adjustment system for health insurers (*Risikostrukturausgleich*, RSA) was introduced in 1994. While the parameters of the system were subject to consecutive adjustment, the RSA is not yet equipped to effectively equalise risks across insurers, rendering risk selection a profitable strategy for the individual firm. At present, transfers between insurance schemes are based on demographic characteristics of the insured that explain only a small fraction of the variance in health care costs. Moreover, since January 2002 a reinsurance scheme compensates for

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cases with health expenditures exceeding a certain threshold.[103] Incorporation of clinical information is vital, however, to adequately capture the risk structure of the insurance pools. Hence, the system would need to be developed further so as to make risk selection by insurers less profitable.[104]

A major reform effort was the introduction of so-called disease management programs (DMP) for chronic disease care in 2002. Under those programmes, an advisory body of the corporatist actors defines requirements for care standards that are then issued as regulations by the federal government. Those regulations serve as guidance for funds to develop contract based comprehensive service arrangements with health care providers, setting a floor for such arrangements. Participation in DMPs will be voluntary for qualifying patients. So far, DMPs for two conditions have been proposed and up to five may be added. The DMPs are meant to address several problems of the health care sector in Germany. First, they intend to encourage better integration of care for chronically ill patients. Second, they aim to foster selective and quality-based contracting. Third, they are also designed to deal with the risk selection problem in choice of health funds, as participation in those programmes will be used as additional parameter in the RSA. Furthermore, data gathered through the DMPs will constitute part of the data necessary to develop a morbidity-based RSA by 2007.

The reform constitutes a first step towards addressing those problems. The financial incentives linked to DMP participation will lessen the risk selection incentives and have the potential to encourage better care integration for the included conditions. Two challenges remain. First, DMPs are a relatively novel tool in health policy reform and their effect will largely depend on the actual content of the program. It is positive that the law mandates an evaluation of the programmes so that their continuation is made dependent on evidence.[105] Second, care needs to be taken that the prominent role of the DMPs in German health policy reform does not lead to unwanted distortions, as resources and attention might become diverted to the conditions covered.

Summary of recommendations

A number of reforms have been implemented to improve the allocative efficiency of public sector spending. Most of the action taken is steps in the right direction. However, more needs to be done to put public sector finances on a sustainable path, improve incentives in inter-governmental financing schemes to contain costs and demand, and improve outcomes in various expenditure programmes. Box 5 provides a synopsis of recommendations for further reform.

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Box 5. Recommendations for enhancing public sector spending reform

Increase the efficiency of public resource management

- Continue introducing more performance-related approaches of personnel management and allow for functional and regional flexibility of public sector pay schemes.
- Subject public sector spending projects to outcome-oriented *ex ante* and *ex post* evaluation methods that assess the costs of a project its benefits over a long-term horizon.

Improve the incentive structure of inter-governmental funding

- Consider widening the scope for regional government to set tax rates, for example as a supplement to the income tax.
- Improve the incentive structure of the inter-state revenue equalisation system by making more use of lump sum transfers from the federal government to the states that are fixed over a multi-annual period. Base transfers on notional rather than actual revenues of the states such that variations in local tax parameters leave transfer positions unaffected.
- Scale down co-funding between different levels of government.
- Internalise spill-overs of local services via co-operation between local governments or public institutions or the merging of local authorities.
- To the extent co-financing is utilised, assess the full costs associated with spending programmes in the selection, planning and evaluation stages.
- Within the Solidarity Pact II (SP II) funding arrangements for the new states require outcome-oriented *ex ante* and *ex post* evaluations of infrastructure investment projects to be included in the Progress Reports as a condition for funding.
- Make sure that an adequate share of the SP II funds are passed on from the *Länder* governments to the communities.

Establish a medium-term consolidation rule

- Support fiscal consolidation and the working of the automatic stabilisers by establishing a binding stability pact between the federal government and the states so as to balance the federal and *Länder* budgets over the cycle.
- Enhance the control of the general government budget by securing that consolidated budgets by government level are timely available in terms of the national accounts standard.

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Box 5. Recommendations for enhancing public sector spending reform (*cont.*)

Make the social assistance system more incentive compatible

- Combine unemployment assistance and social assistance into one single means-tested social assistance benefit to be paid and administered by one unified public sector authority.
- Remove the potentially indefinite duration of the more generous former unemployment assistance.
- Increase incentives from the funding side to integrate the unemployed into the primary labour market.
- Lower withdrawal rates of social assistance and or introduce wage subsidies such that overall remuneration increases with gross earnings.
- Reform the tax and benefit system so as to support job search of the unemployed and labour force participation. This includes in particular monitoring of job search activities, redirecting active labour market measures in favour of job search, tightening of eligibility criteria for unemployment insurance benefits, cuts in benefits in case of non-compliance with job search rules and abolishment of effective early retirement schemes.

Establish a more outcome-oriented regulatory framework for secondary and tertiary education

- Set nation-wide standards for schooling attainments against which the students' performance is to be evaluated on a regular basis.
- Leave the schools more freedom in determining suitable ways of how to reach the educational targets.
- Widen the flexibility of educational paths in secondary education while introducing more full-time schooling and individual coaching of students.
- Introduce compulsory language tests for students with immigrant background and foster their integration into the school system.
- Establish tertiary studies with a shorter duration as the rule.
- Give universities the right to levy tuition fees. To mitigate the potential adverse distribution impact, Germany's student support system might have to be adjusted.
- Implement a higher degree of competition between universities.
- Link funding of universities out of the general government budget to the universities' performance and re-enforce quality control.

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Box 5. **Recommendations for enhancing public sector spending reform** (*cont.*)

Intensify competitive pressures and the use of price signals in the health care system

- Introduce more efficient pricing mechanisms for health care services, such as diagnosis related fees (DRGs) in the hospital sector.
- Consider dropping the dominant role of the doctors' association in determining the terms of health care provision.
- Strengthen the peer review of the physicians' prescribing practice.
- Consider a more systematic approach to demand side cost sharing to discourage precipitate demand for services.
- Support the development of active purchasers of health care by widening the scope for health funds to compete on the basis of managed care and provision of voluntary health care services.
- Further develop the risk sharing system between insurances so as to avoid adverse selection of risks.

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IV. Implementing structural reform: a review of progress

Progress in structural reform should be seen in perspective of the challenges that need to be addressed. Chapter I of the present Survey concludes that the weak growth of labour utilisation over the past several years is the primary contributor to Germany's present weak economic performance. Service sector employment has been growing for some time but not enough to compensate for the fact that employment in both the manufacturing and the construction sectors have been declining. Structural adjustment in the new states is under way, but self-sustained growth in the east has not yet been achieved. High unemployment and relatively generous social transfers put the social security system under stress and imply a heavy tax burden on labour. International competition from lower-cost producers is likely to intensify placing increased pressure on economic institutions and the labour market in particular. The old-age dependency ratio will substantially increase over the next 10 to 15 years burdening Germany's elaborate social system. Hence, the challenges policy makers face remain to improve economic framework conditions so as to facilitate the reabsorption of the unemployed, the restructuring of employment and the adaptability to change more generally. Important issues concerning unemployment-related transfers, education and the health care system have been considered in Chapter III above. This chapter highlights further policy initiatives relevant for labour, product and financial markets and concludes with a discussion of some issues of sustainable development.

Labour market reforms to tackle high structural unemployment

Past initiatives of labour market reform proved insufficient to tackle Germany's persistently high structural unemployment, and in some cases they appeared to be at variance with the proposals of the OECD *Jobs Strategy* as set out in previous *Surveys*. Against the background of weak economic growth in 2001 and with unemployment rising again, the path of labour market reform has been stepped up in recent months. In January and April 2002 new legislation (Job-AQTIV-Act) came into force designed to improve the activation of the unemployed and to increase the efficiency of the public employment service (PES). Policy action was reinforced

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by the fact that a report of the Federal Court of Auditors identified irregularities in the operation of the PES' placement activities. Already before, empirical work repeatedly suggested that job provision and training schemes had contributed little to re-integrating the unemployed into the primary labour market. Responding to the auditor's report, the federal government announced fundamental reform of the PES and installed a commission (*Hartz-Kommission*) to make suggestions for a reorganisation of the PES and labour market reform more generally. Many of the commission's proposals, which were released in August 2002 (see Box 6), are in line with the suggestions that have been made in previous Germany Surveys, while some appear to be less suited to tackle Germany's unemployment problem. Several initiatives already implemented in 2002 are steps in the right direction but are incomplete yet. More comprehensive reform is required to widen the capacity of the German labour market to increase employment and raise the adaptability of the German labour market.

Box 6. Recent legislation and commission proposals for labour market reform

A. The Job-AQTIV-Act

The Job-AQTIV-Act – denoting Activation, Qualification, Training, Investment, Placement, whose parts came into force in January and April 2002 – contains various measures designed to improve the efficiency of labour market policies. The law aims at better activation of the unemployed, by offering aid by the labour office in exchange for more active job search of the unemployed. Key features are:

- *Profiling of job seekers*: newly-unemployed persons as well as applicants for vocational training are profiled by the Labour Office with respect to their professional strength and their perspectives. Following profiling, an "agreement of integration" (*Eingliederungsvereinbarung*) specifies support of the PES (guidance with respect to job opportunities and training) on the one side and the rights and duties of the job seeker on the other side. Agreements of integration are not mandatory.
- *Job-rotation*: companies filling temporary vacancies on account of vocational training spells of their employees with unemployed receive a government grant of between 50 and 100 per cent of the compensation of the substitute worker.
- *Temporary work agencies* (*Zeitarbeit*): the maximal admissible continuous time period for sub-contracting is extended from 12 months to 24 months. After the first 12 months, the temporary worker needs to be given the same working conditions (including pay) as a comparable worker in the user enterprise.

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Box 6. **Recent legislation and commission proposals for labour market reform** (*cont.*)

– *Employment-promoting measures in infrastructure* (*Beschäftigung schaffende Infrastrukturmassnahmen*): regional and local investment in infrastructure is subsidised under the condition, that the firms employ registered unemployed, to up to 35 per cent of their regular staff.

– *Promotion of a low-wage sector*: since April 2002 social security contributions of the employees in low-income jobs are subsidised ("Mainz Model"). The scheme applies for employment with least 15 working hours per week over an earnings band of between € 325 and € 897. The maximal support period amounts to 36 months. (See Chapter III above.)

– *Modification of unemployment benefits*: sanctions for unemployed in case of refusing an employment offer have been sharpened. Unemployed who miss purposely job interviews or behave so as to impede job offers are barred from receiving unemployment insurance benefits or unemployment assistance for 12 weeks.

B. The proposals of the "Hartz-Kommission" for labour market reform

Following the discovery of irregularities in the operation of the Federal Employment Office, the federal government initiated a working group ("Hartz-Kommission", named after its chairman) with 15 representatives of the social partners and the government. This commission was asked to make proposals for a comprehensive reform of the Federal Labour Office including a reorganisation of functions, a new organisational structure and a strategy of implementation. Existing labour market policies and placement strategies should be assessed more generally and proposals should be made. The final report of the Hartz-Commission was released in August 2002. Major recommendations are:

– *Reorganisation of labour administration*: local labour and social welfare offices should be merged organisationally to "Job Centres" and provide "one-stop-services" for all unemployed so as to remove overlaps of responsibilities. The budget responsibility and flexibility of job centres shall be widened. A performance-orientated bonus system for placement officers shall be introduced. An electronic signature card for all employees shall be introduced which tracks elementary information about the individual work and income history. In case of unemployment this information is to be used for a simplified benefit calculation. The placement of the unemployed shall be outsourced to newly-created autonomous "Personal Service Agencies (PSA)" which are commissioned by the job centres. Employable job seekers generally are obliged to be placed by the PSA which operates as a temporary-work agency and is exclusively in charge of placing unemployed via job coaching in companies or in non-profit/community services, including the provision of qualification measures where necessary. The refusal of placement by the job-seeker via the PSA would imply financial sanctions.

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Box 6. **Recent legislation and commission proposals for labour market reform** (*cont.*)

– *More responsibility and commitment of the unemployed*: employees who have been given notice about their redundancy should immediately register at the labour office to start placement activities, otherwise their benefits will be cut. The burden of proof for co-operation in searching for a job shall be assigned to the unemployed. Sanctions for non-co-operation, such as temporary suspension of benefit payments, shall be widened. Moreover, acceptability ("*Zumutbarkeit*") of work for an unemployed should be expanded, in particular with respect to the mobility of the unemployed.

– *Promotion of temporary work agencies* (TWA): legal restraints of TWAs should be further reduced. Moreover, the prohibition of synchronisation between the TWA contract and the placement contract and of repeated placement of one person in the same company should be dropped.

– *Promotion of employment of disadvantaged groups*: young job seekers shall be made an "activating offer" such as a traineeship, PSA contracts or training measures. With the introduction of an individual "training security" ("A*usbildungsZeit-Wertpapier*") financed by a rebate system, public grants and corporate and private contributions, new apprenticeship places should be created and the personal vocational training shall be guaranteed.

– Older unemployed, aged 55 years and above, who have been made redundant and who take up a job that is paid lower than their last one should receive the difference out of social security benefits. Additionally, employers' social security contributions should be temporarily subsidised and the age limit for temporary employment lowered. In contrast, a "bridge system" should provide older unemployed willing to quit the labour market with benefits until they retire at 60 on a "cost neutral" basis, *i.e.* the equivalent of their expected unemployment benefits considering their probability of re-employment.

– *Promotion of self-employment and small jobs in the low wage sector*: to reduce illegal work, unemployed (and family members) can earn up to € 25 000 per year as self-employed ("I*ch*-AG", "F*amilien* AG") with this income being taxed at a flat rate of 10 per cent and subject to social security contributions which will be partly subsidised for up to three years. Moreover, the allowance for small jobs should be expanded to up to € 500 (from € 325) for household-related services and for income between € 501 and € 1 000 social security contribution rates should be reduced. Private households demanding home services should be allowed to deduct the cost from their tax base.

– *Promotion of job creation in* SMEs: small and medium-sized companies creating new permanent jobs should obtain equity and loan capital subsidies to up to € 100 000 for each newly employed. The funds should be financed by bond issues ("Job-Floater"). This measure aims especially at companies with low equity capital resources in Eastern Germany.

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Increasing labour cost and working time flexibility

Opening clauses continue to foster wage and working time flexibility…

Over the past several years, the scope for more flexible wage determination at the company level has been widened by incorporating opening clauses into collective wage agreements. This trend has been reinforced in the 2002 wage round by an agreement in the important chemical sector that allows basing part of the workers' remuneration on the level of a company's profits. Often opening clauses for agreements at the company level are conditional on adverse business conditions and trade off lower wages for employment security of the company's workforce (Table 13). Frequently, exercising the clause requires approval of the employer's association and the union, and in many cases this requirement also applies for the company agreement itself. Comprehensive information is not available to what extent opening clauses are applied in practice.

Table 13. **Opening clauses for wages and hours worked in selected sectors**

Sector or branch	Area of specific regulation	Coverage of opening clauses
Chemical industry (Old states)	Pay	Temporary reduction of regular wages up to 10 per cent; variable determination of the level or date of annual bonus payment in order to maintain employment or increase competitiveness. Agreement of employers' association and union required.
	Working time	Working time corridor between 35 to 40 hours per week without over-time charges (regular working hours per week: 37.5).
Iron and steel industry	Pay (New states)	Suspension of one-off payment; sustained reduction of regular wages in case of economic difficulties (danger of insolvency, employment safeguarding). Agreement of employers' association and union not required.
	Working time (North Rhein-Westphalia)	Prolongation of working time up to 40 hours per week for max. 18 per cent of the company's employees without over-time charges (regular working hours per week: 35).
Paper producing (West Germany)	Pay	Variable determination of the level or date of annual bonus payment and cut of leave pay to 50 per cent in case of economic difficulties and under the condition of no lay-offs. Agreement of employers' association and unions required.
	Working time	Working time corridor between 35 to 39 hours per week; to safeguard employment working time can be cut by 6 hours per week for 6 months in a row (regular working hours per week: 38).

Source: WSI-Tarifarchiv, BDA Tarifarchiv.

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Employment in Germany has reached a substantial degree of working time flexibility. According to the household survey (micro-census) of the Federal Statistical Office, about 40 per cent of all employees in enterprises with a workforce of 50 or more use working time accounting that allows shifting working time individually between days, weeks or months.[106] Working-time accounting is most significant in manufacturing and services, notably banking, insurance companies and public services (one-third of the employees).

Surveys among enterprises point to the need to increase labour time flexibility as a means to better absorb changes in demand and reduce the cost of labour. Indeed, one survey among medium-sized companies (*Mittelstand*) identified a lack in working time flexibility as a more serious shortcoming of collective bargaining outcomes than agreed wage levels.[107] Similarly, recent research suggests that achieving a higher degree of working-time flexibility constitutes a major motive for employers to exit the employers' association.[108] According to this study, more than one-fifth of the companies leaving the collective bargaining system increase the weekly working time thereafter, particularly in western Germany. Against this background, over the last years opening clauses have increasingly been built into collective contracts that allow for firm-specific working time agreements between the management and the workforce, as has been highlighted in previous *Germany Surveys*. According to the micro-census, for about 10 per cent of all employees the annual working time is fixed in intra-firm agreements, and for three per cent the working time is distributed according to operational considerations of the firm. Typically, opening clauses specify working time extensions or corridors that are feasible without over-time charges, or they allow companies facing adverse economic conditions to reduce the number of hours worked under the provision that redundancies are to be avoided (Table 13). In most large industrial sectors opening clauses allow shifting working time over a balancing period of one year. In the chemical industry the balancing period totals three years under certain conditions. Evidence on the application of such clauses is sparse, however. In the chemical industry, 51 company-based agreements utilising working time corridors are in operation covering some 15 000 employees.[109] Initially, company based agreements were mainly designed to reduce hours worked in order to avoid dismissals, but more recently they are predominantly utilised for the purpose of expanding hours. This feature might reflect the comparatively favourable economic situation of the chemical industry and cannot be generalised to other sectors.

… but policy initiatives to support more differentiated collective bargaining outcomes are lacking

Overall, the adjustment observed does not reach far enough to cope with Germany's high level of structural unemployment, and legal framework conditions did not improve so as to support more differentiated bargaining outcomes. In contrast,

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some policy initiatives were aimed at strengthening the role of collective agreements. A draft law linking public procurement with collective bargaining (*Tariftreuegesetz*) stipulated that public procurement contracts would exclusively be granted to companies setting wages in line with collective agreements, even if firms were not a member of the employers' association. The law was rejected, however, by the second chamber of Parliament (*Bundesrat*). The law on the posting of workers (*Arbeitnehmerentsende-gesetz*), which subjects foreign workers on construction sites in Germany to collectively agreed German minimum wage conditions, has been challenged by the EU Commission at the EU Court. The EU argues that the law disadvantages foreign workers.

Making some slice of the workers' pay dependent on the evolution of the companies' profits – as has been agreed as an option in the chemical sector – would add an important element of additional flexibility. Moreover, as has been suggested in previous *Surveys*, consideration should be given to widening the scope for wage determination at the company level so as to better align collective wage contracts with labour market conditions. In such a system collective bargaining outcomes would continue to serve as the default if company-based agreements do not come about. Indeed, the importance of collective bargaining is apparent from the fact that a significant share of enterprises in both the new and the old states apply the provisions negotiated collectively between the social partners although they are not members of the employers' association.[110]

While part-time employment is trending upward, pertinent legislation needs to be monitored

Part time employment has also increased, with its share in total employment standing at around 20 per cent after 14 per cent at the beginning of the 1990s (Figure 29). Since January 2001 employees in companies with more than 15 employees have the legal right to claim part-time employment. According to an establishment poll by the Institute for Labour Market Research (IAB), between January and autumn 2001 9 per cent of the firms in western Germany and 6 per cent of the firms in the east were subject to claims for part-time employment.[111] Only 8 and 4 per cent, respectively, of the enterprises rejected the claims. Another survey by the German Chamber of Industry and Commerce (DIHK), however, shows a higher rate of rejections totalling one-third of the firms with claims.[112] In one-fifth of the rejections, the employee took legal action or considered to do so. As a consequence, 12 per cent of companies declared that they tend to be more cautious in hiring persons that they presume to have specific preferences for part-time work. Hence, there might be a risk of higher labour costs, impacting negatively on hiring as expressed in the last *Survey*. This could disadvantage female job seekers in particular. The implications of the law therefore need to be monitored further, and if necessary it should be abolished.

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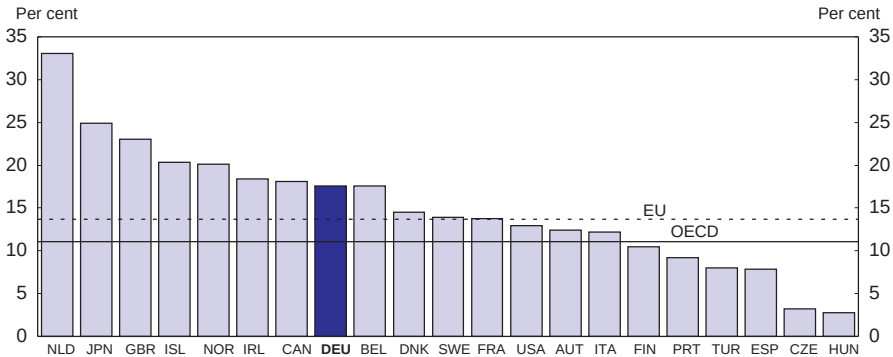
Figure 29. **Part-time employment in international comparison**[1]
As a percentage of total employment, 2001



1. Part-time employment refers to persons who usually work less than 30 hours per week in their main job. For Japan, less than 35 hours per week. Data include only persons declaring usual hours except for Finland and Japan where data are based on actual hours worked. For the United States, data are for wage and salary earners only. For Germany and United Kingdom, data refers to 2000.

Source: OECD, *Employment Outlook*.

The complexity of dismissal procedures needs to be reduced

As past Surveys have emphasised, within the OECD German employment protection for regular employment has been found to be relatively strict (see Figure 30), with both legal provisions and significant rulings of labour courts making redundancies costly.[113] While statutory time periods to be observed for dismissals to become effective are not outstanding, the system appears to be complex in procedural terms. Severance pay may be high and associated with considerable uncertainty. Indeed, in 2000 labour courts decided 246 000 legal cases relating to dismissals.[114] More than half of them were settled by monetary compensation to be paid by the employers to the employees. The overall strictness of dismissal protection has been found to be partly responsible for the increased utilisation of fixed-term contracts, and there is a risk that prolonged and costly court procedures instigated by laid-off employees constitute a significant obstacle for employment creation.[115] The system of dismissal protection should be reviewed in order to promote employment while striking a balance between flexibility and security. Removing legal ambiguities and making the legislation more transparent would already go some way to reducing effective labour costs. An example could also be to offer an option to open work contracts to specify severance pay for workers in case of a redundancy in return for a less stringent dismissal protection.

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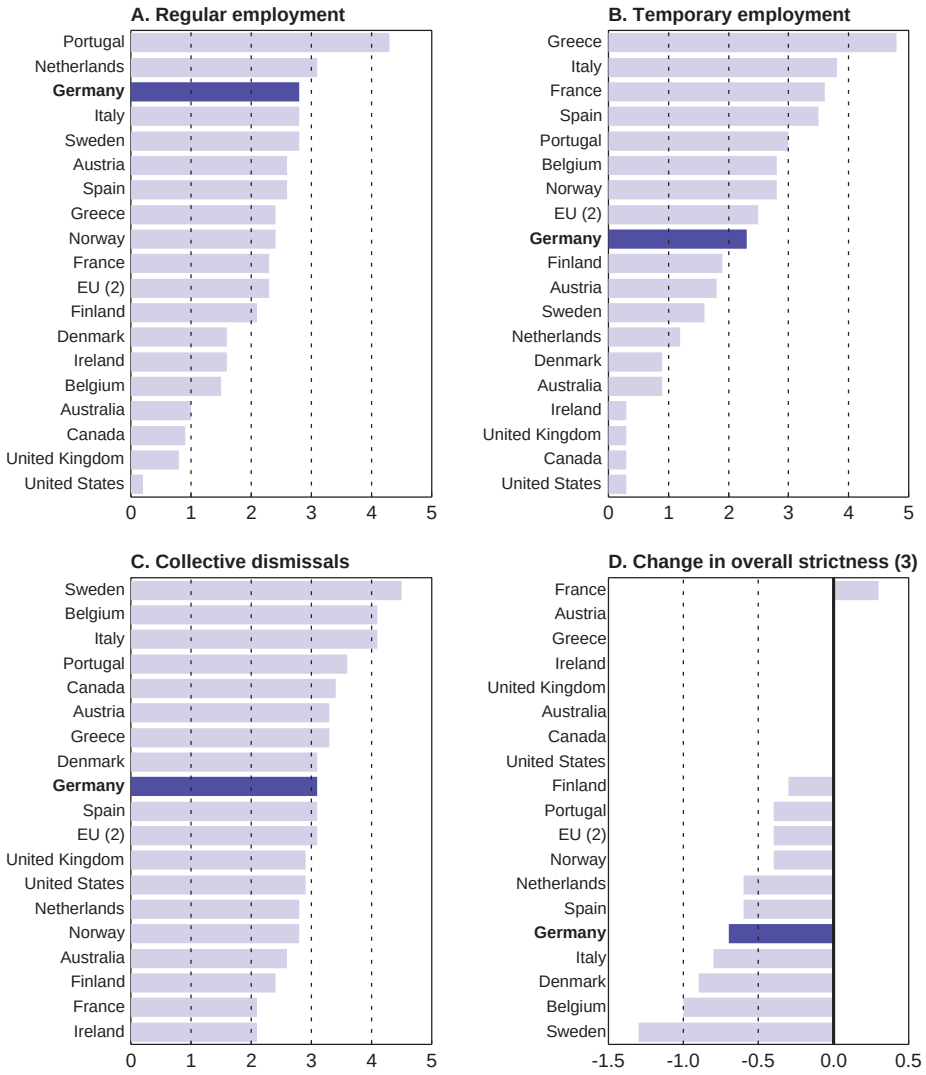
Figure 30. **Employment protection within the OECD**[1]

Late 1990s



1. Indicators of strictness of legislation. Scores can range from 0 to 6 with higher values representing stricter legislation.
2. Weighted average.
3. Average of indicators for regular and temporary contracts since the late 1980s.

Source: OECD, *Employment Outlook,* June, 1999.

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The potential for job creation of temporary work agencies can be more fully exploited

Temporary employment has grown in a number of OECD countries over the past two decades and slots are filled more than proportionately by younger and less-educated persons. While companies often use temporary employment to accommodate fluctuations in product demand, such jobs often serve as "stepping stones" into permanent employment.[116] In Germany, temporary employment grew by 13 per cent between 1991 and 2001, while regular dependent employment stagnated. Since the mid-1980s, the share of temporary employment in total dependent employment rose from less than 10 per cent to more than 13 per cent. While this share stands slightly above the OECD average and is comparable to most other European countries, it largely reflects the fact that it is common practice to employ apprentices on a temporary basis: More than one-third of all employees in Germany on a fixed-term contract are either apprentices or trainees. The probability for a temporary worker to move into permanent employment has been estimated to rise from some 40 per cent in the first year to more than 50 per cent in the second. Younger people aged between 25 and 34 and women are particularly likely to make this transition. Roughly one-quarter of the temporary workers become unemployed thereafter.[117]

Temporary work agencies (TWAs), which sub-contract their employees to other firms for a fixed time period, offer a possibility for the "hard-to-place" unemployed to re-enter the primary labour market.[118] Between 1996 and 2001 the number of TWA-employed persons in Germany rose by more than 50 per cent to 300 388, 0.9 per cent of all dependent employees, the share still being significantly below the one in the United Kingdom, France and the Netherlands. More than 53 per cent of these employees were previously unemployed, and another 7½ per cent were new labour market participants.

New legislation for TWAs came into force in January 2002 extending the maximal continuous time period of sub-contracts with the same user enterprise from 12 to 24 months (see Box 6). Prolongations are restrained, however, in that from the twelfth month onwards the TWA has to provide to the temporary worker the same working conditions – including pay – that prevail for a comparable worker in the user enterprise. Similarly, from 2004 onwards pay conditions need to be generally the same, except a collective agreement specifies differently. While this initiative aims at balancing a higher flexibility of temporary employment with job security, the compulsory equalisation of the pay for regular and temporary workers severely restrains incentives for assignments as the TWAs' profits are predominantly defined by the difference in wages between internal contracts with job seekers and external ones with firms. New impulses for the evolution of temporary employment are therefore unlikely to emanate from this legislation. Moreover, other provisions hamper a more dynamic development of temporary employment. The "prohibition of synchronisation" (*Synchronisationsverbot*) stipulates that

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the contract period between the TWAs and the employee on the one hand must differ from the sub-contracting period of the worker's first assignment on the other hand.[119] The regulation implies that prior to closing the contract between the worker and the TWA after a first temporary work assignment, the agency has to place the employee for at least one quarter with another sub-contractor or – if this is not possible – needs to pay him for this period without an engagement with a borrowing firm. Germany appears to be the only country within the OECD with this type of regulation in place.

To better utilise the capacity of TWAs to generate employment the legislated extension of the maximal sub-contracting period under certain conditions is unlike to suffice. The prohibition to synchronise the contract period between the TWAs and the workers on the one hand with the workers' sub-contracting period on the other hand (S*ynchronisationsverbot*) should be given up. Also, consideration should be given to abolishing the compulsory equalisation of work and pay conditions between temporary employed and regular staff should be given up. Relaxing restrictions for TWAs should be no substitute for more comprehensive labour market reform, however. In particular, such initiatives should go hand in hand with a liberalisation of the general dismissal legislation. Otherwise, relaxing restrictions for TWAs risks creating a segmentation of labour markets in favour of temporary employment.

The labour market reform commission established by the government has recently proposed to link temporary work agencies with the regional labour offices (Box 6). These agencies might be run either by private enterprises or the respective labour offices. Following a spell of unsuccessful job search the unemployed would be obliged to accept temporary placement by such an agency. In case of refusal, unemployment benefits would be cut. The labour office can contribute to covering the cost of the temporary engagement. The initiative is welcome to the extent that it leads to a more efficient regulatory framework for TWAs as placement instruments. It would be counter-productive, however, if private TWAs had to operate under more restrictive rules than temporary placement agencies run by the public employment service. Also, special subsidies for temporary employment could induce significant windfall gains for companies and provoke a crowding-out of regular employment, especially in the low-skilled area. The proposals of the Hertz Commission leave the decision on temporary employment to the TWAs. However, if labour offices channelled all unemployed to TWAs there would be the risk that employment-creation schemes generate a large public employment industry which might have other objectives than placing the unemployed in the primary labour market. The experience in Germany with public work provision and training schemes supports the existence for this risk (see below).

Workers' co-determination has its merits but needs to be seen in relation to its costs

In July 2001 new legislation came into force adapting and widening the workers' co-determination in works councils (B*etriebsrat*).[120] Workers' co-determination

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is widely considered to be a major instrument fostering co-operative industrial relations in Germany, although it has also been criticised to raise labour costs. At present works councils are located at the level of establishments. The new legislation gives the social partners the option to establish works councils at the level of the overall company. Also, the new legislation contains provisions designed to simplify the council's election procedures. These features should imply some potential for cost reductions. However, other features are likely to significantly increase the costs of labour, in particular for small and medium-sized companies. The statutory number of works council members has been increased for companies with more than 100 employees. Similarly, the obligation of establishments has been widened to exempt one or two employees completely from work in order to act as representatives in the councils.[121] Ultimately, part of the cost raising effects would be borne by labour to the extent that the expansion of employment would be hampered. It appears to be necessary to closely monitor the benefits and costs of this amendment in practice, in particular with respect to its impact on employment, and to implement corrections if required. One option would be to make the costs of co-determination more explicit by sharing them between employers and employees. This would also reduce barriers for new hiring.

Improving the efficiency of placement in the primary labour market

Active labour market measures remain largely inefficient

Over recent years work provision measures were reduced in favour of training schemes (see Chapter I above), partly reflecting the fact that the Federal Labour Office obtained increasing freedom in allocating funds to ALMPs. At the same time, the government aimed at redirecting job provision schemes towards employment promotion in the primary labour market. While this reflects a response to adverse empirical outcomes regarding the effectiveness of ALMPs in bringing the unemployed back into employment, some recently introduced programmes remain problematic.

Within the framework of the Job-AQTIV-Act (Box 6), a new programme aims at combining employment promotion with improving the funding of infrastructure investment (*Beschäftigung schaffende Infrastrukturmassnahmen*). The government expects that infrastructure investment generates positive first and second-round employment effects. The Labour Office can subsidise up to 25 per cent of additional infrastructure investment by regional and local governments if private-sector companies hire unemployed (up to 35 per cent of their work force). However, the programme risks crowding out non-subsidised by subsidised labour. Furthermore, as has been argued above in Chapter III, co-financed schemes run the risk of biased resource allocations. Overall, the measure might merely boil down to

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providing budgetary aid for local governments, making the system of intergovernmental funding even more opaque than it is already.

A new job rotation scheme subsidises between 50 and 100 per cent of the compensation for unemployed hired by employers as a replacement for employees on leave for vocational training. This investment aims at both providing incentives for the training of employees and improving the integration of the unemployed. Usually, training schemes for employees last only for some days or weeks. Hence, a sequence of several individual training leaves or long durations of training leaves appear to be a precondition for this measure to be effective in providing valuable training on the job for the unemployed and become warranted for the employers. Adjustment costs for the firm might be comparatively high especially for small enterprises. Indeed, in the first seven months after the measure has become effective, only some 200 persons were placed in a company via the job rotation scheme. No information is available yet on the labour market history of such placements after the temporary assignment terminated.

First steps to improve the profiling of the unemployed need to be put on a more comprehensive base…

Profiling the unemployed with respect to their skills and labour market characteristics is increasingly practised within the OECD, and experience suggests that this is important to identify specific placement risks and to provide adequate support for job search.[122] The Job-Aqtiv Law introduces elements along these lines (Box 6). Newly unemployed persons as well as applicants for vocational training are required to be profiled by the Labour Office with respect to their professional strength and their perspectives. Based on the profile, the job seeker and the labour office can conclude an "agreement of integration" (E*ingliederungsvereinbarung*) about the support of the PES on the one side and the rights and duties of the job seeker on the other side on a voluntary basis. The agreement implies individual guidance by the PES to generate job opportunities and utilise appropriate training schemes ("case management"). The approach can include the support by external placement agencies. In case of non-compliance the unemployed, on the other hand, would be subject to reductions in unemployment benefits. However, the fact that this approach is voluntary casts doubt on its effectiveness. It is not clear what incentives the staff of the PES should have to initiate such contracts. Similarly, the unemployed could avoid sanctions that might arise as a result of the integration contract if such an agreement did not come about. Up to now, only 10 to 20 per cent of the newly-unemployed are subject to an integration agreement.

This low response also reflects the fact that the human resource management of the Labour Office apparently has not yet adapted sufficiently so as to enable effective profiling and placement. Local labour offices are reported to consider their staff over-burdened by bureaucratic tasks leaving little capacity for profiling and

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placement activities.[123] Until recently only little more than 10 per cent of the staff of the Labour Office was concerned with job placement. At present, a concept is being tested in several local offices that aims at providing various PES services by a small team of labour officers ("Labour Office 2000", "A*rbeitsamt* 2000"). Indeed, experiences in other OECD countries suggest that the quality of placement can be improved by a higher integration of the different services of the PES ("One-Stop-Shop").[124] Frictions associated with the re-structuring point to the need to adapt skill profiles: while in the past the unemployed were channelled to vocational counsellors specialised in certain professions, this has been replaced by an alphabetical ordering of cases. Shortcomings in the interaction of local employment services would also have to be addressed: effective placement of unemployed in training schemes is sometimes hampered as information of courses available is not circulated inter-regionally. Lessons can be learnt from other OECD countries, which have made considerable progress in introducing clearly measurable outcome objectives so as to support the effective use of the various instruments of the PES. For example, the Swiss government introduced in 2000 a performance ranking for the public employment services based on a multivariate indicator which serves as a measure for rewarding as well as sanctioning the cantons on the basis of the outcomes achieved by the Regional Placement Offices under their management.[125]

For all benefit recipients the Job-AQTIV-Act aims at increasing job search incentives by sharpening sanctions. Unemployed who miss purposely job interviews or behave so as to impede job offers should be barred from receiving unemployment insurance benefits or unemployment assistance for 12 weeks without exception. However, as is the case for persons subject to integration agreements, evidence for this type of misconduct by purpose has still to be produced by the PES, and this appears to have hampered effective sanctioning in the past. Indeed, various possibilities for sanctions in case of non-co-operation of unemployed benefit recipients exist already, but experience suggests that they are rarely applied in full scope.

… and administrative reform of the public employment service needs to be widened

More fundamental management deficiencies of the Federal Labour Office have recently surfaced. In January 2002, the Federal Court of Auditors (B*undesrechnungshof*) released a report on the effectiveness of job placement in selected labour offices. It revealed that up to 70 per cent of placements that the PES had reported to be successful were false. Apparently, statistics were doctored to prove the effectiveness of the PES activities in bringing the unemployed back into employment. In another forthcoming study, the Federal Court of Auditors is scrutinising the PES policies of channelling the unemployed into training programmes. Such policies have been reported to be arbitrary and little focused on reintegrating the

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unemployed into the labour market. While over the last years the PES has been endowed with substantially more discretion in deciding on how to allocate funds on ALMPs, the – parity – representation of the employers' side and the unions on the governing board of the Labour Office appears to have led to rent sharing between organisations associated with the employers and the unions, respectively.[126] This, in turn, is at variance with allocating funds most efficiently so as to foster the re-integration of the unemployed. Indeed, the massive application of ALMPs in Germany over more than a decade has provoked the development of a sizeable industry of private and semi-private service providers for qualification and retraining programmes. Currently, there exist some 35 000 training providers offering about 400 000 courses per year. These institutions are predominantly run by the social partners, which control about 60 per cent of the market for further education. As has been emphasised in previous *Surveys*, empirical evidence suggests that training programmes have been largely ineffective in re-integrating the unemployed into the primary labour market. Moreover, steps have been made only recently to evaluate such schemes on the basis of individual placement data provided by the employment service (see Chapter III above).[127]

As a consequence of the January report by the Court of Auditors, the governing structure of the PES has been changed so as to reduce the influence of the social partners on current policies (Box 7). Moreover, the potential role for private placement agencies (PPAs) has been widened substantially. PPAs no longer need a license from the Federal Labour Office. Unemployed receiving unemployment benefits or participating in ALMPs, who are not being placed by the PES within three months, can claim a voucher allowing them to consult any PPA for the purpose of placement. The value of the voucher increases in three steps, depending on elapsed unemployment duration (maximum: € 2 500 after nine months). In case of a successful placement of the unemployed in regular employment the PPA can cash in the vouchers in two instalments paid after the person has remained employed for three or six months, respectively.[128] Up to now this system has only been moderately successful, however. From the start of the initiative in March 2002 until end of July 2002 local labour offices issued some 88 000 vouchers of which only some 3 000 have been cashed in. Also, the PPAs are not required to report placing details to the Federal Labour Office. In contrast, experience suggests that a comprehensive integration of the elements of the placement system is necessary for it to operate successfully.

In sum, while important steps have been made recently to raise the efficiency of the PES more is required. The focal point of staff activities needs to be shifted from administration to profiling and placement. For this purpose bureaucratic tasks need to be reduced so as to free staff capacity. Retraining personnel will also be required. The widening of the scope for private placement agencies that became effective in spring 2002, is welcome. An option for continued reform

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Box 7. Features of the Public Employment Service

The tasks of the Federal Labour Office (B*undesanstalt für* A*rbeit*) comprise: job placement and vocational counselling, placement in job provision schemes, organisation and placement in training and re-training schemes, administration of unemployment insurance benefits and unemployment assistance, and provision of labour market statistics. The office is organised in 181 local labour offices (A*rbeitsämter*), 10 regional labour offices (L*andesarbeits-ämter*), and the head office (H*auptstelle*) located in Nürnberg. Its staff totals some 90 000 employees.

The Federal Labour Office is administratively autonomous. It is managed by the Board of Governors (V*erwaltungsrat*), which supervises the management and the administration, decides on the statutes (S*atzung*) and is responsible for budgetary allocations. The Executive Board (V*orstand*) is responsible for the current management. Until recently both of these bodies were composed in equal shares by representatives of the employees, the employers and the public sector. The fact that unemployment insurance benefits and some part of active labour market measures are paid in equal parts by the employers and the employees – out of social charges on earnings – has been a major rationale for the proportionate representation of the social partners in the governing bodies. Following irregularities in the registration of job placements, the federal government reorganised the management structure. Since April 2002 the management of the federal labour office resembles the two-tier structure of an incorporated enterprise. Proportionate representation of the social partners and the public sector is confined to the board of governors whose size was reduced from 51 to 21 members. The executive board now consists of three persons commissioned by the federal government. In contrast to past practice the board's chairman is no civil servant but hired on a private contract.

would be requiring the delivery of PES services to become contestable in a market environment. Indeed, several countries within the OECD have introduced contestability of some PES functions, and this could comprise tendering job brokering and reintegration services, as it is practised in Australia. In this system public and private service agencies compete under equal terms and payments are made to them on the basis of outcomes with placement premiums increasing with the degree of the placement risk. For the PES clearly measurable outcome objectives need to be defined against which the use of the various instruments of the Labour Office can be evaluated. Integration agreements between the public employment service and the unemployed need to be made mandatory. In this context, ALMPs should be used in the first place to facilitate search activities. Sanctioning non-compliance with the re-integration conditions set by the Labour Office should be an important element of the overall strategy to better activate the unemployed. It is of vital importance that sanctions are not only threatened but actually executed

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if the rules are violated. This in turn requires verifiable criteria whether benefit recipients are actively engaged in search. In contrast to past practice in Germany benefit recipients should be required to produce evidence that this is indeed the case. For example, in Switzerland recipients need to document that they contacted enterprises with job openings at a certain rate per month. In this context the acceptability criteria of accepting job offers should be revisited so as to better align reservation wages with market conditions, as has been argued in previous Surveys. The acceptability criteria should be progressively widened with unemployment duration.

Competition enhances potential growth

The economic performance of Germany can be stimulated further by enhancing product market competition through a pro-competitive regulatory framework, with effects materialising through a number of channels and in interactions with other markets. Increasing competitive pressures initially improve allocation of resources, leading to higher productivity. Subsequently, diffusion of product and process innovation is stimulated, resulting in higher multi-factor productivity growth, with for example empirical work for Germany suggesting a strong inverse correlation between productivity growth and market rents at the firm level, indicating that market power reduces efforts to boost efficiency.[129, 130] Moreover, an empirical relationship – at the OECD economy-wide level – has been established between various measures of more intense competition (including inward regulation and non-tariff barriers to trade) and higher R&D intensity.[131] In addition, recent OECD work finds a positive relationship between the R&D intensity and GDP growth per capita.[132] Employment creation is stimulated through increased demand, arising from more competitive prices and an enhanced consumer choice. The implications are that initially employment in individual industries may be reduced, but employment creation is subsequently stimulated, subject to existing labour market regulation.

Controlling for the effects of labour market regulation, empirical investigation for Germany indicates that the more pro-competitive regulation introduced in the 1980s and 1990s had positive employment effects, increasing the employment rate in the order of a couple of percentage points.[133] In the latter part of the 1990s, the overall regulatory stance in Germany was in an international comparison characterised by relatively strict employment protection legislation and about average product market regulation.[134] The implication of such a regulatory stance is that the benefits of product market reforms are likely to manifest themselves faster and more extensively if accompanied by labour market deregulation. Other and less obvious types of regulation may hamper the development of competitive markets, like craft trade regulation which in Germany tends to increase entry costs as well as putting German companies at a cost disadvantages *vis-à-vis* foreign companies (see Box 8).

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Box 8. **The German system of chambers of craft trades and product market competition**

Owners of craft companies in Germany are mandatory fee-paying members of chambers of craft trades. Among other things, these chambers are active – together with other social partners – on the boards setting regulation and standards for apprenticeship and masters' certificates. The latter requires up to 1 600 hours of formal training, including subjects related to technical knowledge and business administration. Various observers have pointed out that this system creates problems in the area of product markets. On the other hand, the system is also a measure to reduce asymmetric information. The council advising on product market issues (*Monopolkommission*) has observed that the requirement of mandatory membership puts German firms at cost disadvantage *vis-à-vis* foreign competitors. The Constitutional Court has ruled that the craft trade regulation, requiring that anyone wanting to establish a company must possess a master certificate, to some extent restrains the free right of practising crafts. The ruling has led to the introduction of some (not legally binding) exceptions from the current rules. When states leave the application of these exemptions in the hands of the regional chambers of craft trades, this may allow present members of the chambers to co-determine entry of newcomers. The effect of this regulation is that entry costs may rise, which according to the European Commission hampers the development of the service sector.

This section on product market competition firstly focuses on some international linkages, before turning to elements that may improve a generally well-functioning framework of *ex post* regulation for promoting competition.[135] Attention will then turn to the opening of competition in network industries, which has presented considerable regulatory challenges. This part will review progress and point to areas where the consolidation of achievements and further progress are likely to require additional measures to promote competition. In the last part of the section, changes in retail laws are discussed.

European regulation has an influence on competition in Germany

The development of competitive product markets in Germany is part of a wider European agenda of creating a well-functioning EU internal market with market liberalisation across the union contributing to improve economic growth and consumer welfare in Germany as well as in the rest of Europe. On the other hand, EU regulation has hampered developments in German product markets. An example has been the EU's block exemption from the general competition rules for the

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automobile industry. The block exemption allowed vertical agreements, resulting in market foreclosure, *i.e.* denying market access to new entrants.[136] As a result, each car company established a proprietary dealer and repairs network in each market. This exemption is likely to have contributed to the fact that prices of cars sold in Germany are among the highest in Europe as well as to have hampered German exporters' access to other EU markets. The scope of the block exemption will be scaled back by 2005, and it will by then prohibit location clauses, allow multi-brand dealer ships, and increase competition in the after-sales sector. These changes are expected to enhance competitive pressures in the market through the lowering of domestic and foreign non-tariff entry barriers.

At times, German specific regulation in seemingly unrelated areas may interact with EU rules to create barriers to entry, such as the introduction of a mandatory return system for non-recyclable containers in Germany as an environmental measure. However, as EU regulation requires spring water to be bottled at source, a mandatory return system imposes relatively higher transportation costs on foreign spring water producers, creating an economic barrier to entry. Hence, conditions attached to the treatment of non-returnable containers should be revised in view of their anti-competitive effects.

The general competition policy framework is well-developed

The well-developed and comprehensive general competition framework, originating from the 1950s, is complemented by sector regulation of newly liberalised network sectors. Some interaction between the general framework and the sector regulators arises from the Federal Competition Authority's merger control activity. The Federal Competition Authority employs about 270 persons (of whom about 10 are employed in a newly-created division for electricity), and the L*ändes*competition offices (L*andeskartellbehörden*) employ about 30 persons and deal with intra-state competition issues, such as unrestricted third party access in the energy sectors.[137, 138] The level of staffing is somewhat higher than that of the sector regulator for telecommunication and postal services (R*eg*TP). The Federal Competition Authority appears reasonably well staffed at levels that are comparable with, for example, the competition authority in the United Kingdom (where there also are sector regulators) which has a staff about 200 persons. However, the Federal Competition Authority is responsible for merger control, which is under strict time limits, so surges in such activity – as recently experienced for example in the energy sectors (see below) – tend to draw resources away from other activities.

The available instruments for deterring anti-competitive behaviour are well developed. In case of competition law infringements, the competition authority is allowed to issue administrative fines of up to € ½ million or up to three times the associated profits.[139] In addition, a recent legal amendment has made it possible for affected customers to sue for damages, further enhancing the cost of

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(uncovered) anti-competitive behaviour. Investigative tools, on the other hand, are somewhat less comprehensive. Fines for non-compliance with information request are comparably much lower at about € 25 000, opening up the possibility that non-compliance costs are considered as just another business expense. Historically, this has probably not been so important as domestic firms have co-operated with the Federal Competition Authority to establish good long-term relationships. However, the increasing internationalisation means that in the future larger numbers of cases are likely to involve foreign firms. In comparison, the telecommunication regulator (RegTP) has the possibility to issue fines up to about € ½ million and the European Commission has the right to issue fines of up to € 50 000 for intentionally or negligently providing incorrect information.

In 2000, the available investigative tools were enhanced with a leniency programme, making firms that divulge the existence of a cartel and co-operate fully with the authorities eligible for reduced fines. The effectiveness of this tool is too early to evaluate, but in mid-2002 the first case of a price-fixing cartel was uncovered under this programme. In general, such programmes are considered as essential in breaking-up hard-core cartels.[140] The leniency programme also aims at providing incentives for employees to participate as they can obtain leniency from the administrative fines that can be imposed upon them by the competition authority, although on occasions such fines have been paid by the employee's firm.[141] In the United States and Canada, among others (and under consideration in the United Kingdom), this issue has been addressed by the possibility of jailing individuals. Another extension of this "stick-and-carrot" strategy would be to offer economic compensation for potentially lost income, which would also attract individuals not liable for fines.

To improve the well-developed framework to promote competition the powers of the Federal Competition Authority should be expanded. The fines for non-compliance with information requests should be increased to secure a deterrence effect. The newly introduced leniency programme is a valuable additional regulatory instrument, although a complete evaluation is still too early. However, it is difficult to make sanctions individually binding when firms may pay fines issued to employees and the latter cannot be jailed. Together with the lack of economic compensation for lost income, this makes the associated costs of non-participation and benefits of participation for individuals relatively modest, pointing to possible avenues to enhance the programme's effectiveness. The resources available to the Federal Competition Authority could be expanded in some areas, like the electricity sector, as the currently available resources may not suffice as a result of intensive merger activities and the need for substantial regulatory activities in sectors where competition is being introduced. Moreover, competition can be further enhanced by stepping up privatisation efforts, including at local levels of government.

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Mergers

The activities of the Federal Competition Authority interact with those of sector specific regulators through the former's responsibility for merger controls. This allows the Federal Competition Authority to affect market structures, whereas sector regulators are charged with introducing competition within given market structures. The interaction in itself so far does not appear to have created regulatory problems. A prominent example of the interaction is in the telecommunication sector, where the incumbent (Deutsche Telecom) wanted to divest its cable TV holdings. The Federal Competition Authority stipulated that a potential buyer (Liberty Media Corporation) had to offer telecommunication services. In the end, the Federal Competition Authority vetoed the sale on the grounds that Liberty would dominate the cable-TV market and parts of the broadband cable networks. Other important merger cases include the merger between two large energy companies RWE and VEW, which was approved subject to strict conditions. The active merger control stance is instrumental in bringing in structural remedies in sectors where the market structure does not promote competition. Following a merger, such structural remedies can only be introduced if the merged company fails to comply with the conditions in the initial merger control decision as neither the Federal Competition Authority nor any of the sector regulators have powers to instigate divestitures or other structural remedies. As a supplement to merger control rules, the introduction of powers to implement structural remedies, such as the possibility for breaking up cartels in the United States, could be considered in cases where abuse of dominant positions has taken place. However, at present such powers are not embedded in EU law. An additional issue is that given the wide-sweeping nature of such instruments such decisions should be subjected to approval by the relevant courts.

Another merger in the energy sector between Germany's largest grid gas company (Ruhrgas) and a large combined gas and electricity company (E.On) was blocked for similar reasons. In the latter case, however, E.On appealed to the Ministry of Economics and Labour for a ministerial clearance to overrule the Federal Competition Authority's decision and allow the merger to be carried through. Such a ministerial clearance can be granted for overriding macroeconomic reasons – for example in terms of securing the national energy supply – although such clearances are very seldom used.[142] In mid-2002, the ministerial clearance was granted with conditions attached.[143]

The competition frameworks for network industries can be improved

As outlined in the previous survey, considerable progress has been made over the past years in introducing more competition in network industries in Germany, often leading to market openings ahead of the requirements in European regulation. The key problem in securing competition in network industries is

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to ensure unrestricted access at non-discriminatory charges, often solved by dynamic *ex ante* regulation capable of adapting to market developments, pointing to the advantages of establishing sector specific regulators. The Monopoly Commission has even suggested a unified network regulator to concentrate available expertise and to lower the risk of regulatory capture. Indeed, existing institutional knowledge concerning network regulation must be exploited to successfully introduce competition in currently regulated network areas such as the provision of water services. Furthermore, the market structures in network industries must be such that they promote competition, which generally points to the benefits of unbundling vertically and horizontally-integrated companies combined with privatisation to avoid conflicts of interest between ownership and regulation.

Telecommunication

Telecommunication prices for baskets of household and business services in Germany have declined more than 15 per cent since the previous survey, which is somewhat more than in other European countries, indicating a relative improvement in the degree of competition. Telecommunication prices for a basket of household services are now comparable with the average of Europe, and somewhat above those in the earliest opened markets in Europe, like Scandinavia and the Netherlands (Figure 31). On the other hand, the same comparison for business prices indicates that prices for this segment are relatively high as compared with other European countries.[144] The above prices reflect the consumption of standard customers using the largest companies and do not reflect that in the German telecommunication market cost-cautious customers may obtain significantly lower prices by shopping around. In addition, prices for mobile phone services were in mid-2002 relatively high in a European context.[145] However, the reported average price developments are reflecting that improvements in fixed-line telephony that have been brought about almost entirely by regulatory measures, as the market is characterised by an inherent risk of abuse of dominant position, and possibly by a high concentration in market for mobile services.[146]

In the fixed-line telephony segment the regulator (RegTP) has forced Deutsche Telekom (DT) to lower barriers for entry and continues to subject the dominant fixed-line provider DT to direct price regulation. In addition, as described in the previous survey, an element-based interconnection system has been implemented. The regulator has lowered interconnection charges for local telephony, access charges to the unbundled local loop and charges for transferring customers that other providers have to pay. The regulator also intends to introduce a system of automatic fines for delays. Moreover, regulatory concerns that DT was dumping charges to prevent entry led to substantial increases for digital connections and installation. In a further move to allow entry in this area (where DT has a virtual monopoly position) the incumbent was ordered to set its line sharing

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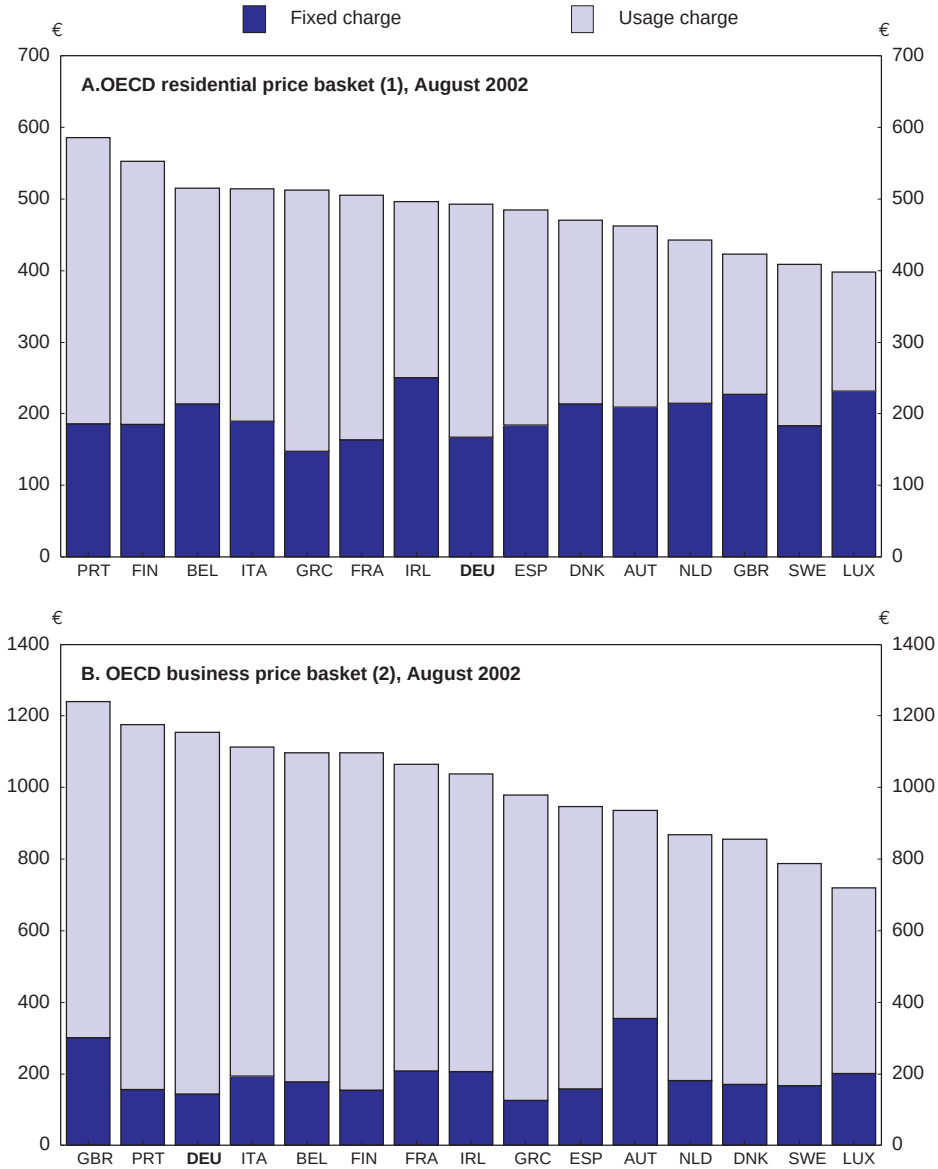
Figure 31. **Telecommunications prices in EU countries**



1. Composite basket including international calls and calls to mobile networks. Prices include VAT.
2. Composite basket including international calls and calls to mobile networks. Prices exclude VAT.

Source: OECD.

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charges for competitors at a third of its preferred level and to offer flat rate access.[147] In addition, the regulator lowered the required number of connection points by 70 per cent for narrow-band flat-rate access, easing access for smaller providers. Following investigations by the regulator, DT lowered retail fees for local calls.[148] In the latter area, price regulation may become less important as the Parliament has passed legislation – in response to the EU Numbering Directive requiring pre-selection by January 2000 – to allow pre-selection and "call-by-call" in the local loop in December 2002. This should enable such service providers to offer a more complete set of services, thus strengthening competition and expanding consumer choice.

In the market for mobile services, there is no single designated provider with substantial market power despite the two largest providers having a combined market share of about 80 per cent, raising the issue of joint dominance.[149] On the other hand, the regulator's latest investigation concluded that the two largest service providers do not have joint dominance. In addition, there is no *ex ante* price regulation in place and regulatory action has aimed at improving framework conditions, such as the introduction of number portability for mobile phones by October 2002, furthering competition through a lowering of customers' switching costs. Nevertheless, competition problems remain in the provision of mobile services, notably in the area of calls between mobile and fixed networks and possibly also in the area of "roaming charges".[150]

The question of non-competitive "roaming" charges (exceeding associated costs) is the subject of a European Commission investigation – initiated in 2001 – into a collusive agreement between DT and providers in the United Kingdom, with the subsequent law case expected to be concluded by end-2002. The main competition problems are that the domestic mobile phone provider has little negotiating power *vis-à-vis* foreign networks and that the user abroad has difficulties in obtaining information about the cheapest foreign provider. As a consequence of EU legislation in this area, national authorities have to assess whether wholesale roaming markets are competitive and, if not, introduce regulation in this area. While some other EU countries start such assessments earlier, the German regulator will only begin its assessment on the day (25 July 2003) that the legislation is in place.

The average per minute price for calling from a mobile network to fixed-line network varies from € 0.07-0.08 per minute on weekends to nearly seven times more for peak time calls. While recognising that the cost of providing the service is an increasing function of capacity utilisation, such a large price differential is unlikely to reflect higher costs. In comparison, Deutsche Telekom's mobile phone subsidiary (Voicestream) in the United States is offering much lower rates, charging the equivalent to the average weekend rate in Germany for connecting mobile customers anytime and anywhere in the United States.[151] Prices for the

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reverse calls are also unlikely to be cost-reflective. The average price of calling from fixed-line to mobile networks varies less over time and is on average a bit more than 2½ times the weekend price for the reverse call, which again does not seem to reflect associated costs. Recognising the lack of competitive forces to lower prices for fixed-line to mobile phones – as the mobile phone customer is mostly interested in the price of outgoing calls and the caller has little choice – the EU has proposed to introduce price regulation in this area. However, despite the high market concentration and the high prices in this area, such charges are unregulated in Germany as no market participant is designated as possessing a dominant market position. On the other hand, German operators reduced their fixed-to-mobile termination charges by 50 per cent following a complaint from a new entrant to the European Commission.[152] Another indication of the problems in this area is that charges for terminating an international call on a mobile phone in Germany is about 10 times higher than the similar charge for termination on a fixed line telephone.

The continued need to regulate a wide set of prices and secure unrestricted interconnection to the fixed-line network in the telecommunication area are reflections of the incumbent's dominant position. Such regulatory efforts are clearly needed as long as DT maintains a dominant market position, implying that the planned 2003 revision of the Telecommunication Act should maintain the present *ex ante* regulatory regime.[153] The establishment of competing networks would erode DT's dominant market position, thus ensuring a competitive market structure.[154] This reinforces the need for ensuring the timely development of UMTS networks (see Box 9). Once installed, these networks will cover all of Germany and their data-transmission capabilities combined with their portability element make them potentially close substitutes to the existing fixed line and mobile phone networks. Other potential candidates for establishing a competing network with high-speed data capabilities are the cable-TV networks in Germany – covering nearly two-thirds of all households, which should be sufficient to make cable-TV networks an effective competitor to DT's network in many areas.[155] However, as described above, the sale of DT's cable TV networks has been stalled, meaning that for the time being little progress has been made in developing the cable-TV networks as competitors to DT's fixed line network.

To sum up, the incumbent's dominance of the local loop means that active *ex ante* regulation will continue to be needed at least until competitive telecommunication structures are secured through the establishment of competing networks. In the area of mobile telecommunication services, developments in joint market dominance, roaming charges and charges for connecting fixed and mobile lines need to be closely monitored by the regulator. If problems in these areas persist, then more regulatory intervention may be needed. Measures could include formally ascribing joint market dominance to the two largest mobile service providers and direct price regulation to bring down prices for voice telephony

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Box 9. Securing the construction of competing UMTS networks

As described in the previous *Survey* the licenses for 12 blocks of spectrum for Universal Mobile Telecommunication System (UMTS) were sold through an auction of non-tradable licenses, which yielded combined license fees of around 2½ per cent of GDP. The non-tradability referred to trade among the participants in the auction and prevented the entering of bidders without the intention of developing such services. Moreover, it assured an outcome with a market structure of either four providers offering advanced services or six providers offering basic services, with the latter being the final outcome. Subsequently, financial pressures have mounted in the sector as market participants struggle to finance the licenses and the construction of the networks at the same time as stock market valuations have plummeted. In order to ease financing costs license holders have been allowed to share some of the UMTS infrastructures.

Until now no financially weak license holder has entered negotiations to sell the license to a non-license holder, possibly because resale prices have been depressed by the exclusion of other UMTS license holders from purchasing additional licenses. The value of a license may even drop to zero if two license holders merge, as one of the licenses must then be returned. The interest of a license holder in acquiring an additional block of spectrum is to offer advanced services. However, a market structure with one company offering advanced services and four companies offering basic services would not be a level playing field, which only would be the case with a market consisting of four providers offering advanced services or the present situation.

If a license holder retires from the market the license would have to be handed over to the government, leading to an increase in market concentration. If the government wants to prevent such an increase in market concentration and sells the license, it appears unlikely that a license fee similar to the original one can be obtained, in which case the new entrant will have a substantial cost advantage, particularly in the current business environment. The original purpose of non-tradable licenses has been fulfilled and combining tradability of individual spectrums with strict merger control – to avoid a concentration of providers offering advanced services – could ease the consolidation process in the sector, preserve competitive markets and help to secure the construction of competing networks.

between fixed-line networks and mobile networks. In the area of roaming charges the anticipated regulatory delay could be bridged by introducing temporary regulation, for example requiring domestic mobile service providers to inform their customers abroad about foreign providers' roaming charges, through SMS text messages. More generally, ways to facilitate the emergence of competing networks, such as cable TV and UMTS networks, should be explored.

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Electricity and gas sectors

The most important developments in the electricity and natural gas sectors since the previous survey have been the ongoing wave of mergers and the expansions of the Associations' Agreements. The latter reflects the German choice of a negotiated third party access approach as opposed to the *ex ante* regulated third party approach in other EU countries. These voluntary agreements between associations of consumers and industry participants contain framework recommendations for their members to conclude freely negotiated contracts for transparent and non-discriminatory access to the transmission and distribution grids (see previous *Economic Survey*). In the absence of an independent sector regulator, the markets are subject to the abuse control provisions of the general competition legislation. The regulatory responsibilities in the sector are divided between the Federal Competition Authority being responsible for cases involving more than one state and the Land competition offices for intra-state cases. Moreover, the enforcement of abuse control has been strengthened with the aforementioned establishment of sector specific division in the Federal Competition Authority for electricity. Further strengthening of the regulatory framework is expected with the adoption of a new energy law proposal, which makes the decisions of the Federal Competition Authority immediately binding. On the other hand, the proposal also gives legal status to the private agreements embedded in the Associations Agreements, which could meddle with the regulatory efforts of the Federal Competition Authority.

Common for the two sectors is the high degree of vertical integration and the fact that the majority of local distribution companies is engaged in more than one network industry, providing a combination of gas, electricity, heat and water services. As a preparation for more competitive markets, there has been a merger wave going through the local distribution companies as many lack minimum efficiency size (in terms of distribution and purchasing power). More worryingly for the promotion of competition, concentration has also increased at the supra-regional level, with for example only four companies left in the electricity sector, generating about 80 per cent of all electricity. Horizontal integration has increased with the mergers between companies in the gas and electricity sector – a prominent example is one of the largest electricity companies RWE AG gaining a strong position in the gas market through its 75 per cent ownership in Thyssengas GmbH.[156] Indeed, the main objections against the ministerial clearance of E. On and Ruhrgas merger were concerns about increasing concentration and integration – both vertically and horizontally – in the energy sector.[157] As raised in the previous survey of Germany, securing competition in network industries hinges on securing non-discriminatory access, which is impeded by vertically-integrated companies. Likewise, horizontal integration may lead to cross-subsidisation between competitive and non-competitive segments.[158] The large number of publicly owned companies only compounds these problems.

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Some progress has been made in achieving competitive energy markets, as measured by how fast prices in Germany have approached best performances in the EU (Figure 32 and Figure 33). Over the past five years, German electricity prices for industrial users have been decreasing more than in most other EU countries, while those for households have developed on a par with the EU average. Nonetheless, the level of pre-tax electricity prices remains higher than the EU average for both industrial and household customers – and for the latter group the price effects of liberalisation have been offset by – environmentally motivated – higher taxes and charges on energy.[159] Reflecting the diverging competition dynamics in the two segments, it has been estimated that 20-30 per cent of large users and 5-10 per cent of small users have changed supplier. In addition, a significant number of consumers have obtained better deals with their existing supplier. Over the same period, gas prices for both industrial and household consumers tended to increase faster than in other countries, leaving Germany with some of the highest gas prices in Europe.

The regulatory framework has not prevented competition problems from surfacing. Network access charges in the electricity sector came under the Federal Competition Authority's scrutiny in mid-2001 as network operators' charges exhibited a large variation and with the highest charges being more than 50 per cent higher than the average. The latter was found by the International Energy Agency to be among the highest in Europe, although international comparisons of charges are difficult because of measurement problems.[160] The investigation resulted in a voluntary lowering of charges, but in early 2002 the Federal Competition Authority instigated new abuse proceedings against 10 network operators. The high access charges reflect, on the one hand, that the charges for high voltage customers are at par with the European average and, on the other hand, much higher charges for consumers connected to lower voltage networks. Arguably the optimal strategy for vertically integrated generating/transmission companies wishing to deter entry is to set the prices for supplied electricity at competitive levels – *i.e.* such that they recuperate variable generation costs – and to set transmission tariffs to at least recoup the integrated company's fixed costs. Empirical research indicates that German companies indeed have high network charges and low profit margins in the competitive segment of supplying electricity, constituting entrant deterrent pricing strategies.[161] For example, the shares of access charges in the end-user prices are estimated to be broadly twice as high as in the United Kingdom for most customer categories, underlining the need to focus regulatory efforts on securing non-discriminatory network access charges. In addition, the Federal Competition Authority has directly criticised the special low loyalty tariffs offered to present customers on the grounds that they hamper new entry.[162] Access problems have led to more than 200 cases being brought to the competition authorities. The Federal Competition Authority has initiated over 20 preliminary investigations concerning access charges, of which 11 cases have led to formal

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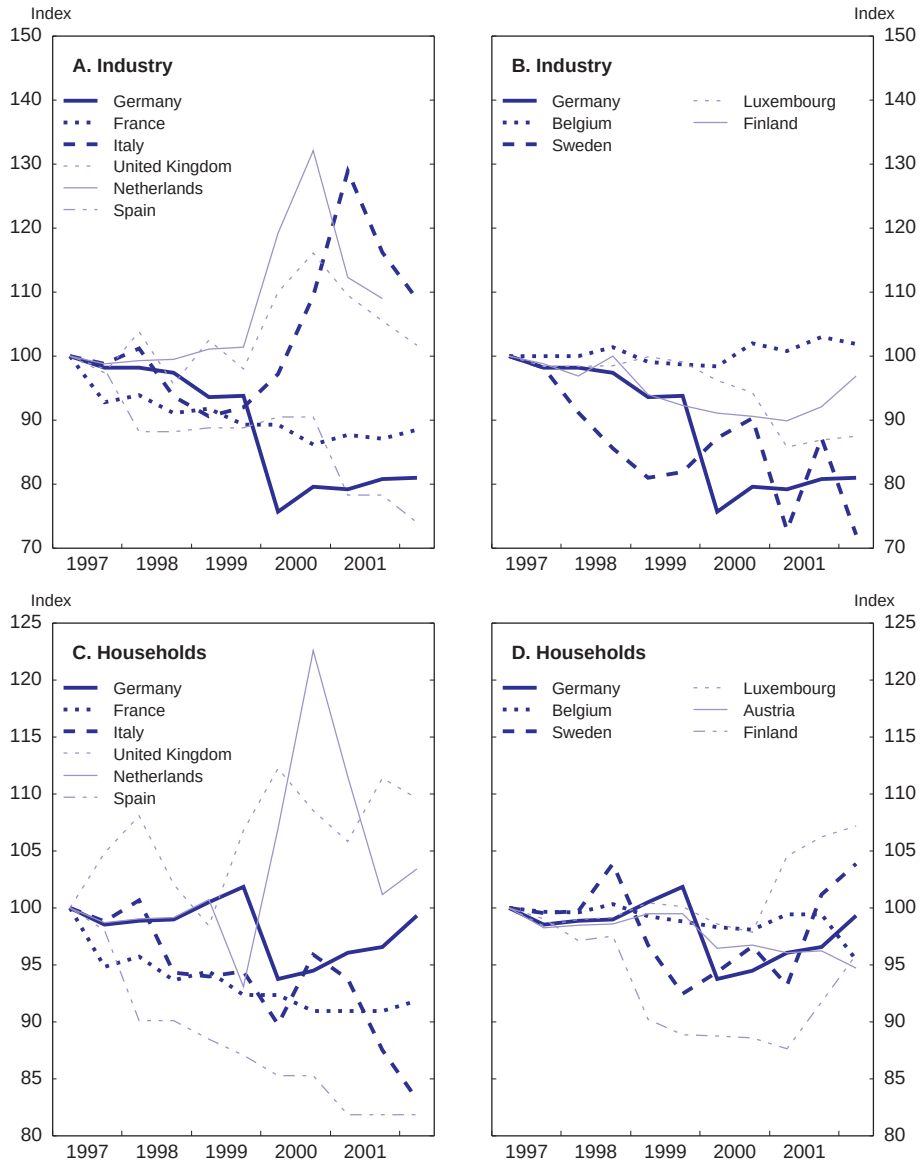
Figure 32. **Electricity prices in the EU**[1]

Index S1 1997 = 100



1. Prices for medium size consumers.

Source: Eurostat.

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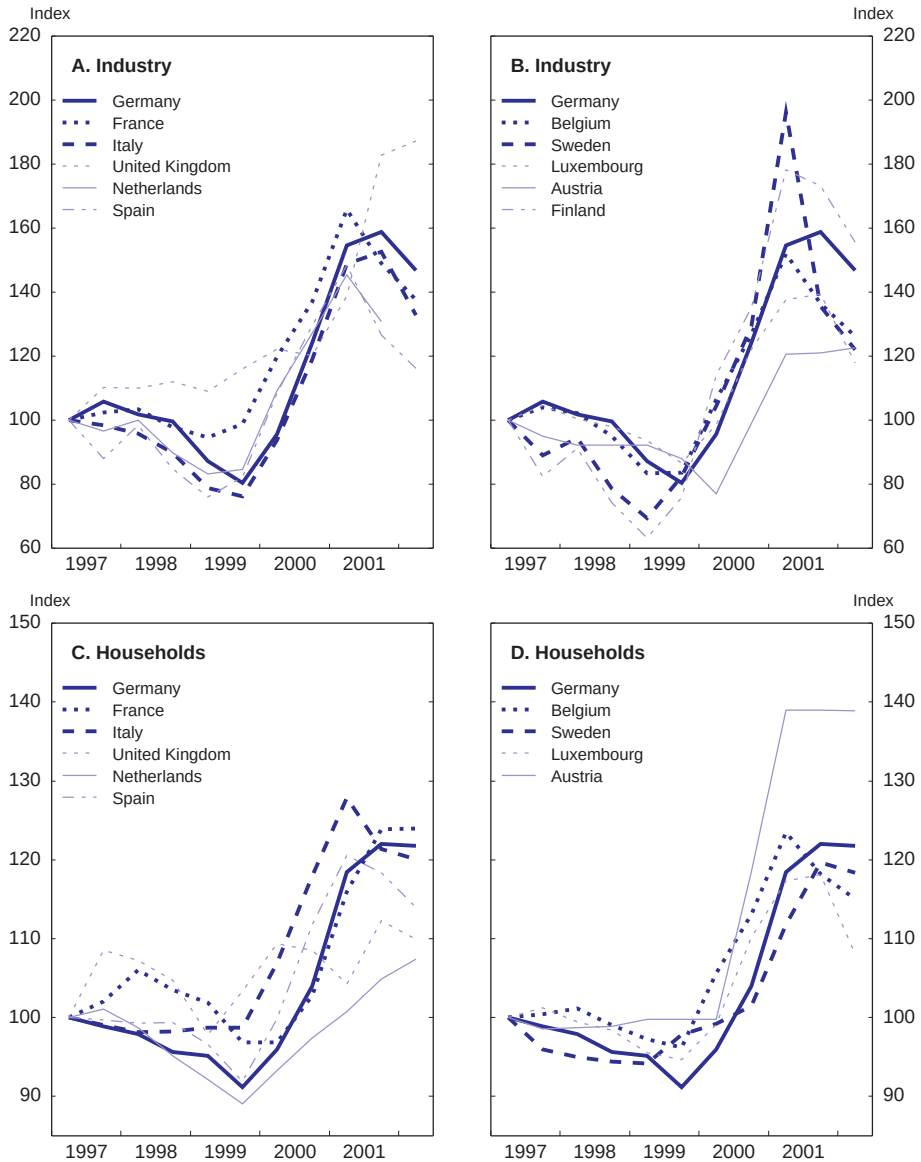
Figure 33. **Gas prices in the EU**[1]
Index S1 1997 = 100



1. Prices for medium sized consumers.
Source: Eurostat.

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investigation procedures. Some of the inquires and informal procedures have led to a reduction of transmission fees in the order of 15-20 per cent, using benchmarking with other German companies, although international benchmarking is planned.

The Associations' Agreement II Plus – the third version and for the first time including the participation of the small consumer association – has simplified the conditions under which smaller consumers can choose their supplier as well as having increased flexibility in balancing regulations. Moreover, the agreed new calculation guidelines and the monitoring mechanism for access fees are expected to lead to a smaller variation in net access fees as differences increasingly reflect only the costs of network operation. However, such developments are unlikely to materialise rapidly. The current wave of mergers has increased concentration (vertically and horizontally) and the negotiated third party process in a complex market structure is time-consuming and can be costly for small market participants and new entrants, possibly leading to discriminatory access conditions.[163] Moreover, dispute settlements are time consuming and access tariffs are difficult to challenge in court. The Ministry of Economics and Labour has set up a Special Task Force to find practical solutions to supplier switching problems for smaller customers.

The natural gas market was liberalised in 1998 and the first Associations' Agreement came into force in mid-2000.[164] The agreement was amended twice in 2001 to include access to storage facilities and networks, rules for ensuring the free choice of gas provider by the final customer, and consumer profiles for smaller customers. The third amendment in 2002 was only agreed upon after the Minister of Economics announced the preparation for establishing a sector regulator due to the lack of agreement. This line of action was abandoned in the wake of the agreement to the Association Agreement Natural Gas II, which reduced the three layered pricing schedule to only a distance related tariff for national and regional transmission and a "stamp tariff" for local distribution, although it is still at the discretion of the network operators to categorise individual pipelines in terms of distance definition.[165] The parties to the agreement have announced additional negotiations to further improve the access system, such as a model for determining third party tariff-setting criteria.[166, 167] Moreover, capacity bottleneck problems have been solved through the introduction of contracts that allows gas to flow until it is physically replaced, preventing the use of "phantom" contracts that reserves capacity without an actual flow of gas.

Despite the progress observed in the various Associations' Agreements, competition in the sector is still not very developed. The number of transmission contracts is still rather low. Concentration in the vertically-integrated industry has increased through mergers and through share holdings in other companies, giving access to the governing boards. Companies have through the latter been able to obtain information about cost structures as well as about the content in contracts

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with new entrants, which allow incumbents to establish similar conditions across the industry to discourage new entry. Moreover, exclusive contracts in the downstream segment of the market are prevalent and include rules concerning minimum gas purchases.[168]

Some of the above competition problems may be addressed as German legislation is brought into compliance with the EU Gas Directive through a legal amendment (as part of the energy law proposal) to secure non-discriminatory third-party access, which was presented to parliament in 2000, but was not passed before the election. The proposal is planned to be re-presented by the new coalition government. The proposal includes the mandatory publications of business and technical requirements for network access, the introduction of separate accounting – although the utilities are not obliged to publish these accounts – and the establishment of a dispute settlement office in the Ministry of Economics.[169] Moreover, the Task Force within the Ministry of Economics and Labour will address practical problems that reduce competition in the sector. Another regulatory issue concerns whether the Federal Competition Authority should establish a gas division equivalent to the new electricity division. An important issue in this respect is how to ensure sufficient staffing to secure non-discriminatory access in a market with 750 networks at the same time as merger control activities are drawing on resources.[170]

The negotiated access approach in the energy sectors was chosen over a regulatory approach as an administratively simpler and less costly regulation. Despite an initial lowering of electricity prices immediately following the opening of the sector, there has since been a lack of better-than-average price performance as compared with the EU area. Indeed, energy prices are relatively high compared with other EU countries. In comparison, the regulatory choice in most other network industries – internationally even in energy industries with similar complex structures has been an *ex ante* regulatory regime, partly reflecting that sector regulators can secure a dynamic and pro-active regulatory stance that reacts to market developments. If the remaining access problems cannot be resolved in the electricity sector, the government should consider introducing a sector regulator. In the gas sector little progress has been observed and given the scale of access problems, the government should reconsider its decision not to install a sector regulator. Vertically-integrated conglomerates are an impediment to non-discriminatory network access. Hence further unbundling of energy generation, distribution and transmission beyond accounting and management separation would reduce regulatory challenges, but is currently hampered by widespread public ownership in local utilities. Moreover, in recently deregulated network sectors purchases of competitors' shares, which are below the threshold for merger control, but which give board representation, should be subject to regulatory control in order to prevent anti-competitive price co-ordination through information sharing of board members.

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Postal services

The development of a more competitive postal market was delayed by the government's decision in 2001 to prolong Deutsche Post AG (DP)'s monopoly on domestic letter delivery to end-2007. The monopoly right will in accordance with the newly agreed EU directive now be gradually reduced from the present 200 grams and less to 50 grams and less by 2006. The prolongation from the original termination date of end-2002 was a consequence of the government making further liberalisation of the postal market dependent on European developments. Besides, the planned full opening of German postal markets will ensure that market opening remains ahead of EU directives.[171] Nevertheless, the decision to postpone the full opening of postal markets should be reconsidered with a view on expanding consumer welfare and improve competitiveness of enterprises.

The playing field where competition has been introduced in the postal sector is far from levelled, as DP is exempt from sales taxes, including on revenues generated on competitive markets, as a compensation for DP's universal service obligation.[172] However, this form of compensation is rather arbitrary and has little consideration of the incumbent's associated costs and benefits from having a nation-wide network. Furthermore, a European Commission investigation showed that DP has used federal transfers (received to compensate its public service obligation) to cross-subsidise activities in competitive market segments in the period 1994-98. As a consequence DP was ordered to repay € 572 million plus interest to the Federal government. In mid-2002 there was a regulatory-induced reduction of the internationally high postal tariff.[173] The postal regulator ordered DP to lower tariffs (affecting about 80 per cent of DP's services) by 4.7 per cent in non-competitive segments and granted the permission to increase tariffs by an average of 0.7 per cent in competitive segments. Insofar as cost of universal service obligation exceeds the benefits of having a nation-wide network the associated compensation should take place as a fiscal transfer from the government. Such a compensation system is rarely used in OECD countries, but would provide for an explicit assessment of the additional costs as well as level the playing field in the competitive segments of the postal market. Moreover, the privatisation of DP should be completed to avoid future conflicts of interest between ownership and regulation.

Railways

There has been some progress in introducing more competition in the railway sector. Effective from mid-2002, the Federal Railways Office (formerly mostly concerned with technical and safety aspects of railway operations) was granted greater powers to prevent discriminatory access, although these do not include powers to regulate route charges.[174] Moreover, the dominant incumbent Deutsche Bahn (DB) has been forced to introduce a non-discriminatory route price

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Box 10. Should the opening of domestic markets be conditioned on the opening of foreign markets?

The opening of German network industries to greater competition has been concurrent with or ahead of similar developments in other EU countries, in part induced by the ongoing creation of the EU internal market. As described in the previous *Economic Survey*, Germany is in the forefront of Europe in the liberalisation process in, for example, the market for telecommunication and the energy sectors. The government, however, has conditioned further market opening in, for example, the postal service on similar developments in other European countries.[1] The arguments for early liberalisation in domestic markets are centred on fostering higher GDP growth and the existence of a first-mover advantage, enabling domestic companies to benefit from the subsequent opening of foreign markets. Arguments for conditioning the domestic liberalisation process on developments in other European countries are based on the possibility that German firms are disadvantaged if only domestic markets are liberalised.

The opening of domestic markets to competition introduces competitive prices and stimulates innovative activity, which simultaneously removes a monopolist rent and increases consumer choice and welfare. However, for these benefits to materialise privatisation and other measures to induce competition must be combined with effective pro-competition regulation, including merger control and a prohibition of predatory pricing strategies, to prevent abuse of dominant positions. A counter-argument against opening the market to foreign companies is that a foreign monopolist (on his home market) entering a newly opened domestic market may have an unfair advantage by the possibility of cross-subsidising competitive market activities with the monopoly rent obtained in its home market, possibly smothering competition.[2] However, such a situation is unlikely to emerge with the presence of an effective pro-competitive regulatory stance that can prevent the use of such predatory pricing strategies. Nonetheless, it should be recognised that the regulatory effort of detecting and controlling such cross-subsidies is substantial, and is a general problem associated with increasing internationalisation. Thus, an effective pro-competitive regulatory stance should insure that the improvement in consumer welfare is not affected by the entry of a foreign monopolist. Another positive effect is the emergence of more efficient domestic companies, irrespective of the presence of a foreign monopoly, which can exploit emerging market opportunities when similar markets in other countries are being liberalised.

1. This, however, has not dented Deutsche Post AG's global aspirations to expand its international operations through mergers and take-overs as outlined in its 2002 annual news conference.
2. More generally, the benefits that the foreign monopolists pursue by expanding abroad are centred on seeking – often elusive – economies of scale effects. The difficulties of such strategies are witnessed by France Telecom's recent € 7¾ billion write down related to its holding in mobile-phone companies in Germany and the United Kingdom. See *www.francetelecom.com/anglais/finances/f_finance.html*

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structure after the Federal Competition Authority began abusive pricing proceedings against the company.[175] However, further market opening is increasingly becoming dependent on developments in other EU countries with an expert group under the Ministry of Transport emphasising that the EU railway directives for opening up to competition should be implemented simultaneously with similar reforms in other countries.[176] The group also concluded that further efforts to improve competition in the railway sector should be centred on ensuring non-discriminatory access to the network infrastructure and on making the relationship within the DB group (comprising of the network operation and the service provider) more transparent by managerial and accounting separation. The government is preparing a bill to secure strict organisational separation and which puts the competence of fixing of access fees and conditions with the network operator. At the more general level, securing non-discriminatory access would cease to be an issue with ownership separation. On the other hand, the need to co-ordinate externalities, such as securing appropriate safety standards, arguably suggests making all stakeholders – providers and purchasers of railway services – owners of the network operator. This would ensure that the interests and concerns of the stakeholders are reflected in the activities of the network operator.[177]

The lack of strict separation between the network operator and the service providers has caused problems and slowed down the emergence of competitive markets.[178] In 2001, the European Commission initiated cartel procedures against DB for refusing to rent rolling stock to a competitor (operating an international line given up by DB) on the same conditions as offered in hiring agreements within the DB group. Moreover, competitors have complained that new entry is being thwarted by the lack of a secondary market for rolling stock as DB dispose of disused material, and infrastructure is only developed on lines where DB wins the tenders for the provision of railway services.[179] These cases add to the argument of securing strict separation. An additional recent measure is the up to 12 years phasing in of public procurement rules to apply for the provision of railway services. While this is in accordance with current EU rules, such a long phasing in period hampers the promotion of more competitive railway markets.

To secure non-discriminatory access to the railway network, the Railways Office should be granted additional regulatory powers concerning network access charges to prevent that the incumbent cross-subsidies competitive activities and raises entry costs. To promote competition in the railway sector there needs to be a strict separation of elements where competition can be introduced, *i.e.* provision of services, and of elements with natural monopoly characteristics. Consequently, ownership unbundling should be considered in the course of privatising D*eutsche* B*ahn*, both as a protection of future shareholders and because of the lack of instruments to do so afterwards.

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Provision of water services

The cost efficiency of German water services has come under scrutiny. In an international comparison, a very high share of the population has access to public water supply and sewage treatment facilities, but the high number of small municipal-owned water utilities – with monopolies for water services ensured by the law – has raised efficiency concerns.[180] Moreover, German drinking water charges are among the highest in the EU.[181] Nevertheless, OECD research indicates that prices may have to increase about 20 per cent if they are going to be based on the full cost recovery principle to avoid cross-subsidisation.[182] Introducing competition in the provision of water services is more difficult than in other network industries. Nevertheless, there may be scope for improving efficiency through greater use of benchmarking and by expanding the geographical coverage of individual water works and the use of competitive tendering of water contracts. For the latter to be effective, outsourcing of water services must be regulated through existing public procurement laws and existing tax advantages in the field of sewage disposal for publicly owned water utilities must be removed.[183]

The retailing sector

The retail laws restricting rebates and the value of promotional offers were abolished in mid-2001, clearing the way for discounts larger than 3 per cent outside the twice annual two-week sales seasons. However, the Law Against Unfair Competitive Practices prohibits time-limited discounts outside the official sales seasons with the exceptions of liquidation and jubilee sales.[184] This led to a combined € 1 million fine to a leading retailer, who offered a 20 per cent discount on all purchases made by credit or debit card – extended to all purchases and with a time limit after a court injunction – in connection with the change-over to the euro.[185] In addition to the general prohibition of predatory pricing strategies, there is regulation in place – only applied when a set of specific factors is in place – which prohibits below-cost pricing, which tends to reduce competitive pressure and consumer choice.[186] Generally, such regulation prevents the use of "loss-leader" strategies to promote sales and of price strategies to eliminate unwanted inventories and weakens larger retailers' negotiating position *vis-à-vis* wholesalers, which otherwise could enhance competitive pressures in the sector.[187] In addition, such regulation raises questions of interpretation like the appropriate distribution of fixed costs and whether such pricing behaviour is just a response to market developments as recently revealed in a court case.

To sum up, the regulatory framework for promoting competition in the retail sector can be improved in a number of areas. To reap the full benefits of recent retail sector reforms the prohibition of discounts outside the sales periods should be revoked. An additional step in this deregulation process should be to remove current provisions against setting prices below costs and let regulation in

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this area rely on the general prohibition of predatory pricing. The effectiveness of such regulatory changes will be enhanced by a further deregulation of shop opening hours as emphasised in previous *Surveys*.

Financial markets

From the beginning of the 1990s, German financial markets are being transformed as the banking sector goes through a cost-related consolidation, partly related to increasing competitive pressures from securities, such as shares and bonds. In the process, the banking sector is losing relative importance to the stock markets. These developments are being driven by internationally common factors, such as new technologies, declining communication costs, product standardisation and the euro, as well as by domestic factors, such as the removal of capital gains taxation for divesting cross-holdings and the introduction of the funded pension layer. These financial market developments and selected policy measures are surveyed in Annex I. The following section reviews selected regulatory responses, including the agreement with the European Commission to phase out public guarantees to publicly-owned banks and measures to improve corporate governance.

The agreement to phase out public guarantees to publicly-owned banks

The EU has long considered the public guarantees to publicly owned savings banks, affecting 12 central savings banks (*Landesbanken*) and 562 savings banks, as being indirect state aid. In 2001, an agreement was reached to phase out the state guarantees (*Gewährträgerhaftung* – guarantor liability – and *Anstaltslast* – state liability) over four years and existing grandfather agreements over 14 years. Two main models have emerged for banks to honour the agreement. The model chosen by West-LB splits the bank into a public-sector parent and a private commercial bank subsidiary coupled with the non-issuance of a "*Patronatserklärung*" – effectively removing all guarantees – by the parent company. The other model, chosen by Bayern LB, consists of creating a holding company, of which 49.98 per cent can be sold to private investors and with all bank activities maintained in a subsidiary. An agreement was also reached with respect to the public guarantees to the state-owned *Kreditanstalt für Wiederaufbau* (KfW) – Europe's fourth largest bond issuer. The institution will be split into a public and a private sector bank by 2007. The state guarantee will be maintained for the public bank, active in areas such as start-up financing, environmental investment and co-operation with developing countries.[188] The private sector bank will not have a guarantee as it competes with private sector banks in areas like export and project financing. The agreement did not include Deutsche Postbank AG – a subsidiary of Deutsche Post – as the Federal government has not issued any guarantees.

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Other aspects of state aid to *Länder*-owned banks have been questioned. In spring 2002, the EU commission initiated formal inquiry proceedings into whether the bailing out of the troubled Land-owned *Bankgesellschaft Berlin* by the city-state of Berlin is in accordance with EU's state-aid rules. The rescue aid granted to *Bankgesellschaft Berlin* has been declared compatible with EU competition law, while the restructuring aid is still under examination. Under the proposed restructuring programme the city-state of Berlin will provide € 1¾ billion financial subsidy as well as a state guarantee to free the bank of existing real-estate risks. In addition, on a number of occasions state banks have received capital injections from public housing loan institutions, allegedly amounting to state-aid, with one case pending at the EU court and with the EU commission initiating investigations into other cases. To avoid all explicit and implicit state support and to further improve the functioning of the banking sector, the phasing out of state guarantees in the sector should be follow up by the privatisation of publicly-owned banks.

Measures to improve corporate governance

A series of regulatory measures are aiming at improving corporate governance, including elements in the *Fourth Financial Market Promotion Act*. The latter pursue three main objectives: improve investor protection; extend and expand the scope for action for market participants; and close gaps in the defences against money laundering. The Act contains measures affecting six areas, including the laws on stock exchanges, securities trading, investment, banking, insurance and mortgages. The integrated financial market regulator (see below) will have greater powers to combat price and market manipulation as well as being allowed to temporarily prohibit short-selling to secure the stability of financial markets. Investor protection will be enhanced by mandatory disclosure of sales restrictions in listing prospectuses, tightening of disclosure rules for ad-hoc information and share dealings of corporate insiders (including a no-delay reporting requirement, but not for shares granted or sold to employees), greater possibilities for obtaining compensation for effects of late, omitted or incorrect disclosure of price-sensitive information. Financial analysts have to observe new rules of conduct in addition to existing "Chinese walls" between various departments within the same financial institutions. In addition, rules related to non-corporate governance issues, such as those for regulating investment funds' exposure in terms of products, instruments and currency exposure will be relaxed. To enhance the market dissemination of relevant information all share transactions that may reflect insider knowledge, such as the granting or selling of shares to employees, should be subjected to an immediate reporting requirement.

The *regulatory framework for corporate take-over* activities has moved from being based on a voluntary code to encoded standards. Take-over activities, mostly under mutual agreement between the involved parties, have been fairly common with a well-established cartel review process in the Federal Competition Authority,

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allowing take-overs to play their usual functions as means to exploit synergies and as disciplinary tools of management of listed companies with dispersed ownership.[189, 190] However, the hostile take-over of Mannesmann AG by Vodafone AirTouch plc and pending EU take-over legislation stimulated a public debate, which resulted in the new Act on the Acquisition of Securities and Take-overs ("*Wertpapiererwerbs- und Übernahmegesetz*") effective from 1 January 2002, replacing the previous voluntary Take-over Code.[191] The Act aims at introducing clear rules for company take-overs by creating common procedural guidelines, improving information and transparency, strengthening the legal position of minority shareholders and meeting international standards (see Box 11).[192]

The new legal framework for take-overs appears to largely observe the principle of proportionality between ultimate economic risk and control.[193] The new take-over legislation contains measures that allow management to take counter action to a hostile take-over bid, although the provision that they must be approved by shareholders should ensure that in general such measures would only be applied if in the interest of the shareholders.[194] The option of adopting advance resolutions concerning defence strategies is unlikely to be widely accepted by shareholders using the share price as an instrument for management supervision.[195] However, some elements in the legislation may create conflicts of interest. The requirement of directors' active participation (through the duty to comments) in the take-over process is a deviation from the neutrality principle, which was originally in the German draft (until March 2001) and in the EU directive rejected by the European Parliament. In addition, the involvement of the supervisory board in the determination of possible defensive strategies may lead to conflict of interest *vis-à-vis* shareholders.[196]

The legal framework for take-overs with stringent information requirements and time schedules is a clear improvement as compared with the previous voluntary approach and enhances the protection of minority shareholders. To address the problems of management applying defensive strategies without prior approval and the potential conflicts of interest embedded in the management' commenting duties strict management neutrality should be adopted. The bidder's obligation of providing legally binding post take-over business strategies is difficult to fulfil, as bidders only rarely have access to all relevant information about the target company, creating potentially open-ended legal costs, possibly preventing some take-overs. Thus, such information requirements should be relaxed insofar as they are economically costly.

A new *voluntary code for company executives* was introduced in early 2002, containing about 50 nationally and internationally recognised standards. These include recommendations to publish board members' remuneration (including performance-related items such as stock options) and a Transparency and Publicity Act for annual declarations and contravention. To ensure independence of supervisory boards

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Box 11. The Act on the Acquisition of Securities and Take-overs

Procedural guidelines require timely publication and schedules for submitting, examining and publishing bids as well as for the subsequent period for accepting bids, including cases with competing offers. *Information duties* include specifications concerning the intended bid, financing, future company location, financial indemnities for management of the targeted company, and planned changes in employment conditions in the target company. The bid documentation must be submitted in German. *Written comments by the directors of the target company* are mandatory and shall address questions concerning the compensation offered, consequences for the company and the bidder's business objectives as well as whether management will recommend to accept the offer. The comments must be published and submitted simultaneously to shareholders, the target company's work council and employees. If any of the two latter bodies submits comments these should be enclosed. *Statutory periods for convening shareholders' meetings* are shortened and procedures simplified following the publication of bids.

Financing must be secured and confirmed by a financial institution. *Compensation* may be any combination of cash or liquid shares of companies listed in the European Economic Area and cannot be lower than the highest price granted or paid by the bidder or the average market price within the three months prior to the bid. *Controlling thresholds* are set at 30 per cent and if reached a mandatory bid must be submitted for all remaining shares, although the regulatory authorities can exempt the bidder from this obligation, if for example full control cannot be achieved due to the opposition of a major shareholder. If the bidder obtains a controlling position in a subsidiary of the targeted company a mandatory bid to the subsidiary's other shareholders must generally be submitted. *Squeeze-outs* are possible when the bidder obtains at least 95 per cent of shares, obliging remaining shareholders to accept cash offers.

The basic principle for possible *defence strategies* that the management can adopt is that only shareholders should decide whether to accept a bid. A number of defensive actions are allowed, of which some are also parts of normal and conscientious management, such as the acquisition or divestment of companies and business. Other actions include defence strategies adopted at specially convened shareholders' meetings, such as the acquisition of external companies to persuade the competition authorities to block the take-over. Similar measures may be adopted as advance resolutions at shareholders' ordinary meetings (maximum 18 months before) with three-quarter majority. Measures approved by the supervisory board are also allowed (within the management authority of the executive board), such as agreements with work councils with possible consequences for the target company in case of a take-over, share buy-backs or the use of authorised capital (maximum half of the company's share capital) to submit a take-over bid for a third company.Another defence measure is a search for a white knight in the form of an acceptable alternative offer or counter-bid from a domestic or foreign bidder.

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they are recommended to not include more than two former managing board members, that managers should not serve on the supervisory boards of competing companies, and that no supervisory board members should serve more than five boards.[197] Conflicts of interest related to consulting tasks (including auditors) must be revealed. The code is voluntary and includes no sanctions and no changes to present liability standards, although the underlying "comply or explain" principle does create incentives for compliance. According to media surveys a number of leading German firms (six out of the DAX's 30 companies) are not prepared to follow the voluntary code. The code is an additional measure to strengthen corporate governance, but insofar as the "comply or explain" principle does not work there may be a need for codifying some elements – such as the reporting of management remuneration, which is already mandatory in take-over cases.

Financial sector supervisor

The increased integration of financial markets with the formation of complex financial groups increases the need for information sharing, co-ordination and co-operation among supervisory authorities with responsibilities for different institutional components to facilitate the formation of a group-wide perspective of the regulated financial institution.[198] Reflecting these trends, the sector specific regulators were merged into the Federal Authority for Financial Services Supervision (BAFin) on 1 May 2002, integrating supervision of banking, insurance and securities markets into one body, making the new authority the single regulator, licenser, supervisor and enforcer of financial market regulation. Previously, banking supervision was shared between the Federal Banking Supervisory Office and the Bundesbank. With the reform the Bundesbank lost some of its previous involvement with respect to banking supervision, although the bank will play a role in advising and supporting the new authority, reflecting the bank's proximity to the market and specialised knowledge. The new framework for financial market regulation follows the approach taken in a number of other EU countries (Table 14).

The integration of financial market supervision at the federal level concentrates supervision powers. However, prosecution remains a *Länder* responsibility and has only slowly adapted to the changes in financial markets. The state of Hessen, where Frankfurt and Deutsche Börse are located, only recently established a special prosecution office for insider trading and stock market manipulation cases at the Deutsche Börse. As other *Länder* have not established similar offices, the effectiveness of prosecution varies across states. In addition, appeal possibilities are not unified either as they are presented at the respective regional *Länder* courts. The integration of financial market supervision at the Federal level is an acknowledgement of the increasing sophistication of modern financial markets, which requires the concentration of specialist knowledge and should be extended to the prosecution and court levels.

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Table 14. **Orientation of national supervisors**

	Orientation of supervision	Supervision of conglomerates
Austria	Integrated supervisor for banking, securities, insurance and pension fund supervision. Central Bank participates in banking supervision.	Identity of lead regulator for a financial conglomerate is determined on the basis of principal activity.
Belgium	Sector-based but steps have been taken to give a co-ordination role to the Central Bank, *inter alia* through the creation of a Financial Stability Committee.	Financial conglomerates operate without a single or lead regulator.
Denmark	Integrated supervisor.	..
Finland	Separate legal entity responsible for the supervision of banks and securities companies, shares the support services of the Central Bank. Insurance companies and private sector pension funds in a separate authority.	Financial conglomerates operate without a single or lead regulator.
France	Sector-based (with Central Bank involvement).	No specific regulations on activities of financial groups, but an agreement to promote co-operation between relevant authorities exists.
Germany	Integrated supervisor (Central Bank co-operation in some cases). Recently established authority integrates supervision of insurance, banking and securities sectors.	Cross-sector oversight is based on a forum in which the Ministry of Finance, the Bundesbank and the three sectors of the integrated supervisor are represented.
Greece	"Largely" sector-based. Central Bank exercises prudential supervision of credit institutions and other financial institutions.	Identity of lead regulator for a financial conglomerate is determined on the basis of principal activity.
Ireland	Single regulatory authority for financial services in the Central Bank.	..
Italy	Largely sector-based. A division of competencies by objectives is adopted between the Central Bank (stability of financial intermediaries) and the securities commission (transparency and rules of conduct).	The Central Bank has responsibility for bank-dominated groups. Arrangements for information sharing between supervisory authorities have been established.
Luxembourg	One authority for supervision of banking, investment firms and financial asset markets.	Financial conglomerates operate without a single or lead regulator.
Netherlands	Objectives-based (Central Bank for banks).	Financial conglomerates operate without a single or lead regulator. Cross-sectoral aspects handled by the Central Bank and the Insurance Board in co-operation (Board of Financial Supervisors). A new legal framework is being developed.
Portugal	"Largely" sector-based.	Board of Supervisory Authorities, chaired by the Central Bank, has responsibility for bank-dominated groups. Working groups of representatives from each supervisory authority for information sharing have been established.
Spain	Sector-based (Central Bank participation).	A lead regulator is appointed in the case of financial groups or conglomerates.
Sweden	Integrated supervisor.	..
United Kingdom	Integrated supervisor (Central Bank co-operation in some cases).	..

Source: Institute of International Bankers (2000), *Global Survey* 2001, September, New York; European Commission and OECD.

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The integrated financial market supervisor should provide for a more coherent supervision of cross sector institutions and products as well as beings more adequate for increasing international co-operation in the supervision of financial markets. However, to secure the potential economics of scope in supervision, avoiding that the reform only leads a continuation of sector supervision under a common umbrella, the government should monitor progress in securing such benefits.

Recommendations

The policy actions reviewed in this chapter and the proposed recommendations, together with actions taken since 1996, are summarised in Box 12 below.

Box 12. **A review of progress on structural reform**

Since the last review a number of policy measures have been implemented. This box summarises actions taken over the entire period since the original review in the OECD *Economic Survey* of 1996.

Previous proposal	Action taken	OECD assessment/ recommendations
A. Progress in labour market reform		
I. Increase wage and labour cost flexibility		
• Encourage wage differentiation, greater plant-level bargaining and opening clauses.	Minimum wage in construction was declared binding in mid-1997. The scope for the government to extend wage agreements to non-contracting parties was widened in 1999. Scope of co-determination in work councils (*Betriebsrat*) enhanced in 2001.	Eliminate administrative extension of wage agreements. Allow firm level wage agreements to override collective agreements. Monitor benefits and costs of the enhanced scope for co-determination and correct if required. Consider whether cost of co-determination should be borne by all social partners
• Facilitate the employment of older workers and reduce incentives for early retirement.	Early retirement age to be increased; part-time work of older workers is subsidised under some conditions; eligibility conditions for invalidity pensions tightened. Actions reversed in 1999. Preferential job-search requirement for older unemployed was terminated.	Consider further means to raise the retirement age (stricter eligibility criteria, actuarial discounts for pension benefits) and keep transitional arrangements to a minimum.

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Box 12. A review of progress on structural reform (*cont.*)

Previous proposal	Action taken	OECD assessment/ recommendations
II. Increase working-time flexibility and ease employment security provisions		
• Reform regulations underpinning inflexible working practices.		Review effects of regulations.
• Liberalise terms for renewing fixed-term contracts.	Renewal period and admissible frequency of renewals increased; scope for secondments extended. In 2000 liberalisation of renewals for older workers, but tightening otherwise.	Monitor employment effects and continue with reforms.
• Facilitate part-time work.	More flexible work arrangements in the public sector; subsidies to part-time work for older workers. In 2000 workers given legal right to switch to part time.	Avoid further subsidisation of part-time work for older workers. Monitor rights for part-time employment and abolish if necessary.
• Reform dismissal protection.	Exemption from general dismissal law widened; social criteria for dismissals relaxed. Both reversed in 1999.	Liberalise dismissal protection. Allow work contracts to offer severance pay in return for less stringent dismissal protection.
• Exploit temporary work agencies' employment generating potential	Doubling of temporary work agencies' sub-contracting periods in 2001 to 24 months. Compulsory equalisation of work and pay conditions between temporary and regular employed after 12 months of employment was introduced in 2001.	Abolish compulsory equalisation of work and pay conditions together with restrictions (*Synchronisationsverbot*) on temporary work contracts. Avoid establishing public temporary work agencies with preferential conditions and refrain from introducing special subsidies.
III. Reduce the distortions arising from unemployment insurance and related benefits		
• Improve the incentive structure of benefits.	Minimum ages for receiving unemployment benefits for more than one year increased by three years; redundancy payments credited against unemployment benefits, reversed in 1999. In 2001 more generous unemployment benefits for older unemployed were not renewed.	Increase the gap between social assistance payments and the earnings of low-income families. Integrate unemployment assistance with social assistance. Lower high effective marginal taxes in social assistance.

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Box 12. A review of progress on structural reform (*cont.*)

Previous proposal	Action taken	OECD assessment/ recommendations
• Restrict the criteria for not accepting job offers, strengthen penalties, and tighten controls.	Qualification no longer reason for rejecting job offers; definition of acceptable wage narrowed; internal auditing introduced of criteria; controls tightened, but reversed in 1999. Unemployment support can from 2001 onwards be lost for up to 12 weeks if refusing employment offers. In 2001, increases in unemployment benefits are to be reduced after one year to account for de-qualification.	Require proof of active job search for recipients of unemployment benefits. Sanction non-compliance with job search rules.
• Lower social security contribution rates.	Temporary curbing of certain social transfers; increase in VAT to avoid further rises in rates; pension reform legislation passed (reversed in 1999); pension contribution rates lowered by introducing new eco-taxes; obligation to pay contributions extended to casual employment and "seemingly" self-employed. In 2000 experimental schemes for low-qualified workers introduced. Subsidies for social security contributions for low-income jobs was introduced in 2002.	Curb social spending; consider further measures to secure financing of the pension system (retirement age, basic pension, funding). Keep the flexibility of casual employment and self-employment by redefining the pertinent rules for paying contributions. Reform subsidies in promoting low-income jobs so as to reduce marginal effective tax rates.
• Simplify the personal tax system and lower marginal taxes.	Income tax reform laws implemented in 1999, 2000 and 2001. In 2002, the 2003 step of the tax reform was postponed one year.	Proceed with tax reform and lower tax rates.
• Reduce the generosity of sickness benefits and strengthen incentives to avoid abuse.	Statutory sick pay lowered from 100 per cent to 80 per cent, reversed in 1999.	Reduce the generosity of minimum sickness benefits.

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Box 12. A review of progress on structural reform (*cont.*)

Previous proposal	Action taken	OECD assessment/ recommendations
IV. Enhance active labour market provisions		
• Exclude measures from the requalifying period for unemployment benefits.	Training periods will not re-establish eligibility for another entitlement period.	Continue reform.
• Target ALMPs at at-risk groups.	More flexible work contracts for formerly unemployed. Communes given greater powers to introduce job-creation programmes and new loan facility given for associated non-wage costs; ALMPs extended (both training and work provision) and eligibility conditions relaxed.	Monitor the efficiency of work contracts for the unemployed, in particular in relation to employment provision measures. Tighten new powers of communes to organise job-creation programmes to avoid simply transferring benefit recipients from one level of government to another. Target ALMPs on at-risk groups and reallocate funds in favour of training and other measures aiming at supporting job search activity. Evaluate all programmes.
• Base employment subsidies on wages below market rates.	Admissible wage base lowered; special conditions for the new *Länder* phased out.	
• Improve job placement	The scope for private placement agencies was widened in 2002, including the introduction of a voucher system. Subsidies for job-rotation where temporary vacancies are filled with unemployed were introduced in 2002. Steps were taken to improve the profiling of unemployed and voluntary integration agreements with the placement office came in force in 2002.	Shift personal resources from administration to profiling and placement tasks. Make the provision of placement services more contestable. To facilitate search activities, increase the scope for sanctioning non-compliance and widen the acceptability criteria. Introduce mandatory integration agreements.

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Box 12. A review of progress on structural reform (*cont.*)

Previous proposal	Action taken	OECD assessment/ recommendations
B. Improving skills, entrepreneurial dynamism and competition		
V. Improve labour force skills		
• Preserve and restore the attractiveness of the dual vocational training system, clarify its relationship to higher education.	Curricula for apprenticeships revised and new occupations introduced. Public support scheme to create additional training introduced. In 2000 mobility grants for apprentices widened.	Continue to revise vocational training schemes and preserve benefits from training apprentices. Do not introduce a training levy; monitor the efficiency of public support schemes; avoid attaching subsidies to training.
• Shorten higher education and focus it on more occupational-oriented studies.	Universities have been given new scope to introduce shorter studies leading to degrees that are internationally comparable.	Shorten higher education and make it more occupationally oriented. Make universities more performance-oriented. Greater use of fees or student loans. Link financing to performance.
		Consider introducing full-day schooling.
		Introduce nationwide standards for schooling attainments and evaluate regularly. Leave schools more freedom in organising ways how to reach educational targets.
VI. Enhance creation and diffusion of technological know-how		
• Foster venture capital and reduce regulatory barriers.		Monitor development of risk capital. Avoid introducing tax disincentives for venture capital in the tax reform.
VII. Support an entrepreneurial climate		
• Facilitate the establishment of new enterprises.	Accelerated depreciation allowances for newly-founded enterprises; wage subsidies for small new-founded firms.	Continue to reduce regulatory impediments.
• Increase the knowledge of the new owners of enterprises in the new *Länder*.	No major changes.	Support projects to diffuse knowledge.

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Box 12. A review of progress on structural reform (*cont.*)

Previous proposal	Action taken	OECD assessment/ recommendations
• Reduce the overall tax burden.	Abolition of wealth tax and the business capital tax; reduction of the inheritance tax; reduction of household tax but temporary increase in business tax. In 2001 major income and business tax reform. Business taxes increased temporarily in 2003	Implement a tax reform with sharply reduced tax expenditures and lower marginal tax rates.
VIII. Increasing product market competition		
• Strengthening the general competition framework	A leniency programme introduced in 2000.	Monitor if implied cost and benefits for participants are sufficiently high to make the programme effective. Increase administration fines for non-compliance with information requests.
• Simplify planning approvals.	Approval procedures simplified and entry into trades made somewhat easier.	Monitor the effectiveness of the new regulations; continue reform and modernisation of law on trades.
• Liberalise shop-closing hours. • Liberalise retailing regulation.	Statutory shop opening hours extended in 1997. Laws restricting rebates and promotional offers were abolished in 2001.	Liberalise further retail regulation by revoking the prohibition of time-limited discounts outside sales periods, removing existing provisions against below cost pricing and relaxing statutory shop opening hours.

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Box 12. A review of progress on structural reform (*cont.*)

Previous proposal	Action taken	OECD assessment/ recommendations
• Encourage competition in the network sectors.	Telecommunication, energy and postal service sectors liberalised, and competition in the first sector is developing rapidly. Competition law widened to cover network sectors and transport. Telecom regulation established. Creation of an electricity division within the Federal Competition Authority in 2001. Ministerial clearance in 2001 of energy merger will increase concentration and higher vertical and horizontal integration. EU regulation to assess whether roaming charges are set competitively will be implemented in 2003. Full opening of markets for postal services was in 2001 postponed five years to 2007. The railway regulator gain powers to ensure non-discriminatory access in technical terms in 2002.	Monitor whether regulations to enforce the opening up of networks are effective and promote contestable markets. Consider establishing regulatory authorities for electricity and gas. Support restructuring of community generators and resist demands to protect them. Resist demands to protect the interests of the incumbent provider. Monitor if staffing suffices within selected divisions of the Federal Competition Authority. Facilitate the construction of competing networks in the telecommunication sector and monitor if the two largest mobile service providers have joint dominance as well as whether roaming charges and charges for connecting fixed and mobile lines are competitively determined. Accelerate post liberalisation and remove differential tax regimes to level playing fields. Extend the railway regulatory powers in the area of price regulation. Consider ownership unbundling in the course of privatising Deutsche Bahn.
• Services by communities need to be more closely market-tested.	No major changes.	Abolish preferential tax treatment of public utilities.
• Pursue privatisation.	Telekom and Deutsche Post partly privatised.	Step up privatisation and complete before the opening of markets.
• Reduce subsidies.	Subsidies benefiting construction reduced overall but reduction of coal subsidies slowed. Subsidies to east placed on new basis.	Avoid new industrial subsidies; set up schedule for reducing existing subsidies; close tax expenditures. Reform regional support.

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Box 12. **A review of progress on structural reform** (*cont.*)

Previous proposal	Action taken	OECD assessment/ recommendations
C. Financial markets		
• Level the playing field in the bank sector.	EU accord to remove public guarantees to state banks in 2002.	Full privatisation.
• Securing clear take-over rules.	Replaced in 2002 the voluntary take-over rules with a codified system.	Introduce strict management neutrality and remove excessive information requirements.
• Improving the functioning of financial markets.	Creation in 2002 of unified financial regulator to improve supervision.	Secure regulatory economics of scope by unifying location of the regulator and monitor progress.
	Introduction in 2002 of a voluntary code of conduct to improve corporate governance.	Consider codifying elements of the voluntary code of conduct if expected results fail to materialise.
	Successive laws implemented to improve flexibility in financial markets and ease access of companies to equity capital.	Require immediate reporting of share transactions that reflects insider knowledge.

Sustainable development in Germany

There is growing concern that long-run sustainable development may be compromised unless OECD countries take measures to achieve balance between economic, environmental, and social outcomes. This section looks at three concrete issues that are of particular importance for Germany and which, between them, touch all the dimensions of sustainable development. In each case, indicators are presented to measure performance, and an assessment is made of government policies that affect the issue. The section also considers whether institutional arrangements are in place to integrate policy-making across the different elements of sustainable development (see Box 13). The section considers first the policies that have been put in place to reduce air pollution. It then looks at waste management policies. Finally, the section looks at the policies that have been adopted or are required for fiscal sustainability in the long run without compromising the goal of income adequacy of older persons.

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Box 13. The integration of environmental concerns into government policy

Sustainable development is an important aspect of German policy making. In 2002, following widespread consultation, the federal government adopted the national strategy "Perspectives for Germany: Our Strategy for Sustainable Development". This strategy highlights 21 areas of concern with specific indicators and targets, which will be monitored on a regular basis. The targets range from a commitment to running budget surpluses from 2007 to reducing the burglary rate by 10 per cent by 2010. In three areas (climate change, transport policy, and agriculture and food safety) the strategy develops broad policy measures, including ecological tax reform. The strategy promises future development of similar programmes for the remaining 18 areas.

The Chancellery is the institutional centre of sustainable development policy, with the Committee of State Secretaries for Sustainable Development responsible for cross-cutting issues and the development of the national strategy. Ministries are required to undertake Environment Impact Assessments for major policy initiatives. However, there is no legal obligation to carry out cost-benefit analysis of environmental and social policy initiatives, and such analyses are rarely used in practice.

As in any federal system, the distribution of authority between levels of government can complicate the legislative process. In the fields of air pollution and waste management, the federal government enjoys concurrent power with the *Länder*. Concurrency requires the approval of the *Länder* representatives in the upper house of parliament, the Bundesrat, for the adoption of legislation. Other levels of government and civil society are also included in the legislative process through hearings held before and after the preparation of draft legislation. Their interests are further institutionalised in the Council for Sustainable Development, consisting of 16 experts representing sub-national governments and other stakeholders. Taking into account the interests of other levels of government and civil society can mean that forging the necessary consensus for new legislation may be time consuming. On the other hand, the federal structure may be an advantage when a (typically environmental) concern is localised both in the cause and effect.

Air pollution

Main issues

While the problem of air pollution has greatly diminished over recent years and is projected to fall further as a result of strict emission requirements

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already in place, local problems for several pollutants remain and exceedences of critical loads for land and water are of concern. The regulations already in place have not imposed particularly high costs on polluters relative to the benefits, though it is unlikely that improvements have been achieved at the least cost. Given that Germany is committed to further abatement that is likely to be more expensive (OECD, 1999a), a key issue is how to implement such reductions in the most cost-effective manner, given that low cost options are largely exhausted.

Performance

As with other OECD countries, there have been substantial reductions of certain air pollutants (Table 15, Figure 34). Fuel switching from high sulphur content coal to natural gas and improvements in abatement technology, particularly in the new *Länder*, were important contributory factors. These developments led to marked reductions in CO_2 and SO_2 emissions. Emissions of SO_2 fell substantially faster, relative to GDP, than in the OECD as a whole. Indeed, emissions per euro of GDP are only one quarter of those in the area as a whole. Together with declining net imports,[199] the cuts in domestic emissions translated into concentrations of SO_2 in the atmosphere falling below those seen as harmful by the World Health Organisation, even in the previously highly polluted new *Länder*. Ambient concentrations of nitrogen oxides across Germany, however, have fallen less, largely as a result of rising traffic volumes.[200] Nonetheless, the share of ecosystems that experience acid depositions that result in serious damage to plant life may fall from 80 per cent in 1990 to 16 per cent in 2010 if current emission projections are realised (Amann *et al.*, 1999a). Peak level concentrations of ground level (tropospheric) ozone have declined during the 1990s, though average concentrations have risen and episodes of high concentrations remain a problem in certain parts of Germany, especially during the summer.

Policy

Within the framework of the UNECE Convention on Long-range Transboundary Air Pollution and the growing importance of EC directives, the German government is committed to substantial emission reductions of various air pollutants by 2010 (Table 16). Policies have been mainly command-and-control in nature, relying heavily on requiring the installation of best-available technology. This has entailed significant investment in abatement technology in the 1990s. Although this may not have been the least-cost solution, the benefits of abatement are likely to have justified these costs.[201]

Continuation of existing command-and-control policy is likely to become increasingly costly in the future. Indeed, the total annual cost for achieving EU targets with such policy could rise to € 16 billion (Amann *et al.*, 1999b, European Commission, 1999), which represents an eight-fold rise from current expenditures, at a time when

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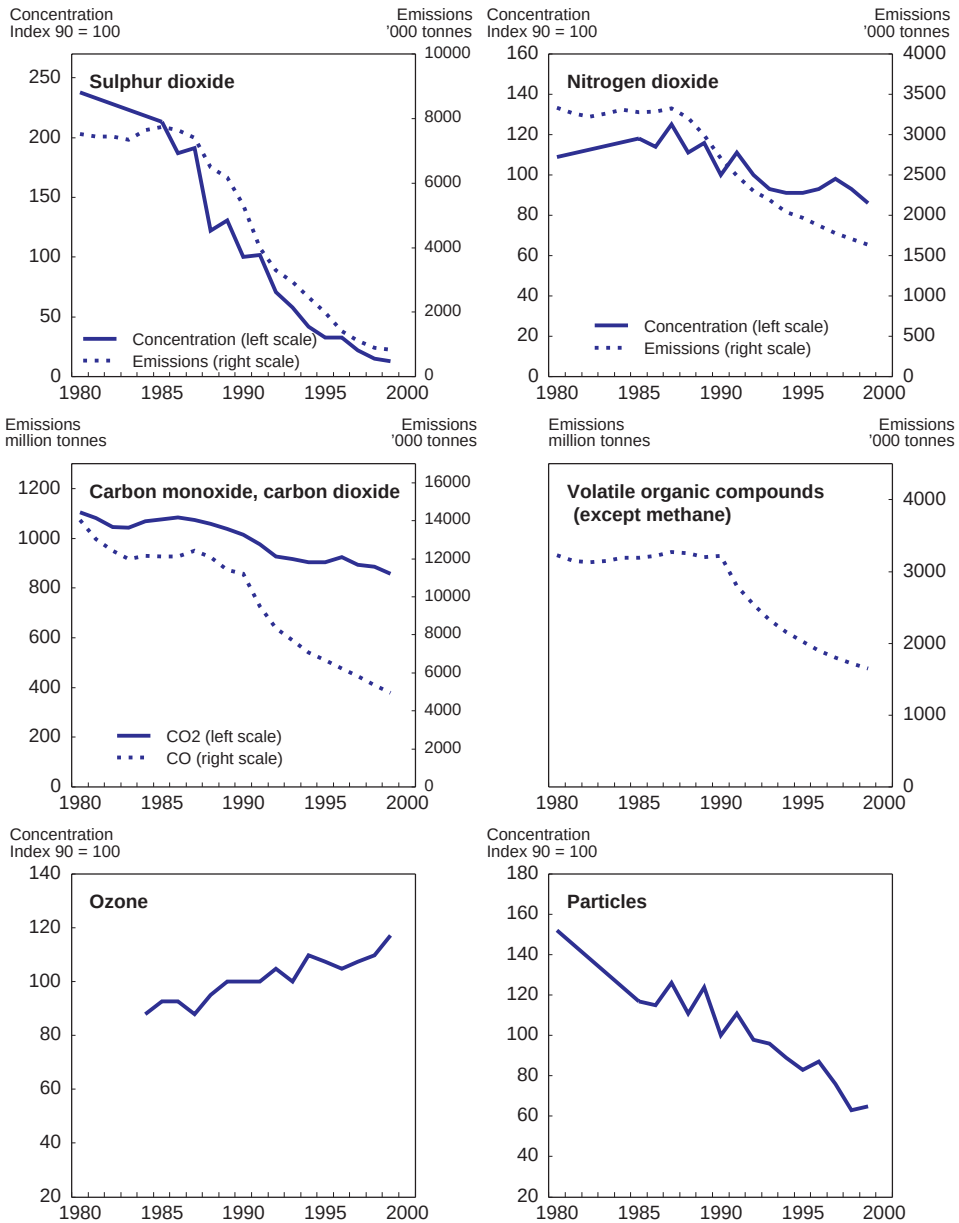
Figure 34. **Emission trends for air polluants**



Source: EMEP, OECD Environmental data 2002 and Federal Environment Agency, 2000.

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Table 15. **Performance indicators: air pollution**

	Change in emissions per unit of GDP, 1990-99[Note:]			Level of emissions , 1999[1]			Improvement in productive efficiency, 1990-1999	
	Sulphur dioxide	Nitrogen dioxide	VOC	Sulphur dioxide	Nitrogen dioxide	VOC	Sulphur dioxide per unit of electricity output	Nitrogen dioxide per vehicle
	Per cent per year			Grams per dollar of GDP, (1995 US$ PPP)				
Australia	3.1	–1.9	–2.9	3.9	5.5	4.1	n.a.	n.a.
Austria	–10.2	–3.5	–6.5	0.2	0.9	1.2	–77.2	–32.5
Belgium	–9.2	–3.6	–4.8	0.8	1.2	1.1	–64.4	–22.6
Canada	–5.3	–2.8	–2.9	3.2	2.6	3.5	n.a.	n.a.
Czech Republic	–19.1	–6.6	–5.7	2.1	3.0	1.9	n.a.	n.a.
Denmark	–14.2	–4.9	–5.2	0.4	1.6	1.0	–79.8	–38.9
Finland	–13.0	–3.8	–4.1	0.7	2.1	1.4	–65.3	–28.7
France	–8.3	–3.8	–5.1	0.5	1.2	1.4	–52.0	–37.8
Germany	**–20.1**	**–7.1**	**–8.8**	**0.4**	**0.9**	**0.9**	**–85.2**	**–43.7**
Greece	–1.4	–0.3	–0.2	3.6	2.5	2.6	–4.6	–22.3
Hungary	–6.1	–1.1	–3.8	5.5	2.1	1.4	n.a.	n.a.
Iceland	–1.1	–1.7	–4.9	3.7	3.9	1.4	n.a.	n.a.
Ireland	–8.1	–6.3	–7.9	1.7	1.3	1.0	–35.8	–19.0
Italy	–7.6	–4.3	–4.4	0.8	1.2	1.4	–52.8	–31.3
Japan	–1.7	–1.3	–2.4	0.3	0.5	0.6	n.a.	n.a.
Korea	–8.9	–3.0	n.a.	2.1	2.0	n.a.	n.a.	n.a.
Luxembourg	–18.7	–9.1	–7.9	0.2	0.9	0.9	n.a.	n.a.
Mexico	–0.4	–0.9	–9.3	1.6	1.6	1.4	n.a.	n.a.
Netherlands	–10.1	–6.5	–8.8	0.3	1.1	0.7	–50.2	
New Zealand	–0.4	–0.5	–1.6	0.7	3.0	2.6	n.a.	n.a.
Norway	–9.7	–2.8	–1.8	0.2	2.0	3.0	n.a.	n.a.
Poland	–10.0	–6.6	–4.8	5.1	2.8	2.2	n.a.	n.a.
Portugal	–2.1	–0.9	0.1	2.4	2.3	3.1	–17.7	–18.3[2]
Slovakia	–13.7	–7.4	–5.6	3.3	2.4	2.0	n.a.	n.a.
Spain	–5.8	–2.1	–3.5	2.2	1.7	3.6	–39.4	–14.4
Sweden	–8.3	–4.3	–3.9	0.3	1.3	2.1	–26.0	
Switzerland	–6.0	–5.4	–6.3	0.1	0.5	0.9	n.a.	n.a.
Turkey	3.2	1.0	0.0	3.4	2.3	1.6	n.a.	n.a.
United Kingdom	–13.9	–7.9	–6.6	1.0	1.3	1.4	–72.8	–51.0
United States	–5.5	–2.5	–4.7	2.0	2.7	1.9	n.a.	n.a.

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Table 15. **Performance indicators: air pollution** (*cont.*)

	Change in emissions per unit of GDP, 1990-99[Note:]			Level of emissions , 1999[1]			Improvement in productive efficiency, 1990-1999	
	Sulphur dioxide	Nitrogen dioxide	VOC	Sulphur dioxide	Nitrogen dioxide	VOC	Sulphur dioxide per unit of electricity output	Nitrogen dioxide per vehicle
	Per cent per year			Grams per dollar of GDP, (1995 US$ PPP)				
European Union	–11.1	–5.0	–5.3	0.8	1.2	1.5		
OECD Europe	–10.1	–4.8	–5.0	1.2	1.4	1.5		
OECD	–6.7	–2.9	–4.4	1.5	1.9	1.6		

Note: For the OECD average data for countries with missing data for either 1990 or 1999, data for the latest year has been substituted. Estimated data for 1999 represents about 5 per cent of the area total.

1. Australia: 1995-99 for sulphur dioxide; Korea and New Zealand: 1990-98; Mexico: 1994-98 for sulphur dioxide and VOC and 1990-98 for nitrogen dioxide; Slovakia: 1990-98 for sulphur dioxide and nitrogen dioxide and 1990-97 for VOC.
2. 1998 for Korea, Mexico, New Zealand and sulphur dioxide and nitrogen dioxide in Slovakia; 1997 for VOC in Slovakia.
3. Between 1990 and 1997 for Portugal.

Source: Cooperative Programme for Monitoring and Evaluating of Long-Range Transmission of Air Pollutants in Europe (EMEP); World Health Organisation; OECD (2002a).

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Table 16. **Selected international commitments related to air quality**

Kilotonnes

	UNECE	EU	Emissions	
	2010		1990	1999
Sulphur dioxide	550	520	5 321	831
Nitrogen oxides	1 081	1 051	2 706	1 637
Volatile organic compounds	995	995	3 221	1 653
Ammonia	550	550	765	624

Source: Directive 2001/81/EC of the European Parliament and of the Council of 23 October 2001 on national emission ceilings for certain atmospheric pollutants, UNECE Convention on Long-range Transboundary Pollution: The 1999 Gothenburg Protocol to Abate Acidification, Eutrophication and Ground-level Ozone.

the benefits from further pollution have likely fallen by more than 50 per cent, bringing costs and benefits into much closer balance than in the past.[202] Moreover, the use of the best available technology criterion does not necessarily result in the optimal allocation of abatement efforts. Best-available technology requirements impose different abatement costs for different emission sources, which are unlikely to be related to the externality. For example, the best technology for removing nitrogen from emissions from coal fired power stations may result in greater emissions than switching to a less polluting gas-fired plant. A broad-based economic instrument such as emission taxes or tradable permits would equalise marginal abatement costs and encourage emission reduction where low-cost technology is available. Such a policy would likely make coal an uneconomic fuel for electricity generation in Germany.

Government policy has changed the way it finances the use of coal as a fuel in the electricity sector, with subsidisation of the consumption of hard coal via higher electricity tariffs and payments to the coal industry being terminated in 1997. Rather, the producers of coal are now subsidised directly from the federal and some *Länder* budgets, but electricity generators are now free to purchase on world markets. As a result the subsidy no longer has direct adverse environmental effects but is designed to make domestic coal producers competitive *vis-à-vis* their counterparts abroad. On the other hand, the continued output of uneconomic pits does, to some extent, reduce the world coal price and hence stimulates the use of coal. The scale of this distortion is being reduced: a law sets a diminishing ceiling on subsidies to the industry, so that by 2005 payments will have fallen by almost 40 per cent in nominal terms from their 1997 level to € 2.2 billion. This drop is expected to result in a marked shrinkage in the industry with employment falling to 35 000 by 2005 (0.08 per cent of the labour force). In contrast to the hard coal market, there are restrictions in the brown coal market (lignite) market, due to expire in 2003, that tend to increase consumption of this highly polluting fuel in the new *Länder*.[203]

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In addressing nitrogen oxide pollution and other ozone precursors,[204] federal law has targeted large point source emitters, but further reductions may have to rely increasingly on cutting emissions from mobile and diffuse emission sources.[205] One approach has been the increasing differentiation of vehicle taxation in line with emission characteristics. A more recent move is the introduction of a distance-based toll for heavy goods vehicles from 2003. The proposed toll is designed to help accelerate the adoption of denitrification technology in commercial vehicles by taking into account the emission characteristics of the vehicle. These policies should help reduce nitrogen oxides emissions, thereby attenuating problems with ground level ozone. However, due to the transboundary and long-range characteristics of ozone precursors, substantial reductions would be required at the national and especially local level to meet EC ozone limit values unilaterally. Episodes of ozone exceedences are difficult to control at the local level. For example, simulation results for Stuttgart suggest that local reductions of nitrogen oxides emissions of around 75 per cent are necessary to have an appreciable effect on ozone peaks in the Stuttgart area (Moussiopoulos *et al.*, 1998). The best approach to deal with such problems would be to co-ordinate responses across regions that are responsible for the pollution.

Fuel and car taxation and road pricing are some of the economic instruments that are and could be used for reducing air pollution. Indeed, the ecological tax reform of 1999 raised energy (including fuel) taxes with the view to reducing consumption and hence emissions of associated pollutants. In addition, a tax supplement on high sulphur content fuel has been in operation since 2001 and will be strengthened in 2003. In the case of car taxation, the annual motor vehicle tax is structured into several bands to account for emission characteristics. Petrol engine cars attract the lowest annual tax, while diesel engine cars are subject to the highest annual fee. Since the mid-1980s, however, fuel taxation has favoured diesel (Figure 35). Such a regime results in petrol-engined cars being subject to higher taxes than diesel vehicles once a car is driven for 7 500 km.[206] The existing fuel tax structure disregards problems associated with currently higher emissions of nitrogen oxides and particulate matter from diesel engines that are ten times more carcinogenic as those from petrol engines (UBA, 2001). Even the EURO 4 standards for diesel vehicles, due to come into force in 2005, have higher nitrogen oxide emissions per kilometre and particulate emissions than petrol engine cars (European Conference of Transport Ministers, 2001). However, technology has advanced faster than regulations and now exhaust filters that virtually eliminate emissions of particulates are being used by several manufacturers.[207] Taxation of fuel remains a blunt instrument to deal with the pollution externality, which varies according to population density and the age of the car. There is, nonetheless, a need to make the taxation of diesel and gasoline reflect the relative external costs of these fuels rather than overtaxing gasoline relative to diesel. However, more tools are needed to reduce pollution efficiently. In

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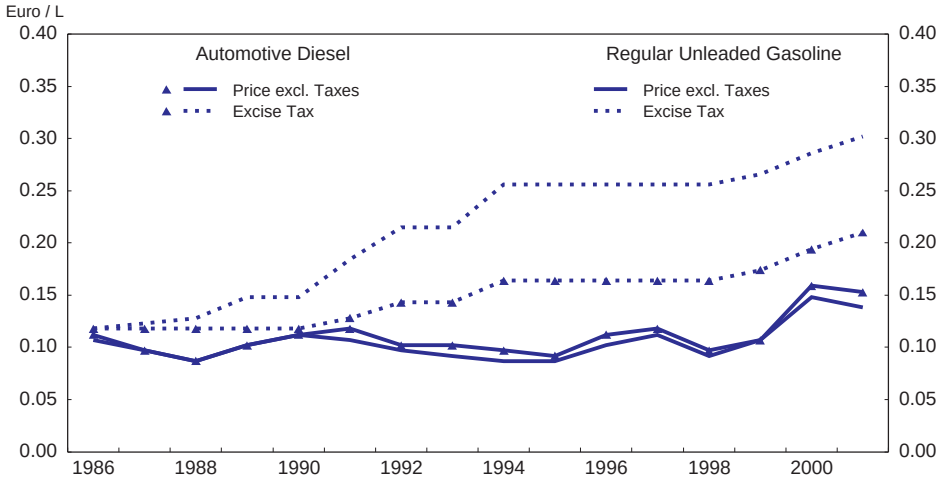
Figure 35. **Relative diesel and petrol taxes**
Household



Source: IEA, *Energy statistics*.

an innovative strategy, the L*and* of Baden-Wurtemberg has introduced a subsidy for new diesel cars that have emissions below the EURO 4 standard in order to encourage the rapid introduction of new technology. Road pricing technology is improving and may also offer a future route that permits better pricing of these externalities.

Conclusions

Regulation based on best-available technology for air quality control has resulted in significant reductions in emissions and some ambient concentrations. The approach has paid little attention to assessing the cost efficiency of alternative policies. As the cost of pursuing further reductions in emissions and reductions of exceedences of critical loads are likely to rise rapidly, it will become increasingly important to use abatement instruments that are cost efficient. In most cases, this will imply supplementing or replacing command-and-control measures with economic instruments. For point sources, taxes on, or tradable permits for, emissions could be adopted. For mobile sources, better alignment of taxes to the externality cost of emissions seems warranted, implying an equalisation of diesel and petrol taxes. The government should also examine the feasibility of road pricing to address localised pollution problems. The introduction of new tighter European standards for emission limits of diesel cars would also be helpful to reduce pollution.

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Waste management

Main issues

Waste disposal, if not handled correctly, can cause long-lasting damage to the environment and human health. In recognition of these risks, the federal government has set demanding targets for reducing the amount of waste generated, incinerating (with energy recovery) all waste sent for landfill, and increasing the proportions of different products that must be recycled. As the system has developed, the main issues include the mounting costs of implementing an elaborate set of policies and balancing them with the derived environmental benefits.

Performance

Germany is one of the few OECD countries that has achieved a reduction in the volume of municipal waste in the 1990s (Table 17). Despite this achievement, the level of municipal waste per capita was still high compared with the OECD average, and especially relative to the average for OECD countries in Europe. In contrast to municipal waste, non-municipal waste increased during the 1990s, though at a modest rate. Recycling is the most common form of municipal-waste treatment in Germany with the recycling rate for such refuse high compared with many other OECD countries. Recycling rates are much higher for non-municipal waste, particularly for construction waste (Table 18).

Policy

As in most other OECD countries, Germany operates a hierarchical approach to waste management, giving priority to avoidance, reuse, and recycling over incineration and landfilling. Policy has targeted waste minimisation during the production process, tightened standards on incineration plants and landfill sites, and set quantitative targets for different waste streams that are often more ambitious than the requirements in the corresponding EU directive. To achieve these targets, the authorities are making increasing use of the principle of extended producer liability and the mandatory sorting of waste. Since 1991, the Packaging Ordinance has made manufacturers, retailers, and importers responsible for the treatment of the packaging waste of their own products and established recycling targets for different waste fractions. Most companies, producing 80 per cent of packaging, have passed their responsibility for dealing with this waste to Duales System Deutschland (DSD), which runs the *Grüner Punkt* (Green Dot) system, in exchange for the payment of item and weight based fees. Households are required to separate packaging material from residual waste, which remains the responsibility of the local authority. In the first five years after the introduction of this system, packaging consumption fell by 13 per cent with recycling increasing

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Table 17. **Performance indicators: waste**

	Waste per capita	Total municipal waste	Municipal waste relative to final consumption	Recycling relative to total municipal waste
	1998	Annual growth 1990-1998	Annual growth 1990-1999	Per cent of total (1998)
	Kg/capita	Per cent per annum		
Austria	525	3.6	1.2	55
Belgium	529	3.2	1.7	47
Canada	820	4.4	2.2	29
Czech Republic				0.2
France	486	1.2	0.1	27
Germany	**547**	**–1.3**	**–3.2**	**41**
Greece	358	3.3	1.1	8.0
Iceland	600	1.5	–0.9	13
Ireland	541			9
Italy	464	3.7	2.2	15
Japan	406	0.2	–1.4	3.9
Korea	333	–7.6	–11.4	26.2
Luxembourg	665	2.2	–0.6	31
Mexico	307	4.8	2.3	0.7
Netherlands	576	2.7	0.0	46
Norway	635	4.6	1.3	22
Poland	306	0.8	–4.2	1.8
Portugal	430	4.6	1.2	28
Sweden	449	2.8	2.2	33
Switzerland	607	0.8	0.0	39.8
Turkey	365	2.7	–1.0	0.7
United Kingdom	532	2.1	–0.1	10
United States	711	1.1	–1.9	32

Source: OECD.

Table 18. **Waste treatment**
1998

	Total waste		Waste treatment				
	Tonnes	Share of total	Landfill	Incineration	Recycled	Incineration with energy recovery	Other
			Per cent				
Municipal	44 094	11.0	36.7	20.7	41.2	0.0	1.4
Mining	56 155	14.1	100.0	0.0	0.0	0.0	0.0
Manufacturing	47 963	12.0	41.2	5.8	46.9	6.0	0.1
Construction	230 997	57.8	12.9	0.0	87.1	0.0	0.0
Special waste	19 102	4.8	31.8	9.3	50.1	4.6	4.2
Total	398 311	100.0	32.1	3.4	63.1	1.0	0.4

Source: Federal Statistical Office.

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markedly. Subsequently, however, packaging consumption has grown by over 2¾ per cent annually (Figure 36).

The costs of the Green Dot system are a major concern. From an economic perspective it is possible to assess whether or not the cost of a recycling system is appropriate. The appropriate cost of the system should be limited to the private cost of an alternative means of disposal plus the cost of any environmental damage caused by this alternative method.[208] In practice, there are no such constraints in the Green Dot system with prices being driven by the arbitrary targets and the costs of the DSD that dominates the market.[209] The costs for recycling plastics, tinplate, and aluminium are particularly high. For plastics, recycling costs nine times the cost of incineration (Taylor Nelson Sofres Consulting, 2000). Furthermore, the costliest recycling of plastics, particularly the mixed lightweight fractions typically collected from households, may have a questionable environmental gain,[210] while alternative disposal methods such as incineration and the use of plastics as feedstock for oil refining[211] can result in energy recovery (Environmental Council, 2000, OECD, 2001).[212] The recent decision to introduce mandatory deposits on some beverage containers from 2003 may also be questionable in promoting more environmentally friendly outcomes as it can result in substantial additional transport costs with the movement of containers around the country. One trip plastic bottles are light and allow much longer distance transport of beverages for the same environmental cost (Environmental Council, 2002).[213] Indeed, the total internal and external costs of reusable glass bottles are approximately 6.5 times greater than the equivalent costs of using one-way plastic bottles (RDC – Environment and Pria International ,2001). One estimate of the average marginal cost of meeting the Packaging Ordinance recycling targets for the principal waste streams with DSD was over € 2 000 per tonne (Staudt and Schroll, 1999), which is more than ten times higher than the total private and environmental costs of landfilling and incineration.

Although the Packaging Ordinance shifted responsibility for recycling to the producers, the municipalities remained responsible for the collection of residual waste, incineration, and landfilling. The volume of residual waste has fallen in the 1990s, partly due to the 20 per cent increase in real terms in fees for household waste collection in the second half of the 1990s.[214] Part of the reason behind these price rises has been the impact of tightening regulations governing landfills and incinerators. While costs have risen, there are also marked differences in the collection fees levied by the local authorities, which can vary by a factor of ten – a far greater differential than can be explained by whether local authorities can use landfills or whether they must incinerate residual waste.

Conclusions

The achievements of waste management policy have come at considerable cost, particularly to households. In this regard, the federal government's commitment

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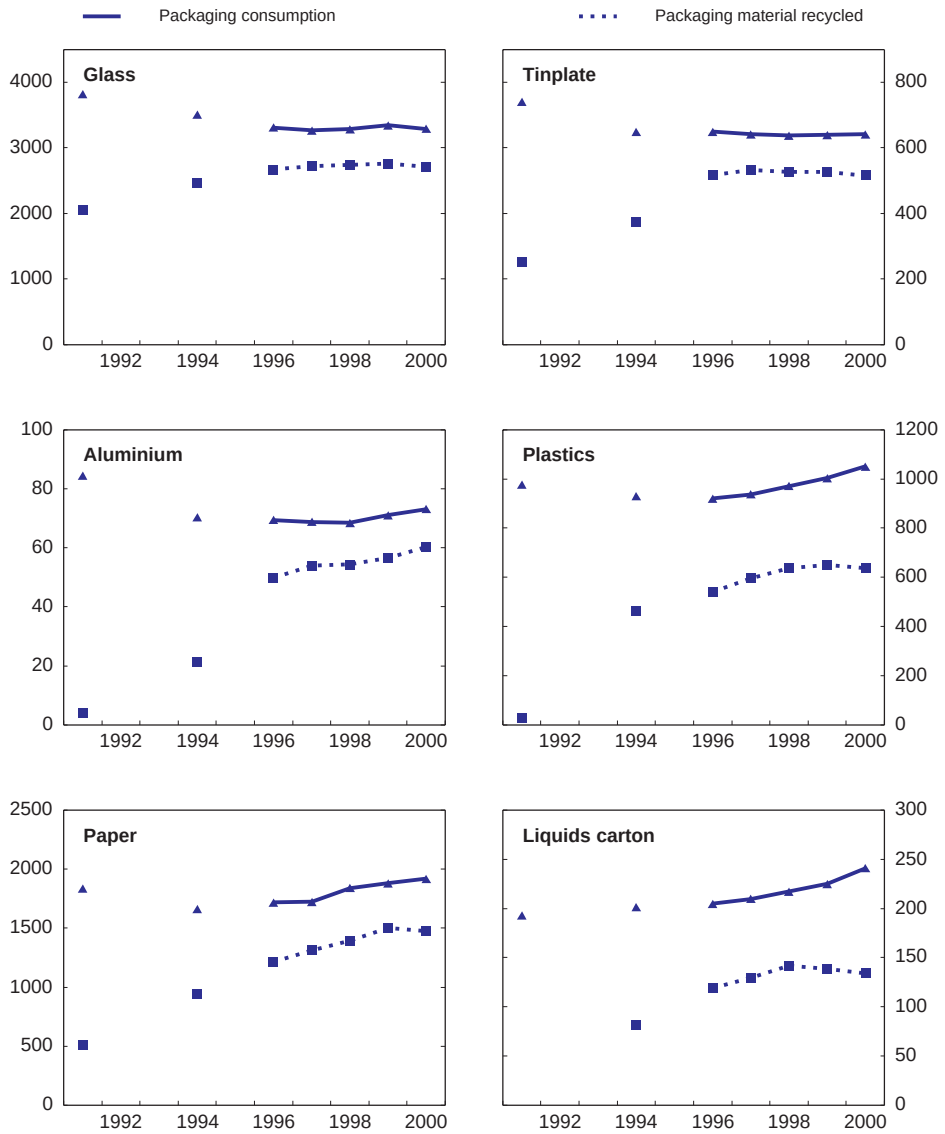
Figure 36. **Trends in packaging consumption and recycling**
Thousand tonnes



Source: Umwelt Bundesamt, Environmental Data, Germany 2002.

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to review the costs of DSD and consider ways to introduce more competition into waste handling is welcome. If the current Green Dot system were to be maintained then it would be appropriate to limit the packaging fees to the cost of the cheapest alternative method of disposal plus an estimated external cost of the alternative treatment method. The DSD would then have to optimise the mix of recycling and alternative treatment in order to balance its costs and revenues. Moreover, DSD should be obliged to call for tenders from alternative providers of recycling services. Such a price-based system would mean that arbitrary quantitative recycling targets would have to be discontinued. The quotas on the use of multiple-trip beverage packaging should also be replaced by price-based incentives that reflect externalities and that do not discriminate against the use of one trip plastic bottles. Finally, the federal government should also examine why municipal waste collection costs are so variable.

Sustainable retirement income

Main issues

Projections suggest that Germany faces one of the sharpest increases in public pension expenditure, rising by around 5 per cent of GDP to 17 per cent of GDP by mid-century (OECD, 2001h). This is largely driven by demographic pressures, with a near doubling of the old-age dependency ratio from the beginning of the century by 2035 (Dang, Antolin, and Oxley, 2001). In this context, the government has recently started the process to put retirement income on a sustainable footing, by reforming the PAYG system and creating a framework for a multi-pillar pension system. Given the long lead times needed for people to adjust their saving in the light of changes to public pensions systems, there is a need to address quickly the needed fundamental pension reforms in a way that ensures retirement income adequacy. The longer the delays in creating the foundations of a sustainable retirement income system, the greater the burden younger cohorts will ultimately have to bear.[215]

Performance

Overall replacement rates for retired people in Germany are among the highest in the OECD area, largely as a result of income from the state pension, which accounts for over 80 per cent of average retirement income. The Pay-As-You-Go (PAYG) pension system has contributed to the relatively low incidence of poverty among the elderly. It currently offers a net replacement rate of 70 per cent of average lifetime earnings[216] for a worker with 45 years of contributions.[217] When taking into account income from housing assets and work-related expenditures, the net income of pensioners aged 65-74 years is over 84 per cent of mean disposable income of individuals aged 51-64 years and even higher relative to all people of working age (OECD, 2001h). When measured in terms of consumption, the position of

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retirees is even more favourable: people aged 60 to 80 years consume only 4 per cent less than those of working age (*i.e.* the consumption-based replacement rate is 96 per cent) – without allowing for the consumption of collectively-provided medical services. On average retirees continue to save and leave relatively high bequests to their children. The near universal coverage of state pensions and the high replacement rate has meant that private pension savings for retirement are among the lowest in the OECD area (Table 19). In addition, the rules of the social security system (pensions, sickness and unemployment insurance) have induced workers to leave the labour market at ever earlier ages with little financial disadvantage (Figure 37), with the result that the male participation rate for all ages above 55 years is now over 12 percentage points lower than in the United States.

Policy

The pension reforms of 2001[218] were the latest, and most radical, in a series of reforms of the public pension system. The main elements of the reforms included:

- A cap on pensions contribution rates at 20 per cent until 2020 and a gradual rise in the ceiling to 22 per cent by 2030.
- A reduction in the full pension net replacement rates from 70 per cent at present to not less than 68 per cent. This will be achieved by changing the system of indexation, so that when contributions to private pension schemes increase, public pensions benefits decline.
- The promotion of private pensions to offset the impact of lower state pensions on income in retirement. This included subsidies and tax breaks for voluntary individual pensions and subsidies, increased portability and reduced vesting periods for occupational pensions. With full take-up of the private pensions, then overall replacement rates could return to over 70 per cent by 2050 (Bonin, 2001).[219]

In the view of some analysts, the 2001 reforms may not be sufficient to ensure the sustainability of the public pension system, despite official 15-year projections, and specially commissioned longer-term projections, that appear to show that the pension reforms could meet the simultaneous targets for replacement and contribution rates. There are a number of reasons for such scepticism. Firstly, the official 15-year projections fail to encompass the period of maximum demographic stress, starting from around 2030. Secondly, the longer-term projections may under-estimate the impact of ageing (Bonin, 2001) and mask the effective contribution rates required (Sinn, 1999).[220] Furthermore, Werding and Blau (2002) find that the 2001 pension reforms will only have a marginal effect on public pension expenditure relative to GDP, reducing it by 0.2 percentage point by mid-century. Indeed, keeping contribution rates at 22 per cent could imply that the replacement rate may have to fall below 63 per cent from 2030 (Figure 38) unless the average effective age of retirement increases more than currently envisaged.

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Table 19. **Key indicators for sustainable retirement income**

	Projected increases in old age pension spending	Disposable income of those 65 and over	Private pension funds 1999	Age of withdrawal, 1994-1999	
	Change in per cent of GDP	First decile[1]	Per cent of GDP	Male	Female
Australia	1.6		62.1	59.7	61.3
Austria	2.2		3.4		
Belgium	3.3		5.7		
Canada	5.8	148.0	48.0	62.6	61.1
Czech Republic	6.8		3.7		
Denmark	2.7		22.7	62.4	61.5
Finland	4.8	101.0		59.8	60.0
France	3.9			59.3	59.8
Germany	**5.0**	**102.0**	**3.0**	**60.5**	**60.8**
Greece				61.7	62.2
Hungary	1.2		2.0		
Iceland			83.3		
Ireland					
Italy	–0.3	128.0	4.2	59.3	58.4
Japan	0.6	72.0	20.8	69.1	66.0
Korea	8.0		3.3	67.1	67.5
Luxembourg					
Mexico			2.4		
Netherlands	4.8	83.0	112.6	61.6	60.1
New Zealand	5.7				
Norway	8.0		7.4	64.2	64.7
Poland	–2.5				
Portugal			10.9	65.3	66.5
Slovak Republic					
Spain	8.0			61.1	61.1
Sweden	1.6	89.0	2.9	63.3	61.8
Switzerland					
Turkey					
United Kingdom	–0.7	76.0	85.1	62.0	61.2
United States	1.8	80.0	74.7	65.1	64.2

1. First decile of pensioners' income relative to first decile of all incomes.
Source: OECD.

Policies predating the 2001 reforms have targeted increasing the average effective age of retirement by eliminating some existing disincentives to continue working at older ages and increasing the age of pension entitlement. Eligibility criteria for pensions on account of unemployment and disability benefits have been tightened. At present, the unemployment and disability schemes still give incentives to retire early, (OECD, 2002e, Antolin and Scarpetta, 1998) (Figure 39) but the

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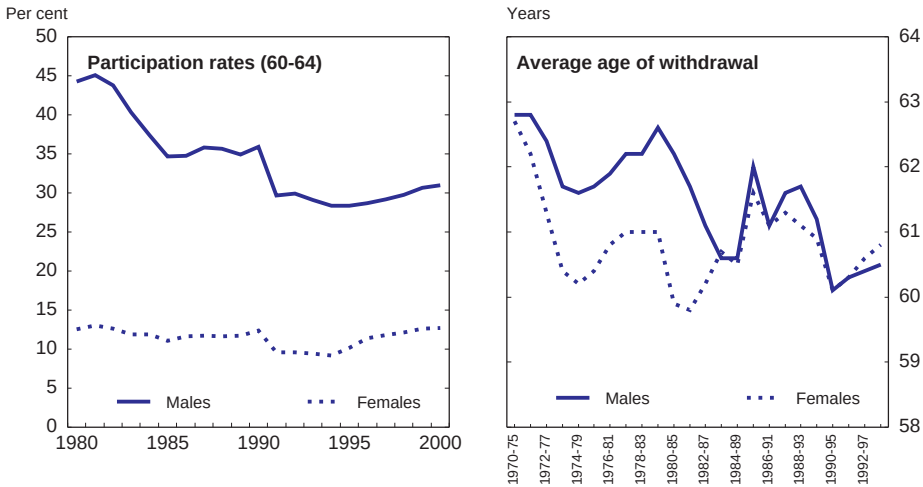
Figure 37. **Age of withdrawal and labour force participation**



Source: OECD, *Labour force statistics*.

incentives offered by the unemployment scheme will be ended in 2011. Furthermore, a previous policy promoting part-time work as a "bridge to retirement" (*Altersteilzeitgesetz*), which also offered a route to early retirement, will expire in 2009.[221] Following reform in 1992, the standard age of entitlement to old-age pensions is in the process of being raised for women to that of men by 2005. Also, the lower age limit for pensions and the minimum number of contribution years for early retirement have been raised.[222] However, although pension adjustments for early and late retirement have been introduced, the pension system is not actuarially neutral in that it may provide some incentives for retirement before the standard entitlement age.

The 2001 reforms also introduced the possibility for individuals to contribute to privately funded pensions schemes. In the first few months after their introduction in 2002 has take up was slow but this has now changed. The government expects that a maximum of 80 per cent of the labour force could be covered by voluntary occupational pensions. While some collective agreements have been agreed with pension providers,[223] there is currently no information about the take up by individual members. In the realm of individual private pensions, however,[224] over one-quarter of the expected 10 million workers that could be covered by this type of pension have started to contribute to such schemes.[225] The system for voluntary private pensions has a number of features that may limit the eventual take-up of individual, as opposed to collective plans. These include: the complexity of the

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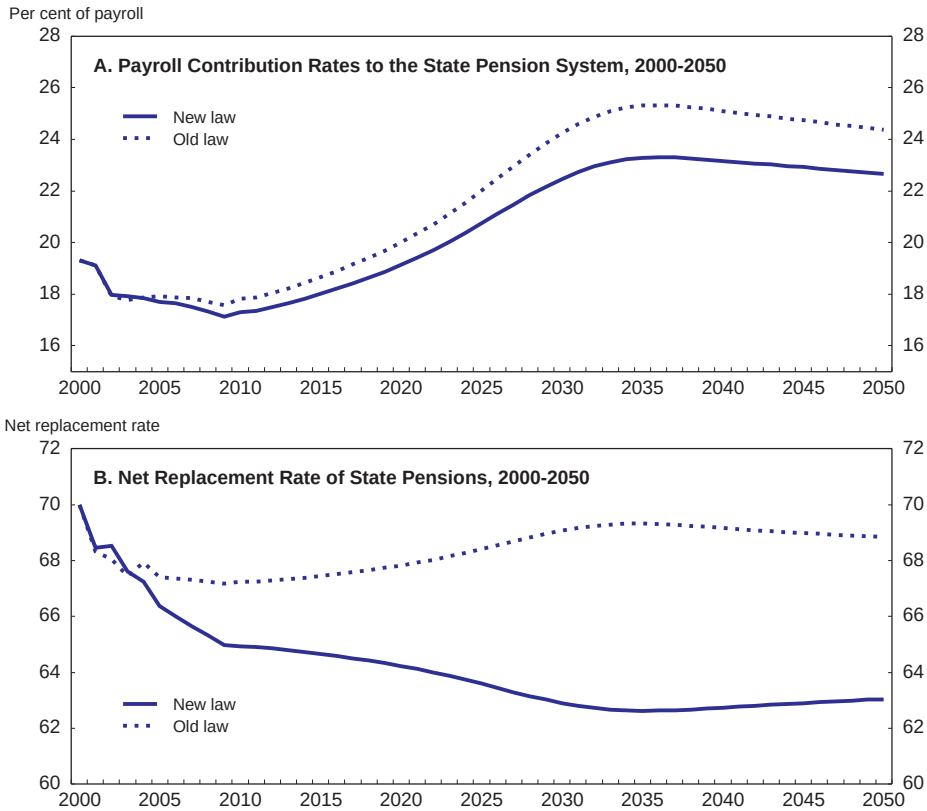
Figure 38. **Net replacement rates and contribution rates**



Source: Bonin (2001).

system; almost total illiquidity of individual private pensions due to the obligation to convert capital saved into an annuity on retirement, with only limited withdrawals allowed to finance house purchases;[226] and uncertainty in the wake of a Constitutional Court ruling requiring the government to review the taxation of pensions.[227]

In attempting to address poverty among the elderly recent pension reforms also introduced some changes to social assistance, which is the responsibility of local authorities. From 2003 social assistance for the elderly will be transformed into a new, less strict, individually means-tested benefit. The responsibility of children to reimburse this benefit has been removed in contrast to social assistance, and the federal government will instead compensate local authorities for

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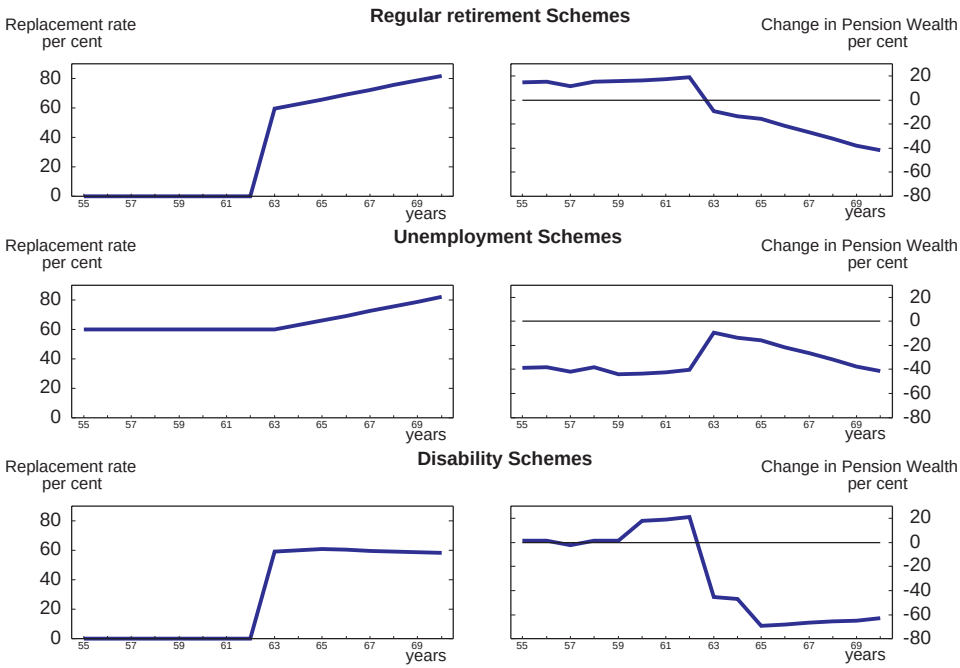
Figure 39. **Incentives for early retirement: replacement rate and change in pension wealth**



Note: The two measures given in the figure give an assessment of the incentives to retire for a hypothetical average worker with 40 years of employment at different ages for the various retirement schemes currently available in Germany. These estimates take into account the pension reforms introduced in 2001 and the incentives embodied in disability and unemployment programmes in the period up to 2011, when the possibility of early retirement through this latter programme will be terminated.

Source: OECD (2002), *Economic Outlook*, December 2002.

such outlays. The number of elderly claimants for the new scheme is expected to double from the current level of 180 000 individuals as a result of this change.

Conclusions

Recent reforms of the German pension system will not reduce the current very high standard of living for retirees in the short term, but there exists uncertainty whether they can meet the joint cap on contribution rates and floor on benefit rates in the longer term without a rise in the average effective retirement age. The current regular 15-year projections should be lengthened considerably to give a better assessment of the future challenges. In addition, in order to ensure that

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the retirement system is on a sustainable basis, strict monitoring and early action on any revealed problems are needed. Furthermore, the government should consider phasing in an increase of the standard age of retirement age, which should then be automatically linked to projected life expectancies at retirement. Finally, more should be done to encourage individuals to ensure that they have adequate alternative sources of retirement income to reduce the risk of lower income from the public system. Towards this end, retired people could be given the right to withdraw the accumulated capital from their individual retirement saving accounts with appropriate penalties, bringing German practice more in line with that in most OECD countries. Moreover, in the context of a sound regulatory framework income ceilings on contributions could be raised.

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Notes

1. At its peak in 1991 the growth differential between (west-) Germany and the EU 15 net of Germany totalled almost 4½ percentage points.
2. At the beginning of 1990s, the steep drop in eastern German manufacturing employment was largely offset by the rapid increase in employment in the construction sector. Also, losses in employment were cushioned by the massive creation of work provision schemes financed by the labour office. While utilisation of such schemes was uneven over time, it has significantly declined in recent years. Since 1995 employment in the construction sector dropped by 20 per cent, reflecting the recession in construction investment and the phasing out of special subsidies for construction in the new states. Moreover, already since the beginning of the 1990s employment in the government sector is has been cut, in particular in eastern Germany. While this reflects to some extent a shift of public sector functions into the enterprise sector – including government-owned units – on balance the secular reduction in public sector employment is significant. By contrast, public sector employment appears to decline more moderately in some other European countries and increases in France. As in other countries, the main source of job creation in Germany has been the service sector with the number of employees, net of general government employment, increasing by some 4.8 million since 1991. For more detail on the restructuring of the eastern German economy see OECD (2001a).
3. The tax wedge denotes the difference, due to income taxes and contributions paid, between the gross wages paid by employers and the net wages received by the employees.
4. Private consumption grew vigorously at the beginning of the 1990s, when the purchasing power of households in the new states was boosted on impact by the conversion of the East-Mark to DM, steep increases in eastern German wages, the extension of the west German social security system to the east, rising west-to-east transfers and a demand push from the new states for western German products. By 1993 the demand boost had faded, however.
5. For a more comprehensive account see also Wurzel (2001).
6. Germany's investment in machinery and equipment boomed in 1990 and 1991, with cumulative growth over the two years totalling 24.2 per cent. In contrast, the European investment cycle in machinery and equipment peaked already in 1989-90 and consolidated in the following years as the capacities created proved to be excessive. The following consolidation process turned out to be much steeper and longer lasting in Germany than in the other European countries. While investment in the EU area without Germany recovered relatively fast and reached pre-recession levels already in 1996, in Germany the 1991 level of investment was only reached in 1999. To a considerable extent these differences reflect the fact that at the beginning of the 1990s

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capacities were created – with the help of considerable subsidies – in eastern Germany, that enterprises considered increasingly unlikely to be fully used in later years (see OECD 2001a).

7. Regarding investment in ICT, ICT using industries in Germany appear to have invested less in ICT equipment than some other OECD countries. However, caution is necessary with respect to the comparability of ICT-related investment data, given the fact that different methods of price and quality adjustment are applied by the national statistics offices. Indeed, applying the US-deflator for ICT goods (and adjusted for exchange rate effects), Colecchia *et al* (2001) estimate that German ICT equipment in the business sector grew at an annual rate of 22.9 per cent since 1991 rather than of 10.1 per cent officially reported. See Colecchia *et al.* (2001). See also, Deutsche Bundesbank (2001a). Since the mid-nineties, the United States – and some other European Countries – use the hedonic approach to calculate quality adjusted price indices for ICT goods as well as for other selected goods. The effect of applying corresponding price adjustment methods on measuring GDP growth overall should be quite small, however.

8. Between 1991 and 1994 German construction investment boomed, with accumulated growth accounting for more than 20 per cent. In fact, over this period the contribution of construction to overall real GDP growth was highest since the middle of the 1960s. At the same time, – between 1990 and 1993 – construction investment in the rest of the EU receded. With the reunification boom fading and phased reductions in subsidies for construction and capital intensive production in favour of the new states, construction in Germany slid into recession. The current weakness is further aggravated by dwindling demand for residential construction in western Germany, which is related to both demographic developments and weak growth in personal disposable incomes since the second half of the 1990s. The current phase of construction recession is less steep and lasts longer than in previous episodes. Since 1995 the growth contribution of construction investment fell on average short of that in the rest of the EU by about ½ percentage point.

9. The size of absorption of West German goods and services exports by eastern Germany is particularly visible in 1991. In this year real exports of the old states totalled € 507.4 billion while exports of Germany overall totalled € 413.6 billion. The difference of € 93.8 billion – 6 per cent of West German GDP – marks the export absorption of the new states. To the extent western German production was diverted to the new states rather than exported the reduced growth contribution of exports is mirrored by higher growth contributions of consumption and investment, respectively.

10. Recent stock data tend to be subjected to relative large revisions as preliminary national accounts data, in addition to measuring stocks, also reconcile differences in production and demand side data.

11. The holding period for investors before realised capital gains become tax exempted was extended from two to ten years in 2001.

12. In October 2002, the stock market corrections even led the Deutsche Börse to announce the closure by early 2003 of the German "Neuer Markt" "technology" segment.

13. Bundesbank (2001b).

14. In 2001, oil and food prices added about 1 percentage point to the inflation rate, which was boosted further by the price effects of the higher indirect taxes on energy (and levies), adding almost ½ percentage point.

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15. According to the association of the automobile industry the share of passenger cars older than ten years has increased from below 22 per cent in mid-1999 to about 28 per cent in early 2002.

16. The market capitalisation of the German owner-occupied housing market is below 50 per cent of GDP, which is lower than most other OECD countries. See IMF (2002).

17. See Boone *et al.* (2001) and Hassler (2001). Focusing on liquid and illiquid financial wealth, Byrne and Davis (2001) shows a similar qualitative result, but of a lesser magnitude. Even if the marginal propensity to consume out of various forms of wealth is similar across countries, the relatively low financial wealth of the Germany household sector indicates a lower wealth effect in Germany than in other countries. See Bertaut (2002).

18. Estimating a private consumption function with wealth effects for Germany has proved difficult, among other reasons for problems related to data. However, the correlation between per capita private consumption and real financial and non-financial wealth is generally lower in Germany than in other countries. See Case *et al.* (2001).

19. Two-thirds of the ½ million additional small hours jobs created since mid-1999 have been occupied by members of the age cohorts less than 25 years or older than 60 years. The initial reaction to the change in social security contribution was a sharp reduction in small hours jobs held as secondary employment. The ISG institute in Cologne has estimated that about one-third (or 550 000) of all secondary held small hours jobs disappeared within less than half a year.

20. In the period 1999-2001, the demographic effect reduced the age cohort 55-59 years by nearly 1 million while the next cohort 60-64 years increased by a bit more than a quarter of a million. At the same time, the employment rates for both the cohort 55-59 years and 60-64 years increased by 2½ and 1½ percentage points, respectively, partly reflecting the increase in small hours jobs occupied by older workers as both unemployed and pensioners holding such jobs can maintain benefits.

21. See Deutscher Bundestag (2001), Rentenversicherungsbericht 2001.

22. Participants in the "Altersteilzeitgesetz" that have retired are accounted for as early retired.

23. Since the instigation of the programme in 1996 more than 100 000 applications have been accepted with an accumulated fiscal cost of DM 1½ billion.

24. See Deutsche Bundesbank (2001b).

25. The outbreaks let to markedly lower beef prices and higher prices on pork and poultry, with the latter two having a relatively higher share in German meat consumption.

26. At the present conjuncture, there are some concerns that the relatively smaller equity capital endowment among east German companies may make them more susceptible to bankruptcies if there is no quick improvement in the economic situation.

27. Statistical revisions of national accounts aggregates accounted for some 0.3 percentage point of the deficit increase.

28. See OECD (2001a) and (1999a).

29. Federal government transfers to the Federal Railways Funds (B*undeseisenbahnvermögen*), which was established when the German Rail was incorporated, comprise administrative costs that cannot be covered by the fund's own revenues. The government also reimburses extra costs for personnel associated with the restructuring of the former D*eutsche* R*eichsbahn* in eastern Germany.

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30. Both the spending as well as the revenue side of the federal budget are affected by a restructuring of aid to the new states that leaves the balance unaffected. Beginning in 2002 earmarked investment aid for eastern Germany is replaced by non-earmarked "Federal Supplementary Transfers" (*Bundesergänzungszuweisungen*), see Chapter III below. In 2002 this reduces both expenditures and revenues by the same amount.
31. The federal government expects the measures against VAT fraud to generate additional revenues of some € 2½ billion from 2002 onwards. Estimates on the size of VAT fraud are surrounded by a high degree of uncertainty. One such estimate stipulates that revenues foregone on account of fraud might total some € 10 billion (0.5 per cent of GDP) annually. The German VAT taxes total consumption, and is levied from enterprises. Double taxation at the various stages of production is excluded by prior-tax deduction (*Vorsteuerabzug*). It refunds firms, paying VAT on their own turnover, the amount of VAT they paid already to other enterprises as part of the bill for intermediary goods. For VAT fraud, chains of virtual enterprises produce faked bills that are presented to the tax authorities for VAT refund. In addition to implementing a bundle of measures to combat VAT fraud, the government has also established a working group that investigates the option of changing the tax regime to final taxation. Such a system can be subject to other forms of fraud, however.
32. The statutory health funds are obliged by law to set contribution rates such that the funds are kept in balance. Health contribution rates have increased from 13.6 per cent of the wage base in the old *Länder* and 13.7 per cent in the new *Länder* in 2001 to 14 per cent in mid-2002.
33. In 2000 and 2001 some 96 per cent of the receipts from the mineral oil and energy tax have been utilised to finance transfers from the federal budget to the statutory pension insurance. This has helped to restrain the increase in the pension contribution rate. The remainder was used, almost exclusively, to subsidise the use of renewable energy sources.
34. See also OECD (2001a).
35. See OECD (1999a). Van den Noord, Paul (2000).
36. EU Commission (2001).
37. The major exception is the business tax, whose receipts mainly accrue to the communities, which set local rates as a multiple of a basic rate.
38. See OECD (2001a).
39. See OECD 1996a and the following issues.
40. Applying the somewhat broader concept of the public sector, which in addition to the general government sector also includes government-owned companies, the reduction in employment is less pronounced.
41. The reduction in infrastructure is less pronounced, however, as the national accounts figures suggest. Certain entities, which traditionally belonged to the government sector are now allocated in the enterprise sector, reflecting privatisations, spin-offs from the budget and revisions in national accounts standards. Hence, a considerable share of infrastructure investment is made outside the government sector.
42. For example: public sector employment regulations imply that university graduates having studied law are automatically advantaged in their career advancement over other graduates. This even holds for administrations such as the Ministry of Finance, the Ministry of Economic Affairs or the Deutsche Bundesbank.
43. See also Shim (2002).

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44. See Bundesministerium der Finanzen (2002a).

45. See OECD (1999a) and (2001a).

46. Existing evaluation studies by academic researchers on the effectiveness of active labour market measures in Germany are almost exclusively based on surveys stemming from other sources.

47. See OECD (2001a). Also, Deutsches Institut für Wirtschaftsforschung, Institut für Weltwirtschaft an der Universität Kiel, Institut für Arbeitsmarkt- und Berufsforschung, Institut für Wirtschaftsforschung Halle, Zentrum für Europäische Wirtschaftsforschung (2002).

48. For an example of replacement investment required in a medium sized eastern German city see Pohl (2002). In the case under consideration, it has been estimated that 23 km of the road network would have to be renewed each year to meet road security standards. At present, replacement investment proceeds at a renewal rate of 7 km per annum.

49. By comparison, general government investment spending accounts for 1.7 per cent of GDP (2001).

50. See Bundesbeauftragte für Wirtschaftlichkeit in der Verwaltung (1998).

51. See Hessischer Rechnungshof (1999), Bemerkungen 1999.

52. For an outline of the system see OECD (1998a). For a summary of the reform see Bundesministerium der Finanzen (2002).

53. For investigations regarding the link between agglomeration and financial requirements see OECD (2001a). More generally, the Constitutional Court also demanded to clarify the principles on which transfers within the equalisation system are based.

54. Studies show that the marginal effective taxation of additional income tax revenues implied by the revenues equalisation system can come close to or exceed 100 per cent. See Huber and Lichtblau (1998). Fenge and von Weizsäcker (2001). Baretti *et al.* (2000) found evidence that high effective tax outflows implied by the equalisation system impact negatively on the states' tax collection. Berthold *et al.* (2001) found a negative relation between equalising transfers and the states' relative growth performance. See Berthold *et al.* (2001). Theoretical analysis by Fritz Söllner (2001) suggests that a high marginal burden of the inter-state transfer system might imply disincentives for growth-oriented policies of the states even if policy makers focus on maximising votes rather than revenues. See also OECD (1998a) and (2001a).

55. At present the equalisation system adds in only 50 per cent of the communities' tax revenues for the purpose of establishing the area states' finance endowments prior to redistribution. (Among Germany's 16 *Länder* there are the three "city-states" Hamburg Bremen and Berlin. The remaining states are "area-states".) This will be increased to 64 per cent. Indeed, empirical evidence suggests, however, that this rule disadvantaged low-density areas because they produce smaller revenues per inhabitant than agglomerations. Moreover, in the new system the weighting scheme for inhabitants of the communities will be changed. See OECD (2001a); Fehr (2001).

56. The *Länder* to which the premium accrues deduct 12 per cent of their above-average increase in revenues from their revenues assessment base.

57. See OECD (1998a). A "lump sum" system of this type of vertical transfers to the regions has recently been adopted by Spain.

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58. This has been repeatedly suggested in the German policy debate. See for example the statement of the Scientific Council of the Ministry of Finance: Wissenschaftlicher Beirat beim Bundesministerium der Finanzen (1992).

59. The earmarked part is granted by the federal government within the framework of the Investment Promotion Law Recovery East (*Investitionsfoerderungsgesetz Aufbau-Ost*).

60. In the first year (2005) the lump-sum payment equals the level of the predecessor scheme (SP I). Thereafter the transfers are mildly digressive until 2008. From 2009 onwards transfers are reduced by € 0.8 billion (DM 1.5 billion) annually, until the scheme exhausts in 2019 (see Figure III.SP). See also Ragnitz (2001).

61. See Rosenfeld and Kronthaler (2002).

62. See OECD (2001a). See also Seitz (1999).

63. On a national accounts basis, for seven out of 11 years since 1991 the general government deficit has exceeded gross investment. Also, the rule does not cover investment in human capital, which importantly impacts on an economy's longer-run growth potential.

64. See OECD (2001b).

65. For Germany, estimates by the OECD Secretariat suggest that an increase in the output gap by 1 percentage point leads to a counter-cyclical fluctuation of the general government deficit by some ½ per cent of GDP. Hence, with the structural budget held in balance GDP growth would have to fall by some 6 percentage points for the actual balance to hit the 3 per cent deficit limit of the Maastricht treaty.

66. While national accounting aims at making transactions explicit domestic administrative accounting systems are more oriented towards managing budgets in terms of cash flows received and paid. Hence, both systems differ in several respects. Some spending or revenue items – such as privatisations – affect the deficit of public sector households in administrative terms while leaving them unaffected in the national accounts. Furthermore, some items – such as tax allowances for families with children – appear (implicitly) on the revenue side in administrative budgets but are explicitly recorded as transfers in the national accounts, and *vice versa*. Also, administrative budgets allocate expenditures and revenues to certain periods in accordance with the associated cash payments or receipts while national accounting is based on the "accruals" principle.

67. Besides social assistance for income support (*Laufende Hilfe zum Lebensunterhalt*) there is also support in special circumstances (*Hilfe in besonderen Lebenslagen*), which is mainly paid for the re-integration of handicapped persons and for exceptional health care needs. Over 2000 some 1.5 million received this type of social assistance. In addition, at the end of 2000 some 0.35 million persons received assistance benefits as asylum seekers.

68. Not employed persons who are not able to work comprise those caring for children or sick relatives, sick persons, and persons which cannot work on account of disability. See Haustein (2002).

69. Benefits are usually paid in cash and comprise a basic component, assistance for housing expenses and aid for one-off expenses such as household effects or removals. The number of HLU recipients trended upwards from some half a million at the beginning of the 1970s to some 2.9 million in 1997, excluding asylum seekers.

70. In 1999 the federal government abolished the eligibility of certain groups of unemployed to receive unemployment assistance benefits without prior receipt of

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unemployment insurance benefits. This implied higher social assistance outlays for the communities.

71. See Huber and Lichtblau (2002).
72. The net incomes have been derived from the ifo data-base of the tax and benefit system.
73. See Sinn *et al.* (2002).
74. See OECD (2001a).
75. See Steiner (2002).
76. For a comprehensive account of the German social transfer system more generally see Von Loeffelholz, *et al.* (1997).
77. Various micro-econometric studies – while not focusing explicitly on sanctions – found that exits from unemployment cumulate around the exhaustion point of unemployment benefits. See Wurzel (1993); Hagen and Steiner, (2000). There are some recent studies available that focus on sanctions. Recent econometric work for Switzerland, which exploits information by K*anton* on how strict potential sanctions in cases of non-compliance with job search requirements are implemented in practice, suggests that unemployment duration falls with the credibility of announced action. This result holds even for unemployed who are not yet on social assistance benefits because their unemployment insurance benefits did not yet exhaust. See Lalive *et al.* (2002). Simulation experiments within a macro-framework for the Netherlands suggest that benefit sanctions might have welfare effects that might be more pronounced than reducing replacement rates. The main effect originates from the threat of penalties, not because the penalty is actually imposed. See Boone and van Ours (2000).
78. Survey information suggests that 20 per cent of the registered unemployed do not want to accept employment. In the first place this group consists of older persons. See Brenke (2002).
79. See OECD (1999a) and OECD (2001a).
80. See Martin (2001).
81. See Sinn *et al.* (2002).
82. A significant share of spending accrues to schooling within the dual vocational training system.
83. OECD (2001c). For a summary of key results and a discussion of various aspects of the PISA study in Germany see also Schleicher (2002); Prenzel (2002); Hohlmeier (2002); Lüdeke (2002); Stange (2002) and Ochel (2002).
84. The variation found in students' average competence levels across the L*änder* amount to about one to one and a half school years in high schools. The German PISA study is supplementary to the international one, and has been commissioned by the education ministers of the German L*änder*. Its goal is an assessment of pupil performance individually by states. Overall, 50 000 pupils were tested. For 14 out of the 16 federal states test results are available for the full programme. For two states, Hamburg and Berlin, the response rates were too low for these two to be included into the general evaluation. They have been included, however for the high school section, where the response rates were sufficiently high.
85. There is a standing conference of the states' ministers for education (K*ultusministerkonferenz*), whose tasks *include inter alia* setting common standards that secure the congruency of school certificates across the states.
86. See OECD (2002a).

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87. The programme aims at supporting the establishment of 10 000 full-day schools by federal aid totalling €4 billion between 2003 and 2007.

88. The German PISA-*Konsortium* has carried out supplementary analysis of the PISA data sets with respect to the socio-economic composition of the test participants. See Deutsches PISA-Konsortium (2001). For a summary see: *www.mpib-berlin.mpg.de*

89. In the OECD's classification tertiary type A programmes are more theoretically oriented, such as the German *Diplom* or the Master's degree in the United States. Tertiary type B programmes are more occupationally oriented leading to direct labour market access. In Germany they comprise polytechnical tertiary education (*Fachhochschulen*).

90. See OECD (2001d).

91. See "Investment in human capital through post-compulsory education and training" in OECD *Economic Outlook* 71.

92. See OECD (2001a).

93. In contrast to the pension system the health insurance system did not benefit from the proceeds of the ecological tax levied since 1998.

94. The measure "potential years of life lost" (PLYL), represents the shortening of life expectancy due to avoidable disease.

95. In 2000 the number of practising physicians per thousand of population was 3.6 in Germany as compared to the OECD average of 2.9. In acute-care hospitals beds per thousand of population stood at 6.4 as opposed to 4.3 on average, and the average length of stay in acute care totalled 9.6 as opposed to an average length of 7.4. The number of countries for which averages are computed varies with data availability. See OECD (2002c).

96. Under the system of sectoral budgets total service volume increased by an estimated 32 per cent for the "blue collar" (RVO) funds and 51 per cent for the substitute funds between 1988 and 1995 (global budgets came into effect 1989). See Mattke (2001). A surge in physicians opening up private practices in anticipation of a law that limited the possibility of opening new practices during that period may have contributed to that increase.

97. Until recently, the regional doctors' associations (*Kassenärztliche Vereinigungen*, KVs) have been subject to statutory regional collective budgets for spending on pharmaceuticals (*Arznei- und Heilmittel*). The volume of the budgets was adjusted annually in accordance with certain rules fixed by law. If collective spending on pharmaceuticals exceeded the budget, the health insurers reduced the remuneration for the doctors, paid to the relevant KV.

98. See *Pharmahersteller hebeln das Sparpaket aus*, Frankfurter Allgemeine Zeitung 2 April 2002.

99. See the special chapter on the German health care system in OECD (1997).

100. See for example OECD (2001b). In Austria, the average duration of hospital stays has declined since performance-related remuneration for hospitals was introduced in 1997. While this has contributed to dampening the cost increase in the hospital sector, the trend increase in the number of hospitalisations was reinforced with the share of day admissions rising.

101. Indeed, reservations have been raised by hospital associations that it would hardly be possible to establish the necessary infrastructure by 2003.

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102. New, mainly enterprise-based insurance funds, which traditionally insured a low-risk population, used their historically low contribution rates to attract better risk.

103. In the Netherlands empirical analysis suggests that risk compensating transfers to insurers based on global variables such as age, gender and location explain some 2½ per cent of the annual variation in health care expenditures. This is less than one-fifth of the predictive power that has been estimated to be attainable. In contrast, limited inclusion of additional information on the health status would have raised the predicted part of the variance to three-fifths of the attainable maximum. See Van Vliet and van de Ven (1992).

104. Alternatively, the insurance system could be moved to a risk based insurance market system However, the market approach alone would not be compatible with the equity goals inherent in Germany's solidarity based health care system. Hence, shifting to a full market system would require that redistribution would have to be left to the fiscal tax and transfer system. Moreover, this approach would raise further issues with respect to general access to health care services.

105. It should be cautioned that some design features of the particular programme will make a rigorous evaluation technically difficult: The actual content will differ by fund, region and provider network, pre-intervention data is unavailable and no comparison group to compare this self-selected sample of DMP participants exists.

106. In companies with 20 to 50 employees the share totals 23 per cent and for small firms with no more than 10 employees it stands at 15 per cent. See Statistisches Bundesamt (2002). The 2001 micro-census covered for the first time questions on working time schedules, which makes comparisons with earlier years not feasible.

107. See Kayser (1997). The Survey comprised 13 800 medium-sized companies.

108. The study by Kölling and Lehmann (2001) covers 13 800 companies embodied in the IAB firm panel. According to this study, 22 per cent of all firms which left the collective wage agreement between 1996 and 1999 increased their weekly working time by at least one hour, 15 by more than one hour. However, some 11 per cent of the companies decreased their regular weekly working time. These developments occur almost exclusively in the western countries, whereas in eastern Germany changes in working time are hardly identifiable. Surprisingly, there is no empirically significant time lag between the date of quitting the wage agreement and the change in regular working time, which would have been assumed under legal conditions. That means, that employers rearrange quickly which workers' representatives, sometimes in avoiding statutory periods of agreement terms.

109. Thelen (2001).

110. See Franz (2001); and Boockmann and Hagen (2001). There is a risk, however, that opening clauses that are conditioned on adverse business conditions are not exercised out of fear that this would disclose a company's unfavourable business condition to the market.

111. Boockmann and Hagen (2001); Magvas and Spitznagel (2002).

112. Deutscher Industrie- und Handelskammertag (2001).

113. See OECD (1994a) and OECD (1998b). Also, Hunt (2000). For features of the German dismissal law see Eger (2001). See also Walwei (2001).

114. Dismissal cases constitute the largest single item of the legal proceedings the labour courts are facing. In 2000 the inflow of all labour related cases combined totalled 569 161 of which 542 337 were instigated by employees.

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115. A recent empirical study, following 4 000 firms from the IAB establishment panel between 1993 and 1999, found a significant positive relationship between the institutional redundancy costs for regular workers and the demand for fixed-term contracts. See Boockmann and Hagen (2001). Another study for the service sector has found that the demand for fixed-term contracts increases if the firm applies a collective wage agreement. In contrast, in this case freelance work decreases. See Kaiser and Pfeiffer (2000).

116. Upward revisions in expected or actual sales have been found to be positively associated with rising temporary employment, suggesting its role as an instrument for adjustment to varying demand conditions.

117. OECD (2002d). Unemployment in the previous five years significantly reduces the probability of moving into permanent jobs.

118. Econometric analysis by Lechner *et al.* (1999) suggests that active labour market measures in Rhineland-Palatine in 1994-96 based on temporary employment significantly increased reintegration probabilities in the primary labour market.

119. See also OECD (1999b); OECD (2001a).

120. The share of employees in the private sector represented by workers council decreased since the early 1980s from about 50 per cent to currently less than 40 per cent. The share of firms, which have no worker's council, increased accordingly from 51 per cent in the mid-1980s to currently 61 per cent. See Hans-Böckler-Stiftung (1999). The Establishment Constitution Act (*Betriebsverfassungsgesetz*) was not modified essentially for the last 30 years. Employees' co-determination in establishment councils involves mainly matters of human resource management and employment protection. It is not to be confused with the representation of employees on the board of incorporated companies.

121. The statutory number of workers' council members in firms with 101 to 200 employees was increased by two persons to seven, and in large firms with more than 7 000 employees by four persons to 35. Companies with 200 employees (before: 300) have to release one person fully from work to run the workers' councils. A second person must be released in companies with more than 500 employees (before: 600). The employer is obliged to support the council's work "on a sufficient scale" with modern communication and information equipment.

122. See for example Rudolf and Müntnich (2001).

123. See Wirtschaftswoche heute Service, 15 Minuten für jeden Arbeitslosen, 9 August 2002.

124. See OECD (1994b) and OECD (2001e).

125. This indicator consists of four different weighted components, such as the average number of compensation days as indicator for the rapidity of the return to the labour market of the unemployed or the ratio of long-term unemployment to total unemployment as a measure for the reduction of long-term unemployment. See OECD (2002d).

126. In 2001, the volume of funds with discretion of the Federal Labour Office amounted to about €14 billion, which is one-quarter of the Labour Offices budget. A larger scope for the Labour Office in assigning funds was introduced in 1998 with the adoption of the third Social Security Code (*Sozialgesetzbuch*, SGB III).

127. Until 2005 the government provides about e6 million for the independent consumer protection association *Stiftung Warentest* to examine 20 training courses per year. First results this exercise pointed to the insufficient quality of these courses.

128. After up to six continuous months of unemployment the premium amounts to € 1 500, between six and nine months € 2 000 and from nine months on € 2 500. The premium

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is paid in two steps: If the job seeker enters regular employment with a weekly working time of at least 15 hours for at least three months, the PPA receives a premium of € 1 000. If he is employed for more than six months the PPA receives the rest of the voucher premium.

129. Firm and industry-level econometric studies have established empirically the effects of innovation and diffusion of new technology as engines of multi-factor productivity growth. See for example Ahn (2002).

130. There are some indications that the strengths of the results are dependent on corporate governance issues, such as ownership diffusion. See Januszewski *et al.* (2001) and Köke (2001).

131. Bassanini and Ernst (2002).

132. Bassanini *et al.* (2001).

133. Nicoletti *et al.* (2001).

134. Nicoletti *et al.* (2000).

135. E*x ante* regulation implies that the incumbent needs the authorisation of the regulatory authority to be able to change prices. E*x post* regulation means that the regulatory authority may review the actions of the market-dominating enterprises.

136. See OECD (1996b).

137. The energy divisions and the L*änder* competition offices will be responsible for enforcing the ban on abusive, obstructive and discriminatory practices, guaranteeing access to the network, removing hindrances to transmission and investigating fees for network use and energy prices.

138. An additional layer to the structure of competition authorities is the Monopolies Commission. The commission has direct regulatory powers, but carries out research into competition issues with a view to provide recommendations to improve the regulatory framework. The commission consists of five members with time-limited government appointments of four years. The main tasks are to prepare opinions, assessing the level and foreseeable development of business concentration, to evaluate the application of provisions, concerning the control of concentrations, and to comment on other topical issues of competition policy.

139. The fines issued by the Federal Competition Authority can be appealed to the Higher Regional Court with the Federal Supreme Court as the next and ultimate appeal instance.

140. OECD (2001f).

141. The competition authority has issued administrative fines to individuals up to € 350 000 and such fines can reach a maximum of € ½ million.

142. The Ministry must ask for an opinion from the Monopolies Commission and in mid-May 2002, the Commission recommended nearly unanimously that a clearance should not be granted under any circumstances. Since 1973 the ordinance has been applied 16 times and the previous time in the late 1980s.

143. The conditions included divesting an East German subsidiary, legal unbundling of Ruhrgas transmission network, and a substantial release of gas supplies to competitors. The ministerial clearance has been subjected to legal challenges, which were still unresolved by September 2002.

144. The German price structure reflects that service providers are providing subsidies for handsets. Thus including this element would lower the cost of the overall mobile

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phone package. See for example Oftel (2002), which suggests that German prices for an overall mobile phone package are similar to those in France and Sweden, while lower than those in Italy and the United Kingdom.

145. See OECD (2003).

146. The RegTP (2001) explicitly notes the regulatory efforts to lower telecommunication prices in the fixed-line segment.

147. The DSL services are being promoted as an add-on to existing telephone services (including ISDN). Competitors would like to offer DSL services both as an Internet connection and for telephony.

148. The European Commission initiated anti-competitive proceedings against DT in spring 2002 on the grounds that competitors are charged more than DT's retail customers are.

149. Recent EU legislation in this area has replaced the notion of Substantial Market Power with one of dominance, which also can be applied in the form of joint dominance, emphasising the competition element.

150. "Roaming" charges refer to the financial compensation foreign mobile phone providers receive for connecting non-resident mobile phone users.

151. The offer includes unlimited number of minutes during weekends. See *www.voicestream.com*

152. Termination charge refers to the charge a receiving provider network demands for terminating (connecting) a call originating from another network.

153. Recently, there has been an increase in the concentration among providers of telecommunication services to the household sector. The extent of this withdrawal and possible consequences for competition are too early to evaluate.

154. A regulatory problem in stimulating the development of competing fixed line networks is that a lowering of interconnection charges to the local-loop reduces entry barriers as well as incentives for construction competing networks. However, the most costly aspect such networks tend to be the right of access, explaining why competing land-based networks normally are constructed on an existing backbone of a non-telecommunication network. Once such right of access has been obtained the additional capital costs are relatively limited and as long as the associated interconnection charges are higher than the very low marginal cost of using a network, incentives to construct competing networks remain in place.

155. It has been estimated that only 12-18 months are required to upgrade cable-TV for telecommunication.

156. IEA (2002).

157. See Monopolkommission (2002).

158. OECD (2001g) Volume 3. No. 3 states that State owned vertically integrated firms should be broken properly up both vertically and horizontally prior to privatisation.

159. Taxes and charges levied on electricity include VAT and the electricity tax from 1999 as well as charges for financing subsidies for renewable energy and Combined Heat and Power generation introduced in 2000. It has been calculated that over the period 1998 to 2000, electricity prices net of taxes and charges fell about 20 per cent and less than half of that when including the taxes and charges introduced in the period. See Auer (2002).

160. See IEA (2002). As a consequence, the competition authority issued a warning letter to one network operator on 18 July 2001 concerning suspicion of abusively excessive fees. By end-2001 fee variation had been reduced, although the highest was about 35 per cent higher than the industry average. See *www.vea.de*. Moreover, in defining the relevant

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market for merger control the competition authority has switched from regional to nationwide markets.

161. See Newbery (2002) and Brunekreeft (2002).

162. Other aspects of the electricity market have also been subject to regulatory concern, such as the Federal Competition Authority's investigation into whether the charges for balancing supply and demand are cost reflective.

163. IEA (2002).

164. German produced natural gas covers about a fifth of domestic demand. Nearly all import of natural gas is undertaken by six of the supra-regional companies long-term take-or-pay contracts with Russia (45 per cent of all imports), Norway (27 per cent) and the Netherlands (22 per cent).

165. The need to modify the Associations' Agreement was highlighted by the European Commission's investigation into whether the Norwegian company Marathon had non-restricted access to the German natural gas market, or more specifically to Thyssengas' net work.

166. Local monopolies to varying degrees are prone to appear in a deregulated natural gas market unless sufficient transmission capacity is in place, which indicates that ensuring sufficient or even excess transmission capacity may be a suitable regulatory response. See Cremer and Laffont (2002).

167. The German natural gas industry covers more than 750 companies, which are engaged in producing (16 companies), long-distance gas transmission companies (14 companies), regional distribution (15 companies), local distribution (more than 700 companies of which many are publicly-owned *stadtwerke*) and gas dealers (11 companies), who have emerged since the 1998 liberalisation of the gas market. In addition, there is considerable vertical integration through direct ownership as well as through cross-holdings among the gas companies.

168. See Monopolkommission (2002).

169. The unbundling of accounts concerns transmission, distribution and storage activities. IEA (2002).

170. In comparison, within the regulated third party access regime in the United Kingdom the natural gas regulator has a staff of around 500 people supervising 13 networks.

171. While Economic Ministers from the EU member states have proposed to accelerate the liberalisation process and the EU parliament has endorsed such a draft, the current plans still foresee a removal of incumbents' monopoly positions only in 2009.

172. The exemption is accordance with EU rules and is leading to nearly € ½ billion in tax expenditures on the sales generated on competitive market segments. Less than 10 per cent of DP's revenues arise from activities in competitive segments of the postal market.

173. As reported in the previous *Economic Survey of Germany*, postal tariffs are relatively high in an international comparison and in mid-2002 the German tariffs for a standard letter were nearly 30 per cent higher than the average in the EU.

174. The measures include the right to intervene on the authority's initiative and not only after complaints as well as more precisely defined rules concerning information rights and fines abilities similar to those of the RegTP.

175. The previous system price system was based on a fixed and a variable component, favouring large railway providers. The non-discrimination is ensured through flat rates that only vary with quality of rails and types of transport. See Bundeskartellamt (2001).

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176. Bundesministerium für Wirtschaft und Technologie (2001).
177. A case for not requiring unbundling can be made if the vertically integrated companies can demonstrate that the co-ordination of externalities is sufficiently strong to warrant integration. See also Monopolkommission (2002).
178. See OECD (2001a) for a closer description.
179. The financing of railway infrastructure (as well as motorway and water infrastructures) is going to be financed through the revenues arising from the newly introduced motorway toll on heavy lorries (more than 12 tonnes).
180. Some 4 500 out of 6 700 water utilities supply about 8 per cent of all water consumption and, for example, Germany has 20 times more water utilities relative to the population than the Netherlands.
181. Boscheck (2002).
182. OECD (1999c)
183. BMWi-Forschungsvorhaben (2001).
184. The Federal Competition Authority has no competence for investigations and failures to comply with the Act and legal redress can only be sought from the civil courts.
185. A leading retailer offered a 20 per cent discount on the use of cards in connection with the change over to the euro to cut queues as shoppers and cashiers adjusted to the euro. After a complain by the Centre for Action Against Unfair Competitive Practises, a state court in Düsseldorf (the base of the company) issued a temporary injunction preventing the retailer from going ahead with the discount. In response the retailer extended the discount to all shoppers and with a time limit, leading to a second injunction, as the discounts would only be legitimate if no time limits were set.
186. Below-cost pricing may have an anti-competitive effect when used as a part of a wider predatory pricing strategy.
187. Predatory pricing refers to the setting of prices below some measure of costs that is unprofitable unless it forces one or more competitors to exit the market, thereby opening an opportunity to more than recoup the lost profits plus losses. For a detailed discussion of the involved issues see OECD (1989).
188. The Commission has stipulated that there needs to be a law specifying the promotional tasks that German special-credit institutions such as KfW can perform while still receiving state support. The deadline for the drawing up of the relevant legislation is the end of March 2004.
189. Deutsch (2002).
190. The motives underlying take-over activities are manifold, but the underlying business logic tends to be reaping economics of scale and scope in production, sales and purchasing, or to develop business strategies for expansion and ensure survival in the marketplace. In addition, take-overs also function as an element of sound corporate governance through the disciplinary effects on management.
191. The successful take-over of Mannesmann AG by the UK based Vodafone continues to be the world's most expensive transaction with a volume of € 179.9 billion, aiding in generating a substantial amount of media interest as the hostile bid (*i.e.* against the management wishes) eventually succeeded.
192. In mid-2002, the European Court of Justice issued a ruling that effectively restricted member government's possibilities for preventing foreign take-overs. While it is unclear how the ruling influences the new take-over legislation, it at least raises question marks

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concerning the Land Lower Saxony's "golden share" in VW, which restrict voting rights to a maximum of 20 per cent.

193. Ultimate economic risk refers to share capital with an unlimited right, which has a right to participate in the associated company's profit or residue on liquidation. See European Commission (2002).

194. In contrast the boards of American companies generally have broad discretion for defensive devices under the business judgement rule, although the laws of the various states differ to a substantial degree. However, such measures are applied in a legal and capital market environment that differs widely from the similar European context. See European Commission (2002).

195. Such resolutions may also signal that the company considers itself a take-over candidate, leading to a possible discount on the shares, which in itself makes the company a more likely take-over candidate.

196. Nonetheless, there is a risk that foreign investors will interpret the stance on the take-over code as a sign of ambivalence towards foreign direct investment (FDI). The German record in attracting FDI is relatively poor – net FDI inflows in 2001 are expected to have accounted for around 7-8 per cent of the EU total, despite the economy accounting for over 20 per cent of EU GDP

197. Under the German system of co-determination, employees have a right to be represented on supervisory boards for companies with more than 500 employees and to have parity representation on supervisory boards for companies with more than 2 000 employees.

198. The German regulatory reform is a reflection of EU's Financial Services Action Plan, which was endorsed by the European Council meeting in Cologne in June 1999 and its implementation deadline of 2005 at the March 2000 meeting in Lisbon. The plan aims at integrating capital markets to further economic growth, job creation, financial stability and lower prices, enabling consumers to reap the full benefits of the euro. The regulatory framework for EU's financial markets is based on the principles of home country control and mutual recognition, which is expected to eventually yield convergence in the regulatory environment. At present there are about 40 supervisors just for security markets in the EU. See OECD (2002e).

199. As a result of a domestic performance better than its upwind neighbours, transboundary deposition, although falling in absolute terms, has accounted for a larger share of total deposition in Germany over time.

200. Emissions per unit of GDP have fallen more rapidly in Germany than in the area as a whole and their level is also lower.

201. The external cost of sulphur dioxide and nitrogen oxides pollution was estimated as being € 40 billion in 1990 (Friedrich and Droste-Franke, 2001).

202. Since 1990 emissions of the principal pollutants have fallen by 63 per cent bringing a proportionate reduction in the economic losses due to poor health outcomes. Against that fall, wages, the principal determinant of the economic losses, have risen by 18 per cent. Consequently, the cost of emissions from the principal pollutants has fallen to around € 15 billion at 1999 prices.

203. In the new L*änder* the right of access to the grid can be limited if this threatens electricity production based on lignite. Parliament can extend this restriction in 2003 for a further two years.

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204. Ozone is formed by complex photochemical reactions mainly from the precursors of nitrogen oxides and volatile organic compounds. With intense sunlight lasting several days high levels of ground level ozone can occur.
205. With the adoption of EURO 4 standards, emissions from petrol engine vehicles will be virtually free of air pollutants. However, the EU requirements for diesel engine and commercial vehicles are less strict, although also contributing to projected falls in ozone precursors and particulate matter emissions.
206. This calculation compares a 2-litre diesel to a 1.6-litre petrol car and assumes 25 per cent less fuel is being used by the diesel.
207. Operational catalytic filters reduce particulate emissions from diesels to a level 10 000 times lower than required by the EURO 4 standard, at which point the pollution is barely detectable.
208. The fee structure of the Green Dot system is problematic, in that it reflects the cost of collection, sorting, and recycling rather than any externality, together with *a priori* judgements of the value of each material.
209. The DSD has attracted periodic criticism, most recently by the Federal Cartel Office, for abusing its dominant position in restraining the emergence of competition.
210. Recycling of household packaging waste in Germany may even generate more waste for plastics, paper, and tinplate than production from virgin material (Taylor Nelson Sofres Consulting, 2000).
211. A process known as feedstock recovery in which the polymer chains in plastics are broken down into their constituent components (hydrogen and carbon monoxide).
212. Until the amendments to the Packaging Ordinance in 1998, incineration was prohibited, though limited "feedstock recovery" was permitted. The majority of "feedstock recovery" in Germany involves converting plastics into methanol or introducing waste plastic directly into blast furnaces. The amendments now allow feedstock recovery or incineration of 40 per cent of plastics recovered.
213. Mandatory deposits for beer and mineral water containers will be introduced from the beginning of 2003, after failing to meet recycling rates of 72 per cent directed in the Packaging Ordinance.
214. Total waste disposal costs are around e150 annually per household, rising to e180 once the indirect costs of recycling and treating packaging waste (such as time spent on sorting) are allowed for.
215. This is a point stressed in OECD (1999).
216. Earnings prior to the year of retirement are revalued by the growth of average earnings.
217. The replacement rate is measured after the deduction of income tax and social security contributions.
218. See OECD (2001a).
219. On the assumption of a real rate of return (net of management costs) of 1½ per cent.
220. Bonin (2001) notes that these projections include smaller increases of life-expectancy than the projections from the Federal Statistical Office, while Sinn (1999) argues that although contribution rates may be capped other taxes and charges on labour income will have to be increased to compensate for this.
221. The employee has the right to work full time during the period designated as part time and so gains a contribution credit that allows the permissible retirement date to be brought forward.

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222. At present, from the age of 58 years, people can effectively withdraw from the labour market and from the age of 60 years receive an unemployment pension if they have been unemployed for over one year in the previous 18 months.

223. Collective agreements have been reached covering the construction industry, Volkswagen, and metal workers unions.

224. The so-called Riester pensions (Pension Provision Contracts) are voluntary private pensions, attracting subsidies and tax breaks. A grant is available, which is inversely related to income and positively related to the number of children. If, however, a tax break is more favourable for the individual, contributions to private pensions are tax deductible, which rise to 4 per cent in 2008 of the previous year's income.

225. The take-up of these pensions may reflect survey evidence that points to a widespread belief that the pension reforms were insufficient, with 85 per cent of respondents agreeing that the pension system will face a crisis in the next 10-15 years (Boeri, Boersch-Supan, and Tabellini, 2002)

226. Mandatory annuitisation of voluntary private savings is relatively unusual in OECD countries, though individuals may receive less favourable tax treatment if they opt for a lump sum payment on retirement (Yermo, 2001). In the German system, pension saving providers must guarantee that the capital accumulated is equal to the contributions made and provide an annuity on retirement. The mandatory annuitisation and guaranteed zero nominal rate of return, which is unusual in an international context, are likely to lead to lower rates of return on invested assets and smaller annuities.

227. The ruling requires the government to standardise taxation of pensions for public servants and the rest of the population. At present PAYG pensions are not taxed, while public servants pensions are. The ruling could potentially lead either to the phased introduction of deferred taxation on pension contributions or taxation of public pension contributions. These rules will also have bearing on the tax treatment of other pension vehicles.

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Annex I
Chronology of main economic events

2001

January

In the area of product market competition a number of measures come into force. These include abolishing the law restricting rebates and promotional offers, and the creation of an electricity division within the Federal Competition Authority. The full opening of postal service markets is postponed to 2007.

The 2001 step of the tax reform and the next step of the ecological tax reform come into force.

February

The government enters an accord with the European Commission to phase out public guarantees to state banks.

June

The Federal and Länder governments reach an agreement on the restructuring of the inter-governmental fiscal equalisation scheme. Special transfers to the new states will be prolonged between 2005 and 2019 ("Solidarity Pact II").

August

Workers' co-determination in work councils is widened through an expansion of representation and additional resources.

September

A ministerial clearance is issued for the merger of two large energy companies following the refusal of the Federal Competition Authority to allow the merger to go through.

November

The Federal budget for 2002 is adopted by parliament.

December

The government updates its stability programme. The objective of a balanced general government budget by 2004 is maintained, although the deterioration of the economic

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situation requires a greater deficit reduction towards the end of the programme's time horizon, which is planned to be achieved through a relatively larger reduction in spending. Parliament adopts legislation introducing road pricing on German motorways for heavy vehicles.

2002

January

The new pension reform becomes effective, opening up for funded private pensions provision. Moreover, the next stage of the "ecological tax reform" raises indirect taxes and levies, and measures to reduce VAT fraud are introduced. In addition, taxes on tobacco and insurance are increased to finance additional security expenses arising in the wake of the 11 September 2001 terrorist attacks in the United States.

Several health insurers raise their contribution rates to restore balance between spending and revenues following the deficits that emerged the year before. A reinsurance scheme for health funds is established to compensate cases where health expenditures exceed certain thresholds.

The first part of the Job-AQTIV-Act comes into force with a second part becoming effective in April. The act aims at improving the efficiency of job mediation and activation of unemployed.

The regulatory framework for corporate take-over activities moves from a voluntary code to encoded standard, providing for clear and transparent rules for hostile take-over activities. Moreover, a voluntary code of conduct to improve corporate governance is introduced the following month.

February

The government commits itself to tighten the path of budgetary restraint, to reinforce fiscal consolidation and avoid an early warning from the EU Council of Economics and Finance Ministers. This commitment is underpinned by the domestic stability pact's stipulation of balanced budgets and limited permissible spending increases for all levels of government.

March

The "Mainz" Model is implemented nation-wide, introducing subsidies for social security contributions for low-income jobs.

May

The Federal Agency for Financial Market Supervision is created, integrating supervision of banking, insurance and securities markets into one body, making the new authority the single regulator, licenser, supervisor and enforcer of financial market regulation.

June

The Associations' Agreement in the gas sector is amended following the Minister of Economics' announcement that unless an agreement was reached a sector regulator would be established. The powers of the railway authority are expanded to ensure non-discriminatory access.

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July

The Fourth Financial Market Promotion Act comes into force.

August

The 2003 step of the tax reform is postponed one year as a financing measure of damages arising from the widespread summer flooding. In addition, the corporate tax rate will temporarily be increased in 2003.

The "Hartz" commission presents its proposals for a comprehensive reform of the Federal Labour Office as well as for improving existing labour market policies and placement strategies. I*nter alia*, the proposals include measures to increase job search incentives, to widen the scope for temporary work agencies, and to activate difficult-to-employ groups.

July

Measures to dampen outlays for pharmaceuticals are introduced. To avoid an increase in pension contribution rates the required financial buffer stock of the statutory pension system is reduced from a full month's spending to 80 per cent.

October

The re-elected government presents the first elements of a fiscal consolidation package, which is further expanded the following months. The package consists largely of revenue-raising measures, but also with some spending restraint. The revenue increasing measures include higher capital gains and energy taxation and a removal of tax exemptions.

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Annex II

Financial market developments in Germany

Financial markets are being transformed, nationally and internationally, under the pressure of common factors, such as new technologies, lower communication costs, product standardisation and the euro. In Germany, this transformation process has been furthered by domestic factors. The banking sector is going through cost-related consolidation and state guarantees to publicly owned banks are being phased out. Stock exchanges are gaining in relative importance. The removal of capital gain taxation of divesting cross-holdings is expected to further the restructuring of German industry. The introduction of funded labour market pensions is expected to boost financial market developments. The regulatory responses have been an integration of supervisory responsibilities of various market segments into one regulator and a continued adaptation of financial market regulation.[1] The transformation of financial markets should improve Germany's growth prospects, as well-functioning financial markets mobilise and allocate savings by diversifying risks as well as being a tool for improved corporate governance, eventually boosting potential growth as shown in recent empirical work.[2] This annex reviews developments in financial markets and forces acting.

The consolidation of the banking sector is likely to continue

The German ("H*ausbank*") banking system is characterised by a strong relationship between banks and customers with the former being the predominant provider of capital to companies and taking actively part in their corporate governance.[3] During the 1990s small and medium sized enterprises became more dependent on bank lending and only larger enterprises (with a balance sheet of DM 100 million and more) became more reliant on other source of finance.[4] The German banks' active role in corporate governance is often contrasted with the more predominant stock market financing in the Anglo-Saxon system, providing for a more market-based system of corporate governance.[5] Thus, bank financing is relatively more important as a source of capital to the corporate sector than in the United States and the United Kingdom, although less so in comparison with southern Europe (Table A1). In addition, a certain degree of over-banking can be observed in terms of banking infrastructure relative to the population.[6]

The fragmented banking industry has consolidated over the past decade, reducing the number of banks by a quarter. The consolidation is driven by the need to solve cost-related profitability problems with a dense network of branches and personnel cost relative to earning being higher than in other European countries.[7] An additional factor was the relative loss of importance *vis-à-vis* the stock exchange as a source of finance for non-financial corporations with saving flows being redirected towards investment funds – growing annually by 20 per cent in the 1990s. Many of these funds, though, are the results of outsourcing from banks and insurance companies, partly in a response to the re-direction of non-financial companies' long-term pension liabilities into funds.[8] The consolidation process is likely to continue,

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Table A1. **Selected bank statistics, 1999**[1]

	Bank loans to the corporate sector, in per cent of GDP	Per 100 000 inhabitants			Assets[2]
		Banks[3]	Branches	Employees	Euro million per employee
Austria		12.0	57	854	6.9
Belgium		1.1	142	743	10.1
Finland		6.7	25	474	4.6
France	37.2	2.0	45	672	8.8
Germany	39.8	3.5	50	881	6.8
Greece		0.2	19	437	1.8
Ireland		1.5	42	1 409	4.7
Italy	49.8	0.4	42	545	5.0
Luxembourg		48.6	..	4 902	28.2
Netherlands		1.0	43	821	9.4
Portugal		0.4	47	571	4.2
Spain	43.1	0.7	99	614	4.0
Denmark		1.8	41	808	4.9
Sweden		0.2	20	429	6.7
United Kingdom		0.1	19	689	4.8
United States	12.6	3.2	..	602	3.2
Euro area	45.2	2.3	55[4]	715	6.3
European Union		1.9	49[4]	706	6.0

1. 1998 for Austria, Finland, Greece, Ireland, Netherlands, the euro area and the European Union.
2. Converted to euro using fixed conversion rates for euro area countries and current exchange rates for the others.
3. Commercial banks for Greece, Luxembourg, Portugal, Sweden, United Kingdom and United States; commercial and savings banks for Denmark.
4. Excluding Luxembourg.

Source: OECD (2001), *Bank Profitability* 2000. Ehrmann, M., L. Gambacorta, J. Martínez-Pagés, P. Sevestre and A. Worms (2001), "Financial systems and the role of banks in monetary policy transmission in the euro area", ECB *Working Paper* No. 105.

both in a further effort to improve margins and as the phasing out of public guarantees to publicly owned banks could trigger another merger wave in the sector. Consolidation is taking place through bank failures, often leading larger banks to take over failed banks, mergers with other banks or other financial institutions, or the merging of activities.[9] An example of the latter being three major banks merging of their mortgage activities into one institution (*Eurohypo* AG).[10] In addition, financial conglomerates have been created with the merging of insurance companies *Allianz* and *Münchener Rück* with the banks *Dresdner* and *Bayerische Hypotheken- und Vereinsbank*, respectively.

The consolidation process in the banking sector, however, has so far not let to changes in terms of competition in traditional deposit and lending business.[11] On the other hand, the investment funds and insurance companies have been started to compete more fiercely for the financial assets of households with the emergence of hybrid financial products, combining the properties of financial assets such as investment fund certificates and insurance products. Consequently, banks have been compelled to offer similar financial products, and the introduction of funded labour market pensions in 2002 has been accompanied by a move towards banks offering other financial institutions' products.[12] Competition has been increased by the introduction of new technology – lowering entry costs – with more than 15 million accounts carried online over the Internet.

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Current capital adequacy regulations as laid down in the international Basel I accord are being revised, as they are not considered to properly reflect banks' risk positions.[13] A reflection of this is an increasing number of bank failures in Germany as bankruptcies jumped about 20 per cent between 1999 and 2001. Reflecting the shortcomings of the Basel I the BIS Committee on Banking Supervision has proposed a new set of rules – Basel II – for consultation with a planned finalisation of the new accord during 2002 and an expected implementation by 2005. Basel II relies on banks' own sophisticated risk models to assess their risk exposure. The use of such internal risk models to gauge the risk of market price positions is already allowed by the German banking act, provided that the Federal Banking Supervisory Office has confirmed the suitability of the models. Some concerns have been expressed that the Basel II is not very suitable to assess German banks' risk exposure to the large "Mittelstand" (SME) sector. The German relationship banking is reducing information asymmetries *vis-à-vis* customers and thus agency costs of bank lending. If banks instead were to use equity to assess the risks associated with the largely non-incorporated SMEs, this would lead to a less favourable assessment, increasing the borrowing costs of SMEs and requiring banks to set aside excessive amounts of capital.[14] Alternatively, SMEs could obtain external credit ratings, but at a considerable cost. Moreover, the implementation costs for small banks of adapting Basel II may be substantial. These concerns, however, are largely addressed in the present Basel II proposal with the provisions of a special retail risk category (with a lower risk attachment) applicable to SMEs and of a fallback rule to Basel I, if smaller banks consider the installation of a risk-modelling system to be excessively expensive. However, one consequence of the Basel II accord is that greater variation in the cost of capital should be expected as individual lending increasingly reflects the associated risks, leading to a better allocation of capital.

Stock markets

Stock markets are relatively less important in Germany than in other countries. In an international context, stock market capitalisation as a share of GDP and debt issuance by the corporate sector are low, although the absolute size of Deutsche Börse makes it one of the main international stock exchanges (Table A2). In line with international developments and as investment funds have invested a substantial part of their portfolios in German equities the importance of domestic stock exchanges has grown. This development has been accompanied by increasing importance of venture capital – including the creation of the "N*euer* M*arkt*" market, which grew rapidly in the latter part of the 1990s.

The new market for high growth and technology companies – N*euer* M*arkt* – began operating in 1997 and had by 2001 raised € 25 billion in around 350 emissions, serving particular ICT (information and communication technology) and media related companies. At its peak in March 2000, the market capitalisation reached almost € 200 billion. Subsequently the bust of the international ICT cycle and the correction of international stock markets have smothered activity on the N*euer* M*arkt*, leaving market capitalisation at about a third of its peak. Contributing to the sharp correction has been over-optimistic sales and profit projections (including fraudulent accounting) and large undeclared share sales by the original shareholders.[15] An additional factor may be generous public venture capital support programmes, removing a large part of the associated risk, inducing an excess supply of risk capital (see Box A1). The emergence of "business angles" should ease some of the problems concerning lacking management skills, failures in due diligence and tendencies for premature IPOs. Nevertheless, D*eutsche* B*örse* has announced a reorganisation in 2003, which entails the closing down of the "N*euer* M*arkt*" and the reorganisation of existing segments into a "prime standard" – reserved for internationally recognised companies – and a "general standard". Companies listed at the latter will be governed by rules similar to those currently applicable to companies registered on the "N*euer* M*arkt*". The reorganisation also creates a

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Table A2. **Stock market capitalisation and corporate debt**

	Stock market capitalisation as per cent of GDP[1]	Corporate debt as per cent of GDP[2]
Austria	13.3	3.6
Belgium	97.1[3]	10.7
Finland	157.2	12.4
France	97.1[3]	21.8
Germany	57.3	4.5
Greece	73.6	3.2
Ireland	78.5	13.1
Italy	49.1	6.4
Luxembourg	125.3	9.7
Netherlands	97.1[3]	27.5
Portugal	43.7	11.4
Spain	83.3	9.7
Denmark	53.1	12.3
Sweden	103.1	16.9
United Kingdom	151.1	25.8
United States	140.1	26.6
European Union	86.9	13.9

1. Capitalisation of main markets in ultimo 2001. Domestic companies excluding investment funds, rights, warrants, convertibles, foreign companies; including common and preferred shares plus shares without voting rights.
2. September 2001.
3. Measured as the stock market valuation of Euronext, which comprises the stock exchanges of Belgium, France and the Netherlands.

Source: Economic Survey of Euro Area (2002).

less rigorously regulated "Domestic" segment reserved for companies mainly seeking national investors.

Listing requirements for the *Neuer Markt* and the DAX (*Deutsche Börse*'s main exchange) were converging. Firms seeking a listing in the former had to issue prospects subject to international accounting standards (IAS or US-GAAP) and to accept the voluntary take-over code. Prospectuses for firms seeking listing on the *Deutsche Börse* were subjected to German accounting standards and the acceptance of the voluntary take-over code was optional. However, recent regulation is introducing the IAS as the common accounting standard in Europe and the voluntary take-over code has been replaced with a legal act. On the other hand, each *Neuer Markt* listing needed at least two "designated sponsors" to ensure a minimum of market liquidity. *Neuer Markt* and DAX reporting requirements were similar with respect to quarterly reports and ad-hoc disclosures. Neither is subjected to encoded standards, but *Deutsche Börse* has issued guidelines to clarify the contents of quarterly reports and a code of conduct for companies and banks involved in IPOs.[16]

The transition of German financial markets is accelerated by internal factors

The introduction of funded pensions – the so-called Riester pensions – is expected to add further impetus to the development of financial markets by channelling savings into these types of financial vehicles. Such pension schemes can either be introduced at the industry level, at the company level or individually.[17] The pension funds' portfolio rules,

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Box A1. Venture capital support programmes

Technology intensive start-ups face problems in raising finance. They often have little collateral and need long-term capital during the seed (concept development) and the start-up (foundation) phases. Investors have two problems to overcome, one of patience – that the return of investment in such firms is often only payable after a substantial time-lag – and one of uncertainty concerning the viability of the start-up. Most OECD countries provide public financial support for such firms.

The main federal government programme to promote venture capital for – particularly high technology orientated – small and medium-sized enterprises (SMEs) is entitled "Venture Capital for small technology companies" or BTU by its German abbreviation and was launched in 1995, extending and replacing previous mostly loan-based programmes. The BTU programme offers support in the form of a refinancing model and a co-investment model. The former entails that the public bank *Kreditanstalt für Weiderauf*bau (KfW) provides long-term capital (combined with a default guarantee of up to 70 per cent) to venture capital companies and banks for financing SMEs. The co-investment model involves the participation of the *Deutsche Ausgleichsbank*, a public support bank, as a co-investor along side private venture capital providers, acting as lead investors. The co-financing can be a maximum of 50 per cent of the total venture capital and may include a (partial) investment guarantee to the lead investor. The majority of private funds in both models is provided to venture capital companies. The eligibility criteria include companies with up to 250 employees and a turnover of € 20 million per year and less than five years old.[1]

The venture capital industry has grown to more than 250 venture capital firms, as well as spawning firms offering support services, such as legal counsel, accounting, etc. In addition, about 40 "business angles" have established themselves, providing expertise in areas like management skills and IPOs (Initial Public Offerings) as well as venture capital. Currently, the outstanding capital in various federal support programmes for SMEs amounts to about € 1½ billion. About 50 out of the more than 750 firms participating in the BTU programme have been listed on the Neuer Markt and with the majority of the rest planning an IPO. However, little IPO activity has taken place since mid-2001, probably reflecting weak capital markets. The prospect of using the Neuer Markt or the new "Prime" segment as exit channels to recuperate the initial investment has improved. Since 2002, liquidations of small holdings of stocks held for more than a year is tax free and the use of stock options as an entrepreneurial reward instrument was enhanced as they became only taxable when excised.[2]

1. There is some empirical evidence that start-ups receiving public support have faster employment growth over a six-year period than non-receivers. See Almus, M. (2002) "Evaluation the Impact of Public Start-up Assistance – Results from an Econometric Approach" ZEW working paper, Mannheim.
2. Taxation of capital gains related to financial wealth is dependent on ownership and duration of holdings. If investor is a natural person investing his personal wealth the capital gain taxation will be zero if ownership of the shares has lasted more than one year and if the value of the share over the past five years have remained below 1 per cent of the company's subscribed capital ("*Nennkapital*"). Otherwise the shares must be hold for seven years to become tax exempted. For incorporated companies the associated capital gains in connection with portfolio investment are exempted from taxation, although sales of subsidiaries are taxed subject to the seven years rule. Realised capital gains in connection with real wealth, on the other hand, are taxed for investors unless ownership duration has been more than 10 years. Generally, for owner-occupied houses there is no capital gains taxation.

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pursuing prudential investment objectives, come across as relatively liberal, allowing for up to 40 per cent of funds' capital to be invested in non-fixed instruments. The pension schemes must guarantee to pay back the equivalent of the accumulated contributions, although temporary losses are permitted. The pension payments must take the form of annuities, although 20 per cent of the savings can be paid out as a lump sum upon entering retirement. Existing labour market pension schemes have only a minor effect on the stock markets. Existing company based pension schemes are typically taken up by larger companies – with the size to exploit advantages such as temporary tax savings as earnings are transferred to pension provisions – exploiting the possibility of using the pension funds as an internal source of finance.[18] Pension funds for civil servants were established in early 1999 with investment guidelines stipulating marketable Federal debt securities.[19]

The German system of relationship banking has furthered the holdings of shares by financial and non-financial companies in other companies, which often are engaged in unrelated lines of business. Untangling this cobweb of cross-holdings – to allow companies to focus on core business competencies – has been hampered by capital gains taxation rules. Since the beginning of 2002, however, capital gains in connection with incorporated companies selling cross-holdings have become tax exempted. The new rules do not apply to non-incorporated companies and sale of subsidiaries. The measure is expected to further the restructuring of German industry, as particular financial institutions off-load their share holdings in non-financial companies.[20] So far, however, relatively little of such activity has taken place, probably because of the current subdued stock markets.

The regulatory framework for corporate governance has improved

The restructuring process is also being stimulated through a series of regulatory measures aimed at improving corporate governance. Besides those reviewed in Chapter IV, these include the bankruptcy code from 1999 and changes to accounting rules.

Bankruptcy rules play an important element in corporate governance to ensure the orderly transfer of corporate governance responsibilities. The bankruptcy code was changed on 1 January 1999. The previous code was centred on the principle of creditors' protection, with the effect that banks had preferential rights for secured loans, increasing their incentives for liquidating companies. The new law's reduction of preferential rights enhanced the rights of the borrowers as a measure to secure the transfer of the troubled company to new management and owners before liquidation.[21]

The move towards the EU-wide application of the *International Accounting Standards* (IAS) by 2005 should further integration of financial markets and investor protection with the IAS's focus on providing useful and more current market based information for investors.[22] Since the EU adaptation of the IAS, the standards are in the process of being strengthened. Standardised accounting is seen as an effective tool to improve corporate governance, to enhance the use of take-overs as a means of disciplining management inefficiency, and generally strengthen the efficiency of capital markets.[23] Companies with a dual listing in the United States are allowed to continue for another two years with the American US-GAAP standard. The transition period with multiple accounting standards creates additional accounting costs and possible less transparency for shareholders.[24] However, once the new standards are implemented, transparency and investor protection will have improved, furthering the functioning of financial markets. After the full implementation of the IAS, companies with a dual listing in the United States will have to apply two accounting standards, pointing to the need for a harmonisation of European and American standards.

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Additional aspects of the Bundesbank *reform*

Part of the Bundesbank reform has been to move government debt management to an independent agency to unify debt management.[25] In September 2000, the Federal Finance Ministry established a legally independent government-owned "*Bundesrepublik Deutschland-Finanzagentur GmbH*" in which debt-management duties hitherto performed by the Federal Government, the Federal Debt Administration and the Bundesbank were combined. The reorganisation is in line with international developments and is not expected to lead to fundamental changes in the public debt management policy, although plans for shortening the maturity of government debt have been criticised for involving greater refinancing risks arising from exchange rate movements and budget changes (although the distribution of risks is difficult to foresee). In addition, the move could possible endanger Germany's benchmark status in medium- and long-term debt instruments, although the relatively small spread between German and other euro area government bonds indicates a relatively limited value of the benchmark status.

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Notes

1. The effects of international integration are mostly on the price side as portfolios are still characterised by a relatively large element of home-bias with relatively large shares of domestic securities.
2. See for example Leahy, M., S. Schich, G. Wehinger, F. Pelgrin and Th. Thorgeirsson (2001), "Contributions of Financial Systems to Growth in OECD countries" OECD *Economics Department Working Paper*, No. 280 and Bassanini, A., S. Scarpetta and P. Hemmings (2001), "Economic growth: the role of policies and institutions, panel data evidence from OECD countries" OECD *Economics Department Working Paper*, No. 283.
3. The SMEs ("Mittelstand") limited use of equity markets should probably also be seen in connection with associated high transaction costs in comparison with bank financing. Most German banks are universal in their provision of services, although relatively few of them can be described as financial conglomerations in terms of being engaged in non-traditional banking activities through affiliates and having a substantial share of total revenues originating from non-interest income. Compared with most other European countries, Universal Banks in Germany are relatively predominant and normally conduct their merchant banking and securities operations in house, while insurance, mortgage banking and investment funds are supplied through affiliated companies. See Vander Vennet, R. (2002) "Cost and Profit Efficiency of Financial Conglomerates and Universal Banks in Europe", Journal of Money, Credit and Banking, Vol. 34, No. 1 (February 2002).
4. The structure of the financial sector has importance in terms of the monetary transmission mechanism. For Germany, it has been estimated that 1 percentage point increase in short-term interest rates lead to a similar contraction of loans – an estimated effect that is similar to that found in Italy and somewhat lower than those found for France and Spain. See Ehrmann, M., L. Gambacorta, J. Martínez-Pagés, P. Sevestre and A. Worms (2001) "Financial systems and the role of banks in monetary policy transmission in the euro area", ECB *Working Paper* No. 105.
5. About two-thirds of the about 2 700 German banks are in the co-operative sector and about one-fifth in savings banks sector. Three-quarters of all the banks are small and cover about 7 per cent of total assets. Both co-operative and the savings banks sectors are two-tiered with a group of large banks serving as head institutions and a lower tier of banks entertaining a close relationship to the head institutions. The result is an internal liquidity management system where the lower tier deposits short-term funds with the head institutions and in return receives long-term loans. See Deutsche Bundesbank (2001) "Bank Balance Sheets, Bank Competition and Monetary Policy Transmission", *Monthly Report*, September 2001, pp. 57
6. The private statutory deposit insurance system, including funds and cross-guarantee arrangements for both savings banks and co-operative banks, effectively amounts to a

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full insurance of all non-bank deposits. See Ehrmann, M., L. Gambacorta, J. Martínez-Pagés, P. Sevestre and A. Worms (2001) "Financial systems and the role of banks in monetary policy transmission in the euro area", ECB Working Paper No. 105. The voluntary scheme became a supplement system with the introduction in mid-1998 of the Deposit Guarantee and Investor Compensation Act, which harmonised German legislation with EU legislation, securing deposit insurance and creating a mandatory system for investor protection and compensation. The scope of protection covers funds, securities, derivatives and money market instruments and come into force to compensate investors if an institution no longer can pay back funds it owns investors or to return financial instruments that belong to investor. Deutsche Bundesbank (2000) *Monthly Report*, July 2000. The deposit insurance scheme has an upper limit of 90 per cent of the non-fulfilled claims and a maximum of € 20 000 for each creditor.

7. Vander Vennet, R. (2002) finds that German universal banks are operating less efficiently than other European universal banks, possible as the result of large non-exploited scale economies for small banks, and concludes that the continued expansion of financial conglomerates and universal banks in Europe may lead to a more efficient financial system.

8. See Deutsche Bundesbank (2001) *Monthly Report*, June.

9. At the overall level the number of mergers and acquisitions relative to the population is relatively low in an international comparison. See R. Fehn and C-P. Meier (2001) "The Positive Economics of Labor Market Rigidities and Investor Protection" Kyklos, Vol. 54. In a similar vein shareholder rights (in terms of legal protection of minority shareholders) are very low (original source La Porta, R., F. Lopez-de-Silanes, A. Shleifer and R.W. Vishny (1998) Law and Finance, *Journal of Political Economy*, 106. The article also points out the strong inter-linkages between the organisations of labour and capital markets, rendering the effects of reforms in only one of the markets less effective than a dual approach.

10. Some consolidation efforts have failed. The three largest banks tried to merger their back room operation into one transaction bank, but failed as one of them insisted on the adaptation of its proprietary operating system.

11. See H. S. Hempell (2002) "Testing for Competition Among German Banks", Discussion Paper 04/02, Economic Research Centre of the Deutsche Bundesbank. Low profit margins in retail banking may reflect both problems associated with publicly owned banks and substantial competition in a sector with a predominance of smaller banks. Some specific competition problems have emerged, such as when a number of German banks in 2001 came under an EC commission investigation for fixing exchange commissions. See Buch, C. M. and S. M. Golder (2002), "Domestic and Foreign Banks in Germany: Do They Differ", Kredit und Kapital, Heft 1/2002 for an empirical investigation.

12. Deutsche Bundesbank Monthly Report, September 2001.

13. Criticism of the current rules includes that the definition of the different risk classes is rather rude and easily manipulated; do not include interest rate risks and portfolio risk; do not have an adequate treatment of netting as the rules do not try to match assets and liabilities; do only poorly accounts for the market value of banks' assets; and assume that banking risks are similar across countries. See U. Vollmer (2001) "Minimum Capital Adequacy Ratios for Banks" in Journal of Economics and Statistics, Vol. 221/No. 5 + 6, Stuttgart.

14. See Friderichs, H. and B. Paranque (2001) "Structures of Corporate Finance in Germany and France", Journal of Economics and Statistics, Vol. 221/No. 5 + 6, Stuttgart.

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15. Stapf, J. and C. Upper (2002) "Asset price misalignments on the Neuer Markt in Germany: Did information disclosure policies and taxes play a role", *Draft working paper*, Bundesbank.

16. On the other hand, *Deutsche Börse* plans to relax rules for de-listing companies so they are not obliged to buy back free-float shares and shareholders will instead have six months to sell their shares over the stock market. However, the de-listing announcement will undoubtedly depress share prices and hurt shareholders, particular smaller ones. De-listing is normally of interest for companies with a limited free float and a large shareholder, making the company's cost of providing information to shareholders unreasonable high as compared with benefits. De-listing is allowed if the company becomes listed on another exchange. The Deutsche Börse's plans are subject to approval by the Land Hesse's financial regulator.

17. The insurance company Allianz won the contract to manage the occupational pensions for the metals industry (Metall Rente pensions framework negotiated between IG Metall and Gesamtmetall (employer)). Allianz will manage the accounts for individual workers, while its subsidiary Dresdner bank will be responsible for sales and marketing. The contract potentially includes the management of the pensions for the industry's 3.6 million workers, although the union expects 400 000 to take out occupational pensions in 2002 and rising to 800 000 by the end of 2008 with the annual premium volume totalling several billions of euros. Already during the summer 2002, the commission fees were doubled, as the original estimates for the administrative costs were too low.

18. See Friderichs *et al. Op. cit.*

19. The Bundesbank and the Land Central banks are entrusted with the management of the civil servant funds. By end-2000, the accumulated funds were nearly 1 billion. Bundesbank (2001) Annual Report.

20. At the same time, the investment guidelines for life-insurance and health-insurance companies were relaxed by raising the upper limit for shares in the portfolio from 30 to 35 per cent, the share of non-European stocks from 6 to 10 per cent, and the share of non-stock products, such as index certificates, from 5 to 10 per cent.

21. Größl, I., P. Stahlecker and E. Wohlers (2001), "An Empirical Investigation of German Firms' Financial Structure and Ensuring Risks" in Journal of Economics and Statistics, Vol. 221/5 + 6.

22. In practical terms, one of the major differences is that the IAS uses market valuations as compared with the German standard of using historical cost valuation for determining the value of assets. For a full description of differences see Deutsche Bundesbank (2002) *Monthly Report*, June 2002.

23. OECD Journal of Competition Law and Policy (2001), Volume 3, No. 3.

24. In 1999, nearly three-quarters of all companies listed in the DAX used IAS or US-GAAP, indicating that such temporary effects are likely to be rather small. See Spanheimer, J. and C. Koch (200) "Internationale Bilanzierungspraxis in Deutschland – Ergebnisse einer empirischen Untersuchung der Unternehmen des DAX und MDAX sowie des neuen Marktes" in Die Wirtschaftprüfung, Vol. 53.

25. At the same time, the management structure of the Bundesbank was streamlined, so the previous 15 member management board will be reduced to eight – with four, including the governor – appointed by the Federal government and the other four nominated by the *Länder* with their appointments subject to approval of the Federal government.

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